

Read the Label

Climate-Attention Timing of Brown Industries,
Audited and Tested Out of Sample

MFE 230GA Active Asset Management, Final Project

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September 2026

Executive Summary

Our team’s halfway project proposed a state-dependent climate trade, and this report is my solo extension of it. The strategy shorts a factor-hedged Brown leg of five industries the team ranked high-emission (utilities, shipbuilding and railroad equipment, aircraft, steel, construction materials; two rank mid-table on EPA factors) for three or six months after “climate-transition attention” crosses its past-only 80th percentile, sized to 5% residual volatility. For that to be climate alpha, three things had to hold: the signal measures climate concern, the gain comes from timing rather than being short, and the rule works on data it was not designed on. I tested each on Fama–French industry and factor returns, FRED news and macro series, two climate-concern indices, media climate change concern (MCCC) and climate policy uncertainty (CPU), and EPA supply-chain emission factors. Every threshold and hedge uses only past data, and independent code rebuilt each module’s headline numbers. The recommendation is **Do not implement**: the label is wrong, and under any label the returns do not pay. First, the “attention” series is FRED’s equity-volatility news tracker for energy and environmental regulation (identical in 500 of 500 months). Its z-score is uncorrelated with MCCC ($r = -0.0004$), its weak link to CPU runs through overall volatility news, and the same rules fed either climate index earn 0 of 12 validation alphas with $t \geq 1.96$. In plain terms, the rule shorts hedged utility and industrial stocks after stock-market volatility makes the news. Second, COVID cannot certify the timing: the always-short position, an untimed short of the same leg, had an alpha of 4.54% a year ($t = 1.96$), and the rule’s edge over it is +1.84% ($t = 0.87$), or 3.0% over an exposure-matched short ($t = 1.67$). Third, timing earns nothing detectable on windows no rule was designed on. A cleaner version frozen before its test (the regulation share of volatility news) has a timing alpha of -0.18% a year over 1994–2009 ($t = -0.27$; $p = 0.48$ against random re-datings), and the team’s signal, lagged, earns $+0.69\%$ there ($t = 1.00$) and -1.11% in the holdout ($t = -1.13$); pooled, these windows detect only edges above about 1.5% a year with 80% power. Fourth, every timed rule lost over the 2022–2026 holdout, before costs as well as after (6-month rule: FF3 alpha -1.97% a year, $t = -1.30$). The always-short position lost too (FF3 alpha -1.72% , $t = -1.09$), the timing lost a further 1.0% to 1.7% a year against an exposure-matched short, and rate factors or dropping any one industry leave the alphas negative. Of the alternatives, climate indices on EPA-ranked legs add no timing edge, and shrinkage timing collapses to the historical mean under pre-specified cross-validation. An optimized industry-momentum book passed a pre-registered 1931–1969 test (alpha 1.95% a year, $t = 2.45$), but it fails the deflated appraisal ratio and its holdout alpha is -0.34% , which earns it a paper-trading pilot, not capital. In short: read the label, benchmark timing against the always-short position, and spend the only window left, the live record, on one frozen rule.

1 What Did I Try?

1.1 The inherited strategy and where the idea came from

Our team’s thesis was falsifiable, and it failed its own holdout. The halfway write-up argued that Green-minus-Brown (GB) has no permanent premium but that its factor-neutral part pays after spikes in climate-transition attention. Green and Brown are the five lowest- and highest-intensity Fama–French 49 (FF49) industries in the team’s emissions file (Brown: Util, Ships, Aero, Steel and BldMt), each an equal-weighted mean of value-weighted industry returns. The rule shorts the Brown leg for 3 or 6 months after the signal’s 60-month z-score crosses its past-only 80th percentile, hedged with rolling 60-month Fama–French three-factor (FF3) betas lagged one month and sized to 5% annualized residual volatility. The six rules are Original and Pure (the raw signal and a macro-purified one) with 3- and 6-month holds, written Original 3m to Pure 6m, plus Continuous raw and Continuous pure, which scale the position with the signal. In validation (2010-01 to 2022-07) the 6-month rules earned 2.45% and 2.59% a year net (FF3 alpha Newey–West $t = 2.23$ and 2.46), and in COVID 2020–21 the 3-month rule earned 5.94% net (FF3 alpha 6.03%, $t = 3.21$). Over the 2022-08 to 2026-07 holdout every discrete rule lost 2.1% to 3.8% a year, and our team wrote “Do not implement”.

The idea descends from HW1’s comparison of high- and low-emitting industries and from a literature that reads realized green returns as responses to concern shocks rather than as a premium. Pástor et al. (2022) explain 2012–2020 green outperformance by climate-concern shocks (mean return after removing concern shocks about -4 bp a month, HML loading -0.26), Eskildsen et al. (2026) find no green-minus-brown alpha that survives Benjamini–Hochberg, and Bolton and Kacperczyk (2021) and Zhang (2025) disagree on the sign of the carbon premium. In the Pástor et al. (2021) model green outperforms only when concerns strengthen unexpectedly, and a spike that is already known carries no positive expected green-minus-brown return. A rule that shorts Brown after a known spike is therefore a bet that concerns keep surprising upward rather than a bet on a known premium.

For our team’s result to be climate alpha, three things had to hold: the signal measures climate concern, the gain comes from timing rather than exposure, and the rule survives a window it was not designed on. The study tests them simplest first, in the order each result demanded (Table 2; the module codes are labels, not the run order). After the replication, M1 asks what the signal is. M1b feeds the same rules climate-concern indices and volatility placebos. M2 runs Cristhian’s four overlooked tests (8-and-8 legs, momentum and commodity controls, the source of the HML loading, uniform costs), M3 takes up Charishma’s question of alpha versus beta, M4 audits the emissions file, and M8 scores one frozen rule once on 1994–2009. The brief’s other drivers became alternatives on the same data: an EPA emissions tilt (M4), carbon-aware industry momentum (M5, which the brief motivates with Zarattini and Antonacci, 2024) and macro-driven shrinkage timing (M6). M7, run last, puts every test in the project into one multiple-testing census and runs the checks that decide the verdict. The design rests on a primary test fixed in code before each run, real-time inputs, and an untimed benchmark for every timing claim.

1.2 Data, and what the data turned out to be

The study uses only public data (Table 1). Returns and macro controls enter only after publication and the EMV trackers are lagged one month; the emissions snapshots are the exception (Section 2.7). The data produced findings of their own that changed the study, so they are reported here, and

the tests they led to are in Section 2.

Table 1: Data. Each series, its source, span and role. Returns are monthly decimals at month-end; in every real-time test the EMV trackers are lagged one month and macro controls enter only after publication.

Series	Source	Span	Role
FF49 value-weighted industry returns; FF3 factors and RF	Team files (Ken French data)	1926-07 to 2026-07	Legs, hedge
FF5 (2×3), UMD, industry BE/ME and firm counts	Ken French library, Aug 2026	from 1963-07 (UMD 1927)	Evaluation
Emissions intensity, one cross-section	Team file (the problem set 1 course file)	undated; 41 of 49	Team legs
EPA supply-chain GHG factors v1.3; USEEIO v2.0.1 (built on BEA input-output accounts); SIC-NAICS maps	US EPA; US Census Bureau	2022 USD; 1,016 NAICS codes	Rebuilt legs (M4)
EMVENRGYENVREG, EMVOVERALLEMV; VIX	FRED (Baker et al., 2019)	1985-01 (VIX 1990-01) to 2026-08	Signal, placebos
MCCC aggregate; CPU index	Ardia et al. (2023); Gavriilidis (2021)	2003-01 to 2025-06; 1987-04 to 2025-09	Climate concern
GS10, TB3MS, BAA, AAA, CPI, CFNAI, WTI, IMF commodities; par yields; Damodaran bond returns	FRED; US Treasury; Damodaran	to 2026-08	Controls, instruments, BOND

The “attention” series is an equity-volatility news count, and since late 2021 it is mostly exact zeros (Figure 1). Our team’s column equals FRED EMVENRGYENVREG in 500 of 500 months. That series is the Baker et al. (2019) Equity Market Volatility (EMV) tracker for energy and environmental regulation: a count of newspaper articles on stock-market volatility that mention the category, scaled to the VIX. It is exactly zero in 12 of 441 months before 2021-10 and in 39 of 59 since (Fisher $p = 1.3 \times 10^{-32}$), including 31 of the 48 holdout months, and a zero month cannot trigger the rule. The zeros did not change which months the rule traded: under our team’s timing, with the z-score’s scaling frozen at 2021-09 or with zeros treated as missing, the rule enters in the same five holdout months, and every team strategy still loses. The label was wrong and the zeros are a data problem, but neither explains the holdout.

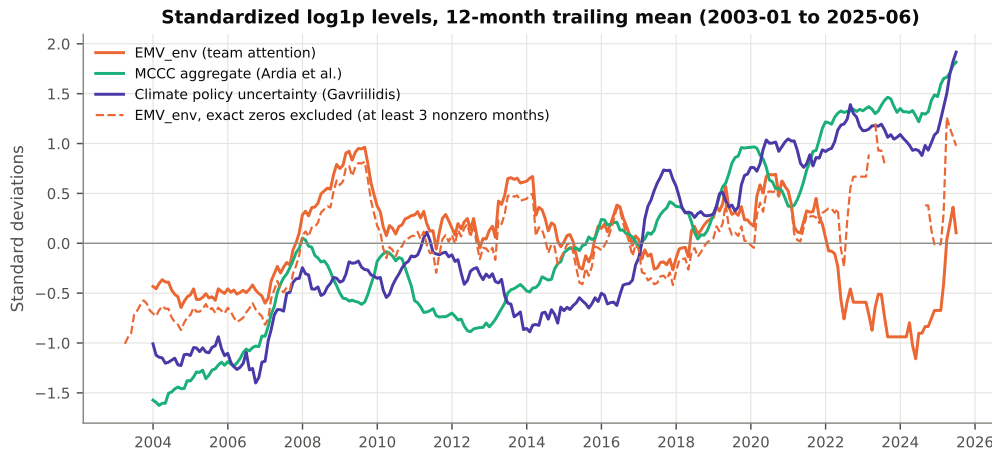


Figure 1: What the signal measures. Twelve-month trailing means of standardized log levels, 2003-01 to 2025-06: our team’s “attention” (FRED EMVENRGYENVREG, orange), media climate concern (MCCC, green; Ardia et al., 2023) and climate policy uncertainty (CPU, purple; Gavriilidis, 2021); the dashed line drops the exact zeros. Climate concern rose from 2022 until MCCC ends in mid-2025, while the team series fell into its zero regime.

The Brown leg rests on an undocumented emissions file (Figure C.7 in Appendix C.5). The team file is our first problem set’s course file and records no source, units, scope or vintage. It has three groups of exactly equal values and a ratio of 6,051 between Util and Fun (entertainment), against a 33-fold range of EPA supply-chain intensities across FF49 means. Against a full EPA rebuild the ranking agrees broadly (Spearman 0.700) but not where it matters: Ships and Aero rank 2nd and 3rd of 41 in the team file and 20th and 31st of 49 on EPA factors. Their values are 14 to 47 times what their manufacturing codes imply and 1.9 to 6.0 times air- and water-transport levels, consistent with, not proof of, a transport mis-mapping. It matters because Aero, Util and Ships carry 86% of the 3-month rule’s COVID gross P&L under our team’s timing (my fact-check of exchange 01).

1.3 Evaluation framework, fixed before testing

The corrected baseline fixes two input problems before any new test ran. Our team traded at the close of month t on month- t EMV, which I could not verify is published by then, so the corrected baseline lags EMV and every purification control one month. I call this real-time timing and the team’s version same-month timing (Original 3m’s 2010–2026 net return moves from 0.07% to 0.73% a year, and no verdict changes). The October 2025 CPI print, never published because of the federal government shutdown, silenced the purified signal for two months and explains the whole holdout gap between the Pure and Original rules. With the print interpolated, the two coincide in the holdout. Every number below uses the corrected baseline unless it says otherwise.

Every timing claim is tested against the always-short position, an untimed short of the same leg with the same hedge and volatility target. Two versions of the comparison appear. The paired alpha is the factor alpha of the rule’s return minus the always-short return. The second, $D_t = R_t^T - \pi R_t^{AO}$, scales the benchmark by π , the ratio of mean absolute positions, so that a rule that is flat part of the time is compared with the same average exposure, and the frozen test uses it. Unless stated otherwise, every t -statistic is Newey–West (Newey and West, 1987) with 6 lags, NW(6). In the tests of our team’s rules and the frozen rule (for the climate-index versions, on windows under 60 months) p-values come from $t(n - k)$ rather than the normal, and with them our team’s “2 of 32 claims survive Holm” (Holm, 1979) becomes 0 of 32.

One rule was frozen before its window was touched. I adopted exchange 01’s follow-up rule (the regulation share of overall EMV, zeros as missing, the 6-month hold, a four-part pass bar) plus a one-month publication lag. The share divides out overall volatility news, treats the zeros as missing and, unlike MCCC (from 2003) and CPU (no signal before 1995), can trade before 2010. I saved a date-stamped pre-registration of 32 implementation choices before writing its code, and its SHA-256 hash shows the file never changed (Appendix C.8). The proposed window was 1993–2009, and the first usable month after the burn-in is 1994-03. That window is out of sample for returns, not for design: the rule was chosen after 2010–2026 had been seen, and our team’s macro-state notebook had used 192 pre-2010 months of forward Brown residuals.

Each module’s headline numbers were produced twice, by the module’s code and by a second implementation that shares none of it (128 of 128 match for the frozen test), and every inferential statistic is logged: 23,923 tests in eight module ledgers, 155 of them primary. Only the two frozen tests have timestamped pre-registrations; the other modules fixed their primary tests in code without a timestamp, and the few fact-check numbers computed only once are listed in Appendix D. An AI coding agent (Anthropic’s Claude) wrote each module’s code under my direction, and a second agent wrote the re-implementation.

Table 2: The research program: each module’s question, its primary test with the headline statistic, the adjusted p-value within its pre-specified family, and the verdict. No module finds a positive alpha that survives its own family, and none survives the project-wide correction (M7); the one pre-registered pass is a momentum book tested before 1970.

Module	Question	Primary test and headline result	Adjusted p	Verdict
Replication	Does the team code reproduce the write-up?	14 unit tests; 32 of 32 Table 1 cells; 11 audit findings	–	Reproduced; two input errors
M1 Signal	What does “attention” measure?	Team z on VIX and overall EMV: $R^2 = 0.169$; correlation with MCCC -0.0004 , CPU 0.123; 8 predictive information coefficients (ICs) null	8.8×10^{-12} ; 0.996, 0.074; ICs ≥ 0.62	Volatility news
M1b Signals	Does climate concern reproduce it?	40 alphas and ICs for MCCC and CPU: no significant positive holdout alpha (largest $t = 0.37$); placebo rule fails on its boundary	1.00	EMV-specific
M2 Four tests	Is any defensible alpha left?	14 alphas each for 8-and-8 legs, commodity hedge, 5 bp costs; GB HML -0.233 , -0.298	≥ 0.397 ; HML ≤ 0.017	No alpha; Brown value tilt
M3 Alpha or beta	Alpha or beta? Rates?	GB BOND loading 0.132, 0.088 post-2010; holdout alphas fall with UMD and BOND; beta timing negative	0.019, 0.549; ≥ 0.155 ; 4 of 6	Beta, not alpha
M4 Emissions	Is the ranking sound?	EPA 5-vs-5 GB, FF5+UMD alpha post-2010 4.7% ($t = 1.52$)	0.129	No alpha
M5 Momentum	Carbon-aware momentum alpha?	FF5+UMD alpha 1.74% ($t = 1.09$), 1.86% post-2010 ($t = 0.71$); carbon IR change $+0.010$ at the primary bound $b = 0$	0.823 each	UMD in industry form
M6 Timing	Can a shrinkage model time GB?	$R_{OOS}^2 = 0.000\%$; attention increment 0; timing minus static $+0.84\%$ ($t = 1.17$)	1.00	Nothing to time
M8 Frozen	Does one frozen rule work in 1994–2009?	Timing alpha -0.18% ($t = -0.27$), shuffle $p = 0.478$; 1 of 4 components	single verdict	FAIL
M7 Ledger	What survives project-wide?	73 module primary alphas, one-sided; search-adjusted and deflated tests; frozen 1931–1969 run of the momentum book, alpha 1.95% ($t = 2.45$)	Holm 1.00; book 0.017, 0.199; frozen 0.015	Book: paper-trade; rest: do not implement

1.4 Implementation: risk model, turnover and costs

The risk model is a rolling 60-month FF3 time-series regression of the Brown leg, and the traded object is its residual. Betas estimated on months $t - 59$ to t hedge the return of month $t + 1$, the position is $-\min\{1, 0.05/(\sqrt{12}\hat{\sigma}_t)\}$ while a hold is open ($\hat{\sigma}_t$ the 36-month residual volatility), and only the Brown leg is traded, so the Green leg never enters the P&L. Realized risk sits below the 5% target because the rules are flat part of the time and the position is capped at one. Over 2010–2026 Original 6m ran at 4.3% volatility with an information ratio (IR, annualized net return over volatility) of 0.28 and a worst drawdown of 14.4%, against 5.2%, 0.20 and 17.0% for the always-short position. Its IR was 0.51 in validation and -0.43 in the holdout. A fuzz test that replaces every input after a cut date with noise changes nothing before it (Appendix D).

The wider risk models add two traded exposures and two conditional decompositions. COMEQ is a traded commodity-equity factor (Oil, Coal, Mines and Gold minus the risk-free rate) that correlates 0.66 with the Brown leg. It is a distinct exposure rather than an alpha source, since its alpha on the Fama–French five factors plus momentum (FF5+UMD) is -0.89% a year ($t = -0.24$). BOND is the excess return of a 10-year par Treasury built from GS10 with exact duration and convexity (validated in Appendix C.4). Conditional betas (Ferson and Schadt, 1996) and the Lewellen and Nagel (2006) split of alpha into conditional alpha, static beta and beta timing complete the set,

and the momentum alternative adds a Grinold and Kahn (2000) optimizer with a bound $c'h \leq b$ on the book's carbon exposure (c the industries' standardized log carbon intensities, h the active weights).

Costs are not the binding constraint; the gross return is. The team's costs are 10 bp on the leg, 5 bp on the market overlay and 25 bp on SMB and HML per unit of turnover. In validation the discrete rules turn over 2.4 to 5.8 times a year (cost drag at most 0.63%). The uniform cost that erases the validation alpha is 40 to 100 bp (211 bp for the always-short position), and the richer FF5+UMD+COMEQ hedge cuts it to 19 to 50 bp. Before any cost the holdout alpha is already negative in all 28 cost-free runs (the six rules and the always-short position, both baselines, 5-and-5 legs, the FF3 or FF5+UMD+COMEQ hedge; highest -0.39%).

1.5 ChatGPT as a research assistant: the prompts

Each prompt was written as a specification, and every reply was fact-checked with code before any follow-up. Following the HW2 workflow, each prompt put our numbers and constraints first and fixed the number of items. It asked for falsifiable pass bars and for the assumptions it was least sure of, capped the length, and said "label any number you have not computed as a guess". The four exchanges cover the brief's three uses plus a red team: a design review, code for the frozen test, the multiple-testing framework behind M7, and an attack on the draft's conclusion, run twice. Each follow-up sent the fact-check's corrections and asked for one thing: a frozen rule with an attribution specification (exchange 01), then six corrected functions with unit tests (exchange 02). The prompts changed after each failure. After exchange 01 accepted an unsourced label, exchanges 02 and 03 named the signal's FRED series; after its generic ideas, exchange 03 said "Do not propose new signals" and allowed "no candidate can reach implement" as an answer; exchange 04 asked for overreach toward my own verdict, and its second round sent the best evidence against the verdict as well. The replies came from a language model instructed to answer as ChatGPT, not from a live ChatGPT session, and Appendix A labels them so and reproduces them in full. Table 3 summarizes them and Section 2.8 evaluates them.

Table 3: ChatGPT interactions: key prompt lines, what came back, what the fact-check found wrong, and what I adopted. Each exchange changed the plan; the two scored reviews got 6 of 52 and 11 of 93 checkable claims wrong, mostly about our data rather than the method, and where the specification was exact, the code was exact.

Exchange	Asked for (key lines)	Produced	What was wrong	What I adopted
01 Idea generation	“Treat this as a design review: I want the weak points found before I spend time on them” (993 words)	A 1,609-word review; a follow-up with a frozen 1993–2009 rule and a pass bar	6 of 52 claims (duration, NW lags, IC flip, dates, coverage); never asked what “attention” measures	The always-short benchmark; the frozen pre-2010 test (M8); Aero and Ships as suspect (M4)
02 Coding support	“a pre-registered backtest I will run exactly once. You do not have the data, so report no results” (997 words)	A 985-line script that, run on my data, gave my verdict (-0.19% , $t = -0.25$); six fixed functions with unit tests	Hard-coded burn-in; a broken positive control; a look-ahead test that missed 2 of 6 bugs; a broken interface	A second implementation matching M8 to 3.3×10^{-13} ; a size check on 100 simulated null datasets
03 Robustness design	“design the robustness and multiple-testing framework that decides my project’s final verdict” (997 words)	1,716 words (cap 1,500): test families, a deflated Sharpe table, two candidates	11 of 93 claims: a UMD beta from the wrong book; Romano–Wolf “stricter” than BH; “insufficient evidence” for a holdout that rejects	Test families defined by the decision each test can change; the deflated appraisal ratio; a holdout equivalence reading; the 1931–1969 run (M7)
04 Red team, two rounds	“Break the conclusion before a grader does” (795 words); round 2 added the best evidence against me (798)	1,340 and 1,091 words (caps 1,200 and 1,000) with replacement text	A fix for a rule that never traded; a 2.4-SE claim on the wrong comparator; a pooled edge read as positive twice; costs deducted twice	Thirteen summary rewordings; the holdout interval and timing loss; three alternatives tested and retired

2 What Did I Learn?

2.1 The signal is volatility news

The team signal is overall volatility news plus a topic share unrelated to climate concern. On the z-scores of VIX and overall EMV it has $R^2 = 0.169$ ($p = 8.8 \times 10^{-12}$), all of it through overall EMV over the full sample (slope 0.43, $t = 5.07$). Its correlation with MCCC is -0.0004 ($p = 0.996$) and with CPU 0.123 (Holm $p = 0.074$), a link running through overall volatility news: the topic share, the part that is not volatility news, correlates 0.03 with CPU. Crossings fall in high overall-EMV months 88% of the time, against 64.5% otherwise (Fisher $p = 0.0006$). The index counts articles about market volatility, so it spikes with turmoil, whatever the climate news.

Climate-concern indices fed through the same rules earn no validation alpha and no significant holdout alpha (Table C.6, Appendix C.2). The EMV tracker earns validation alphas with $t \geq 1.96$ in 4 of 6 rules, MCCC and CPU in none; all six MCCC holdout alphas are negative, and the two positive CPU ones are within noise (largest $t = 0.37$). Overall EMV, a pure volatility measure, comes close to the team signal (mean validation alpha 1.07% against 1.45%), so the validation alpha belongs to the volatility-news family. The same holds on EPA-ranked legs (Appendix C.5), so this is a null about industry legs, not a refutation of Pástor et al. (2022).

2.2 Style and risk exposures: a Brown-leg value bet

The raw spread is short value, profitability and investment, and the value tilt lives in the Brown leg. GB's FF3 HML loading is -0.233 ($t = -4.63$) over 1970–2026 and -0.298 ($t = -5.74$) since 2010, of which the Brown side contributes -0.437 , Steel and Ships alone -0.239 . In FF5 it loads -0.11 on HML, -0.15 on RMW and -0.25 on CMA (only CMA significant, $t = -2.80$), and adding COMEQ gives a loading of -0.18 ($t = -8.70$). GB is long financials and short value-priced commodity and industrial producers, with no significant alpha in any long window.

The hedge removes most of this, and what remains is a timing drag. The rules' residual HML loading is -0.015 to -0.053 since 2010, and their Lewellen and Nagel (2006) beta-timing term (-0.57% to -1.07% a year; 4 of 6 survive Holm) is shared by the always-short position (-1.35%), so it belongs to the hedged leg rather than to attention.

2.3 Time patterns: full sample, post-2010 and the last 18 and 12 months

Outside COVID, the only positive t above 2 is in the validation sample the rules were designed on (Table 4). Over seven factor models, both baselines and all six rules, the smallest post-2010 p is 0.053.

The holdout loss is residual Brown performance plus costs, and the 900 variants that share it are one result. Corrected Pure 6m (identical to the table's Original 6m in the holdout) has an FF3 alpha of -1.97% a year ($t = -1.30$; 95% interval -5.0% to $+1.1\%$) from six entries and 32 months in position. One principal component carries 90% of the corrected rules' holdout variance. Adding BOND and UMD lowers each of the 12 primary holdout alphas (six rules, two baselines; smallest Holm $p = 0.155$), so rates do not explain the loss. With centered, ex post betas, Pure 6m's net -1.84% is leakage (the factor return the lagged hedge failed to remove) of -0.24% , residual -1.14% and cost 0.46%. Steel carries the gross loss in that split (-1.37 of -1.38 points),

Table 4: Time patterns. Annualized net return (%) and FF3 alpha (% , NW(6) t) of three corrected-baseline rules, the always-short Brown position and the raw GB spread. With $t(n - 4)$ p-values, 18 and 12 months need $|t|$ above 2.14 and 2.31.

Window		Original 3m	Original 6m	Continuous raw	Always-short Brown	Raw GB spread ^b
Full sample ^a , $n = 402$	Net	0.34	1.08	0.27	0.73	-0.50
1993-02 to 2026-07	α (t)	0.44 (0.75)	1.32 (1.85)	0.36 (1.18)	0.79 (0.92)	0.54 (0.41)
Post-2010, $n = 199$	Net	0.73	1.19	0.32	1.05	-1.79
2010-01 to 2026-07	α (t)	0.79 (1.04)	1.21 (1.36)	0.24 (0.83)	0.73 (0.67)	-1.12 (-0.49)
Validation, $n = 151$	Net	1.78	2.16	0.57	1.71	-0.65
2010-01 to 2022-07	α (t)	1.62 (1.97)	2.09 (2.18)	0.46 (1.40)	1.39 (1.12)	-0.55 (-0.22)
Holdout, $n = 48$	Net	-2.57	-1.84	-0.48	-1.04	-5.38
2022-08 to 2026-07	α (t)	-2.16 (-1.90)	-1.97 (-1.30)	-0.58 (-1.33)	-1.72 (-1.09)	-4.59 (-0.99)
COVID, $n = 24$	Net	8.04	5.95	3.01	6.87	1.78
2020-01 to 2021-12	α (t)	6.38 (3.62)	3.99 (1.79)	1.95 (2.68)	4.54 (1.96)	-1.04 (-0.24)
Last 18 months, $n = 18$	Net	-4.51	-1.35	-0.73	-1.07	-14.70
2025-02 to 2026-07	α (t)	-4.05 (-1.55)	-4.06 (-1.80)	-1.74 (-2.19)	-6.25 (-1.87)	-19.19 (-2.79)
Last 12 months, $n = 12$	Net	-5.21	0.02	-0.47	1.35	-19.94
2025-08 to 2026-07	α (t)	1.20 (0.30)	1.03 (0.18)	-1.25 (-0.72)	-3.39 (-0.52)	-32.80 (-5.65)

^a Continuous raw from 1993-01; the raw spread from 1970-01 ($n = 679$).

^b Unhedged and cost-free; classical OLS $|t|$ of 1.41 and 1.37 over the last 18 and 12 months. Under our team's same-month timing, Original 3m's alphas over those windows are -10.82% ($t = -4.75$) and -11.31% ($t = -2.68$).

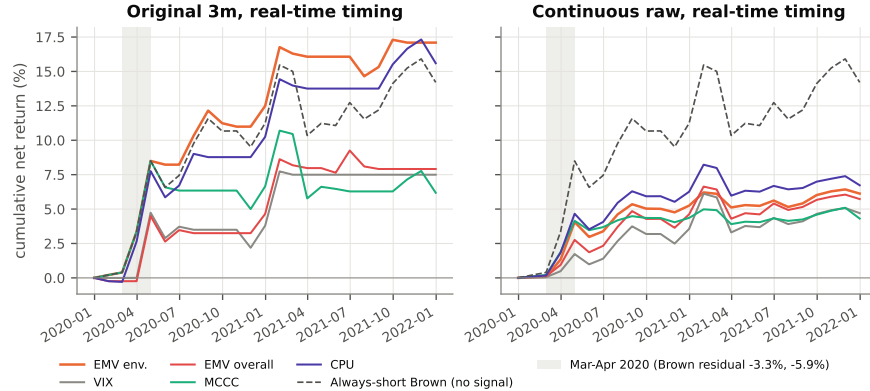


Figure 2: The COVID window. Cumulative net return, 2020-01 to 2021-12, of Original 3m (left) and Continuous raw (right) under real-time timing for each attention measure and for the always-short Brown position (dashed); shading marks March and April 2020. The untimed short tracks the team signal's rule, whose edge over it is within noise.

yet the rule rebuilt without any one industry still loses, and timing lost 1.0% to 1.7% a year against an exposure-matched short ($t = -0.99$ to -1.72 ; Appendix C.4). Over the last 18 months every timed rule lost and the raw spread lost 14.7% a year; twelve months are too few to split a loss into alpha and beta (Original 3m nets -5.21%, alpha +1.20%, $t = 0.30$).

2.4 COVID cannot separate the position from the timing

An untimed short earns 71% of the COVID alpha (Figure 2; Table 4). Original 3m's COVID alpha is 6.38% ($t = 3.62$) and the always-short position's 4.54% ($t = 1.96$). The rule's paired alpha over that position is +1.84% a year ($t = 0.87$; 95% interval -2.6% to +6.2%), or +0.42% without March and April 2020. Against the π -scaled, exposure-matched short the raw edge is larger (5.4% a year, $t = 2.35$) but falls to 3.0% once in-window factors enter ($t = 1.67$, $p = 0.11$). With in-window factors neither edge is significant, and Pure 6m held the always-short position in all 24 months (Figure C.6). By the pre-specified placebo rule the climate reading fails on its boundary: overall

Table 5: The frozen 1994–2009 test: the pre-registered pass bar for the EMV-share rule and, for context, our team’s own signal lagged one month. All four components had to pass; both signals fail.

Component	Bar	Frozen EMV-share rule	Team signal, lagged
(i) Timing alpha, NW(6)	$\alpha > 0, t \geq 2$	−0.18% a year, $t = -0.27$; one-sided $p = 0.605$	FAIL +0.69% a year, $t = 1.00$; one-sided $p = 0.159$
(ii) Calendar shuffle	$p \leq 0.05$	$p = 0.478$ (5,000 draws)	FAIL $p = 0.128$ (5,000 draws)
(iii) Episodes	≥ 8	10 merged episodes	PASS 10 merged episodes
(iv) Drop one industry	all five > 0	2 of 5 positive	FAIL 5 of 5 positive
Verdict	all four	FAIL, 1 of 4	FAIL, 2 of 4

volatility news reproduces the gain in 4 of 6 rules, and for Original 3m, where the volatility placebos fall short, CPU matches it.

2.5 The pre-registered 1994–2009 test

The frozen rule meets one of four components and sits at the center of its own null (Table 5; protocol in Section 1.3). Its timing alpha is −0.18% a year ($t = -0.27$), and 5,000 calendar shuffles of its holding blocks have a median of −0.22% ($p = 0.478$). Power was limited: the t bar needed about +1.4% a year and the shuffle +0.9%, and only an edge near 1.9% passes with 80% probability. But the estimate is negative, and the rule had no in-sample edge to confirm: on the seen 2010–2022 window its timing alpha was already −0.12% ($t = -0.19$).

2.6 Alternatives on the same data

A complete supply-chain ranking flips the raw spread but produces no post-2010 alpha. The EPA-ranked spread’s post-2010 FF5+UMD alpha is 4.7% ($t = 1.52$, $p = 0.129$). Its full-sample alpha (4.5%, $t = 2.58$) comes from short CMA and UMD loadings on a 2022 sort applied back to 1970, and it fails the search correction even there (Romano–Wolf $p = 0.058$).

Equal-weight industry momentum is the UMD premium in industry form. Long the top 8 and short the bottom 8 industries on returns over months $t - 11$ to $t - 1$ (Moskowitz and Grinblatt, 1999), the book nets 8.52% a year, but its FF5+UMD alpha is 1.74% ($t = 1.09$), 1.86% since 2010 ($t = 0.71$), with a UMD beta of 0.99. A carbon limit costs no detectable IR, with low power: the primary bound $b = 0$ cut long-book carbon intensity only 11.6% (IR change +0.010), and $b = -1$ cuts it 60% for an IR change of −0.005 whose interval allows a loss of 0.1.

An optimized momentum book passed a pre-registered test before 1970 and still falls short of capital. The Grinold and Kahn (2000) version (5% ex-ante tracking error) had an exploratory FF5+UMD alpha of 2.17% ($t = 3.07$) over 1970–2026. Following exchange 03, its specification and code were hashed before any pre-1970 return existed, and it ran once on 1931–1969: FF3+UMD alpha 1.95% a year ($t = 2.45$, one-sided $p = 0.007$), positive in both halves. It fails the other bars: the Romano–Wolf search correction (Romano and Wolf, 2005) passes over 1970–2026 ($p = 0.017$) but not over 2010–2026 ($p = 0.199$), its deflated appraisal ratio (Bailey and López de Prado, 2014) is 0.56 and 0.36 against a bar of 0.95, and its holdout alpha is −0.34%. The verdict map fixed before the run sends it to a paper-trading pilot: the pass shows the edge existed before 1970, not that it exists now.

A disciplined shrinkage model refuses to time the spread. Under pre-registered cross-validation the Lehnher et al. (2024)-style ridge picks the largest penalty in all 758 monthly refits, so every forecast is the historical mean; attention lowers out-of-sample R^2 at every fixed penalty.

2.7 Robustness, multiple testing and limitations

No module’s primary alpha survives the project-wide correction, and the holdout is evidence against the attention rules. M7 assigned all 23,923 module tests to families by the decision each could change; 4,184 are nominally significant at 5%, among them 1,624 alphas (783 positive, 841 negative). The 73 distinct primary alpha tests, one-sided, have a best p of 0.030 and a smallest Holm p of 1.00 (BH 0.72). In the robustness grids (9,337 alphas) Benjamini and Yekutieli (2001) keep none and Benjamini and Hochberg (1995) keep one, a CPU variant of the COVID rule that Appendix E retires with every other nominal positive. The holdout rejects a worthwhile alpha for all six rules as our team ran them (every 90% upper bound lies below a margin of a quarter of the rule’s residual volatility), so it is evidence against the rules, not an absence of evidence.

Limitations. One undated (team) or 2022-vintage (EPA) emissions snapshot is applied back to 1970, a look-ahead in leg membership. FRED, EMV and MCCC data are final vintages, and the one-month EMV lag is assumed. COVID and recent-window inference rests on 12 to 24 observations, the NW(6) t over-rejects on synthetic nulls (13 of 100 at 5%), and 1994–2009 predates the concern shocks that Pástor et al. (2022) use to explain 2012–2020 green returns. Pre-specification is timestamped only for the two frozen tests. Costs are linear, and an industry-level test says nothing about firm-level carbon premia (Bolton and Kacperczyk, 2021).

2.8 Critical evaluation of ChatGPT

ChatGPT was a sharp referee on the numbers it was given, and each exchange changed the plan (Table 3). It was wrong on 6 of 52 checkable claims in exchange 01, among them that duration explained the holdout loss (rate beta $t = 0.05$), and on 8 of 72 in exchange 03, mostly about our books.

It never asked what the signal measures, and it did not push back on a confident user: it accepted the “climate-transition attention” label and amplified my wrong claim that the holdout “tested a near-binary signal”. The exchange 04 red team, run twice, prompted thirteen summary rewordings, but its cases against the verdict rested on misread numbers, such as a pooled +0.36% read as positive in both windows.

As a research assistant it is a referee and a second implementation, never a verifier. Where the specification was exact its code was exact: aligned on two open readings, its 985-line script reproduced all 5,000 of my shuffle draws to 3.3×10^{-13} . But it takes its inputs on trust, adopts the framing of a confident prompt, and its checks miss what they were not told to look for: its look-ahead test caught 4 of the 6 timing bugs I injected, and the revised version 7 of 8. These replies came from a model in ChatGPT’s role (Appendix A), so the verdict describes that role.

2.9 Verdict

Do not implement, timed or untimed: the signal is volatility news, the gain is a Brown-leg exposure, and the one pre-registered test failed. The strongest case against the verdict, our team’s own signal lagged one month, has a timing alpha of +0.69% over 1994–2009 ($t = 1.00$) but -1.11% over the holdout ($t = -1.13$). The playbook I would hand a desk has three rules: read the label of every input, benchmark every timing claim against the always-short position, and spend the last unused window on one frozen rule. Each is an afternoon’s check before a desk funds a climate-timing idea.

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A ChatGPT interactions

This appendix documents the four exchanges: for each, a four-line box of what I asked, what came back, what I verified and what I kept, then my evaluation, the prompts and replies verbatim, and the fact-check. The main-body summary is Table 3 and the critical evaluation is Section 2.8. The provenance note below applies to every reply.

Provenance of the replies. The prompts below are the prompts as written, verbatim. The replies were not produced by a live ChatGPT session: each was generated by a separate language model that was instructed to answer as ChatGPT would, was given only the prompt (and, for a follow-up, the earlier turns), and had no files, code execution or web access. In exchange 02 the follow-up reply was generated without the first reply in its context. Every reply is labeled accordingly, and the evaluations in this appendix and in Section 2.8 judge these replies, not ChatGPT itself. Two frozen tests took their design from these replies, the 1994–2009 rule of exchange 01 and the 1931–1969 run of exchange 03; both were fixed before they ran and cannot be re-drawn, because their windows have been spent.

Layout. Each exchange opens with a four-line box and my evaluation, followed by the prompts and replies and the fact-check. Non-ASCII characters in the transcripts are mapped to ASCII so that no glyph is lost, and line breaks inside a listing are marked with an arrow. Word counts are `wc`-style counts of the source file; where Markdown symbols (table bars, bullets) inflate that count, the count without them follows, and that is the count compared with a word cap.

A.1 Exchange 01: idea generation (a design review of the halfway strategy)

Files: `exchange/01_idea_generation/`. Answered by: a model in the ChatGPT role (see the provenance note). Date: 2026-09-26. Purpose: idea generation.

Asked.	A skeptical design review of the halfway strategy: at most six problems ranked by impact on the verdict, a ranking of the seven teammate proposals (A to G), exactly five falsifiable new ideas, a plan, and the three assumptions it was least sure of.
Came back.	A 1,609-word review (cap 1,800); after I sent five corrections and a data fact, a 733-word follow-up with one frozen rule for 1993–2009, a four-part pass bar and an attribution specification.
Verified.	52 checkable claims scored against the data (26 correct, 15 partly correct, 6 wrong, 5 unverifiable); every test it proposed that could run on seen data was run (checks c01 to c11).
Kept.	The always-short benchmark for every timing claim, the pre-registered test on 1994-03 to 2009-12 (M8), and Aero and Ships as suspect Brown members (M4).

My evaluation

The first reply was a competent design review built on a variable it never examined. The fact-check scored 52 checkable claims: 26 correct, 15 partly correct, 6 wrong and 5 unverifiable. Its arithmetic, citations and labeling of guesses were sound.

Three contributions changed the plan. First, critique 1: timed returns had been tested against zero, when a timing claim’s null is always-on Short-Brown with the same hedge. On

seen data it settles the question (timed minus always-on FF5+UMD alpha peaks at $t = 1.09$). Second, Idea 1, a backward test on the one window with no computed strategy returns (pre-2010), was not in my plan. Third, it flagged Aero and Ships as odd Brown members and guessed that Aero, Util and Ships drove the COVID gains. Both held: EPA factors rank Aero 31st and Ships 20th of 49, and those three industries deliver 86% of the Original 3-month rule's COVID gross P&L. Its redirect of test C to the Brown leg was also right (HML +0.44 for Brown vs. +0.14 for Green since 2010).

Six claims were wrong (five corrections in my follow-up). The holdout was not lost to duration: the hedged Brown leg has no holdout rate exposure ($\Delta 10y\ t = 0.05$), and holdout alphas stay at -2.3% to -4.8% annually after FF5+UMD, rate and commodity controls. Its inference fix fails: more Newey–West lags raise the 24-month COVID t from 3.21 to 5.32, and the binding problem is small-sample p-values (with $t(n-4)$, 0 of 32 cells survive Holm). The IC flip is 1.3 to 1.6 SE for the raw and purified signals and 2.8 to 2.9 NW SE for the continuous ones, not “about one”. Idea 1's dates were wrong (first crossings 1993-01 and 1999-01), and the team's macro-state notebook had already used 192 pre-2010 months. It asked where Coal ranks (9th of 41, in the file) and missed that EPA v1.3 lacks electricity codes. Ideas 4 and 5 were generic: climate-premium portfolios with no stated alpha source.

It never asked what “attention” measures. My prompt passed on the team's label without a source, and ChatGPT never asked who builds the index. The series is the EMV tracker for Energy and Environmental Regulation (since 2010, 88.5% of crossings fall in months when overall EMV is above its past-only median, against 56.1% of other months; Fisher $p = 0.0012$), and 31 of 48 holdout months are exactly zero. It put the coverage risk in the wrong period (“early index coverage may be thin”; the pre-2010 zero share is 0.8% to 4.2%). This is a failure of my prompt as much as of the tool.

The follow-up turned the corrections into specifications. It accepted every correction, kept “Do not implement”, and gave a frozen 1993–2009 rule (share of overall EMV, zeros as missing, the 6-month rule, a four-part pass bar) plus an attribution against an exposure-matched always-on benchmark, with a calendar-shuffle null. It rightly declined credit for the holdout: its zero rule silences the signal in all 48 holdout months but was written after seeing them. Three flaws remain. The two zero thresholds are redundant (at most 6 zeros in 60 makes the 48-nonzero minimum inert). The share signal keeps most of the original timing (19 of its 29 post-2010 crossings sit within a month of the team's, against about 10 by chance). Its $k = 14$ undercounts the model's 15 coefficients.

Where each piece went.

- M1b: MCCC and CPU stay the primary alternatives on 2010–2026, with VIX and overall EMV as COVID placebos. The share series was already a robustness measure; the follow-up's point that MCCC starts in 2003 and CPU cannot signal before 1995 is why it carries the pre-2010 test.
- M2 and M4: test C covers both legs, costs are reported as break-evens, and the EPA rebuild patches Util from USEEIO.
- M3: its Ferson–Schadt terms match my planned conditional-beta step, and a GS10 bond return with duration and convexity supersedes its $-8 \cdot \Delta y_{10}$ guess.
- M8: the exposure-matched benchmark, calendar-shuffle null, merged event counts and $t(n-k)$ p-values apply to every timing claim. The 1993–2009 rule is frozen with corrected dates and a one-month publication lag, and is scored once.
- M5 and M6 owe nothing here: neither reply proposed industry momentum or shrinkage timing.

Prompt (verbatim) `prompt.md`, 993 words (980 excluding Markdown symbols).

You are a skeptical buy-side quant refereeing a student research project in climate-driven factor
 ↪ timing. Treat this as a design review: I want the weak points found before I spend time on
 ↪ them.

SETTING. This is the MFE 230GA (Active Asset Management) final project at Berkeley. The report
 ↪ needs a one-paragraph Executive Summary that says "Do not implement" if there is no alpha;
 ↪ "What did you try" (idea, data, risk model, turnover and costs, at least three substantial
 ↪ ChatGPT interactions); "What did you learn" (factor exposures, performance over the full
 ↪ sample, post-2010 and the last 12 to 18 months, robustness and limits, a critical
 ↪ evaluation of ChatGPT). Grading: thesis 40%, execution 40%, originality 20% (novel data use
 ↪ , prompt design). A well-documented "Do not implement" is an acceptable result. Our team of
 ↪ five wrote the first half; I am taking the second half forward.

HALFWAY DESIGN. Thesis: Green minus Brown has no permanent premium, but its factor-neutral part
 ↪ pays after spikes in climate-transition attention. Green is the five lowest emissions-
 ↪ intensity Fama-French 49 industries (Fun, RlEst, Drugs, Telcm, Fin), Brown the five highest
 ↪ (Util, Ships, Aero, Steel, BldMt), each leg equal-weighted across value-weighted
 ↪ industries. Emissions intensity is one static snapshot covering 41 of 49 industries (Oil,
 ↪ Chips, FabPr, Gold, Hshld, LabEq, Other, Toys missing). The traded position is not the
 ↪ spread: it shorts the Brown leg alone, with an FF3 overlay from rolling 60-month betas
 ↪ lagged one month. Signal: climate-transition attention (a monthly index starting in 1985)
 ↪ as a 60-month rolling z-score of $\log(1 + \text{attention})$. A month is extreme when the signal
 ↪ exceeds its expanding, past-only 80th percentile, and each crossing opens a Short-Brown
 ↪ position held 3 or 6 months, sized to 5% residual volatility (cap 1x). Costs per unit of
 ↪ turnover are 10 bp on the leg, 5 bp on the market overlay and 25 bp on SMB/HML, all assumed
 ↪ . Variants: "purified" attention (the residual from a 120-month walk-forward ridge on rate,
 ↪ oil, inflation and activity shocks) and a continuous weight (floor 0.5, 3-month half-life)
 ↪ . Validation ran 2010 to Jul 2022; Aug 2022 to Jul 2026 was a frozen holdout.

RESULTS.

- Validation: the 6-month rules earn about 2.5% a year net (FF3 alpha Newey-West $t = 2.23$ and
 ↪ 2.46); the 3-month rules have t of about 1.5.
- COVID carries it: the 3-month rule earns 5.94% net in 2020 to 2021 ($\alpha t = 3.21$).
- Holdout: every version loses 2.1% to 3.8% a year ($t = -1.7$ to -2.2). The IC flips from about
 ↪ -0.08 (negative is the expected sign for a short-Brown rule) to +0.12 to +0.15.
- Over 2010 to 2026 the best timed version earns 1.3% a year net against 1.0% for always-short
 ↪ Brown.
- The raw spread loads on HML (-0.24 full sample, -0.35 since 2010). After hedging, residual HML
 ↪ is about -0.05 with t of about -2 in every version.
- The continuous variant cuts volatility 55 to 65% and drawdown 60 to 75%, but a calendar-shuffle
 ↪ placebo gives p of about 0.12.
- 2 of 32 strategy-period cells survive Holm, both in COVID.
- A follow-up notebook finds the attention slope turns positive in high-rate months (difference $t =$
 ↪ 1.83, Holm p of about 0.6); 47 of those 74 months are in the holdout.

TEAMMATE PROPOSALS. From Christian: (A) rebuild with 8 Green and 8 Brown industries; (B) add
 ↪ momentum and an explicit commodity exposure to the controls; (C) regress Fin, Telcm, Drugs,
 ↪ Fun and RlEst individually to find where the HML exposure comes from; (D) uniform costs at
 ↪ 5, 10 and 25 bp. From Charishma: (E) a Part B on whether this is alpha or beta at the
 ↪ portfolio level. From the group: (F) macro controls, since exposures look regime dependent
 ↪ (COVID, then inflation and rates); (G) a momentum control, in case the signal is only
 ↪ persistence in past returns.

DATA. In hand: the four team files (FF49 monthly returns, Jul 1926 to Jul 2026; FF3 and rf; the
 ↪ emissions snapshot; a macro file with attention, the 10-year yield, WTI, CPI, CFNAI, the
 ↪ NBER recession flag and GSCPI). Downloadable: Ken French FF5, UMD, short- and long-term
 ↪ reversal, FF49 value- and equal-weighted returns, industry BE/ME, firm counts and average
 ↪ size; FRED Treasury yields (3-month, 2-year, 10-year), BAA and AAA, VIX, commodity indices,
 ↪ activity series, breakevens and TIPS yields; the Gavriilidis Climate Policy Uncertainty

- ↪ index, the Ardia et al. Media Climate Change Concerns index, and EPA supply-chain GHG
- ↪ factors by NAICS. No firm-level data. Every result so far starts in 2010, and the holdout
- ↪ has now been seen, so it is no longer a clean test.

WHAT I NEED, in this order and under these headings.

1. Design critique. At most six problems, ranked by how much each could change the verdict: the
 - ↪ problem, why it matters here, the fix.
2. Proposal ranking. One table for A to G: rank, proposal, the result that would change our
 - ↪ conclusion, what it cannot tell us, overlap with other proposals, effort in hours. One line
 - ↪ of reasoning per row.
3. New ideas. Exactly five, at least two extending the current strategy and at least two
 - ↪ alternatives on the same data, each aimed at a small defensible alpha or a more rigorous "
 - ↪ Do not implement". For each: (a) a falsifiable hypothesis and the result that rejects it; (
 - ↪ b) the exact test: sample, return series, specification, benchmark, pass bar; (c) the data,
 - ↪ and whether it is listed above; (d) main pitfalls (look-ahead, multiple testing, small N,
 - ↪ overlapping holds); (e) effort in hours; (f) what it adds to the report if it fails.
4. Plan. The three items from sections 2 and 3 you would run first, in order, and why.
5. The three assumptions in your answer you are least confident about.

RULES. Work from our numbers, not generic ESG results. Cite a paper only with authors, year and

- ↪ the specific claim. Do not invent backtest results; label any number you have not computed
- ↪ as a guess. Stay under 1,800 words.

Reply (verbatim; model in the ChatGPT role) chatgpt_response.md, 1,772 words
(1,609 excluding Markdown symbols).

```
# Design Review: Climate-Attention Timing of Short-Brown

**Bottom line:** The holdout failed, validation rests on one episode, and timing beats always-on Short-Brown
  ↪ by roughly 0.3% a year (1.3% vs 1.0%, arithmetic on your figures). Make "Do not implement" airtight,
  ↪ and test pre-specified ideas on the one period nobody has looked at, **1990-2009**.

---

## 1. Design critique

**1. The benchmark is wrong for a timing claim.**
- *Problem:* Timed returns are tested against zero. For a timing thesis the null is "always short Brown, same
  ↪ hedge, same vol target."
- *Why it matters:* The signal adds about 0.3%/yr over 2010-2026. Also, the traded position has no Green leg,
  ↪ so "Green minus Brown" isn't what you're testing.
- *Fix:* Make the primary statistic the alpha of (timed - always-on), or regress hedged Brown returns on an
  ↪ in-position dummy. Restate the thesis as "Brown underperforms after attention spikes."

**2. Validation is one episode.**
- *Problem:* Both Holm survivors are COVID; the 3-month rule's 2020-21 alpha (t = 3.21) carries the result.
- *Why it matters:* Aero (Boeing-heavy: 737 MAX grounding plus the travel collapse), Ships and Util took
  ↪ COVID-specific hits unrelated to transition risk. With five equal-weighted industries, each is 20% of
  ↪ the leg.
- *Fix:* Leave-one-industry-out, and rerun excluding Mar 2020-Dec 2021. If the 6-month t-stats drop below
  ↪ -1.5, the Executive Summary should say so.

**3. The holdout is spent; everything after it is post-hoc.**
- *Problem:* The high-rate interaction was found with 47 of its 74 months in the holdout.
- *Why it matters:* It can't be confirmed on the data that suggested it. The holdout also doesn't prove a *
  ↪ reversal*: with ~48 months the IC's standard error is roughly 1/sqrt48 ~ 0.14 (a rough approximation
  ↪ that ignores overlap), so -0.08 -> +0.13 is about one standard error of the difference. The honest
  ↪ reading is "failed to confirm," not "flipped."
- *Fix:* Declare 1990-2009 the new frozen holdout and write down one rule before running it.

**4. The Brown leg may not measure what the thesis claims.**
```

```

- *Problem:* One static snapshot, eight industries missing, including Oil, the most obvious transition-risk
  ↳ industry. Aircraft and Shipbuilding in the top five by emissions intensity is surprising; check the
  ↳ mapping and where Coal ranks.
- *Why it matters:* If the ranking is off, a "climate" result is an "industrials and utilities" result.
- *Fix:* Rebuild intensity from EPA supply-chain GHG factors (NAICS -> SIC -> FF49) covering all 49, and
  ↳ rerun with Oil and Coal in.

**5. Hedge residuals line up with the regime story.**
- *Problem:* Residual HML ~ -0.05 (t ~ -2) in *every* version, and no control for momentum, profitability/
  ↳ investment, rates or commodities.
- *Why it matters:* Util, BldMt and Steel are rate- and commodity-sensitive. A 2022-2026 holdout dominated by
  ↳ rates can lose money through unhedged duration, not climate.
- *Fix:* Attribute ex post against FF5 + UMD + a rate factor (e.g., Delta10-year yield) + a commodity return.
  ↳ Keep the ex-ante hedge unchanged so you aren't re-optimizing.

**6. Inference is weaker than the t-stats suggest.**
- *Problem:* Overlapping 3-6-month holds, few independent crossings, 4+ variants, and a purified signal built
  ↳ from CPI, CFNAI and GSCPI, which are published with a lag and revised.
- *Why it matters:* Too few Newey-West lags overstate t; final-vintage macro data is look-ahead. The
  ↳ continuous variant's lower vol and drawdown is partly mechanical (lower average exposure), which is
  ↳ consistent with placebo p ~ 0.12.
- *Fix:* NW lags >= hold length; report the count of independent events; block-bootstrap by event; lag macro
  ↳ inputs 1-2 months; compare the continuous variant with always-on scaled to the same average exposure.

---

## 2. Proposal ranking

| Rank | Proposal | Result that would change our conclusion | What it cannot tell us | Overlap | Hours |
| --- | --- | --- | --- | --- | --- |
| 1 | **E** Alpha vs beta | Timed - always-on alpha significant after FF5+UMD | Whether the mechanism is
  ↳ climate | B, F, G | 4-6 | The question graders ask, and the one your numbers are closest to answering
  ↳ |
| 2 | **A** 8+8 industries (add leave-one-out) | Timing alpha survives broader legs and dropping Aero |
  ↳ Whether emissions matter vs any 5 industries | C | 3-4 | Cheap; directly tests the COVID
  ↳ concentration |
| 3 | **G** Momentum control | Signal loses a horse race to Brown's own trailing 3-12m return | Whether
  ↳ attention causes the persistence | B | 2-3 | Cheap; kills the simplest alternative story |
| 4 | **B** Momentum + commodity controls | Alpha t falls below ~1.5 with a commodity factor | Climate vs
  ↳ commodity channel (Oil is missing) | G, F | 3-5 | Useful but mostly subsumed by E |
| 5 | **F** Macro controls | Alpha vanishes when conditioning on rate/inflation state | Anything confirmatory
  ↳ ; regimes were picked after seeing the holdout | B, purified variant | 6-10 | Most expensive, most
  ↳ overfit-prone |
| 6 | **C** Green industries one by one | Little: Green isn't in the traded position | Source of the *traded*
  ↳ HML residual (that sits in Brown) | A | 2 | Redirect it to the five Brown industries |
| 7 | **D** Uniform 5/10/25 bp | Only a break-even cost below ~10 bp, unlikely with 3-6-month holds (guess) |
  ↳ Anything about the signal | none | 1 | Costs can't rescue a ~2% to ~4% holdout; report break-even
  ↳ instead |

---

## 3. New ideas

### Idea 1 (extension): Backward out-of-sample test, 1990-2009
- *(a) Hypothesis:* The frozen 6-month rule earns positive alpha over always-on before 2010. *Rejected* if
  ↳ timed - always-on alpha <= 0 or t < 1.
- *(b) Test:* Signal from Jan 1990 (60-month z-score uses 1985-89); purified variant from ~1995 (120-month
  ↳ ridge). Hedged Short-Brown, 5% vol target, validation parameters unchanged. FF3 and FF5+UMD alpha, NW
  ↳ 6 lags. Benchmark: always-short Brown. Pass: t > 2. Pre-register the high-rate interaction as a
  ↳ second hypothesis, Holm across both.
- *(c) Data:* In hand.
- *(d) Pitfalls:* Static snapshot applied backward (classification look-ahead); transition risk was
  ↳ arguably unpriced before ~2005, so failure doesn't fully kill the mechanism; early index coverage may
  ↳ be thin.
- *(e) Effort:* 3-5 h.
- *(f) If it fails:* Twenty more years behind "Do not implement," from the cleanest test left.

```

```

### Idea 2 (extension): Timing increment and event-time profile
- (a) Hypothesis: Hedged Brown residuals are more negative in  $t+1...t+6$  after crossings than
  ↳ unconditionally. *Rejected* if the post-event cumulative residual is within one SE of zero, or the
  ↳ move sits in spike month  $t$  itself.
- (b) Test: 2010-2026 only until Idea 1 is run (don't contaminate the new holdout). Event-time averages  $t$ 
  ↳  $-3...t+6$ ; regression of residual on a post-event dummy, NW 6 lags. Benchmark: always-on. Pass: dummy
  ↳  $t > 2$  excluding 2020-21.
- (c) Data: In hand.
- (d) Pitfalls: Overlapping events (merge crossings within 6 months), small  $N$  (report event count), COVID
  ↳ dominance.
- (e) Effort: 3-4 h.
- (f) If it fails: Evidence attention is priced on impact. Ardia, Bluteau, Boudt & Inghelbrecht (2023)
  ↳ find green stocks rise and brown stocks fall on days of unexpected increases in climate-change
  ↳ concerns; Pastor, Stambaugh & Taylor (2022) attribute green outperformance over 2012-2020 to
  ↳ unanticipated strengthening of climate concerns. If the repricing is contemporaneous, a post-spike
  ↳ rule has nothing left to harvest.

### Idea 3 (extension): Replicate the signal across attention indices
- (a) Hypothesis: If the effect is about climate attention, the frozen rule works with the Climate Policy
  ↳ Uncertainty (CPU) and Media Climate Change Concerns (MCCC) indices. *Rejected* if neither gives
  ↳ timed - always-on  $t > 1$  with the same sign.
- (b) Test: Identical rule and parameters, swap the signal; samples set by index start dates (CPU late
  ↳ 1980s, MCCC ~2003; verify). Pass: same sign in all three, pooled  $t > 2$ .
- (c) Data: Downloadable, listed.
- (d) Pitfalls: Different samples per index; three tests (Holm); the 60-month z-score burns five years of
  ↳ a short index.
- (e) Effort: 4-6 h.
- (f) If it fails: Shows the result is index-specific noise; earns originality credit either way.

### Idea 4 (alternative): Rebuilt carbon-intensity spread across all 49 industries
- (a) Hypothesis: No unconditional premium: an intensity-sorted industry spread has FF5+UMD alpha of zero.
  ↳ Bolton & Kacperczyk (2021) find a positive carbon premium at the firm level; this tests the industry
  ↳ analogue. *Rejected* if  $\alpha_t > 2$  in either direction.
- (b) Test: 1990-2026, monthly; long top-third/short bottom-third intensity, VW industries. FF5+UMD alpha,
  ↳ NW 12 lags; report pre/post-2010 and the last 18 months.
- (c) Data: EPA factors (listed); a NAICS->SIC crosswalk (not listed; public concordances exist).
- (d) Pitfalls: Single-vintage intensity applied backward; crosswalk error; supply-chain factors include
  ↳ upstream emissions, a different concept from your snapshot.
- (e) Effort: 6-10 h.
- (f) If it fails: A rigorous version of the "no permanent premium" half of your thesis, with Oil and
  ↳ Coal included.

### Idea 5 (alternative): Climate-news hedge portfolio on FF49
- (a) Hypothesis: Following Engle, Giglio, Kelly, Lee & Stroebe (2020), who build mimicking portfolios
  ↳ that hedge innovations in a climate news index, an FF49 portfolio mimicking attention innovations
  ↳ hedges out of sample. *Rejected* if its out-of-sample correlation with next-month innovations is  $\leq 0$ .
  ↳
- (b) Test: 1995-2026. Monthly rolling 60-month regressions of each industry's excess return on attention
  ↳ innovations (past-only AR(1) residual) plus FF3; long top-10/short bottom-10 by beta, one-month hold.
  ↳ Pass: OOS correlation  $t > 2$ ; report alpha as the price of the hedge.
- (c) Data: In hand.
- (d) Pitfalls: Noisy betas, only 49 assets, innovation definition choice, full-sample AR fits (look-
  ↳ ahead).
- (e) Effort: 6-8 h.
- (f) If it fails: Reframes attention as a risk to hedge, not a signal to trade.

---

## 4. Plan
1. Idea 2 + leave-one-out (Critique #2). Cheapest, and decides whether timing adds anything over always-
  ↳ on, on data already seen.
2. Idea 1. The only clean out-of-sample test left. Freeze the rule in writing first, and don't touch pre
  ↳ -2010 attention data before then.
3. E with B and G folded in. FF5 + UMD + rates + commodities attribution for full sample, post-2010 and
  ↳ last 18 months; the report needs it whatever the outcome.

---

```

```
## 5. Assumptions I'm least confident about
1. That the attention index has usable coverage in 1985-2009 and the emissions ranking is stable enough to
   ↳ apply backward; if not, Idea 1 is weak.
2. That the COVID gains come from Aero, Util and Ships shocks; this is a hypothesis, not something I computed.
   ↳
3. The index start dates (CPU, MCCC), the NAICS->FF49 mapping effort, and every hour estimate.
```

Follow-up prompt (verbatim) followup_prompt.md, 396 words (392 excluding Markdown symbols).

```
I ran most of your proposed tests on the seen data. Five of your claims are wrong, and you missed
↳ a data fact that changes the review.

The attention series is not what we called it. I traced the team's column to its source: it is
↳ FRED EMVENRGYENVREG, the Baker, Bloom, Davis and Kost Equity Market Volatility tracker for
↳ Energy and Environmental Regulation. It counts newspaper articles on stock-market
↳ volatility that mention the category, scaled to the VIX. Since 2010 its z-score correlates
↳ +0.39 with overall EMV, +0.11 with CPU and -0.06 with MCCC, and 38% of crossing months are
↳ high-EMV months, against 12% of other months (odds ratio 4.5, p = 0.002). Your coverage
↳ worry also points the wrong way. The series is exactly zero in 12 of 441 months from
↳ 1985-01 to 2021-09, but in 39 of 59 months from 2021-10 (31 of 48 holdout months). The
↳ holdout tested a near-binary signal, not the one we validated.

Corrections:
1. Duration. The hedged Brown leg has no holdout rate exposure (Delta10y t = 0.05), and the
   ↳ strategies gain when yields rise. After FF5+UMD+Delta10y+commodity controls, holdout alphas
   ↳ are still -2.3% to -4.8% (t about -2).
2. Newey-West lags. More lags do not shrink the validation t (2.23 at 6 lags, 2.30 at 12); they
   ↳ push the 24-month COVID t to 5.3. The binding problem is small-sample p-values: with t(n-4)
   ↳ , 0 of 32 cells survive Holm.
3. IC flip. It is 1.3 to 1.6 SE for raw and purified, but 2.8 to 2.9 NW SE for the continuous
   ↳ signals.
4. Idea 1 dates. The frozen rule first crosses in 1993-01 (raw) and 1999-01 (purified).

Your tests rescue nothing: timed minus always-on FF5+UMD alpha peaks at t = 1.09, and the Idea 2
↳ post-event CAR is -1.29% (t = -1.08).

Two questions.
(a) Do I salvage this series (as a share of total EMV, or zero-robust) or replace it with CPU or
   ↳ MCCC, given both end in 2025 and burn about eight years under our rule? Give the one rule
   ↳ you would freeze for the unseen 1993 to 2009 test, its pass bar, and how it treats zeros.
(b) Give an exact attribution spec that separates timing alpha from duration, value and volatility
   ↳ exposure: dependent variable, regressors, an exposure-matched benchmark, lags, and what
   ↳ counts as alpha.

Under 600 words. Label anything you have not computed as a guess.
```

Follow-up reply (verbatim; model in the ChatGPT role) chatgpt_followup_response.md, 733 words (675 excluding Markdown symbols).

```
# Follow-up: Signal Choice and Attribution

Your corrections are right, including the ones on duration and the IC. The EMV finding matters more than any
↳ of them. The signal we validated partly tracks news about market volatility, and the holdout ran on a
↳ different, near-binary signal. That explains the holdout but does not rescue the strategy. On the
↳ data you have already seen, timed minus always-on peaks at t = 1.09, so ***"Do not implement" stands***.
↳
```

```

---

## (a) Salvage the series; don't replace it

- **MCCC can't test 1993-2009** because it starts in 2003.
- **CPU would only cover about 1995-2009** once the burn-in is used up (a guess). Its correlation with your
  ↳ series is +0.11, so switching to it tests a new hypothesis rather than a cleaner version of yours.
- Run CPU and MCCC only after the primary test, and label them exploratory.
- **Use the series as a share of total EMV.** As I understand the construction, the category tracker equals
  ↳ overall EMV times the fraction of EMV articles that mention the category. Dividing by overall EMV
  ↳ removes the part scaled to the VIX and leaves the attention part. This goes straight at your odds
  ↳ ratio of 4.5.

**The rule to freeze.** Write it down before you load any pre-2010 data.
1. **Signal:**  $s_t = \text{EMVENRGYENVREG}_t / \text{EMVOVERALLEMV}_t$ .
2. **Zeros:** treat a zero month as missing, not as low attention.
  - A zero month cannot trigger a position.
  - Drop zero months from both the z-score window and the percentile history.
  - The z-score needs at least 48 nonzero months out of the trailing 60.
  - If more than 10% of the trailing 60 months are zero, the signal is off.
3. **Extreme month:**  $z_t$  is the 60-month rolling z-score of  $\log(s_t)$ . A month is extreme when  $z_t$  exceeds
  ↳ its expanding, past-only 80th percentile.
4. **Trade:** the validated 6-month Short-Brown rule, unchanged: lagged 60-month FF3 hedge, 5% vol target,
  ↳ cap 1x, same costs. A new crossing during a hold extends the hold; positions never stack.
5. **Sample:** Jan 1993 to Dec 2009, run once.
6. **Pass bar:** all four of the following. Anything less confirms "Do not implement."
  - The timing alpha from (b) is above zero with  $t \geq 2$ , using  $t(n-k)$  p-values.
  - The calendar-shuffle p is 0.05 or lower.
  - There are at least 8 independent episodes (a guessed threshold).
  - The alpha keeps its sign when each Brown industry is dropped in turn.

**Caveat:** the zero rule (rule 2) would probably have kept you flat for most of the holdout (a guess). It
  ↳ was written after seeing the holdout, though, so take no credit for that period.

---

## (b) Attribution spec

**Dependent variable.**  $D_t = R^T_t - \pi \cdot R^{AO}_t$ , net of costs.
-  $R^T$  is the timed strategy.
-  $R^{AO}$  is always-on Short-Brown with the same hedge and vol target.
-  $\pi$  = mean timed weight / mean always-on weight. The benchmark then has the same average exposure as the
  ↳ timed strategy but no timing.

**Regressors.** These are for ex-post attribution only; keep the ex-ante hedge unchanged.
- FF5 + UMD.
- **Duration:**  $\text{TERM}_t \sim -8 \cdot \text{Deltay10}_t$ , an approximate 10-year Treasury return. The duration of 8 is a guess.
- **Commodity:** the WTI log return.
- **Volatility:**  $\text{DeltaVIX}_t$  (VIX data start in 1990) and  $\text{Deltalog EMV}_t$ .
- **Conditional terms:**  $I_{\{t-1\}} \times \{\text{Mkt}, \text{HML}, \text{TERM}, \text{DeltaVIX}\}$ , where  $I_{\{t-1\}} = 1$  if the strategy is in
  ↳ position. This follows Ferson and Schadt (1996), who let betas vary with lagged public information so
  ↳ that changing exposure is not counted as skill. Here it means any return from loading on value,
  ↳ duration or volatility only while in position counts as beta.

**Model.**  $D_t = \alpha + \beta'F_t + \gamma'(I_{\{t-1\}} \cdot F_t) + \epsilon_t$ .

**Inference.**
- Newey-West standard errors with 6 lags, and 12 as a check.
- p-values from  $t(n-k)$ , with  $k = 14$  parameters.
- The null with matched exposure is a calendar shuffle: move the same hold windows to random dates 5,000
  ↳ times and recompute alpha each time.

**What counts as alpha.** Only alpha.
- Report the decomposition  $\text{mean}(D) = \alpha + \text{Sigmbeta} \cdot \text{mean}(F) + \text{Sigmagamma} \cdot \text{mean}(I \cdot F)$ . That gives duration,
  ↳ value and volatility a number each.
- Timing alpha exists only if alpha passes the bar in (a) and survives the volatility block.
- If alpha drops when DeltaVIX and DeltaEMV enter, the signal was a volatility bet, not a climate one.

```

Fact-check (verbatim excerpt) Sections “Summary”, “Proposed tests that I ran”, “What ChatGPT missed” of factcheck.md (4,299 words in full).

Table A.1: Exchange 01 fact-check, verdict counts by section of the reply (C correct, P partly correct, W wrong, U unverifiable). Counted from the fact-check’s claims table.

Section of the reply	C	P	W	U	Total
Bottom line and design critique	14	11	3	1	29
Proposal ranking (section 2)	2	0	0	0	2
New ideas (section 3)	7	3	3	2	15
Plan and stated assumptions (sections 4, 5) and compliance	3	1	0	2	6
All claims	26	15	6	5	52

Summary

52 checkable claims: 26 correct, 15 partly correct, 6 wrong, 5 unverifiable. The review is well organised,
 → the arithmetic on our figures is right, and its literature citations are accurate. Its main
 → weaknesses are these:

1. **It never asked what "attention" is.** The series is the EMV tracker for Energy and Environmental
 → Regulation, a count of news articles about stock-market volatility. From 2010 to 2026 its z-score
 → correlates +0.39 with overall EMV, +0.11 with CPU and -0.06 with MCCC. Crossings are 4.5 times as
 → likely (odds ratio) in months of high overall EMV ($p = 0.002$).
2. **It got the coverage problem backwards.** It worried that early coverage "may be thin". Before 2010 the
 → zero share is 0.8% to 4.2%. In the holdout, 31 of 48 months are exactly zero, so the holdout tests a
 → different, near-binary signal.
3. **Idea 1 has the wrong start dates.** Under the frozen rule the first raw crossing is Jan 1993 and the
 → first purified crossing Jan 1999, not 1990 and about 1995. Also, the team’s macro-state notebook
 → already estimated attention-to-residual slopes over 1994 to 2026.
4. **The duration story for the holdout is wrong.** The hedged Brown leg has no rate exposure in the holdout
 → ($t = 0.05$). The strategies gain when yields rise. With FF5 + UMD + Delta10y + commodity controls,
 → holdout alphas are still -2.3% to -4.8% (t about -2).
5. **Two statistical diagnoses are off.**
 - The IC flip is 1.3 to 1.6 SE for raw and purified attention, not "about one". For the continuous signals
 → it is 2.8 to 2.9 NW SE ($p < 0.01$).
 - More Newey-West lags do not shrink the validation t (2.23 becomes 2.30 at 12 lags). They inflate the 24-
 → month COVID t to 5.3. The real problem there is small-sample p -values.

Every test it proposed that I could run on seen data fails to rescue the strategy. The timed-minus-always-on
 → FF5+UMD alpha peaks at $t = 1.09$. The Idea 2 post-event CAR has $t = -1.08$ (-0.63 ex-COVID). Momentum
 → and commodity controls do not absorb the signal.

Proposed tests that I ran on seen data (2010-2026), and what they show

Test (ChatGPT’s spec)	Result	Script

| E / critique 1: alpha of (timed minus always-on) after FF5+UMD | Never significant. 2010-2026 t -0.93 to
 → 0.78; Val at most 1.09; HO -1.11 to 1.10. The in-position dummy (unit position) gives Val t of 2.68
 → to 2.75 for 6m holds, but at most 1.14 for 2010-2026 excluding Mar 2020 to Dec 2021, and -1.5 to -2.1
 → in the HO. | c01 |

| Critique 2: drop Mar 2020 to Dec 2021; leave one industry out | 6m Val t 1.42 and 1.65 (ChatGPT’s bar is
 → 1.5); 3m 0.41 and 0.50. Leave-one-out 6m Val t 1.62 to 2.77; COVID 6m t collapses without Aero (0.03,
 → 0.07). HO stays negative in every case (-1.17 to -2.56). | c02 |

| A: 8+8 legs (Hardw and MedEq tie for the 8th Green slot) | 6m Val t 2.16 and 2.31; 3m 1.02 and 1.08.
 → Without Aero: 6m 1.61 and 1.85, 3m 0.67 and 0.69. HO -0.72 to -2.13. The 6m validation edge survives
 → broader legs; nothing else changes. | c11 |

| B / critique 5: FF5 + UMD + Delta10y + commodity attribution | Val 6m alpha 2.2% to 2.5% (t 2.46 and 2.72,
 → slightly higher than FF3); HO -2.3% to -4.8% (t -2.0 to -2.3). Commodity beta t -0.4 to -0.7. It does
 → not "fall below -1.5". | c04 |

| G: horse race with Brown’s trailing 3m and 12m returns | Attention is not absorbed. Val z t is -1.26 alone
 → and -1.84 with the controls; trailing 12m Brown return t is +2.0 (continuation). 2010-2026 z t -0.39
 → alone, -0.90 with controls. The p80 state correlates -0.25 with trailing 12m Brown return. | c08 |

| Idea 2: event-time profile, merged events, post-event dummy, pass if $t > 2$ ex 2020-21 | Rejected by its own
 → rule. CAR over $t+1$ to $t+6$: -1.29% (t -1.08, raw) and -1.01% (t -0.94, pure); ex-2020-21 -0.82% (t

```

    ↪ -0.63) and -0.67% (t -0.57), within one SE of zero. Post-event dummy t -0.40 and +0.54; ex-COVID
    ↪ -0.17 and +0.29. | c07 |
| Critique 6: continuous rule vs always-on at the same exposure | Sharpe over 2010-2026: 0.24 and 0.30 vs
    ↪ 0.20. HO: -0.60 and -0.47 vs -0.21. | c01 |
| Idea 3 premise: do the attention indices agree? | Team z vs CPU z +0.11 and vs MCCC z -0.06 (2010-2026).
    ↪ CPU and MCCC levels correlate 0.70 with each other, but only 0.15 and -0.04 with the team series.
    ↪ Running the frozen rule on CPU and MCCC belongs to module M1b and was not run here. | c06, c10 |

Not run: idea 1 (kept clean on purpose), idea 4 (covered by M4), idea 5, proposal F.

## What ChatGPT missed

1. What the attention variable measures. The prompt gave the variable as "climate-transition attention"
    ↪ with no source, and ChatGPT accepted that label. The series is FRED EMVENRGYENVREG, a count of
    ↪ newspaper articles about stock-market volatility that mention energy or environmental regulation,
    ↪ scaled to the VIX.
    - Its z-score correlates +0.41 with overall EMV and +0.29 with VIX (1993-2026), but +0.13 with CPU and
      ↪ 0.00 with MCCC.
    - Since 2010, crossings fall in high overall-EMV months 38% of the time, against 12% otherwise (odds ratio
      ↪ 4.5, p = 0.002).
    - The rule partly times general market-volatility news. A careful referee would have ranked idea 3 (swap
      ↪ in CPU or MCCC) first. (c10; M1_signal_audit_decomposition)

2. The holdout signal is degenerate. There are 39 zeros in 59 months from 2021-10, against 12 in 441
    ↪ before. In the holdout, zero months map to z between -1.39 and -0.59, and entries require any nonzero
    ↪ print. The IC "flip" and the holdout losses test a near-binary signal, not the validated one.
    ↪ ChatGPT placed the coverage risk in the wrong period. (c06)

3. Same-month look-ahead in the signal itself. Month-t EMV is used to trade at the close of month t.
    ↪ ChatGPT flagged vintage issues in the macro controls but not in the traded series, which moves the
    ↪ headline numbers (row 27).

4. Small-sample inference in COVID. Both Holm survivors are NW(6) t-stats on 24 months. With t(n-4) p-
    ↪ values none survive. Asking for more lags makes that t larger, not smaller.

5. Unequal exposure in the benchmark comparison. The "0.3% a year" compares a rule with mean exposure
    ↪ 0.43 to one with 0.66, and picks the best of six. The comparison should be exposure- or volatility-
    ↪ matched and should correct for selection.

6. Leg weighting. Equal weights give Ships (9 to 10 firms, about $30bn) the same 20% as Util (about $1.1
    ↪ tn). Ships alone accounts for 1.60 pp of the Original 3m rule's 3.36% annual holdout gross loss. (c02
    ↪ )

7. Power of a 48-month holdout. Eskildsen, Ibert, Jensen and Pedersen (2026) give the relevant arithmetic:
    ↪ expected t = 2 x Sharpe over 4 years, so even a true Sharpe of 0.5 gives an expected t of 1.0. This
    ↪ would have sharpened critique 3 more than the IC standard error does. (lit_digest section 7)

8. Emissions-measure gaps. EPA v1.3 has no electricity codes, so Util needs a patch. No intensity-per-
    ↪ output measure captures the downstream emissions that make Oil and Coal transition-exposed. Under EPA,
    ↪ Oil still does not make a five-industry Brown leg. (c05)

9. Alternative indices end early. CPU ends Sep 2025 and MCCC Jun 2025, leaving 10 and 13 holdout months
    ↪ uncovered. With the team's rule each also burns about 8 years at the start. (c06)

10. Not inferable from the prompt, listed for completeness: the missing Oct 2025 CPI print explains the whole
    ↪ purified-vs-original holdout gap (replication.md item 2).

```

A.2 Exchange 02: coding support (a second implementation of the frozen test)

Files: exchange/02_coding_support/. Answered by: a model in the ChatGPT role (see the provenance note). Date: 2026-09-26. Purpose: coding support.

Asked.	A pandas and statsmodels implementation of the frozen 1993–2009 test, one function per step with docstrings, a fixed seed, every silent reading as a parameter, and pre-run checks with stop rules.
Came back.	A 985-line script with 25 numbered assumptions and 13 pre-run checks with stop rules; after a follow-up naming six problems, six replaced functions, each with a unit test.
Verified.	Both versions run on the real data against my own implementation; aligned on two open readings, all 15 coefficients and all 5,000 shuffle draws agree to $3.3\text{e-}13$; mutation tests of its look-ahead audit.
Kept.	A second implementation that confirms the M8 verdict, and its null-seed harness, which showed that the pre-registered NW(6) t over-rejects (13 of 100 seeds) while the shuffle is correctly sized.

My evaluation

ChatGPT wrote a correct backtest inside a flawed test harness. I never used it as the run of record: I ran it on the real data as a second implementation and compared every component with my own (M8), which reproduces the team pipeline bitwise.

The core mathematics matched mine. The 985-line script ran the first time, with only file paths and `fac` supplied, and reached the same verdict (FAIL, 1 of 4; $\alpha -0.19\%/yr$, NW(6) $t -0.25$, vs. $-0.18\%/yr$, $t -0.27$). Two readings explain every numeric gap. With both switched to mine, it reproduces all 15 coefficients, the decomposition, the drop-one alphas and all 5,000 shuffle draws, to $3.3\text{e-}13$ at worst. The timing conventions (lagged hedge, publication lag, extend-not-stack holds, costs paid in $t+1$) needed no correction.

Three of the six problems came from my prompt. It described only the FF5 `fac`, so the code hedged with FF5 SMB (up to 3.5 pp a month off the team FF3 SMB); `b_t` moved by up to 0.09, and ChatGPT’s own check (b)2 stopped at $2.3\text{e-}3$ a month. “Keeps its sign” let (iv) pass a negative alpha, and “the seen 2010-2022 window” let seen mode run to 2022-12, where five holdout months flip the seen alpha from $-0.16\%/yr$ ($t -0.24$) to $+0.26\%/yr$ ($t 0.37$). The other three are ChatGPT’s: a broken instruction, a bug and a blind spot. It hard-coded a 60-calendar-month burn-in and disclosed it (assumption 11), although the prompt asked for every silent reading to be a parameter; the window start moved from 1994-03 to 1995-02, π from 0.729 to 0.809 and the shuffle p from 0.478 to 0.536. Its synthetic GS10, a random walk clipped at 0.5, sat on the floor (76% of 1995–2009 months for seed 0), so BOND went constant and $I \times \text{BOND}$ stood in for an I main effect: the positive control failed its own stop rule (4% of the planted effect in alpha, $t 0.98$), and seed 16 crashed the null-seed check. Its look-ahead test compared end-of-month states only and missed the two return-side bugs among the six I injected.

The follow-up fixed all six and broke one interface. All six unit tests passed the first time without edits. The defaults reproduce my implementation to $3.3\text{e-}13$ and the seen dry run ($-0.12\%/yr$, $t -0.19$), the positive control puts 97–104% of the planted shift in alpha ($t 3.5$ to 6.1), and the new audit flags 7 of 8 injected timing bugs, including both earlier misses. But the new `attention_signal` changed its output layout while the reply said everything else was unchanged. A literal drop-in crashes, and the obvious one-line patch runs without error on a window 14 months too long (from 1993-01, $n = 204$, $\alpha -0.13\%/yr$). It also told me to report a stale seen-window alpha ($-0.16\%/yr$, from its first version). It cannot run code, and the numbers it quotes about its own code are not computed.

Trust its formulas and audit its interfaces. Where the spec was exact, the code was exact. The errors sat outside the formulas: in readings the spec left open, in test scaffolding, and at the seam between old and new functions. None changed the verdict or the frozen rule. The independent reference made the exchange useful: each discrepancy became a switch tied to specific lines, and mutation tests measured what its checks could see. Its own acceptance test

for the fix (“should again match your implementation to $1e-13$ ”) presumes that reference exists. The repaired harness also audited our own inference: on 100 null seeds the calendar shuffle is correctly sized (6 rejections at 5%), but the pre-registered NW(6) t over-rejects (13), a bias that cannot rescue a negative alpha. Run against a reference and never trusted blind, ChatGPT is a cheap second implementation for backtest code, not a verifier of it.

Prompt (verbatim) prompt.md, 997 words (957 excluding Markdown symbols).

Write Python code for a pre-registered backtest I will run exactly once. You do not have the data,
 ↪ so report no results.

CONTEXT. This is my MFE 230GA final project at Berkeley. Our team’s strategy shorts a factor-
 ↪ hedged “Brown” industry leg after attention spikes. It failed its 2022-2026 holdout, and we
 ↪ recommend “Do not implement”. I froze one rule for 1993-2009, the one tradable window
 ↪ nobody has backtested. I will debug your code on synthetic data and the seen 2010-2022
 ↪ window first, so the test window is a parameter and nothing is tuned.

INPUTS. Returns are monthly decimals.

1. ‘ff49_industry_monthly.csv’: a ‘date’ column (month-end, e.g. 1926-07-31) and 49 Fama-French
 ↪ industry columns of value-weighted returns, NaN before an industry exists. Brown = Util,
 ↪ Ships, Aero, Steel, BldMt. The Brown leg R^B_t is their equal-weighted mean minus RF.
2. ‘fac’: a DataFrame parsed from Ken French’s FF5 and momentum files, month-end DatetimeIndex
 ↪ from 1963-07, columns Mkt-RF, SMB, HML, RMW, CMA, RF, UMD.
3. FRED CSVs with columns ‘observation_date’ (first of month; map to month-end) and the series id:
 ↪ EMVENRGYENVREG and EMVOVERALLEMV (monthly from 1985-01; the first has exact zeros, the
 ↪ second none), GS10 (monthly average 10-year yield, percent), MCOILWTICO (monthly average
 ↪ WTI price, from 1986-01), VIXCLS (daily closes from 1990-01-02, blank on holidays).

THE TEAM TRADE (validated, unchanged).

- Hedge: OLS of R^B on a constant and $f = (\text{Mkt-RF}, \text{SMB}, \text{HML})$ over months $t-59$ to t gives $(a_t, b_t$
 ↪ $); \text{residual } e_t = R^B_t - a_{t-1} - b_{t-1}'f_t$.
- Size: $m_t = \min(1, 0.05 / (\text{sqrt}(12) * \text{sd}_t))$, $\text{sd}_t = 36\text{-month rolling std (ddof=1) of } e$ through
 ↪ t .
- Position $w_t = -m_t$ in a hold month, else 0, set at the end of month t ; it earns $w_t * R^B_{t+1} -$
 ↪ $w_t * b_t'f_{t+1}$.
- Costs per unit of turnover: 10 bp on $|Deltaw|$, 5 bp on the Mkt-RF overlay change, 25 bp on SMB
 ↪ and HML overlay changes; a trade at the end of t pays in $t+1$.
- A crossing is an extreme month whose previous month was not extreme. The hold covers the
 ↪ crossing month and the next five. With the publication lag, the hold state at the end of t
 ↪ uses the extreme flag of $t-1$.
- Always-on benchmark R^A_0 : every month is a hold month.

THE FROZEN RULE.

1. Signal: $s_t = \text{EMVENRGYENVREG}_t / \text{EMVOVERALLEMV}_t$, observed one month late (the value for month
 ↪ t is usable at the end of month $t+1$).
2. Zeros: a zero month is missing, not low attention. It cannot trigger a position, and it is
 ↪ dropped from both the z-score window and the percentile history. The z-score needs at least
 ↪ 48 nonzero months in the trailing 60; if more than 10% of the trailing 60 months are zero,
 ↪ the signal is off.
3. Extreme month: $z_t = 60\text{-month rolling z-score of } \log(s_t) \text{ over nonzero months; extreme when } z_t$
 ↪ exceeds its expanding, past-only 80th percentile (team convention: at least 60 prior z
 ↪ observations).
4. Trade: the 6-month rule above. A new crossing during a hold extends the hold; positions never
 ↪ stack.
5. Sample: 1993-01 to 2009-12 (or from the first month the rule can fire, if later), run once.
6. Pass bar, all four required: (i) the timing alpha below is positive with $t \geq 2$ and p from $t(n-$
 ↪ $k)$; (ii) calendar-shuffle $p \leq 0.05$ (move the same hold windows to random dates, 5,000
 ↪ times, recompute alpha); (iii) at least 8 independent episodes (merge holds that touch or

→ overlap); (iv) alpha keeps its sign when each Brown industry is dropped in turn (rebuild
→ leg, hedge, sizing, R^{AO} and π each time).

ATTRIBUTION. $D_t = R^T_t - \pi \cdot R^{AO}_t$, net of costs, with R^T the timed strategy and $\pi = \text{mean} |$
→ $\text{timed position} | / \text{mean} | \text{always-on position} |$ over the test window. Regress D_t on $FF5 + \text{UMD},$
→ BOND , the WTI log return, $dVIX_t$ (change in the monthly mean of daily VIX), $d\log$
→ EMVOVERALLEMV_t , and $I_{t-1} \times \{\text{Mkt-RF}, \text{HML}, \text{BOND}, dVIX\}$, where $I_{t-1} = 1$ if the strategy
→ held a position entering month t . Alpha is the intercept. Newey-West with 6 lags (12 as a
→ check); p from $t(n-k)$, k = number of estimated parameters. Report $\text{mean}(D) = \alpha + \text{sum}$
→ $\beta \cdot \text{mean}(F) + \text{sum } \gamma \cdot \text{mean}(I \cdot F)$.

BOND is the monthly excess return of a constant-maturity 10-year Treasury from GS10 (percent to
→ decimal): $r_t = y_{t-1}/12 - D_{t-1} \cdot (y_t - y_{t-1}) + 0.5 \cdot C_{t-1} \cdot (y_t - y_{t-1})^2$, with
→ D and C the modified duration and convexity of a 10-year semiannual par bond at yield y_{t-1} ,
→ minus RF.

CONSTRAINTS.

- No look-ahead: every input used at the end of month t is known then (the signal a month later).
→ Attribution regressors are contemporaneous by design; do not lag them.
- pandas, numpy and statsmodels only. 'sm.OLS(...).fit(cov_type="HAC", cov_kws={"maxlags": 6},
→ use_t=True)' matches our team's Newey-West t and gives p from $t(n-k)$. Count k from the
→ design matrix.
- One function per step, each with a docstring giving inputs, outputs and timing convention. Fixed
→ seed, no parameter search, no plots.

Where the spec is silent, pick one reading, make it a parameter and say why. Three I see: whether
→ a zero month between two extreme months makes the next one a new crossing; how the shuffle
→ places windows and whether π and I are recomputed per draw; what happens to a hold still
→ open in 2009-12.

OUTPUTS, as DataFrames, also printed.

1. Pass/fail table: one row per component (i) to (iv), with statistic, bar, value, pass/fail; then
→ the overall verdict.
2. Attribution: coefficient, NW(6) t , NW(12) t and p per regressor; n , k , R-squared, π , window
→ dates.
3. Decomposition: $\text{mean}(D)$, α , every $\beta \cdot \text{mean}(F)$ and $\gamma \cdot \text{mean}(I \cdot F)$ term, and the identity
→ gap (zero to machine precision).
4. Diagnostics: first month the rule can fire, zero-month counts, crossing dates, episode list,
→ leave-one-out alphas and t .
5. Shuffle: actual alpha, null quantiles, p .

AFTER THE CODE.

- (a) Number every assumption you made beyond this spec.
- (b) The checks you would run before my single real run, each with the result that would make you
→ stop, including a look-ahead test.

Reply (verbatim; model in the ChatGPT role) chatgpt_response.md, 7,110 words
(6,149 excluding Markdown symbols).

Below is one self-contained script (pandas, numpy and statsmodels only). Each step is a separate function,
→ and each docstring gives the step's inputs, outputs and timing. All choices live in a frozen 'Config'.
→ The test window is a parameter. Nothing is estimated from the test window except the attribution
→ regression itself. The script also contains a synthetic-data generator, a look-ahead truncation test
→ and a planted-effect positive control, so you can debug end to end before the real run.

I have not executed this code. Treat the checks in (b) as mandatory before the single real run.

```
“python
```

```

"""
brown_attention_prereg.py

Pre-registered, run-once backtest of the frozen Brown-attention rule (MFE 230GA).
pandas / numpy / statsmodels only. No plots, no parameter search, fixed seed.

Order of use
1. MODE = "synthetic": debug end to end on random data (+ look-ahead test, planted effect).
2. MODE = "seen": 2010-01..2022-12, compare with the team's existing numbers.
3. MODE = "final": 1993-01..2009-12, exactly once, no code change after step 2.
"""
from __future__ import annotations

from dataclasses import asdict, dataclass, replace

import numpy as np
import pandas as pd
import statsmodels.api as sm

pd.set_option("display.width", 220)
pd.set_option("display.max_columns", 40)
pd.set_option("display.max_rows", 500)
pd.set_option("display.float_format", "{:.6g}".format)

BROWN = ("Util", "Ships", "Aero", "Steel", "BldMt")

# =====
# 0. Frozen configuration
# =====
@dataclass(frozen=True)
class Config:
    """Every number the test uses. Nothing here is estimated or tuned."""
    # test window in RETURN months (inclusive); start moves later if the rule cannot fire yet
    test_start: str = "1993-01"
    test_end: str = "2009-12"
    brown: tuple = BROWN
    hedge_factors: tuple = ("Mkt-RF", "SMB", "HML")
    # team trade (validated, unchanged)
    hedge_window: int = 60
    vol_window: int = 36
    vol_target: float = 0.05
    cost_w: float = 0.0010
    cost_mkt: float = 0.0005
    cost_smb: float = 0.0025
    cost_hml: float = 0.0025
    hold_months: int = 6
    pub_lag: int = 1
    # frozen signal
    z_window: int = 60
    z_min_nonzero: int = 48
    max_zero_share: float = 0.10
    pct: float = 80.0
    min_pct_history: int = 60
    # attribution
    attr_factors: tuple = ("Mkt-RF", "SMB", "HML", "RMW", "CMA", "UMD",
                          "BOND", "WTI", "dVIX", "dlogEMV")
    interact: tuple = ("Mkt-RF", "HML", "BOND", "dVIX")
    nw_lags: int = 6
    nw_lags_check: int = 12
    # pass bars
    t_bar: float = 2.0
    p_bar: float = 0.05
    min_episodes: int = 8
    # readings of the points the spec leaves open -- see notes (a)1-(a)3
    bridge_missing: bool = True # extreme, zero, extreme = one run (no new crossing)
    bridge_limit: int = 6 # ... unless more than 6 consecutive missing months
    end_policy: str = "truncate" # "truncate" or "liquidate" a hold still open at test_end

```

```

shuffle_draws: int = 5000
shuffle_min_gap: int = 1 # shuffled episodes never touch, so K is preserved
shuffle_recompute_pi_I: bool = True
seed: int = 230

# =====
# Calendar / IO helpers
# =====
def _me(x) -> pd.Timestamp:
    """Month-end timestamp (midnight) of any date-like."""
    return (pd.Timestamp(x) + pd.offsets.MonthEnd(0)).normalize()

def _to_month_end(values) -> pd.DatetimeIndex:
    """Vectorised month-end mapping; FRED first-of-month dates map to the same month's end."""
    return (pd.DatetimeIndex(pd.to_datetime(values)) + pd.offsets.MonthEnd(0)).normalize()

def _month_range(a, b) -> pd.DatetimeIndex:
    return pd.date_range(_me(a), _me(b), freq=pd.offsets.MonthEnd())

def _shift_months(ts, k: int) -> pd.Timestamp:
    return (_me(ts) + pd.offsets.MonthEnd(k)).normalize()

def _position_months(window: pd.DatetimeIndex) -> pd.DatetimeIndex:
    """Position months whose positions earn inside the window (return month t <- position t-1)."""
    return _month_range(_shift_months(window[0], -1), _shift_months(window[-1], -1))

def _assert_monthly(idx: pd.DatetimeIndex, name: str) -> None:
    if idx.has_duplicates:
        raise ValueError(f"{name}: duplicate months")
    if len(idx) != len(_month_range(idx.min(), idx.max())):
        raise ValueError(f"{name}: gaps in the monthly index")

def _assert_decimal(frame: pd.DataFrame, name: str) -> None:
    q = np.nanpercentile(np.abs(frame.to_numpy(float)), 99)
    if q > 1.0:
        raise ValueError(f"{name}: 99th pct |return| = {q:.2f}; looks like percent, not decimals")

def _read(src) -> pd.DataFrame:
    return src.copy() if isinstance(src, pd.DataFrame) else pd.read_csv(src)

def _show(title: str, obj) -> None:
    print("\n" + "=" * 100 + f"\n{title}\n" + "=" * 100)
    print(obj.to_string() if hasattr(obj, "to_string") else obj)

# =====
# 1. Loaders
# =====
def load_ff49(src) -> pd.DataFrame:
    """
    Step 1a. Fama-French 49 industries.
    Input : path or DataFrame with 'date' (month-end) + 49 industry columns, monthly decimals.
    Output: DataFrame on a sorted, gap-free month-end index; NaN where an industry does not
            exist; values <= -0.99 (French's -99.99 code) set to NaN.
    Timing: row t = value-weighted return over calendar month t.
    """
    df = _read(src)
    df.columns = [str(c).strip() for c in df.columns]
    idx = _to_month_end(df.pop("date"))

```

```

df = df.apply(pd.to_numeric, errors="coerce")
df.index = idx
df = df.sort_index().mask(lambda x: x <= -0.99)
_assert_monthly(df.index, "ff49")
_assert_decimal(df, "ff49")
return df

def load_factors(fac: pd.DataFrame) -> pd.DataFrame:
    """
    Step 1b. French FF5 + momentum.
    Input : DataFrame with Mkt-RF, SMB, HML, RMW, CMA, RF, UMD (decimals), month-end index.
    Output: those columns on a sorted, gap-free month-end index.
    Timing: row t = return over month t.
    """
    cols = ["Mkt-RF", "SMB", "HML", "RMW", "CMA", "RF", "UMD"]
    out = fac[cols].astype(float).copy()
    out.index = _to_month_end(out.index)
    out = out.sort_index()
    _assert_monthly(out.index, "fac")
    _assert_decimal(out, "fac")
    return out

def load_fred_monthly(src, series_id: str) -> pd.Series:
    """
    Step 1c. Monthly FRED series (EMV trackers, GS10, MCOILWTIC0).
    Input : path or DataFrame with 'observation_date' (first of month) and 'series_id'.
    Output: Series on a complete month-end index; './blank -> NaN; exact zeros kept.
    Timing: the value dated month t describes month t (publication lag handled downstream).
    """
    df = _read(src)
    s = pd.Series(pd.to_numeric(df[series_id], errors="coerce").to_numpy(float),
                  index=_to_month_end(df["observation_date"]), name=series_id).sort_index()
    if s.index.has_duplicates:
        raise ValueError(f"{series_id}: more than one observation per month")
    return s.reindex(_month_range(s.index.min(), s.index.max()))

def load_vix_monthly(src, series_id: str = "VIXCLS") -> pd.Series:
    """
    Step 1d. Monthly mean of daily VIX closes.
    Input : path or DataFrame with daily 'observation_date' and VIXCLS (blank on holidays).
    Output: Series on a complete month-end index = mean of the non-missing closes in month t.
    Timing: known at the end of month t.
    """
    df = _read(src)
    d = pd.Series(pd.to_numeric(df[series_id], errors="coerce").to_numpy(float),
                  index=pd.DatetimeIndex(pd.to_datetime(df["observation_date"]))).dropna()
    m = d.groupby(_to_month_end(d.index)).mean()
    m.index = pd.DatetimeIndex(m.index)
    return m.reindex(_month_range(m.index.min(), m.index.max())).rename("VIX")

# =====
# 2-4. Brown leg, hedge, sizing
# =====
def brown_leg(ff49: pd.DataFrame, rf: pd.Series, industries) -> pd.Series:
    """
    Step 2. Brown leg.
    Inputs : industry returns, RF, list of Brown industries.
    Output : R^B_t = equal-weighted mean of the listed industries available in t, minus RF_t.
    Timing : month-t return.
    """
    missing = [c for c in industries if c not in ff49.columns]
    if missing:
        raise KeyError(f"industries not found in FF49 file: {missing}")
    leg = ff49[list(industries)].mean(axis=1, skipna=True)

```

```

return (leg - rf.reindex(leg.index)).rename("RB")

def rolling_hedge(rb: pd.Series, f: pd.DataFrame, window: int = 60):
    """
    Step 3. Rolling factor hedge.
    Inputs : R^B_t and hedge factors f_t = (Mkt-RF, SMB, HML) on the same complete index.
    Output : a (Series) and b (DataFrame, columns = f.columns).
    Timing : (a_t, b_t) = OLS of R^B on [1, f] over months t-window+1..t, i.e. data through t,
              known at the end of t. NaN unless all 'window' months are complete.
    """
    idx = rb.index
    X = np.column_stack([np.ones(len(idx)), f.reindex(idx).to_numpy(float)])
    y = rb.to_numpy(float)
    coef = np.full((len(idx), X.shape[1]), np.nan)
    for i in range(window - 1, len(idx)):
        Xi, yi = X[i - window + 1:i + 1], y[i - window + 1:i + 1]
        if np.isfinite(Xi).all() and np.isfinite(yi).all():
            coef[i] = np.linalg.lstsq(Xi, yi, rcond=None)[0]
    a = pd.Series(coef[:, 0], index=idx, name="a")
    b = pd.DataFrame(coef[:, 1:], index=idx, columns=list(f.columns))
    return a, b

def hedge_residual_and_size(rb: pd.Series, f: pd.DataFrame, a: pd.Series, b: pd.DataFrame,
                           cfg: Config):
    """
    Step 4. Out-of-sample residual and position size.
    Inputs : R^B, f, (a, b) from step 3.
    Output : e_t = R^B_t - a_{t-1} - b_{t-1}'f_t;
              sd_t = std(e_{t-35..t}), ddof=1, all 36 required;
              m_t = min(1, vol_target / (sqrt(12) sd_t)).
    Timing : e_t, sd_t, m_t use data through t, known at the end of t.
    """
    fitted = a.shift(1) + (b.shift(1) * f[b.columns]).sum(axis=1, min_count=b.shape[1])
    e = (rb - fitted).rename("e")
    sd = e.rolling(cfg.vol_window, min_periods=cfg.vol_window).std(ddof=1).rename("sd")
    m = np.minimum(1.0, cfg.vol_target / (np.sqrt(12.0) * sd)).rename("m")
    return e, sd, m

# =====
# 5-7. Signal, crossings, hold
# =====
def attention_signal(emv_cat: pd.Series, emv_all: pd.Series, cfg: Config) -> pd.DataFrame:
    """
    Step 5. Frozen signal, indexed by SIGNAL month tau.
    Inputs : EMVENRGYENVREG and EMVOVERALLEMV (monthly, month-end index).
    Output : DataFrame with
              s EMVENRGYENVREG / EMVOVERALLEMV
              missing s is zero (or NaN): missing, never "low attention"
              n_missing60 missing months in tau-59..tau (NaN until 60 calendar months exist)
              off month is nonzero but > 10% of the trailing 60 months are missing
              z (log s_tau - mean) / sd over the NONZERO months of tau-59..tau
                  (needs >= 48 nonzero and <= 6 missing; NaN otherwise)
              thr 80th percentile of all valid z strictly before tau (needs >= 60)
              extreme 1 if z > thr, 0 if not, NaN if z or thr undefined
    Timing : row tau uses data dated <= tau only, but tau's data are published at the end of
              tau+1, so row tau may drive a position no earlier than the end of tau+pub_lag.
              Missing months are dropped from the z window and the percentile history.
    """
    start = max(emv_cat.first_valid_index(), emv_all.first_valid_index())
    end = min(emv_cat.last_valid_index(), emv_all.last_valid_index())
    idx = _month_range(start, end)
    s = emv_cat.reindex(idx) / emv_all.reindex(idx)
    missing = ~(s > 0) # exact zero or NaN
    W = cfg.z_window
    max_missing = int(np.floor(cfg.max_zero_share * W + 1e-9)) # = 6

```

```

logs = np.log(s.where(~missing))
n_miss = missing.astype(float).rolling(W, min_periods=W).sum()
n_nz = W - n_miss
mu = logs.rolling(W, min_periods=cfg.z_min_nonzero).mean()
sd = logs.rolling(W, min_periods=cfg.z_min_nonzero).std(ddof=1)
ok = (~missing) & (n_nz >= cfg.z_min_nonzero) & (n_miss <= max_missing) & (sd > 0)
z = ((logs - mu) / sd).where(ok)

zv = z.to_numpy(float)
thr = np.full(len(zv), np.nan)
hist: list = []
for i, zi in enumerate(zv):
    if len(hist) >= cfg.min_pct_history:
        thr[i] = np.percentile(hist, cfg.pct) # past-only: z_tau not yet appended
    if np.isfinite(zi):
        hist.append(zi)
defined = np.isfinite(zv) & np.isfinite(thr)
extreme = np.where(defined, (np.nan_to_num(zv) > np.nan_to_num(thr)).astype(float), np.nan)

return pd.DataFrame({
    "s": s, "missing": missing, "n_missing60": n_miss,
    "off": (~missing) & (n_miss > max_missing),
    "z": z, "thr": thr, "extreme": extreme}, index=idx)

def crossings(extreme: pd.Series, cfg: Config) -> pd.Series:
    """
    Step 6. Crossings.
    Input : extreme flag (1/0/NaN) by signal month.
    Output : bool Series, True in signal month tau if extreme_tau == 1 and the previous month
            was not extreme. With bridge_missing, a missing flag (zero or off month) carries
            the last defined flag forward for up to bridge_limit months, so
            extreme -> zero -> extreme is ONE run. Undefined previous state = not extreme.
    Timing : uses flags dated <= tau; usable at the end of tau+pub_lag.
    """
    prev = extreme.ffill(limit=cfg.bridge_limit) if cfg.bridge_missing else extreme
    prev = prev.shift(1)
    return ((extreme == 1) & (prev != 1)).rename("crossing")

def hold_positions(cross: pd.Series, index: pd.DatetimeIndex, cfg: Config) -> pd.Series:
    """
    Step 7. Hold state in position time.
    Inputs : crossings (signal time) and the master monthly index.
    Output : bool Series on 'index'; True = a short position is SET at the end of month t.
    Timing : hold in signal time covers tau..tau+5 after each crossing (a new crossing extends,
            positions never stack); with the publication lag the hold at the end of t uses
            signal month t-1: hold_pos_t = hold_sig_{t-pub_lag}. So a crossing in tau gives
            positions set at the end of tau+1..tau+6, earning tau+2..tau+7.
    """
    c = cross.reindex(index, fill_value=False).astype(float)
    hold_sig = c.rolling(cfg.hold_months, min_periods=1).max() > 0
    return hold_sig.shift(cfg.pub_lag, fill_value=False).astype(bool).rename("hold")

# =====
# 8. Trade engine (timed and always-on)
# =====
def run_trade(rb: pd.Series, f: pd.DataFrame, b: pd.DataFrame, m: pd.Series,
             hold: pd.Series, cfg: Config) -> pd.DataFrame:
    """
    Step 8. Team trade.
    Inputs : R^B_t, hedge factors, b_t, m_t, hold (bool, set at end of t).
    Output : DataFrame on rb.index:
        hold, m, w (w_t = -m_t if hold else 0, set at end of t),
        h<factor> (overlay = -w_t b_t, set at end of t), trade_cost (trade at end of t),
        w_prev = w_{t-1}, hedged_t = R^B_t - b_{t-1}'f_t, gross_t = w_{t-1} hedged_t,
        cost_t = trade_cost_{t-1} (paid in t), net_t = gross_t - cost_t.
    """

```

```

Timing : row t's net return is earned in month t by the position set at the end of t-1.
"""
idx = rb.index
hf = list(cfg.hedge_factors)
f = f.reindex(idx)[hf]
b = b.reindex(idx)[hf]
hold = hold.reindex(idx, fill_value=False).astype(bool)
mv = m.reindex(idx).to_numpy(float)
w = np.where(hold.to_numpy(), -mv, 0.0)
bv = b.to_numpy(float)
h = np.where(w[:, None] == 0.0, 0.0, -w[:, None] * bv)
dw = np.abs(np.diff(w, prepend=0.0))
dh = np.abs(np.diff(h, axis=0, prepend=np.zeros((1, h.shape[1]))))
unit = np.array([cfg.cost_mkt, cfg.cost_smb, cfg.cost_hml])
trade_cost = cfg.cost_w * dw + dh @ unit

out = pd.DataFrame({"hold": hold, "m": mv, "w": w}, index=idx)
for j, c in enumerate(hf):
    out[f"h_{c}"] = h[:, j]
out["trade_cost"] = trade_cost
out["w_prev"] = out["w"].shift(1)
out["hedged"] = rb - (b.shift(1) * f).sum(axis=1, min_count=len(hf))
out["gross"] = out["w_prev"] * out["hedged"]
out["cost"] = out["trade_cost"].shift(1)
out["net"] = out["gross"] - out["cost"]
return out

def leg_pipeline(ff49: pd.DataFrame, fac: pd.DataFrame, industries, hold: pd.Series,
                index: pd.DatetimeIndex, cfg: Config) -> dict:
    """
    Steps 2-4 + 8 for one set of Brown industries (used for the main run and for each
    leave-one-out rebuild).
    Output : dict with rb, f, a, b, e, sd, m, tr_T (timed book), tr_A0 (always-on book).
    Timing : as in steps 2-4 and 8; the hold series is taken as given.
    """
    f = fac.reindex(index)[list(cfg.hedge_factors)]
    rb = brown_leg(ff49.reindex(index), fac["RF"].reindex(index), industries)
    a, b = rolling_hedge(rb, f, cfg.hedge_window)
    e, sd, m = hedge_residual_and_size(rb, f, a, b, cfg)
    tr_T = run_trade(rb, f, b, m, hold, cfg)
    tr_A0 = run_trade(rb, f, b, m, pd.Series(True, index=index), cfg)
    return dict(rb=rb, f=f, a=a, b=b, e=e, sd=sd, m=m, tr_T=tr_T, tr_A0=tr_A0)

# =====
# 9-10. Attribution regressors
# =====

def par_bond_duration_convexity(y, maturity: float = 10.0, freq: int = 2):
    """
    Step 9a. Modified duration (years) and convexity (years^2) of a par bond.
    Input : annual yield(s) y in decimals, semiannual compounding and coupons.
    Output: (D, C) arrays, from exact cash flows.
    """
    y = np.asarray(y, dtype=float).reshape(-1, 1)
    n = int(round(maturity * freq))
    i = np.arange(1, n + 1, dtype=float)[None, :]
    g = 1.0 + y / freq
    cf = np.repeat(y / freq, n, axis=1)
    cf[:, -1] += 1.0
    disc = g ** (-i)
    price = (cf * disc).sum(axis=1)
    dur = (cf * (i / freq) * disc).sum(axis=1) / g[:, 0] / price
    conv = (cf * (i / freq) * ((i + 1) / freq) * disc).sum(axis=1) / g[:, 0] ** 2 / price
    return dur, conv

def bond_excess_return(gs10: pd.Series, rf: pd.Series) -> pd.Series:

```

```

"""
Step 9b. BOND_t = y_{t-1}/12 - D_{t-1} dy_t + 0.5 C_{t-1} dy_t^2 - RF_t,
y = GS10/100, dy_t = y_t - y_{t-1}, D and C of a 10y semiannual par bond at y_{t-1}.
Timing: month-t excess return (contemporaneous regressor).
"""
y = (gs10 / 100.0).astype(float)
y_prev = y.shift(1)
dy = y - y_prev
dur, conv = par_bond_duration_convexity(y_prev.to_numpy())
r = y_prev / 12.0 - dur * dy + 0.5 * conv * dy ** 2
return (r - rf.reindex(r.index)).rename("BOND")

def attribution_factors(fac: pd.DataFrame, bond: pd.Series, wti: pd.Series, vix_m: pd.Series,
                      emv_all: pd.Series, index: pd.DatetimeIndex) -> pd.DataFrame:
    """
    Step 10. Month-t regressors, contemporaneous by design (NOT lagged):
    FF5 + UMD, BOND_t, WTI_t = log(P_t / P_{t-1}), dVIX_t = VIXbar_t - VIXbar_{t-1},
    dlogEMV_t = log EMVOVERALLEMV_t - log EMVOVERALLEMV_{t-1}.
    All differences are taken on complete monthly calendars before reindexing.
    """
    F = fac.reindex(index)[["Mkt-RF", "SMB", "HML", "RMW", "CMA", "UMD"]].copy()
    F["BOND"] = bond.reindex(index)
    F["WTI"] = np.log(wti).diff().reindex(index)
    F["dVIX"] = vix_m.diff().reindex(index)
    F["dlogEMV"] = np.log(emv_all).diff().reindex(index)
    return F

# =====
# 11. Window, timing difference D
# =====
def test_window(sig: pd.DataFrame, index: pd.DatetimeIndex, cfg: Config):
    """
    Step 11a. Test window.
    Output : (first_signal, first_position, first_return, window).
    first_signal = first signal month with a defined flag (valid z, >= 60 prior z)
    first_position = first month-end a position can be set = first_signal + pub_lag
    first_return = first return month that position affects = first_position + 1
    window = RETURN months [max(test_start, first_return), test_end]
    """
    elig = sig["extreme"].dropna()
    if elig.empty:
        raise ValueError("the signal never produces a defined flag")
    first_sig = elig.index[0]
    first_pos = _shift_months(first_sig, cfg.pub_lag)
    first_ret = _shift_months(first_pos, 1)
    start, end = max(_me(cfg.test_start), first_ret), _me(cfg.test_end)
    if end > index[-1] or start > end:
        raise ValueError(f"window {start:%Y-%m}..{end:%Y-%m} not covered by the data")
    return first_sig, first_pos, first_ret, _month_range(start, end)

def _liquidation_cost(tr: pd.DataFrame, t_end: pd.Timestamp, cfg: Config) -> float:
    """Cost of closing, at the end of t_end, the book set at the end of t_end - 1."""
    prev = _shift_months(t_end, -1)
    h = np.abs(tr.loc[prev, [f"h_{c}" for c in cfg.hedge_factors]]).to_numpy(float))
    unit = np.array([cfg.cost_mkt, cfg.cost_smb, cfg.cost_hml])
    return float(cfg.cost_w * abs(tr.loc[prev, "w"]) + h @ unit)

def timing_difference(tr_T: pd.DataFrame, tr_AO: pd.DataFrame, window: pd.DatetimeIndex,
                    cfg: Config, pi: float | None = None):
    """
    Step 11b. D_t = R^T_t - pi R^AO_t over the window's return months, net of costs.
    pi = mean|w^T_{t-1}| / mean|w^AO_{t-1}| over the same months (positions that earn there),
    unless pi is passed in. end_policy="liquidate" charges both books' closing cost in the last
    window month; "truncate" does not (that trade would be paid after the window).
    """

```

```

Output : D (Series), pi (float), I_prev (0/1: position held entering month t).
"""
T, A = tr_T.loc>window], tr_A0.loc>window]
cols = ["w_prev", "gross", "cost", "net"]
if not (np.isfinite(T[cols].to_numpy(float)).all() and np.isfinite(A[cols].to_numpy(float)).all()):
    raise ValueError("non-finite strategy values inside the test window")
if pi is None:
    pi = T["w_prev"].abs().mean() / A["w_prev"].abs().mean()
RT, RA = T["net"].copy(), A["net"].copy()
if cfg.end_policy == "liquidate":
    RT.iloc[-1] -= _liquidation_cost(tr_T, window[-1], cfg)
    RA.iloc[-1] -= _liquidation_cost(tr_A0, window[-1], cfg)
elif cfg.end_policy != "truncate":
    raise ValueError(f"unknown end_policy {cfg.end_policy}")
D = (RT - pi * RA).rename("D")
I_prev = tr_T["hold"].shift(1, fill_value=False).loc>window].astype(float).rename("I_prev")
return D, float(pi), I_prev

# =====
# 12. Attribution regression and decomposition
# =====
def design_matrix(F: pd.DataFrame, I_prev: pd.Series, cfg: Config) -> pd.DataFrame:
    """
    Step 12a. X_t = [const, FF5+UMD, BOND, WTI, dVIX, dlogEMV, I_{t-1} x {Mkt-RF, HML, BOND, dVIX}].
    No I_{t-1} main effect (the spec lists only the interactions). k = X.shape[1] = 15.
    """
    F = F.loc[I_prev.index, list(cfg.attr_factors)]
    X = F.copy()
    for c in cfg.interact:
        X[f"I*{c}"] = I_prev.to_numpy() * F[c].to_numpy()
    return sm.add_constant(X, has_constant="add")

def attribution_regression(D: pd.Series, X: pd.DataFrame, cfg: Config):
    """
    Step 12b. OLS of D on X; Newey-West (Bartlett) HAC with nw_lags and nw_lags_check.
    use_t=True -> t statistics and p-values from t(n-k), k counted from X.
    Output : coefficient table, NW(6) fit, NW(12) fit. Raises on missing data or rank deficiency.
    """
    bad = X.isna().any(axis=1) | D.isna()
    if bad.any():
        raise ValueError(f"missing regressors/returns in {list(X.index[bad.to_numpy()].strftime('%Y-%m'))}")
    if np.linalg.matrix_rank(X.to_numpy(float)) < X.shape[1]:
        raise ValueError("design matrix is rank deficient (e.g. I_{t-1} never or always 1)")
    f6 = sm.OLS(D, X).fit(cov_type="HAC", cov_kws={"maxlags": cfg.nw_lags}, use_t=True)
    f12 = sm.OLS(D, X).fit(cov_type="HAC", cov_kws={"maxlags": cfg.nw_lags_check}, use_t=True)
    n, k = X.shape
    if int(round(f6.df_resid)) != n - k:
        raise RuntimeError("df_resid != n - k")
    tab = pd.DataFrame({
        "coef": f6.params,
        f"t NW({cfg.nw_lags})": f6.tvalues,
        f"p NW({cfg.nw_lags})": f6.pvalues,
        f"t NW({cfg.nw_lags_check})": f12.tvalues,
        f"p NW({cfg.nw_lags_check})": f12.pvalues,
    }).rename(index={"const": "alpha"})
    return tab, f6, f12

def decompose_mean(D: pd.Series, X: pd.DataFrame, fit) -> pd.DataFrame:
    """
    Step 12c. mean(D) = alpha + sum beta_j mean(F_j) + sum gamma_j mean(I F_j).
    Exact for OLS with a constant (residuals sum to zero); the gap is reported.
    """
    contrib = fit.params * X.mean()
    rows = {"mean(D)": D.mean(), "alpha": contrib["const"]}
    for c in X.columns.drop("const"):

```

```

        rows[f"gamma*mean({c})" if c.startswith("I*") else f"beta*mean({c})"] = contrib[c]
    rows["sum of terms"] = contrib.sum()
    rows["identity gap"] = D.mean() - contrib.sum()
    return pd.Series(rows, name="value").to_frame()

# =====
# 13. Episodes and crossings
# =====
def hold_episodes(hold: pd.Series, pos_months: pd.DatetimeIndex) -> pd.DataFrame:
    """
    Step 13a. Independent episodes.
    Inputs : hold (bool, position time) and the position months that earn inside the window.
    Output : one row per maximal run of consecutive hold months (holds that touch or overlap
              merge), with position/return dates and whether the run was already open before,
              or is still open after, the window.
    """
    cols = ["position_from", "position_to", "months", "return_from", "return_to",
            "open_before_window", "open_after_window"]
    h = hold.reindex(pos_months, fill_value=False).to_numpy(bool)
    before = bool(hold.get(_shift_months(pos_months[0], -1), False))
    after = bool(hold.get(_shift_months(pos_months[-1], 1), False))
    rows, i = [], 0
    while i < len(h):
        if not h[i]:
            i += 1
            continue
        j = i
        while j + 1 < len(h) and h[j + 1]:
            j += 1
        rows.append({"position_from": pos_months[i], "position_to": pos_months[j],
                    "months": j - i + 1,
                    "return_from": _shift_months(pos_months[i], 1),
                    "return_to": _shift_months(pos_months[j], 1),
                    "open_before_window": i == 0 and before,
                    "open_after_window": j == len(h) - 1 and after})
        i = j + 1
    return pd.DataFrame(rows, columns=cols)

def crossing_table(sig: pd.DataFrame, cross: pd.Series, pos_months: pd.DatetimeIndex,
                  cfg: Config) -> pd.DataFrame:
    """
    Step 13b. Crossings whose hold overlaps the window's position months.
    Signal month tau -> positions set at the end of tau+pub_lag .. tau+pub_lag+hold_months-1.
    """
    cols = ["signal_month", "z", "threshold", "position_from", "position_to"]
    rows = []
    for tau in cross.index[cross.to_numpy(bool)]:
        p0 = _shift_months(tau, cfg.pub_lag)
        p1 = _shift_months(p0, cfg.hold_months - 1)
        if p1 < pos_months[0] or p0 > pos_months[-1]:
            continue
        rows.append({"signal_month": tau, "z": sig.at[tau, "z"], "threshold": sig.at[tau, "thr"],
                    "position_from": p0, "position_to": p1})
    return pd.DataFrame(rows, columns=cols)

def diagnostics_summary(sig, hold, leg, ff49, window, first_sig, first_pos, first_ret,
                       eps, xtab, pi, cfg: Config) -> pd.DataFrame:
    """
    Step 13c. Key facts about the run: fire dates, zero counts, hold months, pi, settings.
    Signal months "feeding" the window = the latest flags available at each position month.
    """
    pos_months = _position_months(window)
    sig_months = _month_range(_shift_months(pos_months[0], -cfg.pub_lag),
                             _shift_months(pos_months[-1], -cfg.pub_lag))
    sw = sig.reindex(sig_months)

```

```

trw = leg["tr_T"].loc[pos_months]
hist = _month_range(_shift_months(window[0], -(cfg.hedge_window + cfg.vol_window + 1)), window[-1])
n_brown = int(ff49.reindex(hist)[list(cfg.brown)].notna().sum(axis=1).min())
info = {
    "first signal month with a defined flag": f"{first_sig:%Y-%m}",
    "first month a position can be set (end of)": f"{first_pos:%Y-%m}",
    "first return month the rule can affect": f"{first_ret:%Y-%m}",
    "test window (return months)": f"{window[0]:%Y-%m} to {window[-1]:%Y-%m}",
    "n return months": len(window),
    "EMVENVREG zero months, whole file": int((sig["s"] == 0).sum()),
    "signal months feeding the window": f"{sig_months[0]:%Y-%m} to {sig_months[-1]:%Y-%m}",
    "zero months among them": int((sw["s"] == 0).sum()),
    "signal-off months (>10% zeros in trailing 60)": int(sw["off"].eq(True).sum()),
    "months with a defined flag": int(sw["extreme"].notna().sum()),
    "extreme months": int((sw["extreme"] == 1).sum()),
    "crossings affecting the window": len(xtab),
    "hold months (position months in window)": int(trw["hold"].sum()),
    "independent episodes": len(eps),
    "pi": pi,
    "mean m_t in hold months": float(trw.loc[trw["hold"], "m"].mean()) if trw["hold"].any() else np.nan,
    "min # Brown industries available (hedge history + window)": n_brown,
}
info.update({f"cfg.{k}": v for k, v in asdict(cfg).items()})
return pd.Series(info, name="value").to_frame()

# =====
# 14. Calendar shuffle
# =====
def place_blocks(lengths, n: int, min_gap: int, rng: np.random.Generator) -> np.ndarray:
    """
    Step 14a. Uniform random placement of blocks with the given lengths in n slots, in random
    order, with >= min_gap empty slots between consecutive blocks (stars and bars).
    Output: bool array of length n.
    """
    lengths = np.asarray(lengths, dtype=int)
    K = len(lengths)
    free = n - int(lengths.sum()) - min_gap * (K - 1)
    if free < 0:
        raise ValueError("episodes do not fit in the window with the required gaps")
    lens = lengths[rng.permutation(K)]
    slots = np.sort(rng.choice(free + K, size=K, replace=False))
    starts = (slots - np.arange(K)
              + np.concatenate(([0], np.cumsum(lens)[: -1])))
              + min_gap * np.arange(K))
    out = np.zeros(n, dtype=bool)
    for s, L in zip(starts, lens):
        out[s:s + L] = True
    return out

def calendar_shuffle(alpha_actual: float, leg: dict, hold: pd.Series, F: pd.DataFrame,
                    window: pd.DatetimeIndex, pi_actual: float, I_actual: pd.Series,
                    cfg: Config):
    """
    Step 14b. Calendar-shuffle null.
    Inputs : actual alpha, the Brown pipeline 'leg', actual hold (position time), regressors,
            window, actual pi and I (used only if shuffle_recompute_pi_I is False).
    Method : the K actual episodes (maximal hold runs among the position months that earn in
            the window) are placed at random (place_blocks) inside those position months;
            months before the window keep their actual state. Each draw rebuilds the timed
            book (w, overlay, costs); pi and I_{t-1} are recomputed if shuffle_recompute_pi_I;
            alpha is re-estimated by OLS (point estimate only).
    Output : (summary table, array of null alphas);
            p = (1 + #{alpha_null >= alpha_actual}) / (1 + draws), one-sided.
    """
    pos_months = _position_months(window)
    lengths = hold_episodes(hold, pos_months)["months"].to_numpy(int)

```

```

if len(lengths) == 0:
    raise ValueError("no episodes to shuffle")
loc = hold.index.get_indexer(pos_months)
if (loc < 0).any():
    raise ValueError("position months outside the hold index")
rng = np.random.default_rng(cfg.seed)
base = hold.to_numpy(bool)
Fw = F.loc>window
null = np.empty(cfg.shuffle_draws)
for d in range(cfg.shuffle_draws):
    hv = base.copy()
    hv[loc] = place_blocks(lengths, len(pos_months), cfg.shuffle_min_gap, rng)
    tr = run_trade(log["rb"], log["f"], log["b"], log["m"], pd.Series(hv, index=hold.index), cfg)
    if cfg.shuffle_recompute_pi_I:
        D, _, I = timing_difference(tr, log["tr_A0"], window, cfg)
    else:
        D, _, _ = timing_difference(tr, log["tr_A0"], window, cfg, pi=pi_actual)
        I = I_actual
    X = design_matrix(Fw, I, cfg).to_numpy(float)
    null[d] = np.linalg.lstsq(X, D.to_numpy(float), rcond=None)[0][0]
p = (1 + np.sum(null >= alpha_actual)) / (1 + cfg.shuffle_draws)
q = [0.01, 0.05, 0.10, 0.50, 0.90, 0.95, 0.99]
vals = ([alpha_actual, null.mean(), null.std(ddof=1)] + list(np.quantile(null, q))
        + [p, cfg.shuffle_draws, len(lengths), int(lengths.sum()), cfg.seed])
names = (["actual alpha", "null mean", "null sd"] + [f"null q{int(round(x * 100)):02d}" for x in q]
        + ["p (one-sided)", "draws", "episodes K", "hold months", "seed"])
return pd.DataFrame({"value": vals, index=names}, null)

# =====
# 15. Leave one industry out
# =====
def leave_one_out(ff49: pd.DataFrame, fac: pd.DataFrame, hold: pd.Series, F: pd.DataFrame,
                  window: pd.DatetimeIndex, index: pd.DatetimeIndex, alpha_full: float,
                  cfg: Config) -> pd.DataFrame:
    """
    Step 15. Drop each Brown industry in turn; rebuild leg, hedge, sizing, timed and always-on
    books, pi and D; refit the attribution. Hold timing is unchanged (the signal does not use
    returns). Output: one row per dropped industry with alpha, NW(6) t, p, pi, same_sign.
    """
    rows = []
    for drop in cfg.brown:
        keep = tuple(c for c in cfg.brown if c != drop)
        leg = leg_pipeline(ff49, fac, keep, hold, index, cfg)
        D, pi, I = timing_difference(log["tr_T"], log["tr_A0"], window, cfg)
        _, f6, _ = attribution_regression(D, design_matrix(F, I, cfg), cfg)
        a = float(f6.params["const"])
        rows.append({"dropped": drop, "alpha": a, f"t NW({cfg.nw_lags})": float(f6.tvalues["const"]),
                    "p": float(f6.pvalues["const"]), "pi": pi,
                    "same_sign": bool(np.sign(a) == np.sign(alpha_full))})
    return pd.DataFrame(rows)

# =====
# 16. Pass / fail
# =====
def pass_fail_table(f6, shuffle_p: float, n_episodes: int, loo: pd.DataFrame, cfg: Config):
    """
    Step 16. All four components required.
    Output : (table with one row per component + OVERALL, verdict "PASS"/"FAIL").
    """
    a, t, p = float(f6.params["const"]), float(f6.tvalues["const"]), float(f6.pvalues["const"])
    p1 = p / 2 if t > 0 else 1 - p / 2
    same, nl = int(loo["same_sign"].sum()), len(loo)
    rows = [
        [(i) timing alpha", f"alpha/month, NW({cfg.nw_lags}) t, p from t(n-k)",
         f"alpha > 0 and t >= {cfg.t_bar:g}",
         f"alpha={a:.6f}, t={t:.3f}, p2={p:.4f}, p1={p1:.4f}", bool(a > 0 and t >= cfg.t_bar)],
    ]

```

```

["(ii) calendar shuffle", f"one-sided p, {cfg.shuffle_draws} draws", f"p <= {cfg.p_bar:g}",
 f"{shuffle_p:.4f}", bool(shuffle_p <= cfg.p_bar)],
["(iii) independent episodes", "merged hold runs in window", f">= {cfg.min_episodes}",
 str(n_episodes), bool(n_episodes >= cfg.min_episodes)],
["(iv) leave one industry out", "L00 alphas with the full alpha's sign", f"{nl}/{nl}",
 f"{same}/{nl}", bool(same == nl)],
]
n_pass = sum(r[4] for r in rows)
tab = pd.DataFrame(rows, columns=["component", "statistic", "bar", "value", "pass"])
tab["pass"] = np.where(tab["pass"].astype(bool), "PASS", "FAIL")
verdict = "PASS" if n_pass == 4 else "FAIL"
tab.loc[len(tab)] = ["OVERALL", "all four required", "4/4", f"{n_pass}/4", verdict]
return tab, verdict

# =====
# 17. Run
# =====
def run_backtest(inputs: dict, cfg: Config = Config(), verbose: bool = True) -> dict:
    """
    Step 17. Runs steps 1-16 once; returns and prints the five outputs.
    inputs: ff49 (path/DataFrame), fac (DataFrame), emv_cat, emv_all, gs10, wti, vix
           (FRED CSV paths, or DataFrames in the FRED CSV layout).
    """
    ff49 = load_ff49(inputs["ff49"])
    fac = load_factors(inputs["fac"])
    emv_cat = load_fred_monthly(inputs["emv_cat"], "EMVENRGYENVREG")
    emv_all = load_fred_monthly(inputs["emv_all"], "EMVOVERALLEMV")
    gs10 = load_fred_monthly(inputs["gs10"], "GS10")
    wti = load_fred_monthly(inputs["wti"], "MCOILWTICO")
    vix = load_vix_monthly(inputs["vix"], "VIXCLS")
    index = _month_range(fac.index.min(), min(fac.index.max(), ff49.index.max()))

    # signal and hold: independent of the Brown leg
    sig = attention_signal(emv_cat, emv_all, cfg)
    cross = crossings(sig["extreme"], cfg)
    hold = hold_positions(cross, index, cfg)
    first_sig, first_pos, first_ret, window = test_window(sig, index, cfg)
    pos_months = _position_months(window)
    eps = hold_episodes(hold, pos_months)
    xtab = crossing_table(sig, cross, pos_months, cfg)

    # books, D, regressors
    leg = leg_pipeline(ff49, fac, cfg.brown, hold, index, cfg)
    D, pi, I_prev = timing_difference(leg["tr_T"], leg["tr_A0"], window, cfg)
    bond = bond_excess_return(gs10, fac["RF"])
    F = attribution_factors(fac, bond, wti, vix, emv_all, index).loc[window]
    summary = diagnostics_summary(sig, hold, leg, ff49, window, first_sig, first_pos, first_ret,
                                eps, xtab, pi, cfg)

    if eps.empty:
        pf = pd.DataFrame([["(iii) independent episodes", "merged hold runs in window",
                           f">= {cfg.min_episodes}", "0", "FAIL"],
                           ["OVERALL", "all four required", "4/4", "-", "FAIL (no position in window)"]],
                           columns=["component", "statistic", "bar", "value", "pass"])

        if verbose:
            _show("1. PASS / FAIL", pf)
            _show("4a. DIAGNOSTICS", summary)
        return {"pass_fail": pf, "verdict": "FAIL", "diagnostics": summary,
                "crossings": xtab, "episodes": eps}

    X = design_matrix(F, I_prev, cfg)
    coef, f6, f12 = attribution_regression(D, X, cfg)
    alpha = float(f6.params["const"])
    alpha_ls = np.linalg.lstsq(X.to_numpy(float), D.to_numpy(float), rcond=None)[0][0]
    if abs(alpha_ls - alpha) > 1e-10:
        raise RuntimeError("shuffle alpha path disagrees with statsmodels")
    n, k = X.shape

```

```

meta = pd.Series({"n": n, "k": k, "n - k": n - k, "R-squared": f6.rsquared,
                 "adj. R-squared": f6.rsquared_adj, "pi": pi,
                 "window start": f"{window[0]:%Y-%m}", "window end": f"{window[-1]:%Y-%m}",
                 "HAC": f"Bartlett, maxlags {cfg.nw_lags} ({cfg.nw_lags_check} check), no small-sample
                 ↳ corr."},
                 name="value").to_frame()
decomp = decompose_mean(D, X, f6)

shuf, null = calendar_shuffle(alpha, leg, hold, F, window, pi, I_prev, cfg)
loo = leave_one_out(ff49, fac, hold, F, window, index, alpha, cfg)
pf, verdict = pass_fail_table(f6, float(shuf.at["p (one-sided)", "value"]), len(eqs), loo, cfg)

if verbose:
    _show("1. PASS / FAIL", pf)
    _show("2a. ATTRIBUTION (D_t on FF5+UMD, BOND, WTI, dVIX, dlogEMV, I_{t-1} x F)", coef)
    _show("2b. REGRESSION FACTS", meta)
    _show("3. DECOMPOSITION OF mean(D)", decomp)
    _show("4a. DIAGNOSTICS", summary)
    _show("4b. CROSSINGS AFFECTING THE WINDOW", xtab)
    _show("4c. EPISODES", eps)
    _show("4d. LEAVE ONE INDUSTRY OUT", loo)
    _show("5. CALENDAR SHUFFLE", shuf)

return {"pass_fail": pf, "verdict": verdict, "attribution": coef, "attribution_meta": meta,
        "decomposition": decomp, "diagnostics": summary, "crossings": xtab, "episodes": eps,
        "leave_one_out": loo, "shuffle": shuf, "null_alphas": null,
        "D": D, "X": X, "hold": hold, "signal": sig, "fit_nw6": f6, "fit_nw12": f12}

# =====
# 18. Pre-run checks: synthetic data, look-ahead test, planted effect
# =====
def make_synthetic_inputs(seed: int = 0, end: str = "2025-12") -> dict:
    """
    Debug data only: random inputs in exactly the raw layouts run_backtest expects.
    No effect is planted (use plant_timing_effect for a positive control).
    """
    rng = np.random.default_rng(seed)
    idx = _month_range("1926-07", end)
    n = len(idx)
    mkt, smb, hml = rng.normal(0.006, 0.045, n), rng.normal(0.002, 0.03, n), rng.normal(0.003, 0.03, n)
    rmw, cma, umd = rng.normal(0.003, 0.02, n), rng.normal(0.003, 0.02, n), rng.normal(0.006, 0.04, n)
    rf = np.full(n, 0.003)
    names = list(BROWN) + [f"Ind{i:02d}" for i in range(1, 45)]
    beta_m, beta_h = rng.uniform(0.6, 1.4, 49), rng.normal(0.0, 0.5, 49)
    ind = rf[:, None] + mkt[:, None] * beta_m + hml[:, None] * beta_h + rng.normal(0.0, 0.04, (n, 49))
    ff49 = pd.DataFrame(ind, columns=names)
    ff49.loc[idx < _me("1970-01"), "Ind44"] = np.nan # an industry that starts late
    ff49.insert(0, "date", idx.strftime("%Y-%m-%d"))
    fac = pd.DataFrame({"Mkt-RF": mkt, "SMB": smb, "HML": hml, "RMW": rmw, "CMA": cma,
                       "RF": rf, "UMD": umd, index=idx}).loc[idx >= _me("1963-07")]

def fred(dates, values, sid):
    return pd.DataFrame({"observation_date": dates.to_period("M").to_timestamp().strftime("%Y-%m-%d"),
                        sid: values})

e_idx = _month_range("1985-01", end)
ne = len(e_idx)
x = np.zeros(ne)
for i in range(1, ne):
    x[i] = 0.85 * x[i - 1] + rng.normal(0.0, 0.35)
emv_all = 20.0 * np.exp(rng.normal(0.0, 0.3, ne))
emv_cat = emv_all * np.exp(-3.5 + x)
emv_cat[rng.random(ne) < np.where(e_idx < _me("1996-01"), 0.05, 0.01)] = 0.0
g_idx = _month_range("1953-04", end)
gs10 = np.clip(5.0 + np.cumsum(rng.normal(0.0, 0.2, len(g_idx))), 0.5, 15.0)
w_idx = _month_range("1986-01", end)
wti = 20.0 * np.exp(np.cumsum(rng.normal(0.0, 0.08, len(w_idx))))

```

```

days = pd.bdate_range("1990-01-02", _me(end))
vix = 20.0 * np.exp(rng.normal(0.0, 0.25, len(days)))
vix[rng.random(len(days)) < 0.03] = np.nan
return {"ff49": ff49, "fac": fac,
        "emv_cat": fred(e_idx, emv_cat, "EMVENRGYENVREG"),
        "emv_all": fred(e_idx, emv_all, "EMVOVERALLEMV"),
        "gs10": fred(g_idx, gs10, "GS10"),
        "wti": fred(w_idx, wti, "MCOILWTICO"),
        "vix": pd.DataFrame({"observation_date": days.strftime("%Y-%m-%d"), "VIXCLS": vix})}

def check_no_lookahead(inputs: dict, cfg: Config = Config(), n_random: int = 20,
                       cut_from: str | None = None, cut_to: str | None = None,
                       seed: int = 7) -> pd.DataFrame:
    """
    Look-ahead (truncation) test. For each cut month T keep French data dated <= T and EMV
    data dated <= T - pub_lag (what is published by the end of T), rerun signal, hold, hedge,
    sizing and trade, and compare everything set at the end of T (hold, w, m, overlay, b) with
    the full-sample run. Cuts = every crossing month tau and tau+pub_lag (where a lag bug shows)
    plus random months, restricted to [cut_from, cut_to]. Any row with ok == False = look-ahead.
    """
    ff49 = load_ff49(inputs["ff49"])
    fac = load_factors(inputs["fac"])
    emv_cat = load_fred_monthly(inputs["emv_cat"], "EMVENRGYENVREG")
    emv_all = load_fred_monthly(inputs["emv_all"], "EMVOVERALLEMV")
    index = _month_range(fac.index.min(), min(fac.index.max(), ff49.index.max()))

    def state(ff, fc, ec, ea, idx):
        s = attention_signal(ec, ea, cfg)
        hd = hold_positions(crossings(s["extreme"], cfg), idx, cfg)
        lg = leg_pipeline(ff, fc, cfg.brown, hd, idx, cfg)
        cols = ["hold", "w", "m"] + [f"h_{c}" for c in cfg.hedge_factors]
        return lg["tr_T"][cols].astype(float).join(lg["b"].add_prefix("b_"))

    full = state(ff49, fac, emv_cat, emv_all, index)
    sig = attention_signal(emv_cat, emv_all, cfg)
    cr = crossings(sig["extreme"], cfg)
    taus = cr.index[cr.to_numpy(bool)]
    first = _shift_months(sig["extreme"].first_valid_index(), cfg.pub_lag)
    lo = max(first, _me(cut_from)) if cut_from else first
    hi = min(index[-1], _me(cut_to)) if cut_to else index[-1]
    pool = index[(index >= lo) & (index <= hi)]
    rng = np.random.default_rng(seed)
    cuts = set(taus) | {_shift_months(t, cfg.pub_lag) for t in taus}
    cuts |= set(pd.DatetimeIndex(rng.choice(pool.to_numpy(), size=min(n_random, len(pool)), replace=False)))
    cuts = sorted(t for t in cuts if lo <= t <= hi)

    rows = []
    for T in cuts:
        L = _shift_months(T, -cfg.pub_lag)
        part = state(ff49.loc[:T], fac.loc[:T], emv_cat.loc[:L], emv_all.loc[:L], index[index <= T])
        a, b = part.loc[T], full.loc[T]
        diff = (a - b).abs().max()
        rows.append({"cut": T, "max_abs_diff": float(diff) if pd.notna(diff) else 0.0,
                    "nan_mismatch": bool((a.isna() != b.isna()).any())})
    out = pd.DataFrame(rows, columns=["cut", "max_abs_diff", "nan_mismatch"])
    out["ok"] = (out["max_abs_diff"] <= 1e-12) & ~out["nan_mismatch"]
    return out

def plant_timing_effect(inputs: dict, cfg: Config = Config(), delta: float = 0.03) -> dict:
    """
    Positive control (synthetic only). Lowers each Brown industry's return by 'delta' in every
    month t entered with a position (hold set at the end of t-1). The signal is untouched, so
    the hold series is unchanged. Expected: alpha > 0, roughly
    delta * (1 - pi) * mean_t(I_{t-1} m_{t-1}), with a small shuffle p.
    """
    emv_cat = load_fred_monthly(inputs["emv_cat"], "EMVENRGYENVREG")

```

```

    emv_all = load_fred_monthly(inputs["emv_all"], "EMVOVERALLEMV")
    raw = _read(inputs["ff49"])
    months = _to_month_end(raw["date"])
    sig = attention_signal(emv_cat, emv_all, cfg)
    hold = hold_positions(crossings(sig["extreme"], cfg), _month_range(months.min(), months.max()), cfg)
    on = hold.shift(1, fill_value=False).reindex(months, fill_value=False).to_numpy(bool)
    raw.loc[on, list(cfg.brown)] = raw.loc[on, list(cfg.brown)] - delta
    out = dict(inputs)
    out["ff49"] = raw
    return out

if __name__ == "__main__":
    MODE = "synthetic" # "synthetic" -> "seen" -> "final" (final: once, no edits after "seen")

    if MODE == "synthetic":
        cfg = replace(Config(), shuffle_draws=500)
        syn = make_synthetic_inputs(seed=0)
        _show("LOOK-AHEAD TEST (synthetic)", check_no_lookahead(syn, cfg))
        out = run_backtest(syn, cfg)
        out_planted = run_backtest(plant_timing_effect(syn, cfg, delta=0.03), cfg)
    else:
        fac = pd.read_pickle("fac.pkl") # your parsed French FF5 + UMD DataFrame (decimals)
        inputs = dict(ff49="ff49_industry_monthly.csv", fac=fac,
                     emv_cat="EMVENRGYENVREG.csv", emv_all="EMVOVERALLEMV.csv",
                     gs10="GS10.csv", wti="MCOILWTICO.csv", vix="VIXCLS.csv")
        if MODE == "seen":
            cfg = replace(Config(), test_start="2010-01", test_end="2022-12")
            _show("LOOK-AHEAD TEST (seen data only)", check_no_lookahead(inputs, cfg, cut_from="2010-01"))
        elif MODE == "final":
            cfg = Config()
        else:
            raise ValueError(MODE)
        out = run_backtest(inputs, cfg)
'''

---

## (a) Assumptions beyond the spec

**The three silent points (each is a parameter)**

1. **Zero month between two extreme months** ('bridge_missing=True', 'bridge_limit=6'). Extreme -> zero ->
    ↳ extreme is one run, so the second extreme month is not a new crossing. The last defined flag is
    ↳ carried across up to 6 consecutive missing months (zero or signal-off). Beyond 6, the earlier state
    ↳ counts as "not extreme". Why: the spec says a zero is missing data, not low attention, so it should
    ↳ neither end an episode nor start one. A new crossing would also extend the hold, which would let
    ↳ missing data lengthen positions. The cap of 6 equals the hold length and the largest number of zeros
    ↳ a valid 60-month window can contain. Setting 'False' gives the other reading.
2. **Shuffle** ('shuffle_min_gap=1', 'shuffle_recompute_pi_I=True'). The K actual episodes, at the lengths
    ↳ they have inside the window, are placed uniformly at random over all non-overlapping arrangements, in
    ↳ random order. They are placed within the position months that earn inside the window, with at least
    ↳ one flat month between blocks. Months before the window keep their actual state. Each draw rebuilds w,
    ↳ the overlay, costs, pi and I_{t-1}, then refits the full regression and records the alpha. Why: the
    ↳ null statistic has to be computed exactly like the actual one. Keeping pi and I fixed would pair a
    ↳ shuffled D with the actual hold dummies, which is a different statistic. The gap keeps K and the
    ↳ total number of hold months fixed, so draws differ only in timing.  $p = (1 + \#\{\text{null} \geq \text{actual}\}) / (1 + 5000)$ , one-sided because the bar requires a positive alpha.
3. **Hold still open at 2009-12** ('end_policy="truncate"'). It stops contributing. The position set at the
    ↳ end of 2009-12 earns in 2010-01 and its trade cost is paid in 2010-01, so both fall outside the
    ↳ window. The episode still counts toward (iii). Why: this follows the "trade at end of t pays in t+1"
    ↳ convention and uses no 2010 data, which you have already seen. "liquidate" instead charges both
    ↳ books' closing cost in 2009-12.

**Other assumptions**

4. All inputs are decimals, and the loaders stop if the data look like percentages. FF49 values  $\leq -0.99$  (
    ↳ French's -99.99 missing code) are set to NaN.

```

5. The Brown leg averages the Brown industries available in each month. All five should exist in every month
→ the test uses, and the diagnostics print the minimum count.
 6. Every series is mapped to month-end on a complete monthly calendar. Missing months stay NaN and nothing is
→ forward-filled.
 7. The hedge needs all 60 months present, and sizing needs all 36 residuals present. Both are estimated on
→ the full history from 1963-07, so the results do not depend on where the window starts.
 8. The overlay is $h_t = -w_t b_t$ per factor. Its turnover $|h_t - h_{t-1}|$ includes beta and sizing drift
→ during a hold. There is no drift of weights within a month, no financing or borrow fee, and no cost
→ on the cash/RF leg.
 9. The timed book starts flat, so its first entry pays full cost. The always-on book has been on since the
→ 1970s, so it pays no entry cost at the start of the window.
 10. The z-score uses the mean and the sample standard deviation (ddof=1) of log s over the nonzero months in
→ calendar months $t-59..t$, including t.
 11. The z window must be a full 60 calendar months of EMV history, so the first z is 1989-12. In a full
→ window, "no more than 10% zeros" (at most 6) already implies "at least 48 nonzero", so the 48
→ condition never binds. Both conditions are coded.
 12. A NaN EMV value (none are expected) is treated like a zero.
 13. Signal-off months are treated like zero months: they get no flag, cannot trigger, and are not added to
→ the percentile history. A hold that is already running continues through zero or off months, because
→ the hold is a calendar count.
 14. The percentile uses numpy's linear interpolation, and "exceeds" is strict ($>$). The history is every prior
→ valid z from 1989-12 on, and it is not restarted at the window.
 15. The first month with a defined flag can itself be a crossing, because the undefined month before it
→ counts as "not extreme".
 16. "First month the rule can fire" is the end of (first flagged signal month + pub_lag). The window starts
→ at the first return month that position can affect, or at 1993-01 if that is later. On complete data
→ this should give roughly 1995-02 rather than 1993-01.
 17. pi uses the positions that earn inside the window (w_{t-1}), so it is an ex-post scale for the benchmark.
→ That is fine for attribution, but pi is not a tradable quantity.
 18. I_{t-1} is the hold state set at the end of $t-1$.
 19. As the spec is written, the regression has ****no I_{t-1} main effect****, so alpha is the average timing
→ alpha across all months. If the team intended a main effect, alpha would mean something different.
 20. BOND uses exact cash-flow duration and convexity. GS10 is a monthly average but is used as if it were the
→ month-end yield, as in the spec's formula. RF is French's.
 21. dVIX is the difference between monthly means of the available daily closes. WTI and EMV changes are month-
→ over-month log differences.
 22. HAC uses the Bartlett kernel with no small-sample correction, which is the statsmodels default and your
→ team's convention. Pass (i) uses the NW(6) t, and NW(12) is reported only. Both the two-sided and one-
→ sided $t(n-k)$ p-values are printed.
 23. For (iii), an episode is a maximal run of hold months among the position months that earn in the window.
→ A run cut by the window edge counts once.
 24. For (iv), the hold timing stays fixed because the signal does not use returns. "Keeps its sign" means the
→ same sign as the full-sample alpha, for all five industries.
 25. Current data vintages are used, and French's data are revised slightly each year. As I understand their
→ construction, the EMV category trackers are the overall EMV times a share of articles. So any full-
→ sample scaling cancels in s_t and in the log z-score, and it creates no look-ahead.
- ## (b) Checks before the single real run (and what makes you stop)**
1. ****Units and dates.**** Print the head and tail of every parsed input. EMV 1985-01 must sit at 1985-01-31,
→ and the first monthly VIX value must be 1990-01. ****Stop**** if any series is shifted by a month, if the
→ loaders flag percent units, or if any Brown industry is missing in the window.
 2. ****Reproduce the team trade**** on 2010-2022. Compare m_t , b_t , R^{AO} and your validated timed series against
→ the team's code. ****Stop**** if any month differs by more than $1e-8$.
 3. ****Look-ahead test.**** Run 'check_no_lookahead' on synthetic data and on the seen window (cuts $\geq 2010-01$).
→ It cuts at every crossing month, the month after it, and random months. ****Stop**** if any row has 'ok
→ == False'. Also run a lag probe: set 'pub_lag=0'. ****Stop**** if the hold series does not move exactly
→ one month earlier, whether because nothing changes or because more than a shift changes.
 4. ****Signal unit tests**** on hand-built series:
 - extreme -> zero -> extreme gives one crossing with bridging and two without;
 - 6 zeros in the trailing 60 leaves the signal on, and 7 turns it off;
 - a zero month has NaN z and does not lengthen the percentile history.
- **Stop**** on any mismatch.
5. ****Hold tests.**** One crossing at tau must give exactly six position months, $\tau+1..\tau+6$. Crossings at tau
→ and $\tau+3$ must give one 9-month run. w must stay in $[-1, 0]$. ****Stop**** otherwise.
 6. ****Cost test.**** With all costs set to 0, net must equal gross exactly. For one hand-picked entry, the cost
→ must equal $10bp \cdot m + 5bp \cdot |h_{Mkt}| + 25bp \cdot (|h_{SMB}| + |h_{HML}|)$ and be paid the next month. ****Stop**** on a

```

    ↪ mismatch.
7. **BOND.** At  $y = 5\%$ , the modified duration should be about 7.79, from the analytic value  $(1/y)(1 -$ 
    ↪  $1.025^{-20})$ . The correlation of BOND with  $-\Delta y$  should be above 0.99. **Stop** otherwise.
8. **Regression mechanics.** Check that  $k = 15$ ,  $n - k = df_{resid}$ , the identity gap is below  $1e-12$ , the design
    ↪ matrix has full rank, and the  $lstsq$  alpha equals the  $statsmodels$  alpha (the code raises if not). **
    ↪ Stop** on any failure.
9. **Shuffle validity** on synthetic data:
    - every draw has  $K$  episodes and the same total of hold months;
    - the same seed gives the same  $p$  on two runs;
    - across 20 synthetic seeds with no planted effect,  $p \leq 0.05$  occurs about 5% of the time.

    **Stop** if it happens in more than 4 of the 20.
10. **Positive control.** Run 'plant_timing_effect(delta=0.03)'. Alpha should be positive with  $t$  well above 2,
    ↪ and the shuffle  $p$  should be small. With 'delta=0', you should get the original alpha back exactly.
    ↪ **Stop** if the planted effect is missed or has the wrong sign.
11. **Seen-window run** on 2010-2022, end to end. Numbers should be consistent with the team's existing
    ↪ results. Any change after this point must be a bug fix, and a bug fix means rerunning checks 1-10.
12. **Coverage without peeking.** Before the real run, check only that the 1993-2009 regressors and
    ↪ industries have no NaNs. Do not compute anything from 1993-2009 returns. **Stop** if data are missing,
    ↪ and fix the data, not the rule.
13. **Freeze.** Record the SHA-256 of the script and of every input file, the printed 'Config', the seed and
    ↪ the package versions, then commit. **Stop** if anything differs at run time. Then set 'MODE = "final"'
    ↪ and run once.

```

Follow-up prompt (verbatim) followup_prompt.md, 444 words (437 excluding Markdown symbols).

```

I ran your script on the real data and through your checks. It ran the first time, with only file
    ↪ paths and 'fac' supplied, and reaches the same verdict as my own implementation (FAIL, 1 of
    ↪ 4). The numbers differ, and one of your stop rules fires.

1. 'make_synthetic_inputs'. GS10 is a random walk clipped at 0.5 that sits on the floor (76% of
    ↪ 1995-2009 months for seed 0, every month for seed 16). BOND is then constant,  $I \times BOND$  acts
    ↪ as an  $I$  main effect, and your positive control fails check (b)10: 4% of the planted 3%
    ↪ effect reaches alpha ( $t$  0.98), while  $I \times BOND$  has  $t$  -10.8. Seed 16 crashes the null-seed
    ↪ loop (rank-deficient design).

2. 'attention_signal'. Requiring 60 full calendar months before the first  $z$  is a silent reading
    ↪ you hard-coded. I treat pre-1985 months as unavailable, not zero, so the 48-nonzero minimum
    ↪ is the burn-in. Your reading moves the first threshold from 1994-01 to 1994-12, adds a
    ↪ 2002-12 crossing and starts the window at 1995-02, not 1994-03:  $p$  0.809 vs. 0.729, shuffle
    ↪  $p$  0.536 vs. 0.478.

3. 'leg_pipeline'. You hedge with the Mkt-RF, SMB and HML in 'fac', the FF5 file. The team hedge
    ↪ uses its FF3 file, which my prompt never said (my gap). SMB differs by up to 3.5 pp a month
    ↪ ,  $b_t$  moves by up to 0.09, and your check (b)2 stops at  $2.3e-3$  a month ( $6e-17$  with FF3).

4. 'pass_fail_table'. (iv) passes a negative alpha whose five drop-one alphas are also negative.

With readings 2 and 3 switched to mine, your code matches my implementation to  $1e-13$ , all 5,000
    ↪ shuffle draws included.

Two smaller gaps. 'check_no_lookahead' compares end-of- $T$  states only, so it missed both return-
    ↪ side bugs I injected (month- $t$  return hedged with  $b_t$ ; a position earning its own month).
    ↪ Seen mode ends at 2022-12; its five holdout months flip the seen alpha from  $-0.16\%/yr$  ( $t$ 
    ↪  $-0.24$ ) to  $+0.26\%/yr$  ( $t$  0.37).

Please send corrected versions of these functions only, each with a unit test:
- 'make_synthetic_inputs': BOND varies in every window for seeds 0-20, and the positive control
    ↪ puts at least 80% of the planted shift in alpha with  $t > 2$ .

```

- 'attention_signal': a 'z_burn_in' parameter with both readings; on a zero-free series from
 ↪ 1985-01, the first z falls in month 48 under mine and month 60 under yours.
- 'leg_pipeline': a separate 'hedge_fac' input; changing its SMB moves b_t but not the attribution
 ↪ regressors.
- 'pass_fail_table': a negative alpha fails (iv).
- 'check_no_lookahead': a return-side comparison that flags both injected bugs.
- '__main__': seen mode ends at 2022-07, with 'cut_to' on its look-ahead call.

These fixes are for reproducibility; the frozen rule stays.

Follow-up reply (verbatim; model in the ChatGPT role) chatgpt_followup_response.md, 4,745 words (4,066 excluding Markdown symbols).

The replacements below cover the six functions you listed, and each one comes with a unit test. They share a
 ↪ few small helpers that handle the raw input files, and one adapter, 'run_panel', which is the only
 ↪ place the tests call the rest of the script. I can't run code, so none of this has been tested yet.
 ↪ Please run the tests before anything else, and if one fails, send me the assertion message.

****Changes needed in the driver****

1. 'run_backtest' gets two new arguments, 'hedge_fac' and 'z_burn_in'. Pass 'hedge_fac' to every '
 ↪ leg_pipeline' call: the main run and all five leave-one-out rebuilds. Pass 'z_burn_in' to '
 ↪ attention_signal'.
2. The overlay return '-w_t b_t'f_{t+1}' and the overlay costs must use the leg's 'f_*' and 'b_*' columns (
 ↪ FF3), not 'fac'. If the old overlay used 'fac', the FF5 SMB was getting in there as well as in the
 ↪ hedge. The attribution still uses 'fac'.
3. 'pass_fail_table' now takes plain numbers.
4. 'run_panel' assumes this driver signature: 'run_backtest(paths, fac, start, end, *, hedge_fac, z_burn_in,
 ↪ n_shuffle, run_loo, verbose)'. It also assumes the output keys listed in its docstring. If your names
 ↪ differ, change them in 'run_panel' only.

```

''python
# =====
# Replacements. Everything not defined here is unchanged from the earlier script.
# =====

import os
import sys
import tempfile

import numpy as np
import pandas as pd

# Constants (already in the script; repeated so this block stands alone)
BROWN = ["Util", "Ships", "Aero", "Steel", "BldMt"]
HEDGE_COLS = ["Mkt-RF", "SMB", "HML"]
FRED_IDS = ["EMVENVRGYENVREG", "EMVOVERALLEMV", "GS10", "MCOILWTICO", "VIXCLS"]
FF49_NAMES = [
    "Agric", "Food", "Soda", "Beer", "Smoke", "Toys", "Fun", "Books", "Hshld", "Clths",
    "Hlth", "MedEq", "Drugs", "Chems", "Rubbr", "Txcls", "BldMt", "Cnstr", "Steel", "FabPr",
    "Mach", "ElcEq", "Autos", "Aero", "Ships", "Guns", "Gold", "Mines", "Coal", "Oil",
    "Util", "Telcm", "PerSv", "BusSv", "Hardw", "Softw", "Chips", "LabEq", "Paper", "Boxes",
    "Trans", "Whlsl", "Rtail", "Meals", "Banks", "Insur", "RlEst", "Fin", "Other",
]
LATE_START = {"Soda": "1963-07-31", "Hlth": "1969-07-31", "Rubbr": "1963-07-31", "FabPr": "1963-07-31",
               "Guns": "1963-07-31", "Gold": "1963-07-31", "Softw": "1965-07-31"}
STATE_COLS = ["w", "w_ao", "m", "hold"] + [f'b_{k}' for k in HEDGE_COLS]
POS_OF = {"RT": "w", "RA0": "w_ao"} # return column -> the position that earns it
TEST_WINDOW = ("1993-01-31", "2009-12-31")
SEEN_WINDOW = ("2010-01-31", "2022-07-31") # the holdout starts 2022-08
MONTH_END = pd.offsets.MonthEnd()

# -----
# Raw-input helpers. Checks and tests work on 'raw' (the files as read from disk),
# so synthetic, perturbed and truncated data all pass through your unchanged loader.

```

```

# -----
def read_raw(paths, fac, hedge_fac):
    """Inputs: dict of CSV paths (keys 'ff49' and the FRED ids), fac (FF5+UMD), hedge_fac (team FF3).
    Output: dict of raw frames exactly as on disk plus the two factor frames. Timing: none."""
    raw = {"ff49": pd.read_csv(paths["ff49"], parse_dates=["date"], float_precision="round_trip"),
           "fac": fac, "hedge_fac": hedge_fac}
    for sid in FRED_IDS:
        df = pd.read_csv(paths[sid], parse_dates=["observation_date"], float_precision="round_trip")
        df[sid] = pd.to_numeric(df[sid], errors="coerce") # FRED blanks -> NaN
        raw[sid] = df
    return raw

def write_raw_csvs(raw, folder):
    """Inputs: raw dict, folder. Output: dict of paths in the same layout as the real files. Timing: none."""
    paths = {"ff49": os.path.join(folder, "ff49_industry_monthly.csv")}
    raw["ff49"].to_csv(paths["ff49"], index=False, date_format="%Y-%m-%d")
    for sid in FRED_IDS:
        paths[sid] = os.path.join(folder, f"{sid}.csv")
        raw[sid].to_csv(paths[sid], index=False, date_format="%Y-%m-%d")
    return paths

def truncate_raw(raw, cut_to):
    """Inputs: raw dict, month-end cut_to. Output: copy with every row dated after cut_to removed
    (FRED monthly rows are dated the 1st, so month cut_to itself is kept). Timing: nothing after cut_to
    ↪ survives."""
    cut = pd.Timestamp(cut_to)
    out = {}
    for k, v in raw.items():
        if not isinstance(v, pd.DataFrame):
            out[k] = v
        elif "date" in v.columns:
            out[k] = v[v["date"] <= cut].copy()
        elif "observation_date" in v.columns:
            out[k] = v[v["observation_date"] <= cut].copy()
        else:
            out[k] = v.loc[v.index <= cut].copy()
    return out

def _last_date(raw):
    dates = []
    for v in raw.values():
        if isinstance(v, pd.DataFrame) and len(v):
            col = "date" if "date" in v.columns else ("observation_date" if "observation_date" in v.columns
            ↪ else None)
            dates.append(v[col].max() if col else v.index.max())
    return max(dates)

def _bump(raw, key, T, cols, fn):
    """Copy of raw with fn applied to 'cols' of frame 'key' in the calendar month of T."""
    r = dict(raw)
    df = raw[key].copy()
    p = pd.Timestamp(T).to_period("M")
    if "date" in df.columns:
        rows = (df["date"].dt.to_period("M") == p).to_numpy()
    elif "observation_date" in df.columns:
        rows = (df["observation_date"].dt.to_period("M") == p).to_numpy()
    else:
        rows = df.index.to_period("M") == p
    df.loc[rows, cols] = fn(df.loc[rows, cols])
    r[key] = df
    return r

def _maxdiff(a, b, cols):

```

```

    """Max |a - b| over cols on a's rows; NaN vs NaN counts as equal, NaN vs number as inf."""
    av = a[cols].astype(float).to_numpy()
    bv = b.reindex(a.index)[cols].astype(float).to_numpy()
    if (np.isnan(av) ^ np.isnan(bv)).any():
        return np.inf
    d = np.abs(av - bv)[~(np.isnan(av) & np.isnan(bv))]
    return float(d.max()) if d.size else 0.0

def _ret_err(base, pert, T, expected):
    errs = []
    for col, e in expected.items():
        d = pert.at[T, col] - base.at[T, col] - e if T in pert.index else np.nan
        errs.append(abs(d) if np.isfinite(d) else np.inf)
    return max(errs)

def run_panel(raw, start, end, z_burn_in="nonzero", n_shuffle=0, loo=False, verbose=False):
    """
    ADAPTER to the unchanged driver: the only place the tests touch it. Edit names here if yours differ.
    Inputs : raw dict; window; burn-in reading; shuffle draws (0 = skip); leave-one-out on/off.
    Output : (panel, out).
        panel, indexed by month-end t: w, w_ao (positions set at end of t), RT, RAO (net returns earned
        in t), m, hold, b_Mkt-RF, b_SMB, b_HML (end-of-t states), I (1 if a timed position was held
        entering t).
        out: D (Series), X (design DataFrame incl. constant), alpha, alpha_t (NW(6)), plus the
        driver's printed tables.
    Timing : the window end is clipped to the last month in raw, so truncated inputs run cleanly.
    """
    end = min(pd.Timestamp(end), pd.Timestamp(raw["ff49"]["date"].max()))
    with tempfile.TemporaryDirectory() as d:
        paths = write_raw_csvs(raw, d)
        out = run_backtest(paths, raw["fac"], start, end, hedge_fac=raw["hedge_fac"], # noqa: F821
                           z_burn_in=z_burn_in, n_shuffle=n_shuffle, run_loo=loo, verbose=verbose)
    return out["panel"], out

# =====
# 1. make_synthetic_inputs
# =====
def make_synthetic_inputs(seed=0, out_dir=None, plant=0.0, plant_months=None, end="2026-06-30"):
    """
    Synthetic stand-ins for every input, shaped exactly like the real files.

    Inputs
    seed : fixed RNG seed (nothing is tuned on it).
    out_dir : if given, also write the six CSVs there; their paths are returned under 'paths'.
    plant : positive-control effect (decimal). Every Brown industry return is lowered by 'plant'
            in each month of 'plant_months', so the short Brown leg gains there.
    plant_months : month-ends t whose return is earned by a timed position (I_{t-1} = 1) in an unplanted
                  run with the same seed. The signal files do not depend on returns and the plant draws
                  no random numbers, so the hold schedule and all other data are identical with and without
                  ↪ it.

    Outputs
    dict: 'ff49' (date column + 49 industries, NaN before an industry starts), 'fac' (Mkt-RF, SMB, HML,
        RMW, CMA, RF, UMD from 1963-07), 'hedge_fac' (FF3: Mkt-RF, SMB, HML, RF from 1926-07; its SMB differs
        from fac's, as in the real files), one frame per FRED id (observation_date + id).

    Timing
    Returns at month-ends. FRED monthly rows dated the 1st. GS10 from 1953-04, EMV from 1985-01,
    WTI from 1986-01, VIXCLS daily business days from 1990-01-02 with ~3% blanks.

    Fix
    GS10 used to be a random walk clipped at 0.5, so it could sit on the clip and make BOND constant.
    log(GS10) is now a stationary AR(1): always positive, never clipped, and it moves every month.
    """
    rng = np.random.default_rng(seed)
    idx = pd.date_range("1926-07-31", end, freq=MONTH_END)
    n = len(idx)

```

```

def ar1(nobs, mu, phi, sig, x0):
    eps = rng.standard_normal(nobs)
    x = np.empty(nobs)
    x[0] = x0
    for i in range(1, nobs):
        x[i] = mu + phi * (x[i - 1] - mu) + sig * eps[i]
    return x

y10 = np.exp(ar1(n, np.log(5.0), 0.985, 0.045, np.log(3.5))) # percent, roughly 2-11
rf = y10 * 0.6 * np.exp(ar1(n, 0.0, 0.9, 0.05, 0.0)) / 1200.0 # monthly decimal

zf = rng.standard_normal((n, 7))
mkt, smb3, hml = 0.006 + 0.045 * zf[:, 0], 0.002 + 0.030 * zf[:, 1], 0.003 + 0.030 * zf[:, 2]
rmw, cma, umd = 0.003 + 0.020 * zf[:, 3], 0.003 + 0.020 * zf[:, 4], 0.006 + 0.040 * zf[:, 5]
smb5 = smb3 + 0.004 * zf[:, 6]
hedge_fac = pd.DataFrame({"Mkt-RF": mkt, "SMB": smb3, "HML": hml, "RF": rf}, index=idx)
fac = pd.DataFrame({"Mkt-RF": mkt, "SMB": smb5, "HML": hml, "RMW": rmw, "CMA": cma,
                    "RF": rf, "UMD": umd}, index=idx).loc["1963-07-31:"]

brown_common = 0.015 * rng.standard_normal(n)
rets = {}
for name in FF49_NAMES:
    bm, bs, bh = rng.normal(1.0, 0.2), rng.normal(0.2, 0.3), rng.normal(0.2, 0.3)
    r = rf + bm * mkt + bs * smb3 + bh * hml + 0.05 * rng.standard_normal(n)
    if name in BROWN:
        r = r + brown_common
    if name in LATE_START:
        r = np.where(idx < pd.Timestamp(LATE_START[name]), np.nan, r)
    rets[name] = r
if plant != 0.0:
    if plant_months is None:
        raise ValueError("plant != 0 needs plant_months")
    hit = idx.to_period("M").isin(pd.DatetimeIndex(plant_months).to_period("M"))
    for name in BROWN:
        rets[name] = np.where(hit, rets[name] - plant, rets[name])
ff49 = pd.DataFrame(rets, index=idx)[FF49_NAMES]
ff49.insert(0, "date", idx)
ff49 = ff49.reset_index(drop=True)

def fred(mask, values, sid):
    return pd.DataFrame({"observation_date": idx[mask].to_period("M").to_timestamp(), sid: values})

g = idx >= pd.Timestamp("1953-04-30")
e = idx >= pd.Timestamp("1985-01-31")
o = idx >= pd.Timestamp("1986-01-31")
ne = int(e.sum())
overall = np.exp(ar1(ne, np.log(25.0), 0.8, 0.15, np.log(25.0))) # never zero
env = overall * np.exp(ar1(ne, np.log(0.02), 0.6, 0.35, np.log(0.02)))
p_zero = np.where(idx[e] < pd.Timestamp("1995-01-01"), 0.05, 0.01) # zeros cluster early
env = np.where(rng.random(ne) < p_zero, 0.0, env)
wti = np.exp(ar1(int(o.sum()), np.log(40.0), 0.98, 0.09, np.log(22.0)))
days = pd.bdate_range("1990-01-02", end)
vix = np.round(np.exp(ar1(len(days), np.log(19.0), 0.98, 0.06, np.log(19.0))), 2)
vix[rng.random(len(days)) < 0.03] = np.nan

raw = {"ff49": ff49, "fac": fac, "hedge_fac": hedge_fac,
       "GS10": fred(g, y10[g], "GS10"),
       "EMVOVERALLEMV": fred(e, overall, "EMVOVERALLEMV"),
       "EMVENRGYENVREG": fred(e, env, "EMVENRGYENVREG"),
       "MCOILWTICO": fred(o, wti, "MCOILWTICO"),
       "VIXCLS": pd.DataFrame({"observation_date": days, "VIXCLS": vix})}
if out_dir is not None:
    os.makedirs(out_dir, exist_ok=True)
    raw["paths"] = write_raw_csvs(raw, out_dir)
return raw

def positive_control(seed=0, plant=0.03, window=TEST_WINDOW):

```

```

"""
Check (b)10. Inputs: seed, plant (Brown leg lower by 'plant' in every held month), window.
Output: dict with the planted shift = mean(D_planted) - mean(D_unplanted), the share of it in alpha,
and the planted run's alpha and NW(6) t. Timing: the plant sits in month t when I_{t-1} = 1.
"""
raw0 = make_synthetic_inputs(seed)
p0, o0 = run_panel(raw0, *window)
w0 = p0.loc>window[0]:window[1]]
held = w0.index[(w0["I"] == 1).to_numpy()]
raw1 = make_synthetic_inputs(seed, plant=plant, plant_months=held)
p1, o1 = run_panel(raw1, *window)
shift = o1["D"].mean() - o0["D"].mean()
return dict(seed=seed, same_I=bool(p1["I"].equals(p0["I"])), n_held=len(held), shift=shift,
            share=(o1["alpha"] - o0["alpha"]) / shift, alpha=o1["alpha"], t=o1["alpha_t"])

def test_make_synthetic_inputs(seeds=range(21), pc_seeds=(0, 1, 2)):
    for seed in seeds:
        raw = make_synthetic_inputs(seed)
        g = raw["GS10"]["GS10"]
        assert g.notna().all() and g.min() > 0.5, (seed, g.min()) # no floor
        assert (g.diff().iloc[1:] != 0).all(), seed # moves every month
        assert g.diff().rolling(24).std().dropna().min() > 0.02, seed # and in every window
        for start, end in (TEST_WINDOW, SEEN_WINDOW):
            panel, out = run_panel(raw, start, end)
            X = out["X"]
            I = panel["I"].reindex(X.index).astype(float)
            assert np.linalg.matrix_rank(X.to_numpy()) == X.shape[1], (seed, start) # seed 16 crashed here
            assert (X.std() < 1e-12).sum() == 1, (seed, start) # only the constant
            coef = np.linalg.lstsq(X.to_numpy(), I.to_numpy(), rcond=None)[0]
            r2 = 1 - ((I - X @ coef) ** 2).sum() / ((I - I.mean()) ** 2).sum()
            assert r2 < 0.5, (seed, start, r2) # no regressor combination can stand in for I itself
        for seed in pc_seeds:
            r = positive_control(seed)
            assert r["same_I"] and r["n_held"] > 0 and r["shift"] > 0, r
            assert r["share"] >= 0.8 and r["t"] > 2, r

# =====
# 2. attention_signal
# =====
def attention_signal(env, overall, z_burn_in="nonzero", window=60, min_nonzero=48,
                    max_zero_frac=0.10, pct=0.80, min_prior_z=60):
    """
    Rule steps 1-3: ratio, zero handling, rolling z, past-only percentile threshold, extreme flag.

    Inputs
    env, overall : Series of EMVENRGYENVREG and EMVOVERALLEMV indexed by data month (month-end).
                   Months before the first observation do not exist: unavailable, not zero.
    z_burn_in : "nonzero" (default, your reading): z_t exists once the trailing 60 months hold
                   >= 48 nonzero months, counting only months inside the sample.
                   "calendar" (my previous hard-coded reading): also require 60 calendar months since
                   the first observation.

    Rules shared by both readings: a zero month has no z and cannot be extreme; the month is "off"
    (no z) when the trailing 60 calendar months contain more than 6 zeros (10% of 60) or fewer than
    48 nonzero months; the z window and the percentile history use only months with a z.

    Outputs
    DataFrame by data month t: s, zero, n_nonzero, n_zero, on, z, n_prior_z, thr, extreme.
    z_t = (log s_t - mean) / std(ddof=1) over nonzero months t-59..t.
    thr_t = 80th percentile (linear) of all z before t, needing >= 60 of them. extreme_t = on_t and z_t >
    <math>\hookrightarrow</math> thr_t.

    Timing
    Row t uses EMV data for months <= t only. Publication lag: row t is usable at the end of t+1.
    The lag is applied in the hold step (unchanged), which reads extreme[t-1] at the end of t.
    """
    if z_burn_in not in ("nonzero", "calendar"):
        raise ValueError("z_burn_in must be 'nonzero' or 'calendar'")
    env, overall = env.copy(), overall.copy()

```

```

env.index = pd.DatetimeIndex(env.index) + pd.offsets.MonthEnd(0)
overall.index = pd.DatetimeIndex(overall.index) + pd.offsets.MonthEnd(0)
df = pd.concat({"env": env, "overall": overall}, axis=1).sort_index()
idx = pd.date_range(df.dropna().index.min(), df.index.max(), freq=MONTH_END)
df = df.reindex(idx)

avail = df["env"].notna() & df["overall"].notna() & (df["overall"] > 0)
zero = avail & (df["env"] == 0)
good = avail & (df["env"] > 0)
s = (df["env"] / df["overall"]).where(good)
x = np.log(s)

n_nonzero = good.astype(int).rolling(window, min_periods=1).sum()
n_zero = zero.astype(int).rolling(window, min_periods=1).sum()
max_zero = int(np.floor(max_zero_frac * window + 1e-9))
on = good & (n_nonzero >= min_nonzero) & (n_zero <= max_zero)
if z_burn_in == "calendar":
    on &= pd.Series(np.arange(len(idx)) >= window - 1, index=idx)

roll = x.rolling(window, min_periods=min_nonzero)
z = ((x - roll.mean()) / roll.std(ddof=1)).where(on)

zv = z.to_numpy()
thr = np.full(len(zv), np.nan)
n_prior = np.zeros(len(zv), dtype=int)
hist = []
for i, zi in enumerate(zv):
    n_prior[i] = len(hist)
    if len(hist) >= min_prior_z:
        thr[i] = np.quantile(hist, pct)
    if np.isfinite(zi):
        hist.append(zi)
thr = pd.Series(thr, index=idx)
extreme = on & (z > thr)
return pd.DataFrame({"s": s, "zero": zero, "n_nonzero": n_nonzero, "n_zero": n_zero, "on": on,
                    "z": z, "n_prior_z": n_prior, "thr": thr, "extreme": extreme})

def test_attention_signal():
    idx = pd.date_range("1985-01-31", "2005-12-31", freq=MONTH_END)
    rng = np.random.default_rng(1)
    overall = pd.Series(100 * np.exp(rng.normal(0, 0.2, len(idx)))), idx)
    env = overall * np.exp(rng.normal(np.log(0.01), 0.3, len(idx)))
    a = attention_signal(env, overall, z_burn_in="nonzero")
    b = attention_signal(env, overall, z_burn_in="calendar")
    assert a["z"].first_valid_index() == idx[47] # month 48
    assert b["z"].first_valid_index() == idx[59] # month 60
    assert a["thr"].first_valid_index() == idx[107] and b["thr"].first_valid_index() == idx[119]
    assert np.allclose(a["z"].iloc[59:], b["z"].iloc[59:], rtol=0, atol=0) # same z once the window is full
    t = 100 # hand-checked z
    x = np.log(env / overall).iloc[t - 59:t + 1]
    assert abs(a["z"].iloc[t] - (x.iloc[-1] - x.mean()) / x.std(ddof=1)) < 1e-10
    h = a["z"].iloc[:150].dropna().to_numpy() # past-only threshold
    assert a["thr"].iloc[150] == np.quantile(h, 0.8) and a["n_prior_z"].iloc[150] == len(h)
    env4 = env.copy(); env4.iloc[150] *= 10
    a4 = attention_signal(env4, overall)
    pd.testing.assert_series_equal(a["thr"].iloc[:151], a4["thr"].iloc[:151])
    env2 = env.copy(); env2.iloc[[100, 101]] = 0.0 # zeros dropped, not low
    c = attention_signal(env2, overall)
    assert c["z"].iloc[100:102].isna().all() and not c["extreme"].iloc[100:102].any()
    x2 = np.log((env2 / overall).where(env2 > 0)).iloc[102 - 59:103]
    assert abs(c["z"].iloc[102] - (x2.iloc[-1] - x2.mean()) / x2.std(ddof=1)) < 1e-10
    env3 = env.copy(); env3.iloc[150:157] = 0.0 # 7 zeros in 60 -> off
    d = attention_signal(env3, overall)
    assert not d["on"].iloc[150:210].any() and d["on"].iloc[210]
    assert (a["extreme"] == (a["z"] > a["thr"])).all()

```

```

# =====
# 3. leg_pipeline
# =====
def leg_pipeline(ff49, hedge_fac, brown=BROWN, beta_window=60, sd_window=36, vol_target=0.05):
    """
    Brown leg, rolling hedge, out-of-sample residual, sizing.

    Inputs
    ff49 : industry returns (decimal), month-end DatetimeIndex or a 'date' column.
    hedge_fac : the team's FF3 frame (Mkt-RF, SMB, HML, RF; decimal; month-end). It supplies RB's RF,
                the hedge regressors and the overlay factor returns. The attribution's 'fac' (FF5+UMD)
                never enters this function. An FF5 frame is refused.
    brown : industries in the leg (pass four names for a leave-one-out rebuild).
    Outputs (DataFrame by month-end t)
    RB = mean(brown_t) - RF_t; f_{<k>} = hedge factor returns in t; a, b_{<k>} = OLS of RB on [1, f] over
    t-59..t; e = RB_t - a_{t-1} - b_{t-1}'f_t; sd = std(e_{t-35..t}, ddof=1);
    m = min(1, vol_target / (sqrt(12) sd_t)).
    Timing
    Row t uses data through t only; nothing is shifted forward. A position set at the end of t
    (size m_t, hedge b_t) earns in t+1: w_t RB_{t+1} - w_t b_t' f_{t+1}, with f_{t+1} from THIS frame.
    """
    if "date" in ff49.columns:
        ff49 = ff49.set_index("date")
    ff49 = ff49.copy()
    ff49.index = pd.DatetimeIndex(ff49.index)
    missing = [c for c in HEDGE_COLS + ["RF"] if c not in hedge_fac.columns]
    if missing:
        raise ValueError(f"hedge_fac lacks {missing}")
    if {"RMW", "CMA"} & set(hedge_fac.columns):
        raise ValueError("hedge_fac has RMW/CMA: that is the FF5 file; the team hedge uses its FF3 file")
    idx = ff49.index.intersection(hedge_fac.index).sort_values()
    per = idx.to_period("M").asi8
    if len(per) > 1 and (np.diff(per) != 1).any():
        raise ValueError("gaps in the monthly overlap of ff49 and hedge_fac")

    F = hedge_fac.loc[idx, HEDGE_COLS].astype(float)
    rb = ff49.loc[idx, list(brown)].mean(axis=1, skipna=False) - hedge_fac.loc[idx, "RF"]
    y = rb.to_numpy(float)
    X = np.column_stack([np.ones(len(idx)), F.to_numpy()])
    usable = np.isfinite(y) & np.isfinite(X).all(axis=1)
    coef = np.full((len(idx), 1 + len(HEDGE_COLS)), np.nan)
    for i in range(beta_window - 1, len(idx)):
        lo = i - beta_window + 1
        if usable[lo:i + 1].all():
            coef[i] = np.linalg.lstsq(X[lo:i + 1], y[lo:i + 1], rcond=None)[0]

    out = pd.DataFrame(index=idx)
    out["RB"] = rb
    for k in HEDGE_COLS:
        out[f"f_{k}"] = F[k]
    out["a"] = coef[:, 0]
    for j, k in enumerate(HEDGE_COLS):
        out[f"b_{k}"] = coef[:, j + 1]
    fit_prev = out["a"].shift(1) + sum(out[f"b_{k}"].shift(1) * out[f"f_{k}"] for k in HEDGE_COLS)
    out["e"] = out["RB"] - fit_prev
    out["sd"] = out["e"].rolling(sd_window, min_periods=sd_window).std(ddof=1)
    out["m"] = np.minimum(1.0, vol_target / (np.sqrt(12.0) * out["sd"]))
    return out

def test_leg_pipeline():
    raw = make_synthetic_inputs(seed=3)
    ff49, hf = raw["ff49"].set_index("date"), raw["hedge_fac"]
    leg = leg_pipeline(ff49, hf)
    T = pd.Timestamp("2000-06-30"); i = leg.index.get_loc(T); Tm1 = leg.index[i - 1]
    win = leg.index[i - 59:i + 1]
    Xw = np.column_stack([np.ones(60), hf.loc[win, HEDGE_COLS].to_numpy()])
    yw = (ff49.loc[win, BROWN].mean(axis=1) - hf.loc[win, "RF"]).to_numpy()

```

```

bh = np.linalg.lstsq(Xw, yw, rcond=None)[0]
assert np.allclose(leg.loc[T, ["a"] + [f"b_{k}" for k in HEDGE_COLS]].to_numpy(float), bh, rtol=0, atol=1
    ↪ e-12)
e_hand = leg.at[T, "RB"] - leg.at[Tm1, "a"] - sum(leg.at[Tm1, f"b_{k}"] * hf.at[T, k] for k in HEDGE_COLS)
    ↪
assert abs(leg.at[T, "e"] - e_hand) < 1e-12 # residual uses t-1 coefficients
ff49b, hfb = ff49.copy(), hf.copy() # month T+1 cannot reach row T
ff49b.loc[leg.index[i + 1], BROWN] += 0.2
hfb.loc[leg.index[i + 1], HEDGE_COLS] += 0.2
pd.testing.assert_frame_equal(leg.iloc[:i + 1], leg_pipeline(ff49b, hfb).iloc[:i + 1])
hf2 = hf.copy() # hedge SMB moves b, not X
hf2["SMB"] = hf2["SMB"] + 0.01 * np.random.default_rng(0).standard_normal(len(hf2))
assert (leg_pipeline(ff49, hf2)["b_SMB"] - leg["b_SMB"]).abs().max() > 1e-3
_, o1 = run_panel(raw, *TEST_WINDOW)
_, o2 = run_panel(dict(raw, hedge_fac=hf2), *TEST_WINDOW)
pd.testing.assert_frame_equal(o1["X"], o2["X"])
try:
    leg_pipeline(ff49, raw["fac"])
except ValueError:
    pass
else:
    raise AssertionError("FF5 frame accepted as hedge_fac")

# =====
# 4. pass_fail_table
# =====
def _fin(v):
    try:
        return v is not None and bool(np.isfinite(float(v)))
    except (TypeError, ValueError):
        return False

def _fmt(v, spec=".6f"):
    return format(float(v), spec) if _fin(v) else "nan"

def pass_fail_table(alpha, t_alpha, p_alpha, shuffle_p, n_episodes, loo_alpha,
                    t_bar=2.0, p_bar=0.05, min_episodes=8, n_brown=5):
    """
    Inputs : alpha, t_alpha, p_alpha (NW(6) fit; p from t(n-k)), shuffle_p, n_episodes (merged holds),
            loo_alpha (Series of drop-one alphas indexed by the dropped industry).
    Output : DataFrame indexed (i), (ii), (iii), (iv), overall; columns name, statistic, bar, value,
            passed, result, detail.
    Timing : ex post summary of the single run.
    (iv) : "keeps its sign" is read as "stays positive": the full alpha and all five drop-one alphas
            must be > 0. A negative alpha whose drop-one alphas are also negative fails. So does a
            missing or NaN drop-one alpha.
    """
    loo = pd.Series(loo_alpha, dtype=float)
    all_a = pd.concat([pd.Series({"full": alpha}, dtype=float), loo])
    ok1 = _fin(alpha) and _fin(t_alpha) and alpha > 0 and t_alpha >= t_bar
    ok2 = _fin(shuffle_p) and shuffle_p <= p_bar
    ok3 = _fin(n_episodes) and n_episodes >= min_episodes
    ok4 = len(loo) == n_brown and bool(all_a.notna().all()) and bool((all_a > 0).all())
    rows = [
        ("i)", "timing alpha", "NW(6) t of alpha", f"alpha > 0 and t >= {t_bar:g}", t_alpha, ok1,
         f"alpha = {_fmt(alpha)}/month, p = {_fmt(p_alpha, '.4f')} (t, n-k df)"),
        ("ii)", "calendar shuffle", "shuffle p", f"<= {p_bar:g}", shuffle_p, ok2, ""),
        ("iii)", "independent episodes", "merged hold episodes", f">= {min_episodes}", n_episodes, ok3, ""),
        ("iv)", "drop each Brown industry", "min(full, drop-one alphas)", f"> 0, all {n_brown + 1}",
         all_a.min() if len(all_a) else np.nan, ok4, ", ".join(f"{k}: {_fmt(v)}" for k, v in all_a.items()))
    ]
    n_pass = int(sum(r[5] for r in rows))
    rows.append(("overall", "verdict", "components passed", "4 of 4", n_pass, n_pass == 4, f"{n_pass} of 4"))
    t = pd.DataFrame(rows, columns=["component", "name", "statistic", "bar", "value", "passed", "detail"])
    t = t.set_index("component")

```

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t["result"] = np.where(t["passed"], "PASS", "FAIL")
return t[["name", "statistic", "bar", "value", "passed", "result", "detail"]]

def test_pass_fail_table():
    pos, neg = pd.Series(0.002, index=BROWN), pd.Series(-0.001, index=BROWN)
    t = pass_fail_table(-0.002, -1.0, 0.3, 0.5, 5, neg)
    assert t.at["(iv)", "result"] == "FAIL" and t.at["overall", "result"] == "FAIL" # the reported bug
    t = pass_fail_table(0.004, 2.0, 0.05, 0.05, 8, pos) # boundaries pass
    assert (t["result"] == "PASS").all()
    t = pass_fail_table(0.004, 2.5, 0.01, 0.01, 9, pd.Series([0.002] * 4 + [-1e-5], index=BROWN))
    assert t.at["(iv)", "result"] == "FAIL" and t.at["overall", "value"] == 3
    t = pass_fail_table(0.004, 2.5, 0.01, np.nan, 9, pos.iloc[:4])
    assert t.at["(ii)", "result"] == "FAIL" and t.at["(iv)", "result"] == "FAIL"
    t = pass_fail_table(0.004, 1.99, 0.05, 0.01, 7, pos)
    assert t.at["(i)", "result"] == "FAIL" and t.at["(iii)", "result"] == "FAIL"

# =====
# 5. check_no_lookahead
# =====
def check_no_lookahead(raw, run_fn, start, end, cut_to=None, delta=0.05, n_months=8, min_obs=48,
                      seed=0, tol=1e-10, state_cols=STATE_COLS, pos_of=POS_OF, brown=BROWN):
    """
    Look-ahead audit on both the state side and the return side.

    Inputs
    raw : raw dict (read_raw / make_synthetic_inputs).
    run_fn : raw -> panel by month-end t with w, w_ao (positions set at end of t), RT, RAO (net returns
            earned in t), m, hold, b_k (end-of-t states).
    start,end: window audited.
    cut_to : if given, every input is cut at this month BEFORE anything runs, so later data are never
            loaded (seen mode: 2022-07-31).
    Tests (one report row per test and month T)
    post_window inputs cut at 'end': all states and returns in the window identical.
    truncate inputs cut at T: states through T-1 identical (the old check; T >= start + min_obs).
    rb_perturb every Brown industry +delta in T: states and returns before T identical, and
            RT_T moves by exactly w_{T-1}*delta (RAO_T by w_ao_{T-1}*delta).
    f_perturb hedge-file Mkt-RF, SMB, HML +delta in T: same, with move -w_{T-1}*delta*sum_k b_{T-1,k}.
    signal_perturb EMVENRGYENVREG for T spiked: states through T and returns through T+1 identical,
            because the value for T is usable only at the end of T+1.
    A month-t return hedged with b_t, or a position earning its own month, leaves every end-of-month
    state unchanged. Both do change how RT_T and RAO_T respond to month-T data, and the two perturb
    tests measure that response. D and alpha are not compared: pi is an ex post window constant by design.
    Months T : fixed-seed draw of up to n_months each from hold entries/exits, held months (w_{T-1} != 0)
            and flat months (w_{T-1} = w_T = 0) inside the window.
    Output : (report DataFrame, ok). Stop if ok is False.
    """
    start, end = pd.Timestamp(start), pd.Timestamp(end)
    if cut_to is not None:
        raw = truncate_raw(raw, cut_to)
        end = min(end, pd.Timestamp(cut_to))
    rows = []

    def add(test, T, pre, err):
        rows.append(dict(test=test, month=T, pre_diff=pre, ret_err=err, passed=bool(pre <= tol and err <= tol)
            ⇨ ))

    ret_cols = list(pos_of)
    all_cols = state_cols + ret_cols
    base = run_fn(raw)
    if _last_date(raw) > end:
        add("post_window", end, _maxdiff(base.loc[start:end], run_fn(truncate_raw(raw, end)), all_cols), 0.0)

    w, wprev = base["w"], base["w"].shift(1)
    inwin = base.index.to_series().between(start, end)
    groups = {
        "change": inwin & (w != wprev) & ((w == 0) | (wprev == 0)) & wprev.notna(),

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        "held": inwin & (wprev != 0) & wprev.notna(),
        "flat": inwin & (wprev == 0) & (w == 0),
    }
    rng = np.random.default_rng(seed)
    months = pd.DatetimeIndex([])
    for mask in groups.values():
        ix = base.index[mask.to_numpy()]
        if len(ix) > n_months:
            ix = ix[np.sort(rng.choice(len(ix), n_months, replace=False))]
        months = months.union(ix)
    if len(months) == 0:
        add("coverage", pd.NaT, np.inf, 0.0)

    for T in months:
        pos = base.index.get_loc(T)
        Tm1, Tp1 = base.index[pos - 1], base.index[min(pos + 1, len(base) - 1)]
        before = base.loc[start:Tm1]
        if T >= start + pd.DateOffset(months=min_obs):
            add("truncate", T, _maxdiff(before, run_fn(truncate_raw(raw, T)), state_cols), 0.0)

        p = run_fn(_bump(raw, "ff49", T, list(brown), lambda v: v + delta))
        exp = {r: base.at[Tm1, c] * delta for r, c in pos_of.items()}
        add("rb_perturb", T, _maxdiff(before, p, all_cols), _ret_err(base, p, T, exp))

        p = run_fn(_bump(raw, "hedge_fac", T, HEDGE_COLS, lambda v: v + delta))
        bsum = sum(base.at[Tm1, f"b_{k}"] for k in HEDGE_COLS)
        exp = {r: -base.at[Tm1, c] * delta * bsum for r, c in pos_of.items()}
        add("f_perturb", T, _maxdiff(before, p, all_cols), _ret_err(base, p, T, exp))

        p = run_fn(_bump(raw, "EMVENRGYENVREG", T, ["EMVENRGYENVREG"], lambda v: 5.0 * v + 0.5))
        pre = max(_maxdiff(base.loc[start:T], p, state_cols), _maxdiff(base.loc[start:Tp1], p, ret_cols))
        add("signal_perturb", T, pre, 0.0)

    rep = pd.DataFrame(rows)
    return rep, bool(len(rep)) and bool(rep["passed"].all())

def _fred_month_end(df, sid):
    s = df.set_index("observation_date")[sid].astype(float)
    s.index = pd.DatetimeIndex(s.index) + pd.offsets.MonthEnd(0)
    return s

def _toy_run(raw, bug=None):
    """Minimal pipeline used only to test the auditor. bug: None, 'hedge_bt' (month-t return hedged with b_t),
    ↪
    'own_month' (position set at end of t earns month t), 'signal_lag' (extreme flag of t used at end of t)
    ↪ ."""
    leg = leg_pipeline(raw["ff49"], raw["hedge_fac"])
    sig = attention_signal(_fred_month_end(raw["EMVENRGYENVREG"], "EMVENRGYENVREG"),
        _fred_month_end(raw["EMVOVERALLEMV"], "EMVOVERALLEMV"))
    ext = sig["extreme"].reindex(leg.index, fill_value=False).astype(bool)
    cross = ext & ~ext.shift(1, fill_value=False)
    usable = cross if bug == "signal_lag" else cross.shift(1, fill_value=False)
    hold = usable.astype(float).rolling(6, min_periods=1).max().astype(bool)
    m = leg["m"].fillna(0.0)
    B = leg[[f"b_{k}" for k in HEDGE_COLS]].fillna(0.0)
    F = leg[[f"f_{k}" for k in HEDGE_COLS]].to_numpy()
    panel = pd.DataFrame({"m": leg["m"], "hold": hold}, index=leg.index)
    for k in HEDGE_COLS:
        panel[f"b_{k}"] = leg[f"b_{k}"]
    for wcol, rcol, wpos in (("w", "RT", -m.where(hold, 0.0)), ("w_ao", "RAO", -m)):
        panel[wcol] = wpos
        if bug == "own_month":
            p, bb = wpos, B.shift(1)
        elif bug == "hedge_bt":
            p, bb = wpos.shift(1), B
        else:

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```

        p, bb = wpos.shift(1), B.shift(1)
        gross = p * (leg["RB"] - (bb.to_numpy() * F).sum(axis=1))
        ov = B.mul(wpos, axis=0)
        trade = (0.0010 * wpos.diff().abs() + 0.0005 * ov["b_Mkt-RF"].diff().abs()
                + 0.0025 * (ov["b_SMB"].diff().abs() + ov["b_HML"].diff().abs()))
        panel[rcol] = gross - trade.shift(1) # trade at end of t pays in t+1
    return panel

def test_check_no_lookahead():
    raw = make_synthetic_inputs(seed=0)
    kw = dict(start="1995-01-31", end="2009-12-31", n_months=4)
    rep, ok = check_no_lookahead(raw, _toy_run, **kw)
    assert ok, rep.loc[rep["passed"]]
    for bug, must_fail in (("hedge_bt", {"rb_perturb", "f_perturb"}),
                           ("own_month", {"rb_perturb", "f_perturb"}),
                           ("signal_lag", {"signal_perturb"})):
        rep, ok = check_no_lookahead(raw, lambda r, b=bug: _toy_run(r, bug=b), **kw)
        failed = set(rep.loc[~rep["passed"], "test"])
        assert not ok and must_fail <= failed, (bug, failed)
        assert rep.loc[rep["test"] == "truncate", "passed"].all(), bug # the old state-only check misses all
            ↪ three

def test_seen_cut():
    raw = make_synthetic_inputs(seed=0)
    assert _last_date(raw) > pd.Timestamp("2022-12-31")
    assert _last_date(truncate_raw(raw, SEEN_WINDOW[1])) <= pd.Timestamp("2022-07-31")

def run_unit_tests():
    for f in (test_attention_signal, test_pass_fail_table, test_seen_cut, test_leg_pipeline,
            test_check_no_lookahead, test_make_synthetic_inputs):
        f()
        print(f"{f.__name__}: ok")

# =====
# 6. __main__
# =====
if __name__ == "__main__":
    MODE = "synthetic" # "synthetic" -> "seen" -> "test"
    Z_BURN_IN = "nonzero" # your reading; "calendar" reproduces my previous version
    WINDOWS = {"synthetic": TEST_WINDOW, "seen": SEEN_WINDOW, "test": TEST_WINDOW}
    PATHS = {"ff49": "ff49_industry_monthly.csv", **{sid: f"{sid}.csv" for sid in FRED_IDS}}
    fac = None # <- your FF5+UMD frame, as before
    hedge_fac = None # <- NEW: the team's FF3 frame (Mkt-RF, SMB, HML, RF), decimals

    start, end = WINDOWS[MODE]
    if MODE == "synthetic":
        run_unit_tests()
        raw = make_synthetic_inputs(seed=0)
    else:
        if fac is None or hedge_fac is None:
            sys.exit("Set 'fac' (FF5+UMD) and 'hedge_fac' (team FF3) first.")
        raw = read_raw(PATHS, fac, hedge_fac)

    cut_to = end if MODE == "seen" else None
    if cut_to is not None:
        raw = truncate_raw(raw, cut_to) # nothing after 2022-07 reaches the audit or the run

    rep, ok = check_no_lookahead(raw, lambda r: run_panel(r, start, end, z_burn_in=Z_BURN_IN)[0],
                                start, end, cut_to=cut_to)
    print(rep.groupby("test")["passed"].agg(["size", "sum"]).to_string())
    if not ok:
        print(rep.loc[~rep["passed"]].to_string())
        sys.exit("STOP: look-ahead audit failed.")

```

```

run_panel(raw, start, end, z_burn_in=Z_BURN_IN, n_shuffle=5000, loo=True, verbose=True) # prints the five
    ↪ tables
'''

**Checks to run first.** Each of these should match exactly:
- 'attention_signal(env, overall, z_burn_in="calendar")' should give the same 'z', 'thr' and 'extreme' as the
    ↪ old function on the real EMV files. If it doesn't, the difference is one of the conventions in the
    ↪ docstring, and you should keep the old behaviour (your implementation matches it). Tell me which
    ↪ convention it is.
- 'leg_pipeline(ff49, fac[["Mkt-RF","SMB","HML","RF"]])' should reproduce the old leg exactly.
- Run with '("nonzero", FF3)', the full backtest should again match your implementation to 1e-13. (iv) used
    ↪ to pass when it shouldn't have, so compare each pass/fail row, not just the count of passes.

**Choices I made inside these functions**
1. **Zero rule:** I count zero months in the trailing 60 calendar months. Months before 1985-01 are not
    ↪ counted as zero. The month is off when there are more than 6 zeros. Both burn-in settings use this
    ↪ rule.
2. **Off months:** An off month gets no z and is left out of the percentile history, the same as a zero month.
    ↪
3. **Risk-free rate:** RF for RB comes from 'hedge_fac'. In the French files it is the same series as 'fac'
    ↪ s RF, and the driver can assert that.
4. **Criterion (iv):** I read "keeps its sign" as "stays positive": the full alpha and all five drop-one
    ↪ alphas must be above zero.
5. **Positive control:** The "planted shift" is mean(D) with the plant minus mean(D) without it, using the
    ↪ same seed. The plant lowers the Brown leg by 3 percentage points in every held month. That is
    ↪ deliberately large, because the check tests the pipeline, not statistical power.
6. **Look-ahead audit:** It compares positions and returns, not D, because pi is a constant computed over the
    ↪ whole window after the fact, by design.

The verdict is FAIL in your run and in mine, under either burn-in reading, so the default (your reading)
    ↪ doesn't change any conclusion. It is set to reproduce your pre-registered implementation, not picked
    ↪ on results. The seen-window numbers should be reported with the 2022-07 end: alpha of -0.16%/yr, t
    ↪ -0.24.

```

Fact-check (verbatim excerpt) Sections “Summary”, “6. Bug and issue list”, “R2 Summary”, “R2.9” of factcheck.md (10,321 words in full).

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## Summary

**Verdict: the code is correct and reaches the same verdict as the reference.** Its numbers differ from the
    ↪ reference only through two readings that the prompt left open. "Do not implement" stands under either
    ↪ reading.

1. **Same verdict as the reference.** I ran the code on the real data with glue only; no line of ChatGPT's
    ↪ file was edited.
    - It returns FAIL, with 1 of 4 components met (episodes).
    - Timing alpha -0.19%/yr, NW(6) t -0.25, one-sided p 0.60. Shuffle p 0.536. 10 episodes. Drop-one: 4 of 5
        ↪ alphas keep the sign.
    - The reference gives -0.18%/yr, t -0.27, one-sided p 0.605, shuffle p 0.478, 10 episodes, and 3 of 5.

2. **Two readings explain every numeric gap. Neither is a coding error.**
    - ** (A) z burn-in.** ChatGPT requires 60 full calendar months of EMV history before the first z (lines
        ↪ 274-275). Its first z is therefore 1989-12 and its first threshold 1994-12. The reference (C5) lets
        ↪ the 48-nonzero minimum act as the burn-in, which gives a first z in 1989-01 and a first threshold
        ↪ in 1994-01. Under A:
        - the window starts at 1995-02 instead of 1994-03 (179 months, not 190);
        - the threshold moves by up to 0.076;
        - one extra crossing (data month 2002-12) opens a new hold;
        - pi rises from 0.729 to 0.809.
    - ** (B) Hedge factor file.** ChatGPT hedges with the SMB and HML of the 'fac' it was given (the Ken French
        ↪ FF5 file) and subtracts that file's RF (lines 377-378). The team rule uses the team FF3 file. The
        ↪ prompt described only 'fac', so this is a gap in the prompt. SMB differs by up to 3.5 pp a month
        ↪ between the two files. As a result, b_t moves by up to 0.09 and m_t by up to 0.012.
    - ** With both switched to the reference reading (variant AB), ChatGPT's code reproduces every reference
        ↪ number to 1e-13. ** That covers:
        - z, the threshold and every crossing over 1985-2026;

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- the holds, b_t , m_t , and net R^T and R^A ;
 - π and D ;
 - all 15 coefficients with their NW(6) and NW(12) t-statistics and p-values;
 - the decomposition, the 10 episodes and the five drop-one alphas;
 - all 5,000 shuffle draws, draw by draw (largest gap 8e-15; same seed and the same placement algorithm).
 - On the seen 2010-01 to 2022-07 window, AB reproduces the reference dry run exactly.
3. ****One genuine bug, in the test harness, not the backtest (BUG-1).****
- The synthetic GS10 generator (line 881) is a clipped random walk. It sits on its 0.5% floor in 76% of
 - 1995-2009 months for seed 0, and in all 192 months of 1994-2009 for seed 16.
 - BOND is then constant, so $I \times BOND$ acts as an I main effect.
 - As a result, ChatGPT's own positive control fails its own stop rule: only 4% of the planted effect
 - reaches alpha (t 0.98), and $I \times BOND$ has $t = -10.8$.
 - Seed 16 of its null-seed check crashes on a rank-deficient design.
 - One edit fixes it (vfix). The positive control then passes: 105% of the planted effect reaches alpha, t
 - 4.15, shuffle p 0.005.
4. ****ChatGPT's own battery would have caught (B).**** Its check (b)2 (reproduce the team trade to 1e-8) stops
 - on the as-delivered code at a gap of 2.3e-3 a month. With the team FF3 file, the gap is 6e-17 and the
 - check passes.
5. ****Limits in the test design (not bugs):****
- The look-ahead test compares states only. It misses 2 of 6 injected timing bugs, both on the return side.
 -
 - Component (iv) does not require a positive alpha.
 - The burn-in reading is not a parameter, although the prompt asked for every silent reading to be one.
 - The "seen" mode includes 5 holdout months, and they flip the sign of the seen-window alpha.
- ## 6. Bug and issue list (most severe first)**
1. ****BUG-1, genuine (test harness), line 881.****
- Cause: the synthetic GS10 random walk sticks to its 0.5% clip, so BOND is constant.
 - Consequences: (a) $I \times BOND$ stands in for an I main effect and absorbs 96% of the planted effect, so
 - ChatGPT's positive control, check (b)10, would STOP the protocol before the real run; (b) seed 16
 - gives a constant BOND over the whole window, so the rank check (line 521) raises and the (b)9 null-
 - seed loop aborts.
 - The docstring expectation at lines 948-949 (alpha of about $\delta(1 - \pi)\text{mean}(I \ m)$) holds only if no
 - regressor spans I . That is true on real data (99%) but false for this generator.
 - Fixed in vfix by one edit. There is no effect on real-data numbers.
2. ****D1, spec reading, not a parameter (discrepancy A), lines 274-275.****
- ChatGPT requires a full 60-month calendar window, so the "at least 48 nonzero months" clause can never
 - bind. Its assumption 11 says so explicitly.
 - The prompt asked that each silent reading be a parameter. The three points the prompt named are
 - parameters (lines 66-72), but this fourth one is hard-coded.
 - The reference reading (C5) gives the clause a role as the burn-in.
 - Impact: window 1995-02 instead of 1994-03, π 0.81 instead of 0.73, shuffle p 0.54 instead of 0.48. The
 - verdict does not change.
3. ****D2, input gap in the prompt (discrepancy B), lines 377-378.****
- The hedge and the leg's RF use 'fac', the FF5 file, whose SMB is built differently from the FF3 SMB. The
 - Mkt-RF vintages also differ by up to 8 bp; HML and RF are identical.
 - The prompt never said that the team hedge uses the team FF3 file, so this is not ChatGPT's error.
 - Impact on alpha: 0.004%/yr. ChatGPT's own check (b)2 stops on it.
4. ****D3, test coverage, lines 895-941.**** 'check_no_lookahead' compares only states set at the end of T . It
 - misses the injected same-month hedge (M2) and same-month earning (M6) bugs, and it is conditional on '
 - `cfg.pub_lag` itself being right. ChatGPT's separate lag probe (b)3 covers that last point.
5. ****D4, pass-bar logic, lines 730, 744, 754.****
- (iv) passes when every drop-one alpha has the sign of the full alpha, even if that sign is negative. So
 - a negative alpha with five negative drop-one alphas would print "(iv) PASS".
 - This is a literal reading of "keeps its sign". The reference (C26) also requires $\alpha > 0$; the one-line
 - fix is '`bool(a > 0 and same == nl)`'.
 - No effect here: (i) fails, and as delivered (iv) is 4/5.
6. ****D5, "seen" mode, lines 979-980.****
- The seen window ends at 2022-12, so it includes holdout months 2022-08 to 2022-12. They flip the seen
 - alpha from -0.16%/yr (t -0.24, through 2022-07) to +0.26%/yr (t 0.37).
 - The look-ahead call in that mode sets no 'cut_to', so its cuts reach 2024-08 despite the "seen data only
 - " label.
 - This is cosmetic for the frozen test.

R2 Summary

****Verdict: the corrected code is correct.**** With its defaults ('z_burn_in="nonzero"', hedge on the team FF3

```

    ↪ file), it reproduces the reference to rounding error on every component and reaches the same verdict:
    ↪ FAIL, 1 of 4 (episodes only). "Do not implement" stands.

1. **All six round-1 items are fixed: BUG-1 and D1 to D5.** ChatGPT's six unit tests pass the first time,
    ↪ with no edit to ChatGPT's code. So do its three "checks to run first" and its checks (b)2, (b)9 and (
    ↪ b)10.
2. **The test-window run equals the reference.**
  - Result: FAIL 1/4. Alpha -0.18%/yr (NW(6) t -0.27, NW(12) t -0.29, p 0.790, one-sided 0.605). Shuffle p
    ↪ 0.478. 10 episodes. (iv) FAIL.
  - Largest gaps over all 15 terms: coefficients 3.8e-14, t-statistics 3.3e-13, p-values 1.9e-13.
  - The shuffle p is exact, and the 5,000 null draws agree draw by draw to 8.3e-15.
  - Every pass/fail row matches: (i) FAIL, (ii) FAIL, (iii) PASS, (iv) FAIL, overall FAIL.
3. **The two readings are now switches, and each does what it says.**
  - 'z_burn_in="calendar"' reproduces ChatGPT's old signal exactly. The full run equals round-1 v1 + B to 1e
    ↪ -14.
  - The new 'hedge_fac' input puts the team FF3 file into the leg, the hedge and the overlay. The
    ↪ attribution still uses FF5 + UMD.
  - The verdict is FAIL 1/4 under either burn-in reading.
4. **The seen window equals the reference dry run:** alpha -0.12%/yr, t -0.19, shuffle p 0.510, 6 episodes, 0
    ↪ of 4. ChatGPT's instruction to report -0.16%/yr and t -0.24 for this window is wrong for its own new
    ↪ code. Those are round-1 numbers with the FF5-file hedge (N2).
5. **The new look-ahead audit works.**
  - It passes all 361 rows on four runs (synthetic, test under both readings, seen). Its largest return-side
    ↪ error is 5e-17.
  - It flags 7 of 8 injected timing bugs, including round 1's two misses (M2, M6). The eighth (M5) makes the
    ↪ audited pipeline raise an error, which also stops the protocol.
6. **One new issue matters: an integration gap (N1).**
  - The new 'attention_signal' changes its output layout, and the reply's driver instructions do not mention
    ↪ it.
  - A literal drop-in crashes ('KeyError 'off''). The obvious one-line patch then runs silently on a window
    ↪ 14 months too long (1993-01, n = 204): alpha -0.13%/yr, t -0.21. The verdict is still FAIL 1/4.
  - I bridged the gap with a value-preserving adapter.
7. **Other items are minor:** three inaccurate claims (N2 to N4) and some dead or incompatible code (N5).
8. **Extra observation (not about ChatGPT's code).** On 100 further null seeds, the calendar shuffle is
    ↪ correctly sized, but the pre-registered NW(6) t over-rejects (13 of 100 beyond the 5% two-sided
    ↪ critical value). This affects the reference's component (i) as much as ChatGPT's. It cannot turn a
    ↪ FAIL into a PASS (R2.9).

## R2.9 Extra check: size of the two tests on the synthetic null (not about ChatGPT's code)

R12 ran 100 further null seeds (21-120, 200 draws each, no planted effect; 'out/r2_null_seeds_d200_all.csv').

| Statistic | Nominal | Observed (100 seeds) | Reading |
|---|---|---|---|
| Shuffle p <= 0.05 | 5 | 6 (P(>= 6) = 0.38) | Correctly sized. Kolmogorov-Smirnov against uniform: p 0.23 |
| Shuffle p <= 0.10 | 10 | 11 | Correctly sized |
| \NW(6) t\ above the two-sided 5% t(175) critical value | 5 | 13 (P(>= 13) = 0.0015) | Over-rejects: the
  ↪ sd of t is 1.24 over 120 seeds |
| NW(6) t >= 2 (the (i) bar) | 2.4 | 5 (P(>= 5) = 0.09) | Probably oversized |

This concerns the pre-registered inference (C21), which v2 and the reference share. It is not a coding error
  ↪ in either. The likely cause is that D_t switches regime in blocks of 6 to 63 months, and 6 Newey-West
  ↪ lags do not span that. The calendar shuffle is built from the same blocks and stays correctly sized.
  ↪ The verdict is unaffected: the real alpha is negative, and a PASS needs all four components,
  ↪ including (ii).

```

A.3 Exchange 03: robustness design (the multiple-testing framework behind M7)

Files: exchange/03_robustness_design/. Answered by: a model in the ChatGPT role (see the provenance note). Date: 2026-09-26. Purpose: robustness design.

Asked.	A robustness and multiple-testing framework that decides the final verdict: which test families to correct and how, a deflated Sharpe specification with every input, at most two candidates with exact pass bars, the one check most likely to overturn “Do not implement”, a committee’s remaining objections, and a checklist of at most ten items.
Came back.	A framework of 1,716 words of text (cap 1,500) with a hand-computed deflated Sharpe table, two candidates (the momentum optimizer book and the EPA-ranked spread) and a frozen pre-1970 run of the book as the one decisive check.
Verified.	93 checkable claims scored against the module outputs (67 correct, 15 partly correct, 11 wrong); all ten cells of its deflated Sharpe table reproduced to within 0.005; its power, holdout and multiple-testing arithmetic rerun (checks fc03 and fc03b).
Kept.	Families defined by the decision a test could change, the deflated appraisal ratio, Romano–Wolf and SPA tests over the searched strategies, a one-sided holdout reading against a worthwhile alpha, and the frozen 1931–1969 run of the book, all implemented in M7; not its 155-row family, its post hoc composite, its $\pm 2\%$ margin, its MCCC pass test, ETF proxies or its crash budget.

My evaluation

ChatGPT’s statistics hold up; its reading of our books does not. The fact-check scored 93 claims: 67 correct, 15 partly correct, 11 wrong and none unverifiable. Every formula and citation checks out, all ten cells of its deflated Sharpe table reproduce to within 0.005, and its bottom line (no candidate reaches “implement”) matches mine.

The framework is what I kept. Families are defined by the decision a test could change, so loadings, placebos and exploratory rows sit outside every family. The ratio to deflate is the appraisal ratio: the optimizer book earns 4.53% over FF5 ($t = 5.21$) but 2.17% once UMD enters, so its raw Sharpe flatters it. A best-of-many claim needs a max-t or SPA test over every series I could have presented. And the book’s one untouched window, 1931–1969, deserved a frozen, hashed, single run.

Eleven claims were wrong, and three trace back to my prompt. I printed the equal-weight book’s UMD beta (0.99) beside the optimizer’s numbers, and two of its claims rest on it; the optimizer loads 0.27 ($t = 15.2$). I never named the book’s carbon source, so it guessed “2022 intensities”. The other eight are its own. Romano–Wolf is not stricter than BH: a lone BH rejection on the 8,091-cell grid needs $p \leq 6.2e-6$, while max-t would flag the grid’s best cell ($p = 0.00018$) at an effective count below about 285. Loadings are 32% of the nominal hits, not “most”. M2 and M5 did test 25 bp. “The BOND control helps” contradicts M3. Its window cannot start before 1931-07 (the optimizer needs 60 months), and the 5% tracking target holds prewar residual volatility down (5.14% against 5.44%). “Insufficient evidence, not evidence of no alpha” is false for the attention thesis: each of the six rules rejects a +2% holdout alpha at one-sided $p \leq 0.0016$. It also broke the word limit (1,716 against 1,500).

Several of its pass bars would have misread our data. The DSR clause in its pre-1970 bar is decided by post-1970 data and fails on its own arithmetic beyond 7 trials, so its “one check” could never overturn anything. Its $\pm 2\%$ TOST calls a significantly negative holdout “insufficient evidence”. Its MCCC control raises the EPA spread’s alpha (the shock loading is negative), and the pass is knife-edge ($t = 2.13$ or 1.85, by shock construction). “No significant carbon cost” rewards low power.

It missed the conflict between its own yardsticks. At 42 effective trials a 5% max-t bar is 3.03 and a 0.95 DSR needs about 3.85; the book’s t of 3.07 sits between them, and the reply never says which governs. It ignored decay (2.99% in validation, -0.34% in the holdout) and gave no reading of a fail at roughly 0.6 power.

What I adopted (adopted_checks.md, hashed at 15:49 UTC before any pre-1970 return of the book existed): all 23,923 tests sorted into seven families; one-sided Holm and BH on the

73 distinct primary alphas; BH and BY on the grids, by window; Romano–Wolf and SPA over two search families; the deflated appraisal ratio, required in both windows; a holdout margin of a quarter appraisal ratio; a 25 bp cost stress; the frozen run with three pre-set readings; and a verdict map. The rule it lacked: max-t decides whether an alpha survives the search, and only the DSR can move a verdict. I dropped the 155-row family (M1’s Fisher test, $p = 1e-32$, would “survive” it), the post hoc composite, the $\pm 2\%$ margin, the MCCC pass test, ETF proxies and a crash budget set after 2009.

The frozen run passed and the verdict held. The book earned 1.95% a year over FF3+UMD before 1970 ($t = 2.45$; halves 1.71% and 0.90%), but its deflated appraisal ratio is 0.559 and its holdout alpha -0.34% , so it moves to paper trading at pilot size. Do not implement stands.

Prompt (verbatim) prompt.md, 997 words (994 excluding Markdown symbols).

You are the skeptical member of an investment committee. Your job is to design the robustness and
 ↳ multiple-testing framework that decides my project’s final verdict. Everything you need is
 ↳ below.

SETTING. This is my solo extension of our team’s MFE 230GA (Active Asset Management, UC Berkeley)
 ↳ final project. The brief grades thesis 40%, execution 40% and originality 20%, and requires
 ↳ "Do not implement" when there is no alpha. Our team’s thesis: short a Brown industry leg (
 ↳ Util, Ships, Aero, Steel, BldMt, from the Fama-French 49), hedged with rolling 60-month FF3
 ↳ betas, for 3 or 6 months after "climate-transition attention" crosses its expanding 80th
 ↳ percentile. The six variants use a raw or macro-purified signal with either hold, or a
 ↳ continuous weight. Validation is 2010-01 to 2022-07, the holdout 2022-08 to 2026-07. The
 ↳ verdict is Do not implement. I ran eight modules; an independent verifier reproduced each.
 ↳ Each module names its primary tests in its code; only the 1994-2009 rule (item 7) has a
 ↳ hash and timestamps proving they came first.

EVIDENCE (annualized; Newey-West(6) t , p from $t(n-k)$).

1. Team record. Validation: the 6-month rules earn FF3 alphas of 2.1% to 2.5% (t 2.2 to 2.6).
 ↳ Holdout: all six lose 0.5% to 3.8% a year net (FF3 alpha t down to -2.19), and the IC flips
 ↳ from -0.09 to $+0.12$. Audit: the team’s "2 of 32 cells survive Holm" were two 24-month
 ↳ COVID regressions with normal p -values (0 of 32 with $t(n-4)$); a missing October 2025 CPI
 ↳ print created the whole Pure-versus-Original holdout gap; the emissions file has no source,
 ↳ units or date.
2. The signal is FRED EMVENRGYENVREG, a volatility-news tracker for energy and environmental
 ↳ regulation. Its z -score correlates -0.0004 with the Ardia et al. Media Climate Change
 ↳ Concerns index (MCCC) and 0.12 with climate policy uncertainty (CPU), through overall
 ↳ volatility news (partial r 0.05). It is exactly zero in 31 of 48 holdout months.
3. MCCC and CPU in the identical machinery: 0 of 12 validation alphas reach t 1.96 (EMV: 4 of 6),
 ↳ all six MCCC holdout alphas are negative, and the smallest Holm p over 40 primary tests is
 ↳ 1.00. A signal-free always-short Brown position earns 71% of the COVID alpha.
4. Christian’s tests (8-and-8 legs, FF5+UMD+commodity controls, HML source, 5/10/25 bp costs):
 ↳ 8,091 alpha tests, 0 positive survivors under Holm or BH; 0 of 900 attention-rule holdout
 ↳ alphas are positive.
5. Alpha versus beta. A traded 10-year Treasury factor (BOND) plus UMD makes every holdout alpha
 ↳ slightly more negative; rates moved holdout returns by 0.5 pp a year at most. The Lewellen-
 ↳ Nagel beta-timing term is negative (-0.6% to -1.1%), and the always-short benchmark carries
 ↳ it too. The COVID gain is market and size exposure the lagged hedge left open.
6. Emissions. On EPA supply-chain intensities Aero and Ships rank 31st and 20th of 49 (the team
 ↳ file: 3rd and 2nd of 41). The EPA-ranked Green-minus-Brown spread earns 6.4% post-2010 (t
 ↳ 1.72) and 9.6% in the holdout (t 1.61), but its pre-specified FF5+UMD alpha is 4.7% (t
 ↳ 1.52). Its full-sample alpha, 4.5% (t 2.58), comes from short CMA and UMD loadings and a
 ↳ 2022 sort applied back to 1970.
7. Frozen rule. A pre-registered EMV-share rule, scored once on the unseen 1994-03 to 2009-12
 ↳ window, fails 3 of 4 pass-bar components: alpha -0.18% (t -0.27), calendar-shuffle p 0.48.

8. Shrinkage timing (Lehnherr-Mehta-Nagel style ridge). Cross-validation picks the maximum penalty
 - ↪ in 100% of monthly refits, so every forecast is the historical mean; attention lowers out-
 - ↪ of-sample R2 at every fixed penalty.
9. Carbon-aware industry momentum (8 long, 8 short, t-12 to t-2, 10 bp). Net Sharpe 0.47, but the
 - ↪ pre-registered FF5+UMD alphas are 1.74% (t 1.09) and 1.86% post-2010 (t 0.71), with UMD
 - ↪ beta 0.99. A Grinold-Kahn optimizer book (IC fixed at 0.05, 5% ex-ante tracking; its alpha
 - ↪ was exploratory) has full-sample alpha 2.17% (t 3.07, BH p 0.027) and IR 0.555, post-2010
 - ↪ alpha 3.12% (t 2.18), but holdout alpha -0.34% (t -0.15). Rank IC: 0.054 full sample, 0.030
 - ↪ holdout. Carbon frontier: at bound b = -1 long-leg carbon intensity falls 60% for an IR
 - ↪ change of -0.005 (95% CI -0.110 to +0.096); only b = -2 costs significantly (-0.284, p
 - ↪ 0.038).

TEST COUNT. The eight ledgers hold 23,923 tests: 155 primary, 8,113 robustness, 15,485 exploratory,
 ↪ 170 placebo or reference. About 4,200 have nominal p < 0.05, many of them factor loadings.
 ↪ They overlap heavily: shared holdout months, near-identical variants, one grid of 8,091
 ↪ alphas.

UNSEEN DATA. FF49 returns start in 1926-07 and FF5 in 1963-07, but no module computed a strategy
 ↪ return before 1970.

WHAT I NEED, under these headings.

1. Families. How to control the family-wise error rate or the false discovery rate across a
 - ↪ project this size: which families to define, which get Holm or Romano-Wolf and which get BH
 - ↪ or BY, whether loadings and descriptive tests belong in any family, and how to treat
 - ↪ dependence between near-duplicate variants.
2. Deflated Sharpe. How to deflate the best Sharpe ratios for the number of trials: which ratios,
 - ↪ what counts as a trial, how to estimate the effective number of independent trials, and
 - ↪ every input the formula needs. I have the optimizer book (679 months, annualized Sharpe
 - ↪ 0.555, skew -0.28, raw kurtosis 5.77; post-2010: 199 months, 0.70, 0.29, 3.66) and the EPA
 - ↪ spread (post-2010: 0.52, 0.35, 4.77).
3. Candidates. At most two that deserve the strictest scrutiny, and the exact tests that would
 - ↪ move each to "implement": sample, benchmark, statistic, pass bar.
4. The one robustness check most likely to overturn "Do not implement", and the result that would
 - ↪ do it.
5. What an investment committee would still object to after all of this.
6. A prioritized checklist, at most ten items: test, pass bar, what a fail means.

RULES. Work from these numbers. Cite a paper only with authors, year and the specific claim. Label
 ↪ any number you have not computed as a guess. Do not propose new signals. "No candidate can
 ↪ reach implement" is an acceptable answer if you say why. Under 1,500 words.

Reply (verbatim; model in the ChatGPT role) chatgpt_response.md, 1,907 words
 (1,716 excluding Markdown symbols).

****Bottom line:**** no candidate reaches "implement" on the current evidence. The attention thesis is finished (
 ↪ items 1-3, 7 and 8). Two strategies from outside the thesis deserve strict scrutiny. Both fail today,
 ↪ and only one still has an unseen sample that could change that.

1. Families

Group tests by the decision they could change, not by module.

- ****Confirmatory family, controlled for family-wise error.**** This holds only the tests that could flip the
 - ↪ verdict: each candidate's pre-specified alpha test. Make them one-sided, because only a positive
 - ↪ alpha leads to implementation.
- Use Romano-Wolf stepdown. Romano and Wolf (2005, *Econometrica*) show that a bootstrap stepdown controls
 - ↪ the family-wise error rate while using the dependence between tests, so it has more power than Holm.

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    ↪
- Bootstrap with a stationary block bootstrap that resamples the same months across every series. Report
    ↪ Holm alongside, since Holm is valid under any dependence.
- **The 155 "primary" labels.** Only item 7 can be shown to have been pre-registered. Treat all 155 as one
    ↪ project-level family, not eight module families. A committee cannot check labels a module gave itself.
    ↪
- **Robustness grids, controlled for false discovery rate.** Christhian's 8,091 alphas are a sensitivity
    ↪ analysis.
- BH (Benjamini and Hochberg 1995) holds under positive dependence. Benjamini and Yekutieli (2001) prove
    ↪ this for PRDS test statistics, and near-duplicate one-sided alphas are plausibly PRDS.
- BY, from the same 2001 paper, holds under any dependence but costs a factor of about  $\ln m + 0.58$ . That is
    ↪ about 9.6 for  $m = 8,091$  (my arithmetic, a guess).
- BH already finds zero positive survivors, and BY and Romano-Wolf are stricter, so they will not rescue
    ↪ anything.
- Report the grid as a distribution: the share of positive alphas and the median  $t$ .
- **Exploratory tests (15,485): no family.** They generate hypotheses. Nothing here can be confirmed on the
    ↪ data that produced it, and the optimizer book's full-sample alpha sits in this group.
- **Excluded: loadings, descriptive statistics and placebos.**
- Loadings (UMD beta 0.99, the beta-timing term) are diagnostics of the mechanism. Report them with
    ↪ confidence intervals.
- Placebos calibrate the null; they are not hypotheses.
- Most of the roughly 4,200 nominal hits are loadings, so that count tells you nothing.
- **Near-duplicate variants**, in order of preference:
  1. Pre-specify one composite statistic per strategy family, for example the equal-weight average of the six
    ↪ variants' alphas. That is one test, with more power.
  2. Use a max- $t$  bootstrap such as Romano-Wolf, which handles the correlation automatically.
  3. To count tests, use Nyholt (2004):  $M_{\text{eff}} = 1 + (M-1)(1 - \text{Var}(\lambda)/M)$ , where  $\lambda$  are the
    ↪ eigenvalues of the variants' return-correlation matrix.
- **"Best of many" claims.** Use White's Reality Check (2000, *Econometrica*) or Hansen's SPA test (2005, *
    ↪ JBES*). Both test whether the best strategy beats the benchmark after allowing for the search. SPA is
    ↪ less distorted by poor, irrelevant alternatives.

## 2. Deflated Sharpe

**Formula.** Bailey and Lopez de Prado (2014, *JPM*) define the deflated Sharpe ratio as  $DSR = PSR(SR0)$ ,
    ↪ where:


$$PSR(SR0) = \Phi[(SR^{\wedge} - SR0) \cdot \sqrt{T-1} / \sqrt{1 - \gamma_3 \cdot SR^{\wedge} + ((\gamma_4 - 1)/4) \cdot SR^{\wedge 2}}]$$



$$SR0 = \sqrt{V[SRn]} * [(1 - \gamma) \cdot \Phi^{-1}(1 - 1/N) + \gamma \cdot \Phi^{-1}(1 - 1/(N \cdot e))]$$


**Inputs:**
-  $SR^{\wedge}$  as a monthly figure, not annualized
-  $T$ , the number of months
- skewness  $\gamma_3$ 
- raw kurtosis  $\gamma_4$ 
-  $N$ , the effective number of independent trials
-  $V[SRn]$ , the variance of Sharpe estimates across trials, in monthly units
-  $\gamma \sim 0.5772$  (the Euler-Mascheroni constant)
- pass bar:  $DSR \geq 0.95$ 

**Which ratios to deflate:**
1. The book's Sharpe ratio.
2. The book's appraisal ratio (alpha divided by residual volatility). The claim is alpha, and with a UMD beta
    ↪ of 0.99 the raw Sharpe is mostly momentum.
3. The EPA spread's post-2010 Sharpe ratio.

**What counts as a trial:** every return series you looked at that could have been presented as "the strategy
    ↪ ." That includes the team variants, the MCCC and CPU versions, Christhian's grid, momentum lookbacks,
    ↪ carbon bounds, the optimizer's IC and tracking-error settings, and the choice of sample window (full
    ↪ versus post-2010). Placebos and loadings are not trials.

**Effective number of trials:** cluster the trials' return series. Lopez de Prado and Lewis (2019, *
    ↪ Quantitative Finance*) use clustering of strategy returns to estimate how many trials are effectively
    ↪ independent. Alternatively, use the eigenvalue  $M_{\text{eff}}$  above.

** $V[SRn]$ :** use the variance of Sharpe ratios across clusters. Setting it to the null value  $1/(T-1)$  gives a
    ↪ lower bound on  $SR0$ .

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**Autocorrelation:** PSR assumes independent returns. Your Newey-West t-statistics are below their iid
    ↪ equivalents (EPA: 1.72 against about 2.1). Correct with Lo (2002, *FAJ*), which gives the serial-
    ↪ correlation adjustment for annualized Sharpe ratios and their standard errors.

**Hand arithmetic with  $V = 1/(T-1)$ . These are my calculations from your summary statistics; treat every
    ↪ figure as a guess until you code it.

| Series | PSR(0) | DSR, N=10 | DSR, N=100 | DSR, N=1,000 |
|---|---|---|---|---|
| Book, full sample (0.555, T=679) | ~1.00 | ~0.99 | ~0.94 | ~0.81 |
| Book, post-2010 (0.70, T=199) | ~0.998 | ~0.90 | ~0.62 | ~0.34 |
| EPA, post-2010 (0.52, T=199) | ~0.98 | ~0.71 | - | - |

- **EPA spread:** DSR is only about 0.94 to 0.95 at N = 2 (the team ranking versus the EPA ranking), before
    ↪ the autocorrelation correction. It fails.
- **Book, appraisal ratio:** roughly  $t/\sqrt{\text{years}} \sim 0.41$ . It clears DSR  $\geq 0.95$  only if the effective number
    ↪ of trials is at most about 7 (guess). Your project almost certainly has more, so the full-sample
    ↪ alpha does not survive deflation.

## 3. Candidates

**A. Carbon-aware momentum optimizer book.** It has the strongest statistics, but its alpha was exploratory.
- **Holdout:** -0.34% with a standard error of about 2.3% (backed out from the t). Too short either to
    ↪ confirm the alpha or to kill it.
- **Holdout rank IC:** 0.030, about one standard error below 0.054. I guess that standard error at about 0.02
    ↪ for 48 months x 49 industries. The signal has not clearly died.
- **Sample:** 1927-07 to 1969-12. FF49 starts in 1926-07 and the momentum signal needs 12 months of history.
    ↪ Nothing has touched this period.
- **Freeze before running:** the code, IC 0.05, 5% tracking error, b = -1, 10 bp costs, and the rule for
    ↪ industries with missing early data. Hash and timestamp it as in item 7, then run it once.
- **Benchmark:** FF3 + UMD, because FF5 starts in 1963-07. As a secondary test, FF5 + UMD on 1963-07 to
    ↪ 1969-12. The carbon ranks use 2022 intensities. That is acceptable here because carbon acts as a
    ↪ constraint, not as the source of return.
- **Statistics:** the one-sided NW(6) t on alpha, and the alpha of the book minus the unconstrained book.
- **Pass bar:**
    - alpha t  $\geq 2.0$ ;
    - alpha positive both before and after 1948;
    - no significant carbon cost;
    - project-wide DSR on the appraisal ratio  $\geq 0.95$ .
- **Power (guess):** your t implies residual volatility of about 5% a year, so the standard error is about
    ↪ 0.8% a year over 43 years. A true 2% alpha would clear t = 2.0 about two-thirds of the time. Residual
    ↪ volatility before the war is probably higher, which lowers power.

**B. EPA Green-minus-Brown spread.** This one is weaker.
- Its pre-specified alpha has t 1.52.
- Its full-sample alpha comes from CMA and UMD loadings and a 2022 sort applied back to 1970.
- Pastor, Stambaugh and Taylor (2022, *JFE*) attribute green stocks' outperformance in 2012-2020 to
    ↪ unexpected rises in climate concern, not to higher expected returns.
- **Test:** post-2010 FF5 + UMD alpha with MCCC innovations added as a control. MCCC is an existing series,
    ↪ not a new signal.
- **Pass bar:** one-sided alpha t  $\geq 2.0$  and DSR  $\geq 0.95$ .
- It has no clean unseen sample. Emissions intensities as they stood at the time do not exist before 2010,
    ↪ and a pre-1970 test says nothing about climate. It cannot reach "implement" within this project; its
    ↪ only route is live paper-trading.

## 4. The one check

**The frozen pre-1970 run of the book.** The result that would overturn "Do not implement":
- FF3 + UMD alpha of roughly 1.5% a year or more (guess) with one-sided t  $\geq 2.0$ ;
- the same sign in both halves;
- DSR  $\geq 0.95$  at a documented effective number of trials.

Even that would justify only "implement at pilot size with paper-trading," because the post-1970 evidence is
    ↪ exploratory and the holdout is flat. It would also change the verdict for a different strategy, not
    ↪ the attention thesis. EMV data start in 1985, and the frozen 1994-2009 test has already failed.

## 5. What the committee will still object to

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- **It is just momentum.** The book is industry momentum with a carbon screen (UMD beta 0.99). Why not buy a cheaper momentum factor? The incremental alpha is small and was found by searching.
- **Crash risk.** Daniel and Moskowitz (2016, *JFE*) show momentum suffers rare, persistent crashes in panic states after market declines, when the market rebounds. The full-sample skew is -0.28.
- **Low power.** Forty-eight holdout months cannot tell 0% from a 2-4% alpha. "Do not implement" rests on insufficient evidence, not on evidence of no alpha.
- **Unverified pre-registration.** Only item 7 is provable. With 23,923 tests, every other "primary" label has to be taken on trust.
- **Data provenance.** One missing CPI print created an entire result. The emissions file has no source. The 2022 carbon ranks are applied back to 1970.
- **Implementability.** FF49 portfolios cannot be traded directly. ETF proxies, borrow costs for shorting Aero and Ships, and costs above 10 bp are untested.
- **Regime.** The IC flip and the across-the-board holdout losses coincide with the 2022 rate and energy shock. The BOND control helps, but 48 months cover a single regime.

6. Checklist

#	Test	Pass bar	A fail means
1	Frozen pre-1970 run of the book	$\alpha t \geq 2.0$; positive in both halves	The book is a post-1970 artifact; close it
2	DSR on the book's appraisal ratio, effective trials from clustering	≥ 0.95	The full-sample alpha is a product of selection
3	Project-wide one-sided Romano-Wolf on the 155 primary tests	any positive survivor at 5%	Confirms the verdict
4	Hansen SPA over all strategy return series against zero alpha	$p < 0.05$	The best result from the search is noise
5	Holdout power and equivalence (TOST, +/-2% a year margin, a guess)	90% interval inside the margin	The holdout cannot settle it; say "insufficient evidence"
6	EPA spread with MCCC-innovation control	$\alpha t \geq 2.0$	Its returns came from a shock in climate concern
7	Data audit: ALFRED vintages for CPI and EMV, emissions source	headline figures move < 0.1 pp	Rerun everything downstream
8	Pre-registered crash stress on the book (1932, 2009)	drawdown within the stated budget	Needs smaller size or an overlay
9	ETF proxies with 25 bp costs	net $\alpha > 0$	The alpha exists only on paper
10	Carbon frontier re-run on pre-1970 data	IR change at $b = -1$ not significant	The carbon constraint is not free

This exchange had one round: no follow-up prompt was sent.

Fact-check (verbatim excerpt) Section "Summary" of factcheck.md (7,697 words in full).

Summary

- I extracted 93 checkable claims and recommendations: **67 correct**, 15 partly correct, 11 wrong, 0 unverifiable. The statistics are sound and the arithmetic is right. The errors concern our own strategies, and several of them trace back to the prompt.
- The formulas and hand arithmetic check out.**
 - The PSR, DSR and SRO formulas match Bailey and Lopez de Prado (2014): monthly Sharpe ratio, raw kurtosis in $(\gamma_4 - 1)/4$, and the expected-maximum term with the Euler-Mascheroni constant.
 - The BY constant is right: the harmonic sum is 9.576 at $m = 8,091$, against $\ln m + 0.58 = 9.576$. The Nyholt formula is printed correctly.
 - All ten cells of its DSR table reproduce to within 0.005 (largest gap 0.0047, 'dsr_table.csv').
 - Its appraisal ratio (0.41) and its break-even count of "about 7" trials reproduce. The exact FF5+UMD appraisal ratio is 0.400, and it clears DSR 0.95 for N of 6 or fewer (7 with ChatGPT's 0.408).
 - Two facts about the lead candidate are wrong, and both came from the prompt.**
 - The UMD beta of 0.99 belongs to the EW book (recomputed 0.987, t 30.5). The book's UMD beta is 0.273 (t 15.2, R^2 0.39). The prompt printed 0.99 beside the optimizer's numbers without naming the EW book.
 - The book's carbon constraint does not use "2022 intensities". It uses the team's undated 'emissions_ff_industry.csv' (M5 section 3). The 2022 EPA factors are used only by M4's EPA spread, and the prompt's "2022 sort applied back to 1970" referred to that spread.
 - One statistical claim is wrong.** Romano-Wolf is not in general stricter than BH.

- When one p-value stands alone, BH needs it to be at most q/m , which is $6.2e-6$ on the 8,091 grid. A max-t $\hookrightarrow$ stepdown uses the correlation between tests and can reject when BH rejects nothing.
 - The grid's best positive cell has one-sided p 0.000176 (a 24-month COVID alpha, t 4.30). A max-t test $\hookrightarrow$ would flag it if the grid's effective number of independent tests were below about 285.
 - That cell is a known small-sample COVID artefact (M1b, M3), so the conclusion "nothing implementable is $\hookrightarrow$ rescued" very likely survives. The stated reason does not. The test cannot be run as things stand, $\hookrightarrow$ because M2 did not save its grid's return series (M7 section 4).
4. ****The frozen pre-1970 test was mis-specified, and its pass bar could not be met as written.****
- The optimizer needs 60 months of returns for its covariance and residual volatility, so the window is $\hookrightarrow$ 1931-07 to 1969-12 (462 months), not 1927-07. Only 34 to 37 of the book's 41 industries exist $\hookrightarrow$ before 1963-07.
 - Power at a true 2% alpha is 0.61 on the feasible window at post-1970 residual volatility (normal $\hookrightarrow$ approximation; M7 reports 0.62). ChatGPT's two-thirds is right for its own, infeasible window $\hookrightarrow$ (0.65).
 - The pass bar's "project-wide DSR on the appraisal ratio of at least 0.95" clause is decided by post-1970 $\hookrightarrow$ data, and ChatGPT's own arithmetic already fails it for more than about 7 trials.
 - M7 has since run the test: FF3+UMD alpha 1.95%, t 2.45, halves 1.71% and 0.90%. It passes the first two $\hookrightarrow$ clauses and fails the DSR clause (0.559). By ChatGPT's own bar, the "one check" did not overturn " $\hookrightarrow$ Do not implement".
5. ****The "low power" objection overreaches for the attention thesis.**** With p-values from $t(44)$, as the $\hookrightarrow$ prompt specified, all six rules reject a +2% holdout alpha at one-sided p of 0.0016 or less. Three of $\hookrightarrow$ the four hold rules have 90% intervals wholly below zero; Original 6m's upper bound is +0.03%. The $\hookrightarrow$ low-power point is right only for the book (holdout SE 2.27%, 90% interval -4.16% to +3.48%).
6. ****Its equivalence test would misread the strongest evidence.**** A TOST with a +/-2% margin fails all four $\hookrightarrow$ hold rules because their intervals are too negative (Original 3m: -7.13% to -0.94%). Under ChatGPT's $\hookrightarrow$ reading, that fail means "insufficient evidence". The question for implementation is one-sided (non- $\hookrightarrow$ inferiority against a worthwhile alpha), which is what M7 check 6 adopted.
7. ****Its EPA test and its reading of a fail are backwards on our data, and the result is knife-edge.**** Adding $\hookrightarrow$ a same-month MCCC shock raises the post-2010 alpha, because the shock loading is negative: the $\hookrightarrow$ spread lost when concern rose, the opposite of the Pastor, Stambaugh and Taylor (2022) channel.
- With M1's expanding-window AR(1) shock, the alpha is 6.66% (t 2.13; M7 check 9).
 - With PST's own trailing-36-month AR(1), which I computed, the alpha is 5.88% (t 1.85) and the loading is $\hookrightarrow$ -0.42% per s.d. (t -1.89).
 - So whether the spread "passes" depends on how the shock is built.
8. ****Four smaller factual errors.****
- Loadings are 32% of the 4,184 nominal hits, not "most". Alphas are the largest group at 39% (783 $\hookrightarrow$ positive, 841 negative).
 - Costs above 10 bp were tested: M2 at 25 bp; M5 at 25 bp for the EW book (FF5+UMD alpha -0.14%) and for $\hookrightarrow$ the book (IR 0.49).
 - "The BOND control helps" contradicts M3: adding BOND and UMD leaves every holdout alpha negative.
 - "Residual volatility before the war is probably higher" ignores the book's 5% ex-ante tracking target. $\hookrightarrow$ The realized pre-1970 residual volatility was 5.14%, against 5.44% after 1970 (M7).
9. ****It broke the word limit.**** The reply has 1,716 words with its two tables and 1,481 without them, against $\hookrightarrow$ "under 1,500".

A.4 Exchange 04: red team of the draft report (two rounds)

Files: `exchange/04_red_team/`. Answered by: a model in the ChatGPT role (see the provenance note). Date: 2026-09-26. Purpose: red-team review of the conclusion.

Asked.	Round 1: a red team of the draft summary and its claims (five weakest claims, at most four alternatives with separating predictions, the case against “Do not implement” and its reversing result, three edits, two least certain assumptions). Round 2: the same on the revised summary, now with the best evidence against the verdict, at most three alternatives, and edits no longer than the text they replace.
Came back.	Round 1: 1,340 words (cap 1,200) with a reversal test on the untimed EPA-ranked short. Round 2: 1,091 words (cap 1,000) arguing “not proven” rather than “disproven”, with three alternatives (wrong legs, direction-free regulation news, a crisis hedge) and three replacement texts.
Verified.	Every criticism checked against the module outputs, and the missing tests run (checks fc04 and fc04b): holdout paths, intervals and timing, frozen-test power, a COVID bootstrap, the rules rebuilt without each Brown industry, MCCC and CPU on EPA-ranked legs, a direction split of the holdout entries, a crisis split of the timing gain, and our team’s signal on both windows it was not designed on.
Kept.	Thirteen summary rewordings, the holdout interval and timing loss, and three alternatives tested and retired; not its edits as written, its cost comparison (the timing alpha is already net), its one-industry reading of the holdout, or its case for capital in the momentum book.

Round 1, on the first draft Kept verbatim in `report/transcripts_src/`.

My evaluation of round 1

ChatGPT refereed my wording well and my evidence badly. Every criticism aimed at a sentence landed, but almost every number it needed from our files was guessed, and four of the six guesses I could check were wrong. I checked each point against the modules and ran the missing tests (`checks/fc04_checks.py`, `exploratory`).

All five criticisms of my wording were right, and the executive summary now follows them. “900 of 900” is one result: corrected Original and Pure coincide in all 48 holdout months, so the six rules have four return paths, and the first principal component carries 90% of their variance. The summary and Section 2.3 now lead with Pure 6m (FF3 alpha -1.97% a year, $t = -1.30$, six entries; 95% interval -5.0% to $+1.1\%$). Its power arithmetic for the frozen test is right (standard error 0.68%, so only an edge of about 1.9% a year passes with 80% probability) and exposed an error of mine: “+0.9%” was the shuffle’s bar, while the t bar needed $+1.4\%$. The COVID edge’s interval runs from -2.6% to $+6.2\%$, and a block bootstrap puts the 71% between -36% and 233% (90%), so the ratio left the summary. The omitted CPU correlation and the “real-time inputs” overclaim are fixed in its words.

Three corrections were wrong. Its first fix, lead with the frozen rule’s holdout alpha, is impossible: the frozen rule’s zero clause switches it off from 2022-04 onward, so it never trades there. Its guess that the holdout $|t|$ is below 1 fails for every rule ($t = -1.30$ to -1.90 corrected, -1.66 to -2.19 as our team ran them); only the residual term qualifies ($t = -0.86$). And its second edit would have put a false claim in the summary: “2.4 standard errors below its validation alpha” compares a timing alpha with a mean strategy alpha (1.45%); the frozen rule’s own seen-window timing alpha was -0.12% ($t = -0.19$), so its proposed lesson, that the 1994–2009 test is informative, has no footing either. It flagged this as its least certain assumption. A true 1% edge fails 70% of the time, not half as it guessed.

Three of its four alternatives were already answered by modules my prompt left out. Same-month slopes on MCCC and CPU shocks have the wrong sign ($t = -0.49$ and -1.23). M2’s commodity hedge left every holdout alpha negative; oil in the evaluation trims Pure 6m’s to -1.50% ($t = -0.88$), and breakevens add nothing once oil is in. The loss is concentrated, not

where it guessed: Steel carries -1.37 of the -1.38 -point gross loss, and Util offsets it ($+1.07$). On the EPA leg the frozen rule earns -0.01% over 1994–2009.

Its case against the verdict was aimed at the wrong result. Its reversal test, a positive holdout alpha for the untimed EPA-ranked short, is met ($+1.07\%$, $t = 0.79$) by a position whose 1994–2009 alpha is -1.57% ($t = -0.74$): a sign on 48 months, the counting its own first point warned against. The strongest case against sits in my own results: our team's signal, lagged, has a timing alpha of $+0.69\%$ over 1994–2009 ($t = 1.00$) and $+0.36\%$ ($t = 0.70$) pooled over both windows it was not designed on. My prompt left it out.

ChatGPT is a good referee of prose and arithmetic and a weak one of data it cannot see. Its derived numbers were right to rounding and its citations accurate; it ran 12% over its word cap (1,340 of 1,200). A red team is only as hard as its evidence: the next prompt sends the best result against my verdict, not only the results for it.

Round 1 prompt (verbatim) 04_prompt.txt, 795 words (782 excluding Markdown symbols).

Read my draft twice: first as a skeptical investment-committee member who must sign off on "Do not
 ↪ implement", then as a finance-journal referee. Break the conclusion before a grader does.
 ↪ No summary, no praise.

SETTING. My UC Berkeley MFE 230GA final project (thesis 40%, execution 40%, originality 20%). I
 ↪ extended our team's halfway project alone. "What did you learn" is at its page limit, so an
 ↪ edit there must replace text.

THE EXECUTIVE SUMMARY, AS DRAFTED.

Our team's halfway project proposed a state-dependent climate trade: short the factor-hedged Brown
 ↪ industry leg (Util, Ships, Aero, Steel, BldMt) for three or six months after "climate-
 ↪ transition attention" crosses its past-only 80th percentile, sized to 5% residual
 ↪ volatility. Its best-specified form, which I froze before testing it, triggers on the
 ↪ energy-and-environmental-regulation share of volatility news, lagged one month. I asked
 ↪ three questions of the strategy: what the signal measures, what the legs are, and whether
 ↪ any version survives a window it was not designed on. The study uses Fama-French industry
 ↪ and factor returns, FRED news and macro series, two climate-concern indices (MCCC and CPU)
 ↪ and EPA supply-chain emission factors; every inference uses real-time inputs and small-
 ↪ sample p-values, each module fixes a primary test, and an independent verifier rebuilt each
 ↪ module's headline numbers. The recommendation is Do not implement, for four reasons. First
 ↪ , the "attention" series is FRED's equity-volatility news tracker for energy and
 ↪ environmental regulation (identical in 500 of 500 months), its z-score is uncorrelated with
 ↪ media climate concern ($r = -0.0004$), and the same machinery fed genuine climate measures
 ↪ earns 0 of 12 validation alphas with $t \geq 1.96$. Second, the rule frozen before its test
 ↪ window earns -0.18% a year over 1994–2009 ($t = -0.27$, shuffle $p = 0.48$). Third, the COVID
 ↪ gain belongs to the position: an untimed short of the same leg earns 71% of it, and the
 ↪ rule's edge over that short is $+1.84\%$ a year ($t = 0.87$). Fourth, every timed rule has a
 ↪ negative 2022–2026 holdout alpha in all 900 net-of-cost configurations, all 28 zero-cost
 ↪ runs are negative too, and rates do not explain the loss. Carbon-aware industry momentum
 ↪ and shrinkage-based timing on the same data fare no better. In short: read the label,
 ↪ benchmark timing against the always-on position, and spend the last clean window on one
 ↪ frozen rule.

THE CLAIMS BEHIND IT (Newey-West t , 6 lags; p from $t(n-k)$).

1. Signal. On VIX and overall-EMV z-scores, the team z-score has $R^2 = 0.169$, all through overall
 ↪ EMV; its CPU correlation is 0.123 (Holm $p = 0.074$). Its zeros (39 of 59 months since
 ↪ 2021–10) changed no holdout entry.
2. Placebo. Overall EMV, a pure volatility measure, earns a 1.07% mean validation alpha, against
 ↪ the team signal's 1.45%.
3. Exposures. Green-minus-Brown is short HML (-0.233 , $t = -4.63$), mostly through Brown; the
 ↪ untimed short shares the rules' beta-timing drag.

4. Holdout. Pure 6m nets -1.84% = hedge leakage -0.24% + residual -1.14% - cost 0.46% ; Treasury $\hookrightarrow$ and UMD factors make every holdout alpha more negative.

5. Frozen test. Its pre-registration was hashed before its code existed, but the rule was chosen $\hookrightarrow$ after 2010-2026 was seen; it meets 1 of 4 pass-bar components, and passing needed about $\hookrightarrow +0.9\%$ a year.

6. Legs. The team emissions file is undocumented; Ships and Aero rank 2nd and 3rd of 41 there, 20 $\hookrightarrow$ th and 31st of 49 on EPA factors, and with Util carry 86% of the COVID gross P&L. An EPA- $\hookrightarrow$ ranked spread earns a post-2010 FF5+UMD alpha of 4.7% ($t = 1.52$).

7. Testing. No positive alpha survives its pre-specified family among 23,923 logged tests; only $\hookrightarrow$ the frozen test's pre-specification is timestamped, and NW(6) over-rejects on synthetic $\hookrightarrow$ nulls (13 of 100 at 5%).

Already disclosed: static emissions snapshots, final vintages, an assumed EMV lag, a 48-month $\hookrightarrow$ holdout.

WHAT I NEED, under these headings.

1. Weakest claims. Exactly five, ranked by effect on the verdict or the grade: the sentence, why $\hookrightarrow$ the evidence is weaker than the wording, and the test or rewording that fixes it. Overreach $\hookrightarrow$ toward "Do not implement" counts ($t = 0.87$ does not prove timing adds nothing).
2. Alternative explanations I have not ruled out. At most four, each with a prediction that $\hookrightarrow$ separates it from mine on this data.
3. The case against "Do not implement": the strongest argument that the verdict is wrong or too $\hookrightarrow$ broad, and the one result that would reverse it; if none survives these numbers, say so.
4. The three edits that would most improve the grade, each with its section and replacement text.
5. The two assumptions in your answer you are least sure of.

RULES. Work from these numbers; label any number you have not computed as a guess. Cite a paper $\hookrightarrow$ only with authors, year and the specific claim. Stay under 1,200 words.

Round 1 reply (verbatim; model in the ChatGPT role) 04_chatgpt_response.txt,
1,385 words (1,340 excluding Markdown symbols).

```
## 1. Weakest claims (ranked)

**1. "every timed rule has a negative 2022-2026 holdout alpha in all 900 net-of-cost configurations, all 28
   $\hookrightarrow$  zero-cost runs are negative too"
*Why weak:* All 900 configurations run on the same 48-month path of the same Brown leg. That makes "900 of
   $\hookrightarrow$  900" one observation reported 900 times. You give no t-statistic, no interval and no count of entries.
   $\hookrightarrow$  Since 39 of the last 59 months are zeros, the rule may have traded only a few episodes. Guess: at 5%
   $\hookrightarrow$  residual volatility over 48 months,  $-1.84\%$  a year has  $|t| < 1$ .
*Fix:* Lead with the frozen specification's holdout alpha, its NW t, a 95% interval and the number of entry
   $\hookrightarrow$  months. Report how much of the variance in the 900 return series the first principal component
   $\hookrightarrow$  explains. Call the rest "robustness of sign."

**2. "the rule frozen before its test window earns  $-0.18\%$  a year over 1994-2009 ( $t = -0.27$ , shuffle  $p = 0.48$ )
   $\hookrightarrow$  "
*Why weak:* Failing to reject zero is not evidence of zero. The implied standard error is about  $0.67\%$  a year
   $\hookrightarrow$  ( $0.18/0.27$ , derived), so the  $+0.9\%$  pass bar needed  $t \sim 1.35$ . Guess: a true  $1\%$ -a-year edge would fail
   $\hookrightarrow$  about half the time. The window also comes before transition risk was plausibly priced, and the legs
   $\hookrightarrow$  use today's emissions snapshot.
*Fix:* Test against the effect you are rejecting. If the frozen rule's own validation alpha is near the  $1.45\%$ 
   $\hookrightarrow$  mean (guess),  $-0.18\%$  is about 2.4 standard errors below it. Report that, plus the minimum detectable
   $\hookrightarrow$  effect at 80% power: about  $2.8 \times 0.67 \sim 1.9\%$  a year (derived).

**3. "the COVID gain belongs to the position ...  $+1.84\%$  a year ( $t = 0.87$ )"
*Why weak:* The implied standard error is about  $2.1\%$ , so the 95% interval runs from roughly  $-2.3\%$  to  $+6.0\%$  a
   $\hookrightarrow$  year. The point estimate favours timing. "71%" is a ratio of two noisy P&Ls with no interval.
*Fix:* "Most of the COVID gain came from the position (71%). The timing edge of  $+1.84\%$  a year cannot be told
   $\hookrightarrow$  apart from zero or from  $5\%$  a year." Add a bootstrap interval on the 71%.

**4. "its z-score is uncorrelated with media climate concern ( $r = -0.0004$ )"
*Why weak:* You quote MCCC and leave out CPU ( $r = 0.123$ , Holm  $p = 0.074$ ). CPU is the index closest in concept
```

↪ to "environmental regulation." A Pearson r on a zero-inflated series is fragile. Also, $R^2 = 0.169$
 ↪ means 83% of the signal's variance is not volatility. Counting newspaper articles that tie market
 ↪ volatility to environmental regulation is plausibly a transition-policy measure; the label alone does
 ↪ not rule that out.

Fix: Report both correlations, as Spearman and on innovations. Add the overlap between trigger months and
 ↪ top-quintile CPU and MCCC months. Reword to "weakly related to climate-policy uncertainty, unrelated
 ↪ to media climate concern."

****5. "every inference uses real-time inputs ... each module fixes a primary test"****
Why weak: Your own disclosures contradict both halves. Final vintages and static emissions snapshots are
 ↪ not real-time inputs, and only the frozen test's pre-specification is timestamped. A referee who
 ↪ catches this will discount everything after it.

Fix: "Every threshold, percentile and hedge uses only past data; macro series are final vintages and
 ↪ emissions a static snapshot. Each module names a primary test; only the frozen test's was timestamped
 ↪ before its data were seen."

2. Alternative explanations not ruled out

- **Wrong horizon, not wrong signal.****
 - Pastor, Stambaugh and Taylor (2022) find green stocks outperformed in 2012-2020 mainly because climate
 ↪ concern rose more than expected. Ardia, Bluteau, Boudt and Inghelbrecht (2023) find brown stocks
 ↪ fall on days when climate concern rises unexpectedly.
 - If prices adjust in the same month, a short entered one month later and held 3-6 months arrives after
 ↪ the move.
 - ***Prediction:*** Green-minus-Brown loads positively on lag-0 CPU and MCCC innovations, and at zero or below
 ↪ at lags 1-6. Your explanation predicts no loading at any lag.
- **Industry-specific holdout shocks.****
 - **Guess:** utilities repricing on data-centre power demand in 2024-25, and Ships and Steel in 2021-22, drive
 ↪ the -1.14% residual.
 - ***Prediction:*** splitting the residual by industry and year puts most of it in one or two legs and periods.
 ↪ Your explanation predicts losses spread across legs and years.
- **Unhedged inflation or commodity exposure.****
 - You tested rates, not inflation. Steel, BldMt and Ships are commodity-sensitive.
 - ***Prediction:*** adding FRED WTI returns or 10-year breakeven changes to the hedge pulls the holdout
 ↪ residual toward zero. Your explanation predicts no change.
- **Wrong leg.****
 - Ships and Aero rank 20th and 31st of 49 on EPA factors.
 - ***Prediction:*** the frozen rule applied to the EPA-ranked leg shows a validation timing edge over the
 ↪ untimed EPA short. Your explanation predicts about zero edge there as well.

3. The case against "Do not implement"

****The strongest argument:****

- The evidence rejects one timing rule on one poorly classified leg, but the verdict reads as a rejection of
 ↪ the climate trade.
- The untimed short captured 71% of the COVID gain, and the timing edge's point estimate is positive.
- The EPA-ranked spread earns 4.7% a year after 2010. Its implied standard error is about 3.1%, so the
 ↪ interval runs from roughly -1.4% to +10.8%.
- On that evidence the trade is unproven, not refuted.

****Why it mostly fails:****

- A t of 1.52 does not survive a family of 23,923 tests.
- The EPA spread was probably specified after the post-2010 results were known (my assumption).
- Bolton and Kacperczyk (2021) find that higher-emission stocks earn higher returns, so an always-on Brown
 ↪ short expects to pay a carbon premium.
- Pastor, Stambaugh and Taylor (2022) argue that green stocks' realized outperformance does not mean higher
 ↪ expected returns.

****Verdict:****

- "Do not implement the timed rule" survives. Nothing in the current data can reverse it, because both clean
 ↪ windows are spent.
- The broad version should be narrowed, not defended.
- The one result that would reverse the broad version is a positive 2022-2026 alpha for the untimed EPA-
 ↪ ranked short, which you do not report. If it is negative, say so and the broad verdict stands.

```

## 4. Three edits for the grade

**1. Executive summary, verdict sentence.** Replace "The recommendation is Do not implement, for four reasons
  ↳ ." with:
> "The recommendation is Do not implement the timed rule. The untimed Brown short is unproven rather than
  ↳ refuted; an EPA-ranked Brown leg (post-2010 alpha 4.7% a year, t = 1.52) is the only open lead and
  ↳ does not survive the test family. Four reasons:"

**2. Executive summary, reasons two to four.** Replace with:
> "Second, the frozen rule earns -0.18% a year over 1994-2009 (t = -0.27), about [2.4] standard errors below
  ↳ its validation alpha, though the window cannot exclude an edge near 1.9% a year. Third, the untimed
  ↳ short earns 71% of the COVID gain; the timing edge, +1.84% a year (95% interval about -2.3% to +6.0%),
  ↳ is not distinguishable from zero. Fourth, the frozen specification's 2022-2026 alpha is [x]% (t = [y
  ↳ ], [n] entries); all 900 configurations share that 48-month path and its sign, and rates do not
  ↳ explain it."

Drop "Carbon-aware industry momentum and shrinkage-based timing ... fare no better", or give each one's best
  ↳ alpha and adjusted p.

**3. "What did you learn".** Replace whichever lesson restates a count of tests or configurations:
> "Test the claim, not zero. Counting negative configurations felt decisive, but 900 configurations share one
  ↳ 48-month path. What separates results is how far the out-of-sample alpha sits from the in-sample one
  ↳ in standard errors, and the smallest effect the window could detect. By that standard the 1994-2009
  ↳ test is informative and the holdout only weakly so."

## 5. The two assumptions I am least sure of

1. **The 2.4-standard-error figure.** It assumes the frozen rule's own validation alpha is near the 1.45%
  ↳ mean and that alpha divided by t gives a usable standard error. Your finding that NW(6) rejects 13 of
  ↳ 100 synthetic nulls at 5% suggests the standard errors are understated. If so, every interval above
  ↳ widens and the 2.4 shrinks.

2. **What drove the holdout.** I assumed there are few holdout entries, and that Util, Ships and Steel drove
  ↳ the loss. I inferred both from the zero count and from market history, not from your files.

```

Round 2, on the revised draft The prompt sent the revised summary and the best evidence against it.

My evaluation of round 2

This round did what round 1's evaluation asked for: the prompt sent the best evidence against my verdict as well as the evidence for it. ChatGPT used that evidence as an advocate would, selectively, and that is where it went wrong. I checked every point against the modules and ran the tests it proposed (`checks/fc04b_checks.py`, exploratory).

All five weak claims found real gaps in the wording, and the summary now closes them. "Five high-emission industries" hid that EPA factors rank Ships and Aero 20th and 31st of 49. "Most of the COVID gain came from the position" rested on a point estimate: the rule's edge is +1.84% over the untimed short ($t = 0.87$) and 3.0% over an exposure-matched one ($t = 1.67$), so the window cannot split the gain. " $p = 0.48$ " needed its null, whose median is -0.22% . "Value-tilted" described the raw leg rather than the hedged rule, and a wrong label explains why the thesis is untested, not why the returns fail, so the verdict sentence now says both. "The last unused window" is the live record.

Four of its supporting claims were wrong. The holdout does not indict "the position, not the timing": against an exposure-matched short the timing lost a further 1.0% to 1.7% a year ($t = -0.99$ to -1.72). The loss does not come from one industry either: rebuilt without any one industry, the rule still loses in 10 of 10 cases (FF3 alpha -1.26% to -2.61%). Both results now stand in the summary. The figures it called unreconciled add up in my own claims list. And the residual test it asked for is the frozen rule, which trades the signal with overall volatility news divided out, the same rule it had just dismissed as one I am not proposing.

Each alternative failed its own prediction. On EPA-ranked legs the same-month slopes on MCCC and CPU shocks became more wrong-signed ($t = -2.95$ and -2.17), and CPU timing added no detectable edge over the untimed EPA short (holdout $+0.05\%$ to $+1.22\%$, largest $t = 1.43$; negative in 5 of 6 rules over 1994–2009); Section 2.1 and Appendix C now say so. With the administration as a crude proxy for policy direction, the short lost more after entries under Biden (-4.5 gross points) than under Trump (-1.0). And 80% of the 6-month rule’s 1994–2026 timing gain came in months with VIX at or below 30, 95% outside NBER recessions.

Its case against the verdict misread what I sent. “Positive in both unseen windows” reads a pooled $+0.36\%$ as two positives; the windows give $+0.69\%$ ($t = 1.00$) and -1.11% ($t = -1.13$). Its committee argument set that timing alpha against the 0.46% holdout cost, but the alpha is computed on net returns, so the cost is already inside it. The deflated appraisal ratio is applied to the windows the momentum book was chosen on, and 2010–2026 is one of them, so it is no second out-of-sample pass. None of its three edits went in as written: two carry these errors, and the third runs 18 words against a 17-word original while claiming 17.

ChatGPT referees sentences well and reads numbers carelessly. Sending the counter-evidence bought sharper, testable alternatives, but an advocate quotes the favorable half: 5.4% without the 3.0% , and the 1994–2009 drop-one result without the holdout. As in round 1, the error that reached its replacement text sat in the assumption it flagged as least certain, and it again ran over its cap (1,091 words of 1,000). Read its flags, test its predictions, and never paste its edits.

Round 2 prompt (verbatim) prompt.md, 798 words.

Read my draft conclusion as the skeptical investment-committee member who must sign "Do not
 ↪ implement", then as a finance-journal referee. No summary, no praise.

SETTING. My solo extension of our team’s MFE 230GA final project (40% thesis, 40% execution, 20%
 ↪ originality).

EXECUTIVE SUMMARY (DRAFT).

Our team’s halfway project proposed a state-dependent climate trade, and this report is my solo
 ↪ extension of it. The strategy shorts a factor-hedged Brown leg of five high-emission
 ↪ industries (utilities, shipbuilding and railroad equipment, aircraft, steel, construction
 ↪ materials) for three or six months after "climate-transition attention" crosses its past-
 ↪ only 80th percentile, sized to 5% residual volatility. I tested it on Fama-French industry
 ↪ and factor returns, FRED news and macro series, two climate-concern indices, media climate
 ↪ change concern (MCCC) and climate policy uncertainty (CPU), and EPA supply-chain emission
 ↪ factors. Every threshold, percentile and hedge uses only past data, p-values come from
 ↪ small-sample t distributions, and independent code rebuilt each module’s headline numbers.
 ↪ The recommendation is Do not implement, for four reasons. First, the "attention" series is
 ↪ FRED’s equity-volatility news tracker for energy and environmental regulation (identical in
 ↪ 500 of 500 months). Its z -score is uncorrelated with MCCC ($r = -0.0004$), its weak link to
 ↪ CPU ($r = 0.123$, Holm $p = 0.074$) runs through overall volatility news, and the same rules fed
 ↪ either climate index earn 0 of 12 validation alphas with $t > 1.96$. In plain terms, the rule
 ↪ shorts value-tilted utility and industrial stocks after stock-market volatility makes the
 ↪ news. Second, a cleaner version frozen before its test (the regulation share of volatility
 ↪ news, lagged one month) has a timing alpha of -0.18% a year over 1994–2009 ($t = -0.27$; $p = 0.48$
 ↪ against random re-datings of its holding periods); that window detects an edge of about
 ↪ 1.9% a year with 80% power, so the fail excludes a large edge, not a small one. Third, most
 ↪ of the COVID gain came from the position: the rule’s alpha over an untimed short of the
 ↪ same leg is $+1.84\%$ a year ($t = 0.87$; 95% interval -2.6% to $+6.2\%$). Fourth, every one of our
 ↪ team’s timed rules lost over the 2022–2026 holdout, before costs as well as after (6-month
 ↪ rule: FF3 alpha -1.97% a year, $t = -1.30$), the untimed short lost too (-1.72% , $t = -1.09$), and
 ↪ rates do not explain the loss; the 900 net-of-cost variants share that one 48-month window,
 ↪ so their common sign is one result. Of the alternatives, shrinkage timing collapses to the
 ↪ historical mean, and an optimized industry-momentum book that passed a pre-registered

↪ 1931-1969 test (alpha 1.95% a year, $t=2.45$) fails the deflated appraisal ratio and lost in
 ↪ the holdout (-0.34%), which earns it a paper-trading pilot, not capital. In short: read the
 ↪ label, benchmark timing against the always-short position, and spend the last unused
 ↪ window on one frozen rule.

CLAIMS BEHIND IT (annualized; Newey-West t , 6 lags).

1. Signal: $R^2=0.169$ on VIX and overall-EMV z-scores, all through overall EMV; its non-volatility
 ↪ topic share correlates 0.03 with CPU; same-month Green-minus-Brown slopes on MCCC and CPU
 ↪ shocks are wrong-signed ($t=-0.49$, -1.23).
2. Exposures: Green-minus-Brown loads -0.233 on HML ($t=-4.63$), mostly through Brown; the hedged
 ↪ rules keep -0.015 to -0.053 since 2010.
3. Holdout: the 6-month rule's -1.84% net is leakage -0.24%, residual -1.14% and cost 0.46%, from
 ↪ six entries; Steel carries -1.37 of -1.38 gross points. BOND, UMD or a commodity factor
 ↪ leave every alpha negative.
4. Frozen rule: chosen after 2010-2026 was seen; in position 142 of 190 months; off from 2022-04,
 ↪ so flat through the holdout.
5. Legs: Ships and Aero rank 2nd and 3rd of 41 in the undocumented team file, 20th and 31st of 49
 ↪ on EPA factors. The EPA-ranked spread's post-2010 FF5+UMD alpha is 4.7% ($t=1.52$).
6. Testing: 73 primary alphas, best one-sided $p=0.030$, smallest Holm $p=1.00$; that t rejects 13 of
 ↪ 100 synthetic nulls at 5%.

BEST EVIDENCE AGAINST ME.

- a. Our team's signal, lagged: timing alpha +0.69% over 1994-2009 ($t=1.00$; all five drop-one-
 ↪ industry alphas positive), +0.36% ($t=0.70$) pooled over both unseen windows.
- b. COVID, against an exposure-matched benchmark: 5.4% ($t=2.35$), 3.0% ($t=1.67$) with in-window
 ↪ factors.
- c. Momentum book, 2010-2026: alpha 3.12% ($t=2.18$).

WHAT I NEED, under these headings.

1. Weakest claims: five summary sentences, ranked by effect on the verdict (overreach toward "Do
 ↪ not implement" counts), each with the gap between evidence and wording, and the test or
 ↪ rewording that closes it.
2. Alternatives I have not ruled out: at most three, each with a separating prediction on this
 ↪ data.
3. The case against "Do not implement": the strongest argument that it is wrong or too broad, and
 ↪ the one result that would reverse it (or "none").
4. Three edits that would most improve the report, with replacement text no longer than the
 ↪ original.
5. Your two least certain assumptions.

RULES. Label any number you have not computed as a guess. Cite papers by authors, year and claim.
 ↪ Under 1,000 words.

Round 2 reply (verbatim; model in the ChatGPT role) chatgpt_response.md, 1,108 words (1,091 excluding Markdown symbols).

****1. Weakest claims (ranked by effect on the verdict)****

1. ****Reason 2, the frozen rule.**** This rejects a rule you are not proposing. It was chosen after 2010-2026
 ↪ was seen and was in position 142 of 190 months, so it is mostly the untimed short. Your team's lagged
 ↪ rule earns +0.69% in the same window ($t=1.00$, all drop-one alphas positive), and the 1.9% power
 ↪ caveat applies to that result too. A p of 0.48 for a negative alpha means either the re-dating null
 ↪ is centred below zero or the test is two-sided. Say which. Fix: Edit 1, and name which window is "the
 ↪ last unused" one.
2. ****Reason 4, the holdout.**** "Every one of our team's timed rules lost" suggests breadth. The evidence is
 ↪ one industry (Steel, -1.37 of -1.38 gross points), six entries and $t=-1.30$. The timed and untimed
 ↪ shorts lost similar amounts, so the holdout indicts the position, not the timing the thesis sells.
 ↪ The -1.97% FF3, -1.84% net and -1.38% gross figures are never reconciled. Fix: report the holdout

↪ timing alpha (timed regressed on untimed) and the ex-Steel alpha, and name the model behind each
 ↪ number (Edit 2).

3. **Reason 1, "In plain terms..."**
 - $R^2=0.169$ leaves 83% of the signal unexplained by VIX and overall EMV.
 - The hedged rules keep at most -0.053 HML.
 - So both halves of the sentence overstate.
 - A mislabelled signal can still pay. This reason supports "don't call it climate", not "don't implement".
 - Fix: rerun the rule on the signal's residual after VIX and EMV. If alpha is unchanged, move the label
 ↪ point out of the verdict (Edit 3).

4. **Reason 3, COVID.** "Most of the gain came from the position" rests on a point estimate. The interval
 ↪ (-2.6% to +6.2%) allows a timing share of anything from none to all, and the exposure-matched
 ↪ benchmark gives 5.4% ($t=2.35$). Fix: write "the data cannot split the COVID gain between position and
 ↪ timing", report both benchmarks and say why you prefer one.

5. **Opening, "five high-emission industries."** Your EPA ranks put Ships and Aero 20th and 31st of 49. The
 ↪ summary also leaves out the EPA-ranked spread's 4.7% ($t=1.52$). The verdict therefore covers this
 ↪ construction, not the thesis. Fix: write "five industries our team labelled high-emission (two rank
 ↪ mid-table on EPA factors)", and add the EPA result to the evidence against you.

2. Alternatives not ruled out

1. **Wrong legs.** Your wrong-signed MCCC and CPU slopes contradict two papers:
 - Ardia, Bluteau, Boudt & Inghelbrecht (2023): green beats brown when climate concern rises unexpectedly.
 - Pastor, Stambaugh & Taylor (2022): green outperformance in 2012-2020 came from concern shocks.

Prediction: on EPA-ranked legs the slopes turn positive and the MCCC- and CPU-timed validation alphas
 ↪ improve. If they do not, the thesis fails at industry level whatever the signal.

2. **Regulation news without a direction.** EMV (Baker, Bloom, Davis & Kost 2019) counts articles, so it
 ↪ carries no policy direction. Guess: in 2022-2025 the coverage was mostly IRA subsidies, tariffs and
 ↪ rollbacks that favoured Steel. Prediction: split entries by direction, using the administration in
 ↪ power as a crude proxy or hand-coding the articles. The short should earn after tightening news and
 ↪ lose after loosening news, and the Steel loss should cluster in tariff and IRA months.

3. **Crisis hedge.** Brown cyclicals fall in crises, and the signal loads on overall volatility news. The
 ↪ payoff may be crisis convexity (2008, 2020) rather than alpha. Prediction: timing alpha concentrates
 ↪ in months with VIX above 30 or in NBER recessions, and is near zero elsewhere. If so, price it as a
 ↪ hedge.

3. The case against "Do not implement"

The verdict is right for this rule, but it is written as if the climate thesis was tested. It was not: the
 ↪ signal is volatility news, two Brown legs are mid-emission, and the legs' climate slopes are wrong-
 ↪ signed.

For the rule itself, the evidence reads "not proven", not "disproven":
 - positive in both unseen windows;
 - robust to dropping any one industry;
 - 5.4% ($t=2.35$) against an exposure-matched COVID benchmark;
 - a holdout loss that comes from one industry.

On momentum: the deflated Sharpe ratio (Bailey & Lopez de Prado 2014) is built for in-sample selection.
 ↪ Applying it to a pre-registered pass double-counts trials. Suppose 2010-2026 was not used for
 ↪ optimisation. Then the book has two out-of-sample passes ($t=2.45$ and 2.18), and with the industry-
 ↪ momentum prior of Moskowitz & Grinblatt (1999) that argues for small capital, not only paper trading.

The committee's case for signing is stronger than the one you wrote. Your best unseen estimate (+0.36%) is
 ↪ below the holdout cost (0.46%), so the rule loses money even if that estimate is true. Bolton &
 ↪ Kacperczyk (2021) also find a carbon premium, so the always-short leg the timing sits on has a
 ↪ negative expected return.

Reversing result: none on existing data, because any rule built now has already seen both unseen windows.
 ↪ Only a frozen rule's forward record can reverse the verdict.

4. Three edits

```

*Edit 1, replacing Reason 2 (58 words; the original has 60):*
"Second, on data unseen when rules were set, our team's rule lagged one month earns a timing alpha of +0.36%
  ↪ a year (t=0.70), below its 0.46% annual holdout cost; a cleaner frozen variant earns -0.18% over
  ↪ 1994-2009 (t=-0.27). These tests detect only edges above about 1.4-1.9% a year, so they exclude a
  ↪ large edge, not a small one."

The 1.4% is my arithmetic from your t=0.70, not a computed result.

*Edit 2, replacing Reason 4 (52 words; the original has 57):*
"Fourth, over the 2022-2026 holdout the timed rules lost (6-month rule: FF3 alpha -1.97% a year, t=-1.30)
  ↪ about as much as the untimed short (-1.72%, t=-1.09), so timing added little; Steel accounts for
  ↪ nearly all of the gross loss, from six entries, and the 900 variants' common sign is one 48-month
  ↪ result."

*Edit 3, replacing "In plain terms..." (17 words; the original has 17):*
"In plain terms, the signal counts regulation mentions in volatility news; no test links it to climate
  ↪ concern."

**5. My two least certain assumptions**

1. I read "13 of 100 synthetic nulls" as meaning your nominal 5% test rejects about 13% of the time. If so,
  ↪ every t in the report is overstated, including 2.35, 2.45 and 2.18, and the summary's sentence on p-
  ↪ values needs a caveat. If it means a family-wise max-t p of about 0.13, ignore this point.
2. I assumed "timing alpha" is measured the same way for the team rule and the frozen rule. I also assumed
  ↪ the 0.46% holdout cost reflects the lagged rule's turnover in other windows. The cost comparison in
  ↪ Edit 1 depends on both.

```

Each round was one prompt and one reply; no follow-up was sent.

Fact-check The check scripts are in `exchange/04_red_team/checks/` (`fc04_apply_revisions.py`, `fc04_checks.py`, `fc04b_apply_revisions.py`, `fc04b_checks.py`), with their outputs in `checks/out/`.

B Replication and audit of our team’s pipeline

B.1 How reproduction was checked

I ported our team’s two analysis scripts (`climate_alpha_analysis.py` and `macro_state_dependence.py`, commit 08e7d31) into one library, `lib/team_pipeline.py`, with 14 unit tests that pass in about 20 seconds. Reproduction was checked in four steps.

1. **The team code regenerates the team’s committed tables** (comparison table maximum difference 3.8×10^{-15} ; the $\text{\LaTeX}$ table byte-identical; macro-state tables 1.9×10^{-14}), with numpy, pandas and scipy versions equal to the team lock file.
2. **The library reproduces the committed tables** to 3.8×10^{-15} and 3.1×10^{-14} .
3. **The library reproduces every intermediate object of the team script**: positions, gross and net returns and turnover of all 8 strategies bitwise; rolling betas, hedged returns and residuals bitwise; the 48-row period table, 32 claims, bootstrap and placebo draws exactly.
4. **The write-up is reproduced**: all 32 Table 1 cells at two decimals, the COVID numbers (5.94%, 6.03%, $t = 3.21$; 7.41%, 5.23%, $t = 2.07$), “2 of 32 claims survive Holm”, and raw-spread HML loadings of -0.24 and -0.35 over the write-up’s windows, 1970 to 2022-07 and 2010-01 to 2022-07 (over 1970–2026 and 2010–2026 they are -0.233 and -0.298).

A default run takes 1.9 seconds against 36.5 seconds for the two team scripts, which made the many reruns in M1b and M2 cheap.

B.2 Audit findings, ordered by impact on the conclusions

Table B.1: Audit of the team pipeline. Each item was checked by running code; the status column says how the item is treated in this report.

#	Finding	Effect	Status in this report
1	The signal is exactly zero in 31 of 48 holdout months	Near-binary signal in the holdout	Real, but the zeros changed no holdout trade (M1); not an explanation of the loss
2	The missing Oct-2025 CPI print silenced the purified signal for two months	Explains the whole Pure-versus-Original holdout gap (Pure 3m -2.17% becomes -3.82%)	Fixed in the corrected baseline
3	Normal p-values on 24-month NW(6) t -statistics	“2 of 32 survive Holm” becomes 0 of 32 with $t(n-4)$ (smallest Holm $p = 0.119$); COVID t from 2.36 (OLS) to 4.00 (12 lags)	$t(n-k)$ p-values in the strategy tests
4	The continuous variant’s gain is mostly de-leveraging	Mean position 0.14–0.16 against 0.26–0.47; vol-matched paired p rise from 0.014, 0.072, 0.031, 0.023 to 0.062, 0.829, 0.145, 0.148	Not called an improvement
5	Month- t EMV used at the close of month t	Original 3m 2010–2026 net 0.07% becomes 0.73% with a one-month lag	Lag applied in the corrected baseline
6	<code>active_months</code> counts exit-cost months	Original 3m: 96 reported, 75 held	Months held reported instead
7	Emissions file: 41 of 49 industries, static, tied values, Coal 9th	Mild look-ahead; 8-and-8 tie at Hardw/MedEq	Audited in M4
8	Write-up text numbers slightly off	Volatility cut 52–67% (not 55–65%), holdout loss cut 70–84% (not 75–85%), residual HML -0.040 to -0.061	Not repeated
9–11	IC window boundary, cost timing, data vintage	At most 0.008 in any IC, 0.04 pp in holdout net, 10^{-5} in table cells	Negligible

B.3 Checked and clean

A fuzz test that randomizes every input after 2012-06, 2016-12 and 2020-03 changes no signal or strategy value before the cut (the 2016-12 version is kept as `test_no_lookahead`). The rolling hedge uses betas estimated through t on $t+1$ returns; annualization ($12\times$ mean, $\sqrt{12}\times$ standard deviation), the Holm implementation and the bootstrap p-value inversion are correct; and the FF49 portfolios include dead firms, so there is no survivorship bias at the industry level. The write-up’s 12- and 18-lag robustness holds (validation Pure 6m $t = 2.45$ and 2.44).

B.4 Design notes

Only the Brown leg is traded: the Green-leg model is built but never used in a strategy, so “Green minus Brown” describes the signal’s motivation, not the position. The “buy-and-hold Green-Brown” benchmark row is unhedged, not volatility-targeted and cost-free, so it is not like-for-like with the strategy rows.

C Supporting exhibits by module

Each subsection names the module’s question and verdict and points to its FINDINGS.md and VERIFY.md in modules/<id>/. Tables whose captions are not written for this report are the module’s own output tables, included unchanged apart from plain column labels; exhibits appear in the order the main text uses them. The project-wide checks of M7 are in Appendix E.

C.1 M1: what the signal measures

The team’s “attention” is FRED’s EMV tracker for energy and environmental regulation; it is overall volatility news plus a topic share, and it is unrelated to climate concern. Same-month slopes of GB on MCCC and CPU shocks have the wrong sign ($t = -0.49$ and -1.23), so a faster trade would not have caught a Pástor et al. (2022) repricing on these legs. VIX adds nothing to overall EMV over the full sample ($t = -0.35$), but in the validation window it enters negatively ($t = -2.51$).

Table C.1: Exact zeros in EMV env (FRED EMVENRGYENVREG) by period. Fisher exact test of the zero rate 2021-10 to 2026-08 versus 1985-01 to 2021-09.

period	start	end	n months	n zero	zero share
full 1985	1985-01	2026-08	500	51	0.10
pre 2021-10	1985-01	2021-09	441	12	0.03
post 2021-10	2021-10	2026-08	59	39	0.66
post2010	2010-01	2026-07	199	43	0.22
validation	2010-01	2022-07	151	12	0.08
holdout	2022-08	2026-07	48	31	0.65
pre covid	2010-01	2019-12	120	4	0.03
covid	2020-01	2021-12	24	4	0.17
inflation rates	2022-01	2024-12	36	25	0.69
last18	2025-02	2026-07	18	9	0.50
last12	2025-08	2026-07	12	8	0.67

Table C.2: How much of the team signal $z_t = \text{rolling } z(\log 1p(\text{EMV env}))$ is explained by rolling z-scores of $\log 1p(\text{VIX})$ and $\log 1p(\text{overall EMV})$. OLS, Newey-West (6) t-statistics.

sample	start	end	n	regressors	R2	slope 1	t 1	slope 2	t 2
full	1992-12	2026-08	405	z VIX	0.083	0.25	4.3		
full	1992-12	2026-08	405	z EMV overall	0.168	0.40	7.3		
full	1992-12	2026-08	405	z VIX + z EMV overall	0.169	-0.03	-0.4	0.43	5.1
pre 2021-10	1992-12	2021-09	346	z VIX	0.102	0.26	4.4		
pre 2021-10	1992-12	2021-09	346	z EMV overall	0.161	0.38	6.6		
pre 2021-10	1992-12	2021-09	346	z VIX + z EMV overall	0.161	0.02	0.2	0.36	4.1
post 2021-10	2021-10	2026-08	59	z VIX	0.000	-0.00	-0.0		
post 2021-10	2021-10	2026-08	59	z EMV overall	0.245	0.58	4.0		
post 2021-10	2021-10	2026-08	59	z VIX + z EMV overall	0.281	-0.28	-1.9	0.67	3.8
validation	2010-01	2022-07	151	z VIX	0.021	0.14	1.5		
validation	2010-01	2022-07	151	z EMV overall	0.126	0.37	4.9		
validation	2010-01	2022-07	151	z VIX + z EMV overall	0.166	-0.30	-2.5	0.62	5.4
holdout	2022-08	2026-07	48	z VIX	0.011	0.16	0.6		
holdout	2022-08	2026-07	48	z EMV overall	0.250	0.55	3.6		
holdout	2022-08	2026-07	48	z VIX + z EMV overall	0.267	-0.22	-1.1	0.63	3.2

Table C.3: Crossings and volatility. Share of crossing months flagged as high-volatility against all other months, with Fisher exact, circular-shift and linear-probability (LPM, NW(12)) p-values. A shift p of zero is printed as its resolution, one over the number of rotations, which runs from 25 in the holdout to 382. The link to overall EMV survives the dependence corrections and the VIX link is borderline; the holdout LPM p is not informative, because all five holdout crossings are flagged.

Flag	Window	Months	Crossings	Share, crossings	Share, other	p Fisher	p shift	Rotations	p LPM
High VIX	All	1993-01 to 2026-08	50	0.62	0.49	0.097	0.113	381	0.049
High VIX	Since 2010	2010-01 to 2026-07	26	0.50	0.42	0.524	0.534	176	0.337
High VIX	Pre-2010	1993-01 to 2009-12	24	0.75	0.56	0.083	0.210	181	0.071
High VIX	Validation	2010-01 to 2022-07	21	0.52	0.40	0.343	0.328	128	0.198
High VIX	Holdout	2022-08 to 2026-07	5	0.40	0.47	1.000	1.000	25	0.730
High overall EMV	All	1992-12 to 2026-08	50	0.88	0.65	<0.001	<0.003	382	<0.001
High overall EMV	Since 2010	2010-01 to 2026-07	26	0.88	0.56	0.001	<0.006	176	<0.001
High overall EMV	Pre-2010	1992-12 to 2009-12	24	0.88	0.72	0.140	0.137	182	0.101
High overall EMV	Validation	2010-01 to 2022-07	21	0.86	0.48	0.002	<0.008	128	<0.001
High overall EMV	Holdout	2022-08 to 2026-07	5	1.00	0.81	0.573	0.720	25	0.023

Table C.4: Predictive information coefficients: slope of the standardized next-month outcome on the standardized signal (NW(6) t). GB = Green-minus-Brown; Brown resid. = Brown-leg residual from the team's lagged rolling FF3 hedge. The team hypothesis implies $IC > 0$ for GB and $IC < 0$ for the Brown residual.

Outcome	Signal	Start	IC, full	t	IC, validation	t	IC, holdout	t
GB	z EMV env.	1987-12	0.03	0.6	0.18	1.8	-0.13	-1.3
GB	z EMV overall	1987-12	0.04	0.7	-0.01	-0.1	-0.20	-1.8
GB	z VIX	1992-12	0.06	1.1	-0.12	-1.3	-0.07	-1.0
GB	z EMV share	1987-12	0.03	0.6	0.22	2.6	-0.12	-1.0
GB	z MCCC	2005-12	-0.02	-0.4	0.01	0.1	0.11	0.6
GB	z MCCC transition	2005-12	-0.05	-0.8	-0.00	-0.1	0.05	0.3
GB	z CPU	1990-03	0.03	0.6	0.01	0.1	0.05	0.6
GB	shock EMV env.	1988-02	0.06	1.3	0.21	2.4	-0.06	-0.7
GB	shock EMV overall	1988-02	-0.03	-0.5	0.06	0.7	-0.19	-2.1
GB	shock VIX	1993-02	-0.03	-0.4	0.03	0.4	-0.13	-1.7
GB	shock EMV share	1988-02	0.07	1.5	0.23	2.9	-0.09	-0.7
GB	shock MCCC	2006-02	0.00	0.0	-0.02	-0.2	0.08	0.4
GB	shock MCCC transition	2006-02	-0.06	-0.9	-0.06	-0.7	0.05	0.3
GB	shock CPU	1990-05	0.04	1.0	0.03	0.5	0.16	1.9
Brown resid.	z EMV env.	1987-12	-0.02	-0.3	-0.09	-1.2	0.12	1.1
Brown resid.	z EMV overall	1987-12	-0.03	-0.7	-0.05	-0.6	0.10	1.0
Brown resid.	z VIX	1992-12	-0.09	-2.2	-0.04	-0.4	-0.04	-0.2
Brown resid.	z EMV share	1987-12	-0.00	-0.0	-0.08	-1.5	0.13	1.1
Brown resid.	z MCCC	2005-12	0.07	1.3	0.06	0.8	0.02	0.1
Brown resid.	z MCCC transition	2005-12	0.06	1.1	0.04	0.6	0.06	0.4
Brown resid.	z CPU	1990-03	-0.07	-1.6	-0.07	-1.0	-0.08	-0.9
Brown resid.	shock EMV env.	1988-02	-0.01	-0.2	-0.10	-1.5	0.16	2.4
Brown resid.	shock EMV overall	1988-02	0.01	0.1	-0.10	-0.8	0.15	1.2
Brown resid.	shock VIX	1993-02	-0.02	-0.3	-0.13	-1.2	0.05	0.4
Brown resid.	shock EMV share	1988-02	-0.02	-0.5	-0.08	-1.7	0.14	1.3
Brown resid.	shock MCCC	2006-02	0.02	0.3	0.04	0.5	-0.11	-0.9
Brown resid.	shock MCCC transition	2006-02	0.02	0.3	0.02	0.3	-0.07	-0.7
Brown resid.	shock CPU	1990-05	-0.08	-1.8	-0.06	-0.8	-0.18	-1.9

Table C.5: Counterfactual versions of the team rule after 2021-10 (validation tail 2021-10 to 2022-07 and holdout; the CSV adds the full validation window). frozen: window mean and sd fixed at the 60 months to 2021-09; zeros missing: zero months dropped before the rolling z-score. Differ = months whose state or crossing flag differs from the team rule.

variant	window	start	end	state	crossings	crossing months					state differs	crossing differs
team	validation tail	2021-10	2022-07	1	1	2022-07					0	0
team	holdout	2022-08	2026-07	7	5	2023-04	2024-06	2025-02	2025-09	2025-11	0	0
frozen 2021-09 scaling	validation tail	2021-10	2022-07	1	1	2022-07					0	0
frozen 2021-09 scaling	holdout	2022-08	2026-07	7	5	2023-04	2024-06	2025-02	2025-09	2025-11	0	0
zeros missing 60m calendar	validation tail	2021-10	2022-07	1	1	2022-07					0	0
zeros missing 60m calendar	holdout	2022-08	2026-07	6	5	2023-04	2024-06	2025-02	2025-09	2025-11	1	0
zeros missing last60 nonzero	validation tail	2021-10	2022-07	1	1	2022-07					0	0
zeros missing last60 nonzero	holdout	2022-08	2026-07	7	5	2023-04	2024-06	2025-02	2025-09	2025-11	0	0
frozen 2021-09 scaling team threshold	validation tail	2021-10	2022-07	1	1	2022-07					0	0
frozen 2021-09 scaling team threshold	holdout	2022-08	2026-07	7	5	2023-04	2024-06	2025-02	2025-09	2025-11	0	0

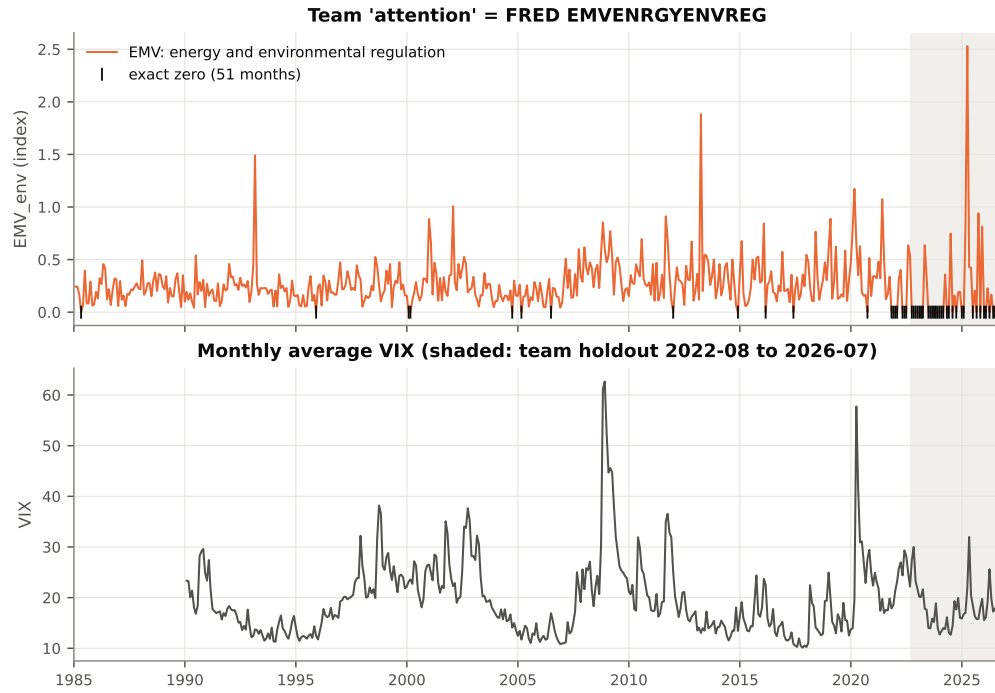


Figure C.1: The team series against VIX, with exact-zero months marked. The zeros cluster after 2021-10: 39 of the 59 months since then are exactly zero.

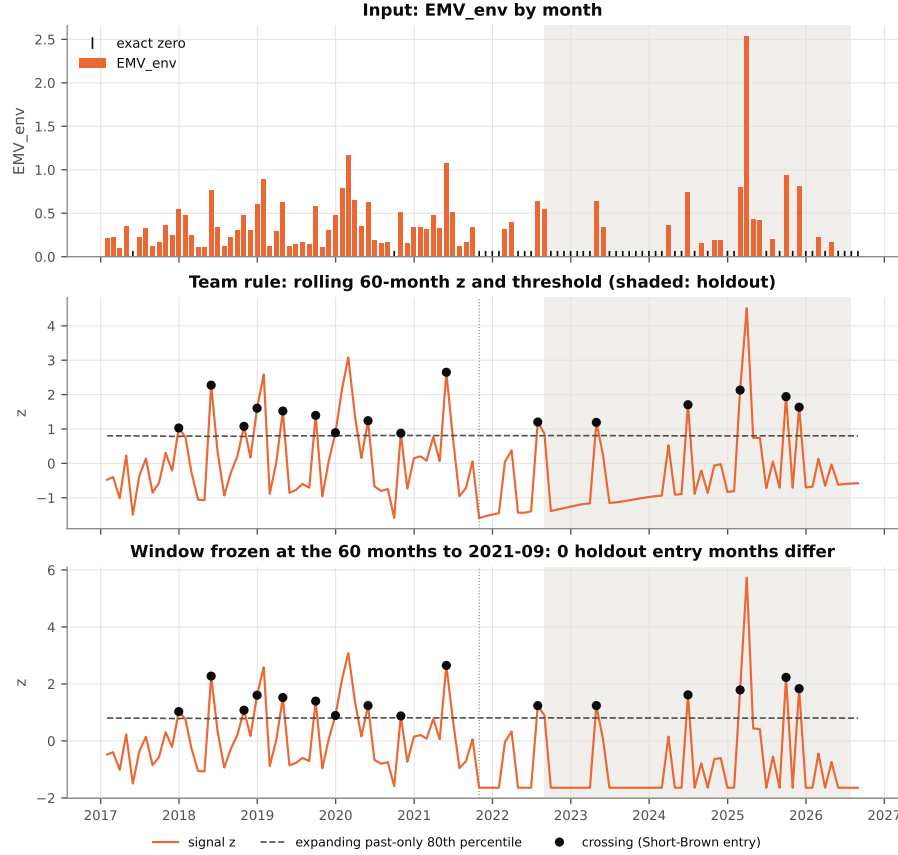


Figure C.2: Rule mechanics, including the counterfactual that freezes the z-score's scaling at 2021-09. No holdout entry month differs, so the zeros did not cause the holdout loss.

C.2 M1b: the same machinery, other signals

Fed MCCC or CPU, the team machinery produces no validation alpha near $t = 2$ and no significant positive holdout alpha (the largest, CPU Original 3m, is +0.36%, $t = 0.37$), and the COVID placebo rule fails on its boundary. The COVID paths of every measure are Figure 2 in Section 2.4.

Table C.6: The same machinery, three signals. FF3 alpha, % a year (NW(6) t), of our team's six rules under real-time timing, fed the EMV tracker, MCCC or CPU. Validation is 2010-01 to 2022-07; the holdout runs from 2022-08 to each measure's last usable month (48, 36 and 39 months). Only the EMV tracker produces validation alphas near $t = 2$, and every MCCC holdout alpha is negative.

Rule	EMV tracker (team)		MCCC		CPU	
	Validation	Holdout	Validation	Holdout	Validation	Holdout
Original 3m	1.62 (1.97)	-2.16 (-1.90)	0.44 (0.53)	-2.91 (-1.88)	1.50 (1.44)	0.36 (0.37)
Pure 3m	1.70 (2.01)	-2.16 (-1.90)	0.37 (0.40)	-2.19 (-1.95)	-0.11 (-0.11)	-0.61 (-0.52)
Original 6m	2.09 (2.18)	-1.97 (-1.30)	1.05 (0.93)	-2.70 (-1.70)	1.81 (1.73)	-1.62 (-0.98)
Pure 6m	2.33 (2.59)	-1.97 (-1.30)	0.04 (0.05)	-1.66 (-1.31)	1.37 (1.28)	0.20 (0.20)
Continuous raw	0.46 (1.40)	-0.58 (-1.33)	0.25 (0.53)	-0.95 (-1.55)	0.41 (0.87)	-0.56 (-0.91)
Continuous pure	0.51 (1.46)	-0.59 (-1.33)	-0.12 (-0.22)	-0.56 (-1.39)	0.28 (0.88)	-0.30 (-0.56)
Count: $t \geq 1.96 / \alpha < 0$	4 of 6	6 of 6	0 of 6	6 of 6	0 of 6	4 of 6
Mean validation alpha	1.45		0.34		0.88	

Table C.7: Primary test: team machinery fed a climate-concern measure, corrected-baseline timing (inputs lagged one month, CPI gap interpolated). FF3 alpha in % per year with NW(6) t in parentheses; IC is the slope of $z(\varepsilon_{t+1})$ on $z(\text{signal}_t)$ (negative favors Short-Brown). Holdout ends 2025-07 for MCCC (36 months) and 2025-10 for CPU (39 months); 2026-07 for EMV env. (48 months). Smallest Holm-adjusted p across the 40 primary tests: 1.00.

Strategy	EMV env. Val.	EMV env. Hold.	MCCC Val.	MCCC Hold.	CPU Val.	CPU Hold.
Original 3m	1.62 (1.97)	-2.16 (-1.90)	0.44 (0.53)	-2.91 (-1.88)	1.50 (1.44)	0.36 (0.37)
Pure 3m	1.70 (2.01)	-2.16 (-1.90)	0.37 (0.40)	-2.19 (-1.95)	-0.11 (-0.11)	-0.61 (-0.52)
Original 6m	2.09 (2.18)	-1.97 (-1.30)	1.05 (0.93)	-2.70 (-1.70)	1.81 (1.73)	-1.62 (-0.98)
Pure 6m	2.33 (2.59)	-1.97 (-1.30)	0.04 (0.05)	-1.66 (-1.31)	1.37 (1.28)	0.20 (0.20)
Continuous raw	0.46 (1.40)	-0.58 (-1.33)	0.25 (0.53)	-0.95 (-1.55)	0.41 (0.87)	-0.56 (-0.91)
Continuous pure	0.51 (1.46)	-0.59 (-1.33)	-0.12 (-0.22)	-0.56 (-1.39)	0.28 (0.88)	-0.30 (-0.56)
IC: raw attention z	-0.02 (-0.23)	0.02 (0.27)	0.04 (0.55)	0.13 (1.31)	0.02 (0.25)	0.07 (0.48)
IC: purified attention	-0.05 (-0.41)	0.03 (0.33)	0.03 (0.47)	0.16 (2.22)	0.02 (0.31)	0.02 (0.12)
IC: continuous weight (raw)	-0.15 (-2.16)	0.06 (0.56)	0.04 (0.45)	0.21 (1.90)	-0.00 (-0.02)	0.03 (0.29)
IC: continuous weight (pure)	-0.16 (-2.19)	0.06 (0.57)	0.09 (1.07)	0.10 (0.90)	-0.00 (-0.08)	-0.03 (-0.29)

Table C.8: COVID window (2020-01 to 2021-12, 24 months): FF3 alpha in % per year (NW(6) t) of the team strategies fed each measure; the last two columns are the FF3 alpha of the return difference between the EMV env. strategy and the volatility placebo. With 24 observations, $|t| > 2.09$ is needed for $p < 0.05$ under $t(20)$. Pre-specified rule: a placebo reproduces the gain when its alpha is positive and at least half the EMV env. alpha; real-time timing: VIX 3 of 6, EMV overall 4 of 6; team timing: VIX 3 of 6, EMV overall 6 of 6. For reference, the signal-free always-short-Brown benchmark earns 4.54% ($t = 1.96$) in the same window.

Strategy	EMV env.	VIX	EMV overall	MCCC	CPU	EMV env. – VIX	EMV env. – EMV overall
<i>Real-time timing (primary)</i>							
Original 3m	6.38 (3.62)	2.41 (2.03)	1.65 (1.29)	1.63 (0.91)	6.44 (4.46)	3.97 (2.67)	4.73 (3.22)
Pure 3m	6.65 (3.95)	0.54 (0.49)	2.69 (1.64)	4.84 (2.54)	4.81 (2.59)	6.11 (2.70)	3.96 (2.75)
Original 6m	3.99 (1.79)	0.40 (0.25)	4.83 (1.98)	3.23 (2.18)	4.29 (2.00)	3.59 (3.18)	-0.84 (-1.16)
Pure 6m	4.54 (1.96)	5.79 (3.06)	3.06 (2.07)	2.89 (2.02)	2.45 (1.66)	-1.25 (-0.39)	1.48 (1.01)
Continuous raw	1.95 (2.68)	1.22 (1.03)	1.61 (1.68)	1.48 (2.58)	2.08 (2.52)	0.74 (0.78)	0.34 (0.53)
Continuous pure	1.85 (2.42)	0.97 (0.53)	1.19 (0.94)	1.20 (2.27)	1.04 (1.12)	0.88 (0.55)	0.66 (0.62)
<i>Team timing (same month)</i>							
Original 3m	6.03 (3.21)	1.95 (1.75)	4.64 (2.18)	1.82 (0.64)	6.33 (3.57)	4.08 (2.34)	1.39 (0.87)
Pure 3m	6.27 (3.09)	0.74 (0.55)	3.30 (2.18)	4.48 (3.98)	4.10 (1.93)	5.53 (2.15)	2.96 (1.29)
Original 6m	5.23 (2.07)	2.57 (1.58)	4.91 (2.41)	3.90 (1.87)	4.43 (1.95)	2.67 (1.58)	0.32 (0.46)
Pure 6m	5.30 (2.09)	3.71 (2.30)	2.83 (1.87)	2.94 (1.18)	2.91 (1.87)	1.59 (0.48)	2.47 (1.44)
Continuous raw	2.14 (3.28)	1.45 (1.37)	1.65 (1.86)	1.25 (2.07)	2.11 (2.55)	0.69 (0.83)	0.49 (0.90)
Continuous pure	2.09 (3.14)	1.31 (0.77)	1.18 (1.00)	1.09 (1.72)	1.01 (1.12)	0.78 (0.55)	0.91 (1.06)

Table C.9: Exploratory: Original 3m in the COVID window (2020-01 to 2021-12). FF3 alpha in % per year (NW(6) t; $|t| > 2.09$ for $p < 0.05$ under $t(20)$); ratio of the alpha to the EMV env. alpha under the same timing; which of March and April 2020 the rule was short in; their share of the summed COVID net return; FF3 alpha re-estimated without those two months (22 observations). The always-short row holds the position in every month.

Measure	COVID α (t)	Ratio to EMV env.	Mar-Apr 2020 held	Mar-Apr share of net	α ex Mar-Apr (t)
<i>Real-time timing (primary)</i>					
EMV env.	6.38 (3.62)	1.00	Mar, Apr	49%	3.48 (2.68)
CPU	6.44 (4.46)	1.01	Mar, Apr	54%	3.81 (2.44)
MCCC	1.63 (0.91)	0.25	Mar, Apr	124%	0.03 (0.01)
VIX	2.41 (2.03)	0.38	Apr	64%	2.20 (1.25)
EMV overall	1.65 (1.29)	0.26	Apr	60%	1.18 (0.67)
Always-short Brown (no signal)	4.54 (1.96)	0.71	Mar, Apr	58%	3.06 (1.05)
<i>Team timing (same month)</i>					
EMV env.	6.03 (3.21)	1.00	Mar	23%	4.25 (2.41)
CPU	6.33 (3.57)	1.05	Mar, Apr	61%	4.22 (1.92)
MCCC	1.82 (0.64)	0.30	Mar, Apr	112%	0.64 (0.22)
VIX	1.95 (1.75)	0.32	Mar, Apr	133%	-0.23 (-0.29)
EMV overall	4.64 (2.18)	0.77	Mar, Apr	67%	2.64 (0.99)
Always-short Brown (no signal)	4.54 (1.96)	0.75	Mar, Apr	58%	3.06 (1.05)

Table C.10: Recent windows, corrected-baseline timing: annualized net return and FF3 alpha in % (NW(6) t) for Original 3m (O3), Original 6m (O6) and Continuous raw (CR). The always-short row repeats the same signal-free position in every column. $p < 0.05$ under $t(n-4)$ needs $|t| > 2.14$ with 18 observations and $|t| > 2.31$ with 12. MCCC and CPU end before these windows.

Measure	O3 net	O3 α (t)	O6 net	O6 α (t)	CR net	CR α (t)
<i>Last 18 months (2025-02 to 2026-07)</i>						
EMV env.	-4.51	-4.05 (-1.55)	-1.35	-4.06 (-1.80)	-0.73	-1.74 (-2.19)
VIX	2.49	0.79 (0.86)	2.06	-0.61 (-0.32)	0.19	-0.22 (-0.64)
EMV overall	1.97	-1.56 (-0.45)	-0.35	-4.51 (-1.45)	0.34	-1.29 (-0.81)
EMV env. share	-4.51	-4.05 (-1.55)	-1.35	-4.06 (-1.80)	-0.83	-1.64 (-2.33)
Always-short Brown (no signal)	-1.07	-6.25 (-1.87)	-1.07	-6.25 (-1.87)	-1.07	-6.25 (-1.87)
<i>Last 12 months (2025-08 to 2026-07)</i>						
EMV env.	-5.21	1.20 (0.30)	0.02	1.03 (0.18)	-0.47	-1.25 (-0.72)
VIX	4.43	0.82 (0.90)	3.78	-3.79 (-1.74)	0.43	-0.35 (-0.59)
EMV overall	4.51	1.30 (0.18)	1.35	-3.39 (-0.52)	1.03	-1.82 (-0.55)
EMV env. share	-5.21	1.20 (0.30)	0.02	1.03 (0.18)	-0.65	-0.84 (-0.56)
Always-short Brown (no signal)	1.35	-3.39 (-0.52)	1.35	-3.39 (-0.52)	1.35	-3.39 (-0.52)

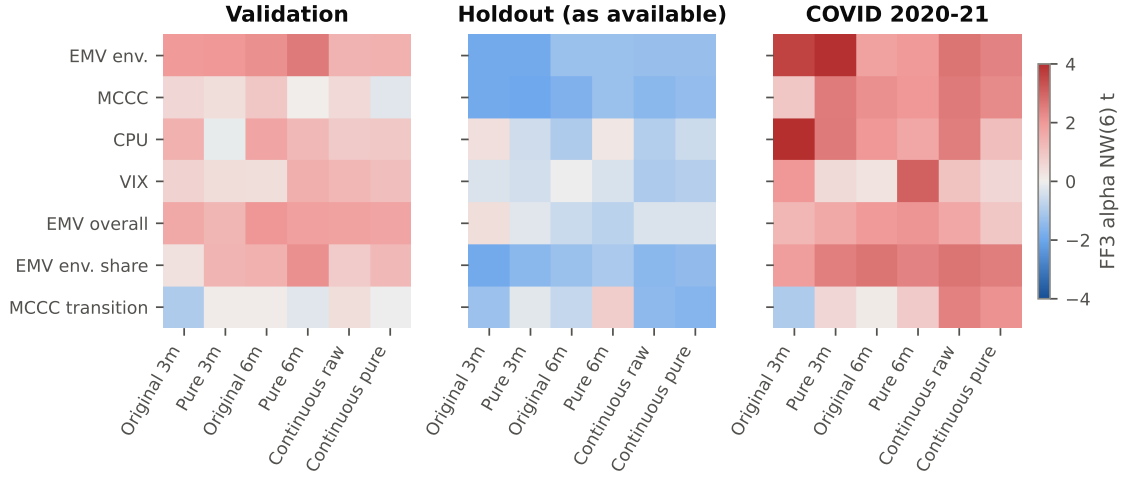


Figure C.3: Alpha t -statistics by measure, rule and window under real-time timing. The validation gains belong to the EMV family (the tracker, its share and overall EMV), every holdout cell of the EMV tracker and MCCC is negative, and COVID is positive for almost every measure, volatility placebos included.

C.3 M2: Cristhian's four tests

Wider legs, momentum and commodity controls and uniform costs leave no positive alpha that survives Holm or Benjamini–Hochberg among 8,091 tests (368 positive and 531 negative alphas have $p < 0.05$ before correction); the HML loading is a Brown-leg value tilt.

Table C.11: FF3 alpha (% p.a., NW(6) t in parentheses) of each strategy with 5-and-5 and 8-and-8 legs, corrected baseline, FF3 hedge and team costs. Val = 2010-01 to 2022-07, Hold = 2022-08 to 2026-07. 8/8 H and 8/8 M differ only in the tied 8th Green industry (Hardw or MedEq).

Strategy	Val 5/5	Val 8/8 H	Val 8/8 M	Hold 5/5	Hold 8/8 H	Hold 8/8 M
Original 3m	1.62 (1.97)	1.30 (1.85)	1.30 (1.85)	-2.16 (-1.90)	-2.51 (-1.81)	-2.51 (-1.81)
Pure 3m	1.70 (2.01)	1.36 (1.74)	1.36 (1.74)	-2.16 (-1.90)	-2.51 (-1.81)	-2.51 (-1.81)
Original 6m	2.09 (2.18)	1.80 (1.87)	1.80 (1.87)	-1.97 (-1.30)	-1.24 (-0.60)	-1.24 (-0.60)
Pure 6m	2.33 (2.59)	2.36 (2.47)	2.36 (2.47)	-1.97 (-1.30)	-1.24 (-0.60)	-1.24 (-0.60)
Continuous raw	0.46 (1.40)	0.36 (1.32)	0.36 (1.32)	-0.58 (-1.33)	-0.44 (-0.78)	-0.44 (-0.78)
Continuous pure	0.51 (1.46)	0.40 (1.39)	0.40 (1.39)	-0.59 (-1.33)	-0.41 (-0.70)	-0.41 (-0.70)
Always-short Brown	1.39 (1.12)	2.05 (1.71)	2.05 (1.71)	-1.72 (-1.09)	-0.39 (-0.19)	-0.39 (-0.19)

Table C.12: Raw 5-and-5 Green-minus-Brown spread: alpha (% p.a.) and HML loading, NW(6) t in parentheses, across control sets. Full = 1970-01 to 2026-07 (from 1992-02 when WTI and IMF controls are included). WTI and IMF are non-traded and demeaned within the window.

Controls	Full alpha	Full HML	Post-2010 alpha	Post-2010 HML	Holdout alpha	Holdout HML
FF3 (team)	0.54 (0.41)	-0.23 (-4.63)	-1.12 (-0.49)	-0.30 (-5.74)	-4.59 (-0.99)	-0.13 (-1.34)
FF3+UMD	1.20 (0.89)	-0.26 (-5.20)	-0.67 (-0.30)	-0.32 (-6.16)	-3.96 (-0.85)	-0.11 (-1.45)
FF5+UMD	2.20 (1.60)	-0.14 (-2.38)	-0.16 (-0.07)	-0.24 (-3.21)	-4.44 (-0.90)	0.02 (0.14)
FF5+UMD+COMEQ	1.91 (1.57)	-0.11 (-2.30)	-0.41 (-0.22)	-0.21 (-2.67)	-2.37 (-0.53)	0.24 (2.01)
FF5+UMD+COMEQ ex-Gold	1.77 (1.45)	-0.09 (-1.72)	-0.67 (-0.34)	-0.18 (-1.99)	-3.35 (-0.77)	0.17 (1.22)
FF5+UMD+COMEQ + WTI, IMF	1.19 (0.79)	-0.09 (-1.56)	-0.40 (-0.22)	-0.19 (-2.55)	-1.90 (-0.42)	0.20 (1.96)
FF5+UMD+COMEQ + WTI, IMF (t and t+1)	0.70 (0.44)	-0.08 (-1.39)	-1.13 (-0.59)	-0.18 (-2.40)	-3.91 (-0.77)	0.23 (1.56)

Table C.13: Contribution of each industry to the Green-minus-Brown HML loading: +loading/5 for Green industries, -loading/5 for Brown industries. Contributions sum exactly to the spread's loading (last row).

Industry	Leg	FF3 Full	FF3 Post-2010	FF3 Val	FF3 Hold	FF5U Full	FF5U Post-2010	FF5U Val	FF5U Hold
Fun	Green	0.01	-0.07	-0.08	-0.04	-0.01	-0.05	-0.01	-0.11
RIEst	Green	0.14	0.09	0.08	0.11	0.11	0.09	0.10	0.06
Drugs	Green	-0.05	-0.02	-0.06	0.10	-0.08	-0.06	-0.10	0.11
Telcm	Green	0.03	0.05	0.03	0.13	0.01	0.00	0.00	0.12
Fin	Green	0.06	0.08	0.07	0.09	0.09	0.10	0.11	0.09
Util	Brown	-0.06	-0.04	-0.02	-0.09	-0.05	-0.02	-0.01	-0.02
Ships	Brown	-0.10	-0.10	-0.10	-0.12	-0.05	-0.07	-0.09	-0.01
Aero	Brown	-0.08	-0.09	-0.08	-0.11	-0.04	-0.07	-0.07	-0.02
Steel	Brown	-0.09	-0.13	-0.14	-0.13	-0.07	-0.11	-0.05	-0.20
BldMt	Brown	-0.09	-0.07	-0.06	-0.08	-0.06	-0.05	-0.05	0.00
Total = GB loading		-0.23	-0.30	-0.35	-0.13	-0.14	-0.24	-0.18	0.02

Table C.14: FF3 alpha (% p.a., NW(6) t in parentheses) at uniform costs of 5, 10 and 25 bp per unit of turnover (asset and every overlay factor), corrected baseline, 5-and-5 legs, FF3 hedge. Turnover is annual validation turnover (asset plus overlay). Break-even is the uniform cost at which validation alpha reaches zero; none = gross validation alpha is not positive.

Strategy	Val 5bp	Val 10bp	Val 25bp	Hold 5bp	Hold 10bp	Hold 25bp	Turnover	Break-even bp
Original 3m	1.92 (2.31)	1.66 (2.02)	0.89 (1.09)	-1.87 (-1.66)	-2.08 (-1.83)	-2.69 (-2.34)	5.1	42
Pure 3m	2.05 (2.37)	1.76 (2.06)	0.89 (1.07)	-1.87 (-1.66)	-2.08 (-1.83)	-2.69 (-2.34)	5.8	40
Original 6m	2.25 (2.36)	2.12 (2.22)	1.74 (1.80)	-1.70 (-1.12)	-1.89 (-1.25)	-2.47 (-1.63)	2.4	93
Pure 6m	2.51 (2.80)	2.37 (2.63)	1.98 (2.13)	-1.70 (-1.12)	-1.89 (-1.25)	-2.47 (-1.63)	2.5	100
Continuous raw	0.54 (1.63)	0.47 (1.44)	0.27 (0.85)	-0.51 (-1.19)	-0.56 (-1.29)	-0.70 (-1.58)	1.3	46
Continuous pure	0.60 (1.69)	0.53 (1.50)	0.31 (0.90)	-0.52 (-1.19)	-0.57 (-1.29)	-0.72 (-1.60)	1.5	47
Always-short Brown	1.47 (1.18)	1.43 (1.15)	1.33 (1.06)	-1.67 (-1.07)	-1.70 (-1.08)	-1.78 (-1.13)	0.7	211

Table C.15: Best-case evidence per strategy over the whole M2 grid of 8,091 alpha tests (7,938 net-of-cost strategy alphas and 153 raw-spread alphas, which are unhedged and cost-free), over all leg designs, baselines, hedge sets, evaluation sets, cost levels and windows. Alpha in % p.a. with NW(6) t in parentheses; grid Holm p is the Holm-adjusted p of the single best test. Q2 primary: corrected baseline, 5-and-5, FF5+UMD+COMEQ hedge and alpha, Holm over 14 tests.

Strategy	Best any window	Best post-2010	Best holdout	Grid Holm p (best)	Q2 primary	Holm p (post-2010 / holdout)	Survives
Original 3m	6.15 (4.36), COVID 2020-21	1.08 (1.41)	–	1.000	1.000 / 0.397		no
Pure 3m	6.48 (4.33), COVID 2020-21	1.20 (1.50)	–	1.000	1.000 / 0.397		no
Original 6m	4.41 (3.05), COVID 2020-21	1.36 (1.98)	–	1.000	1.000 / 0.752		no
Pure 6m	4.27 (2.93), COVID 2020-21	1.41 (1.98)	–	1.000	1.000 / 0.752		no
Continuous raw	1.99 (3.48), COVID 2020-21	0.32 (1.09)	–	1.000	1.000 / 1.000		no
Continuous pure	1.99 (3.46), COVID 2020-21	0.40 (1.56)	–	1.000	1.000 / 1.000		no
Always-short Brown	3.87 (2.76), COVID 2020-21	1.29 (1.36)	0.42 (0.22)	1.000	1.000 / 1.000		no
Raw GB spread (unhedged)	2.52 (2.57), 1970-2026	1.71 (1.17)	1.55 (0.33)	1.000	–		no

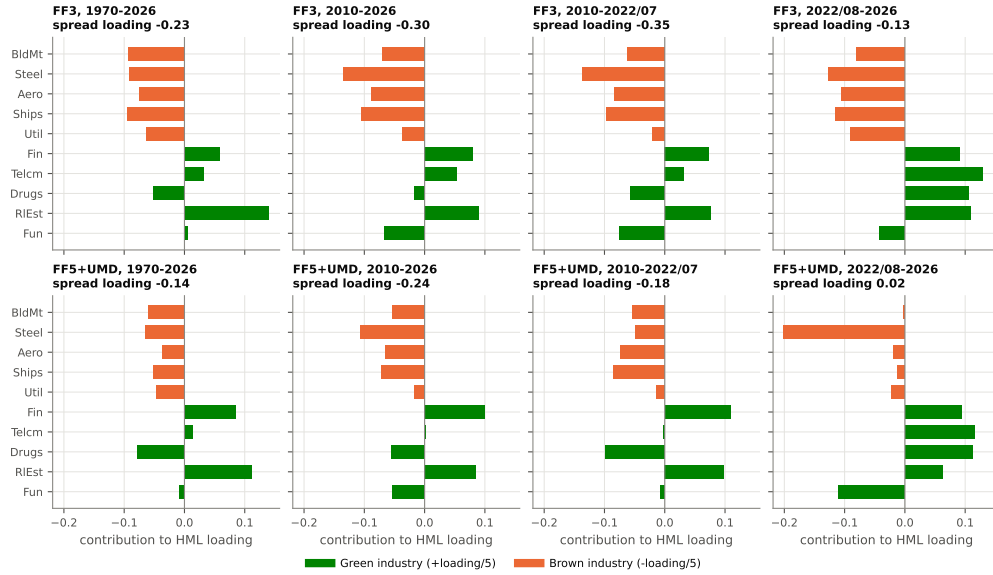


Figure C.4: Where GB's HML loading comes from: industry contributions by leg. The Brown side supplies the negative loading, led by Steel and Ships.

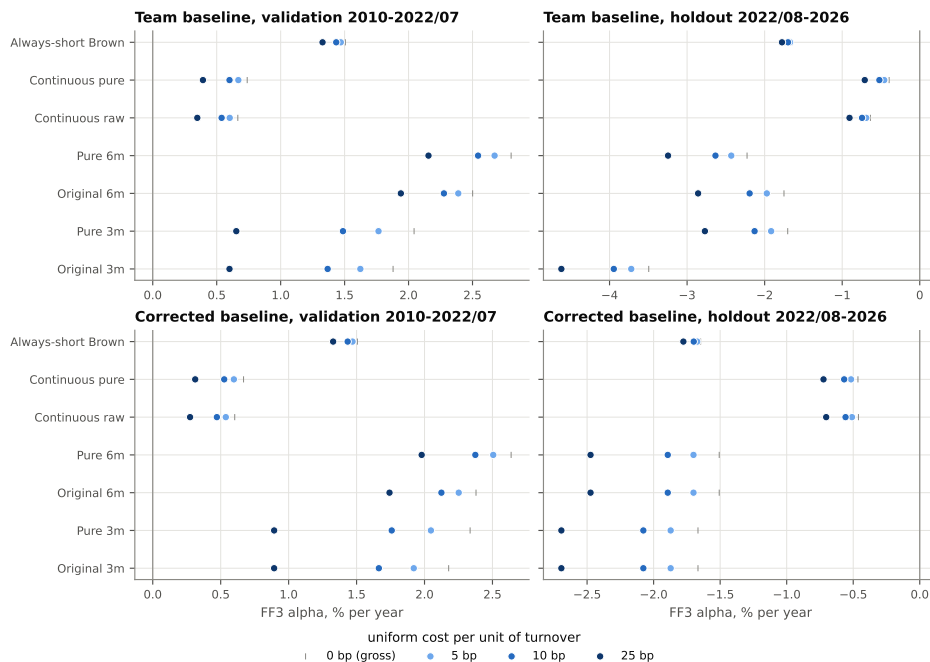


Figure C.5: Validation and holdout alphas at uniform costs of 5, 10 and 25 bp. Costs shrink the validation alphas, and the holdout alphas are negative before any cost.

C.4 M3: alpha versus beta

Green-minus-Brown is static factor beta, the holdout loss is not a rates accident, and the strategies' beta-timing term is a drag that the untimed short shares. The BOND factor tracks Damodaran's annual 10-year Treasury returns with a correlation of 0.991 over 1954–2025 (0.987 over 1993–2025); its 2022 total return is -15.0% against Damodaran's -17.8% , the gap coming from GS10's monthly averaging. Steel carries the corrected 6-month rule's gross holdout loss in the five-industry split (-1.37 of -1.38 points), but that split is accounting, not a counterfactual: rebuilt without any one Brown industry (leg, hedge and volatility scaling re-estimated), the rule's holdout FF3 alpha is -1.45% to -2.61% for 6-month holds and -1.26% to -2.43% for 3-month holds, 10 of 10 negative. Without Steel the untimed short loses more (-2.67% , $t = -1.66$), and only the timing alpha from M8's attribution regression turns positive ($+0.39\%$, $t = 0.38$, against -1.11% with Steel). Against the π -scaled short with in-window FF3, UMD and BOND, the four corrected discrete rules' holdout timing alphas are -1.03% to -1.71% ($t = -0.99$ to -1.72), so the holdout indicts the timing as well as the position (exploratory; `exchange/04_red_team/checks/fc04b_checks.py`).

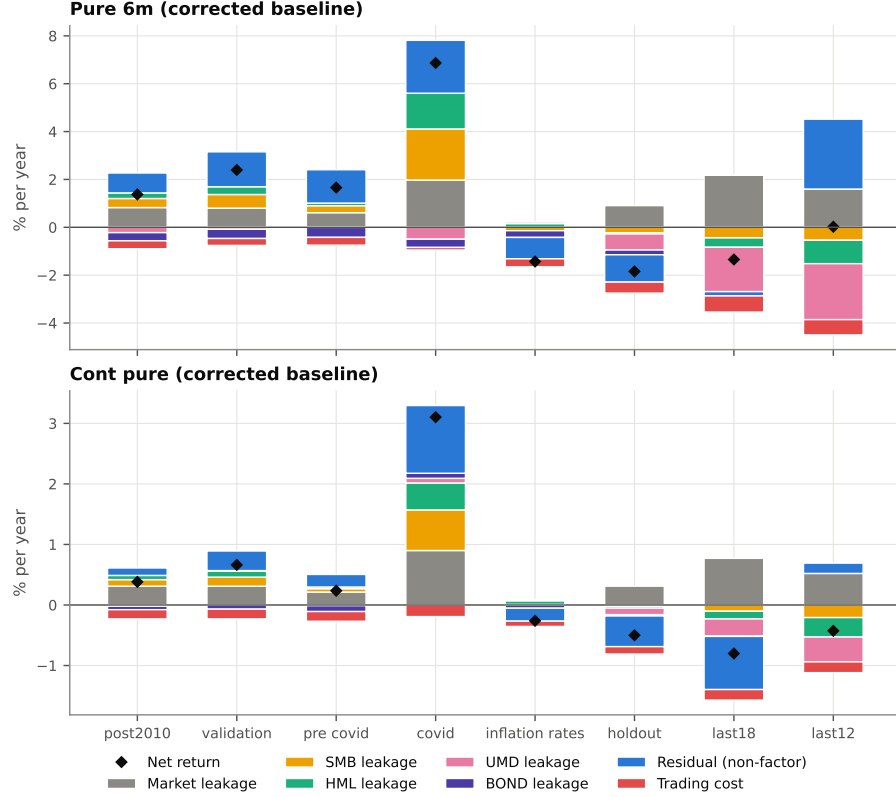


Figure C.6: What the traded position earned, window by window. Each bar splits the annualized net return of corrected Pure 6m (top) and Continuous pure (bottom) into leakage of the lagged FF3 hedge by factor (market, SMB, HML, UMD, BOND), the residual non-factor return and trading cost; diamonds mark the net return. Betas come from centered 36-month windows, so the split is descriptive and ex post. The COVID gain is mostly market and size leakage (4.76 of 6.87 points, with a residual of 2.21, $t = 1.27$), and the holdout loss is residual Brown performance plus costs, not rates.

Table C.16: Holdout alphas before and after adding UMD and BOND (M3 primary test ii), both baselines. Every alpha becomes more negative.

Baseline	Strategy	Net	α FF3	t	α +BOND	t	α +UMD+BOND	t	p	β_{BOND}	β_{UMD}	Wald p
team	Orig 3m	-3.82	-4.03	-2.19	-4.44	-2.25	-4.43	-2.31	0.026	-0.09	-0.02	0.342
team	Pure 3m	-2.17	-2.21	-1.93	-2.45	-1.97	-2.45	-1.97	0.056	-0.05	0.00	0.410
team	Orig 6m	-2.14	-2.28	-1.66	-2.94	-2.14	-2.94	-2.14	0.038	-0.14	0.00	0.248
team	Pure 6m	-3.04	-2.71	-1.94	-3.24	-2.13	-3.22	-2.16	0.037	-0.12	-0.02	0.262
team	Cont raw	-0.65	-0.76	-1.98	-0.86	-2.17	-0.86	-2.21	0.033	-0.02	-0.01	0.394
team	Cont pure	-0.48	-0.54	-1.79	-0.62	-1.99	-0.62	-2.00	0.052	-0.02	-0.00	0.680
corrected	Orig 3m	-2.57	-2.16	-1.90	-2.41	-1.92	-2.40	-1.97	0.056	-0.06	-0.02	0.507
corrected	Pure 3m	-2.57	-2.16	-1.90	-2.41	-1.92	-2.40	-1.97	0.056	-0.06	-0.02	0.507
corrected	Orig 6m	-1.84	-1.97	-1.30	-2.27	-1.40	-2.27	-1.40	0.169	-0.06	0.00	0.728
corrected	Pure 6m	-1.84	-1.97	-1.30	-2.27	-1.40	-2.27	-1.40	0.169	-0.06	0.00	0.728
corrected	Cont raw	-0.48	-0.58	-1.33	-0.65	-1.35	-0.64	-1.36	0.181	-0.02	-0.01	0.342
corrected	Cont pure	-0.50	-0.59	-1.33	-0.64	-1.32	-0.64	-1.33	0.192	-0.02	-0.01	0.377

Table C.17: Lewellen–Nagel decomposition, post-2010, backward 36-month betas (M3 primary test iii). The beta-timing term is negative for the rules and for the always-short position alike.

Asset	Baseline	n	Mean	Cond. α	Static $\bar{\beta}'\bar{f}$	Timing cov	95% low	95% high	p
GB	legs	199	-1.79	0.72	-0.78	-1.72	-3.75	0.57	0.146
Green leg	legs	199	12.33	-1.66	13.40	0.59	-0.80	1.93	0.444
Brown leg	legs	199	14.12	-2.38	14.19	2.31	0.41	4.12	0.014
Brown leg FF3-hedged	legs	199	-1.33	-2.61	-0.66	1.94	0.28	3.41	0.020
Orig 3m	team	199	0.07	0.76	0.23	-0.92	-1.57	-0.20	0.010
Pure 3m	team	199	0.55	1.27	0.25	-0.97	-1.66	-0.24	0.006
Orig 6m	team	199	1.34	2.35	0.23	-1.23	-2.13	-0.18	0.018
Pure 6m	team	199	1.23	2.42	0.05	-1.24	-2.16	-0.18	0.023
Cont raw	team	199	0.34	0.81	0.13	-0.61	-1.06	-0.17	0.003
Cont pure	team	199	0.44	0.92	0.15	-0.64	-1.10	-0.18	0.003
Always-short Brown	team	199	1.05	1.96	0.43	-1.35	-2.48	-0.09	0.035
Orig 3m	corrected	199	0.73	1.55	0.17	-0.99	-1.55	-0.34	0.002
Pure 3m	corrected	199	0.75	1.69	0.12	-1.06	-1.64	-0.42	0.001
Orig 6m	corrected	199	1.19	2.05	0.18	-1.04	-1.87	-0.09	0.030
Pure 6m	corrected	199	1.37	2.40	0.04	-1.07	-1.97	-0.07	0.032
Cont raw	corrected	199	0.32	0.78	0.11	-0.57	-0.99	-0.16	0.003
Cont pure	corrected	199	0.38	0.85	0.13	-0.60	-1.04	-0.17	0.003
Always-short Brown	corrected	199	1.05	1.96	0.43	-1.35	-2.48	-0.09	0.035

Table C.18: Ferson–Schadt conditional alphas and the joint test that betas do not vary with the instruments, with fixed-design and wild block bootstrap p-values.

Asset	Period	n	α uncond.	t	α cond.	t	Fixed-design boot p	Wild boot p	OLS F p	α holdout	p	α COVID	p
GB	full 1970	679	0.99	0.74	0.77	0.57	0.073	0.086	0.000	—	—	—	—
GB	post2010	199	-0.74	-0.32	0.35	0.17	0.159	0.274	0.184	-1.23	0.798	3.77	0.473
Green leg	full 1970	679	0.05	0.06	-0.30	-0.35	0.249	0.626	0.000	—	—	—	—
Green leg	post2010	199	-0.45	-0.31	-0.13	-0.09	0.365	0.703	0.008	0.53	0.871	-3.82	0.241
Brown leg	full 1970	679	-0.94	-0.96	-1.08	-1.03	0.032	0.169	0.000	—	—	—	—
Brown leg	post2010	199	0.29	0.18	-0.47	-0.28	0.506	0.819	0.485	1.76	0.680	-7.60	0.049
Orig 3m	full live 1999	329	0.46	0.68	0.09	0.13	0.110	0.818	0.052	—	—	—	—
Orig 3m	post2010	199	0.67	0.94	0.57	0.76	0.066	0.603	0.229	-1.39	0.274	7.34	0.005
Pure 3m	full live 1999	329	-0.22	-0.28	-0.42	-0.56	0.040	0.590	0.004	—	—	—	—
Pure 3m	post2010	199	0.76	1.03	0.66	0.89	0.038	0.397	0.417	-1.27	0.289	8.18	0.000
Orig 6m	full live 1999	329	0.85	1.06	0.58	0.73	0.213	0.791	0.023	—	—	—	—
Orig 6m	post2010	199	0.99	1.15	1.52	1.67	0.170	0.524	0.462	-0.49	0.796	6.75	0.012
Pure 6m	full live 1999	329	0.70	0.85	0.39	0.48	0.169	0.761	0.016	—	—	—	—
Pure 6m	post2010	199	1.24	1.48	1.67	1.91	0.121	0.485	0.629	-0.21	0.910	7.22	0.002
Cont raw	full live 1999	329	0.20	0.54	-0.01	-0.02	0.026	0.659	0.000	—	—	—	—
Cont raw	post2010	199	0.19	0.69	0.29	0.94	0.034	0.558	0.038	0.00	1.000	2.72	0.008
Cont pure	full live 1999	329	0.12	0.34	-0.10	-0.30	0.038	0.680	0.000	—	—	—	—
Cont pure	post2010	199	0.22	0.77	0.33	0.99	0.039	0.554	0.024	0.07	0.919	2.66	0.017

Table C.19: Holdout net return when the rolling hedge also includes BOND (and UMD). A rates hedge does not turn the holdout positive.

Baseline	Strategy	Net FF3 hedge	t	Net FF3+BOND hedge	t	Net FF3+UMD+BOND hedge	t
team	Orig 3m	-3.82	-2.29	-3.79	-2.16	-4.01	-2.30
team	Pure 3m	-2.17	-2.69	-2.11	-2.25	-2.25	-2.42
team	Orig 6m	-2.14	-1.53	-1.86	-1.30	-1.99	-1.37
team	Pure 6m	-3.04	-2.25	-2.77	-1.96	-2.91	-2.01
team	Cont raw	-0.65	-1.79	-0.60	-1.61	-0.67	-1.75
team	Cont pure	-0.48	-1.76	-0.42	-1.53	-0.47	-1.72
corrected	Orig 3m	-2.57	-2.27	-2.30	-1.93	-2.66	-2.22
corrected	Pure 3m	-2.57	-2.27	-2.30	-1.93	-2.66	-2.22
corrected	Orig 6m	-1.84	-1.30	-1.44	-1.00	-1.86	-1.35
corrected	Pure 6m	-1.84	-1.30	-1.44	-1.00	-1.86	-1.35
corrected	Cont raw	-0.48	-1.09	-0.43	-0.97	-0.54	-1.19
corrected	Cont pure	-0.50	-1.13	-0.45	-1.02	-0.57	-1.23

Table C.20: Validation of the BOND factor against Damodaran’s annual 10-year T-bond returns and month-end par yields.

Year	GS10 Dec (%)	BOND total (%)	Exact reprice (%)	Excess of RF (%)	Damodaran (%)
2022	3.62	−15.0	−14.9	−16.4	−17.8
2013	2.90	−8.0	−7.9	−8.0	−9.1
1994	7.81	−7.5	−7.5	−11.4	−8.0
1999	6.28	−6.7	−6.7	−11.4	−8.3
2009	3.59	−6.7	−6.7	−6.8	−11.1
1969	7.65	−4.9	−5.0	−11.5	−5.0
1987	8.99	−4.5	−4.6	−10.0	−5.0
2021	1.47	−3.7	−3.6	−3.7	−4.4

C.5 M4: the emissions file

The team’s ranking is undocumented and puts Ships and Aero in the Brown leg on values closer to transport than to manufacturing emissions. A complete EPA ranking changes both legs and flips the raw spread, but produces no post-2010 alpha (none of 9 specifications, smallest Holm $p = 0.471$). Its full-sample FF5+UMD alpha (EPA 5-vs-5 4.5%, $t = 2.58$; 7 of 9 variants with raw $p < 0.05$, 5 of 9 surviving Holm) comes from short CMA and UMD loadings on a 2022 sort applied back to 1970.

Exchange 04 asked whether the climate reading fails only because of the legs, so I fed MCCC and CPU through our team’s rules on the EPA-ranked legs (exploratory; `exchange/04_red_team/checks/fc04b_checks.py`). The same-month GB slopes on real-time concern shocks become more wrong-signed, not less: -0.51% and -0.29% a month per standard deviation ($t = -2.95$ and -2.17 ; with FF3 controls $t = -2.50$ and -2.05), against -0.09% and -0.16% on the team legs. All six CPU-timed rules have positive holdout alphas to 2025-10 (0.49% to 1.97% , largest $t = 1.79$), but the untimed EPA short earns 1.41% over the same 39 months ($t = 1.05$), the timing edge over an exposure-matched short (π and FF3 fitted in each window) is $+0.05\%$ to $+1.22\%$ there (largest $t = 1.43$), and over 1994–2009, a window this combination never saw, it is negative in 5 of 6 rules (-2.59% to $+0.15\%$). MCCC timing on the same legs does no better (holdout edges -1.29% to $+0.61\%$). The holdout gain belongs to the EPA position, not the climate timing.

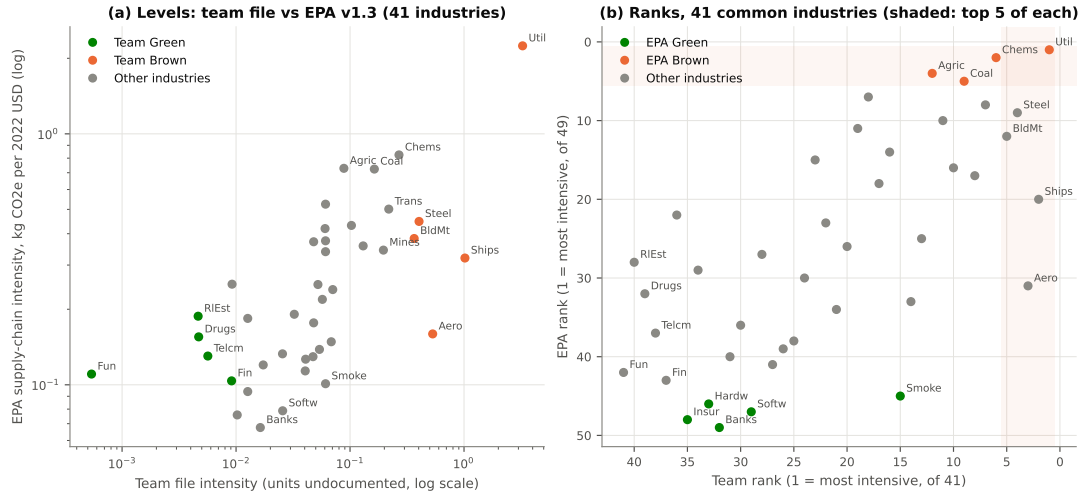


Figure C.7: The Brown leg as delivered. (a) The team's intensity file against EPA supply-chain GHG intensity (kg CO₂e per 2022 dollar, log scales), 41 common industries; (b) ranks, top five of each shaded. The rankings agree broadly (Spearman 0.70), but Ships and Aero rank 20th and 31st of 49 on EPA factors, and EPA's Green leg shares no industry with the team's.

Table C.21: Where the contested industries land. Rank 1 = most intensive. Team ranks are out of 41, EPA and direct ranks out of 49.

ff49	team intensity	team rank from top of41	epa sc mean	epa rank from top of49	epa median rank from top of49	useeio direct mean	direct ghg rank from top of49	n naics
Util	3.264	1.000	2.242	1	1	3.308	1	13
Ships	1.019	2.000	0.320	20	16	0.018	36	2
Aero	0.531	3.000	0.160	31	30	0.011	43	4
Steel	0.403	4.000	0.448	9	12	0.121	14	23
BldMt	0.365	5.000	0.383	12	21	0.338	10	68
Chems	0.269	6.000	0.824	2	3	0.593	6	21
Trans	0.219	7.000	0.501	8	6	0.518	7	60
Mines	0.196	8.000	0.344	17	18	0.499	8	19
Coal	0.163	9.000	0.724	5	4	1.556	2	4
Oil	–	–	0.377	13	11	0.397	9	7
Agric	0.088	12.000	0.729	4	7	0.842	5	65
Other	–	–	0.752	3	2	1.552	3	12
Gold	–	–	0.593	6	5	0.876	4	2
Food	0.061	18.000	0.524	7	9	0.075	19	34
Fun	0.001	41.000	0.110	42	43	0.020	35	44
RIEst	0.005	40.000	0.188	28	22	0.058	25	11
Drugs	0.005	39.000	0.155	32	28	0.117	15	4
Telcm	0.006	38.000	0.130	37	45	0.030	27	10
Fin	0.009	37.000	0.104	43	48	0.023	32	19
Banks	0.016	32.000	0.068	49	46	0.007	48	16
Insur	0.010	35.000	0.076	48	49	0.018	37	13
Softw	0.026	29.000	0.079	47	44	0.009	46	5
Hardw	0.013	33.000	0.094	46	47	0.011	42	5
Smoke	0.061	15.000	0.101	45	41	0.010	45	1

Table C.22: Leave-two-out calibration: regress log team intensity on log measure over the other 39 industries, then predict Aero and Ships using their own manufacturing NAICS or the air/water transportation NAICS. team over predicted = team value / predicted value. holm p 12: Holm-adjusted over these 12 tests.

measure	industry	mapping	measure value	team value	predicted team value	team over predicted	t prediction	p two sided	holm p 12
epa sc mean	Aero	mfg	0.155	0.531	0.025	21.377	2.882	0.007	0.052
epa sc mean	Aero	transport	0.644	0.531	0.192	2.757	0.936	0.355	1.000
epa sc mean	Ships	mfg	0.320	1.019	0.070	14.465	2.516	0.016	0.114
epa sc mean	Ships	transport	0.816	1.019	0.271	3.765	1.210	0.234	1.000
useeio direct mean	Aero	mfg	0.008	0.531	0.014	38.817	2.984	0.005	0.050
useeio direct mean	Aero	transport	0.726	0.531	0.211	2.518	0.742	0.463	1.000
useeio direct mean	Ships	mfg	0.018	1.019	0.022	47.027	3.177	0.003	0.036
useeio direct mean	Ships	transport	0.516	1.019	0.171	5.961	1.446	0.157	0.940
useeio direct co2 mean	Aero	mfg	0.008	0.531	0.014	36.902	2.989	0.005	0.050
useeio direct co2 mean	Aero	transport	0.719	0.531	0.286	1.853	0.496	0.623	1.000
useeio direct co2 mean	Ships	mfg	0.018	1.019	0.024	42.362	3.138	0.003	0.037
useeio direct co2 mean	Ships	transport	0.443	1.019	0.207	4.912	1.298	0.202	1.000

Table C.23: Post-2010 FF5+UMD alpha across the pre-specified EPA specifications, with Holm and BH adjusted p-values (family = these 9 specifications).

spec	n	alpha ann	t alpha	p alpha	holm p	bh p	b SMB	b HML	t HML	b RMW	b CMA	t CMA	b UMD
epa5	199	0.047	1.516	0.129	0.674	0.233	-0.328	-0.024	-0.239	-0.035	-0.380	-2.012	0.066
epa8	199	0.037	1.351	0.177	0.707	0.265	-0.185	0.026	0.238	-0.131	-0.586	-3.116	0.097
epa median5	199	0.019	0.473	0.636	0.877	0.636	-0.112	0.200	1.303	-0.169	-0.557	-2.116	0.109
epa nomargin5	199	0.054	1.737	0.082	0.649	0.233	-0.387	-0.163	-1.433	-0.054	-0.415	-2.007	0.058
epa anylink5	199	0.055	1.744	0.081	0.649	0.233	-0.313	-0.082	-0.791	0.016	-0.400	-2.156	0.085
epa41 5	199	0.052	1.588	0.112	0.674	0.233	-0.370	0.046	0.490	-0.095	-0.445	-2.488	0.046
epa41 8	199	0.051	1.940	0.052	0.471	0.233	-0.213	-0.020	-0.267	-0.136	-0.432	-2.789	0.044
epa5 exOther	199	0.036	0.869	0.385	0.877	0.433	-0.351	0.117	0.872	-0.056	-0.570	-2.309	0.143
epa median8	199	0.029	1.053	0.292	0.877	0.376	-0.173	-0.008	-0.073	-0.126	-0.578	-2.970	0.085

Table C.24: Sensitivity of the primary test (EPA 5v5, FF5+UMD, 2010-01 to 2026-07, NW 6 lags) to leg membership: the 6th-lowest industry in place of the 5th Green member, and each leg member dropped in turn. Added after independent verification; not pre-specified.

variant	n	ann ret	alpha ann	t alpha	p alpha
epa5 (primary)	199	0.064	0.047	1.516	0.129
Chips replaces Smoke in Green	199	0.080	0.050	1.684	0.092
drop Banks (Green, 4 industries)	199	0.068	0.050	1.653	0.098
drop Insur (Green, 4 industries)	199	0.068	0.047	1.466	0.143
drop Softw (Green, 4 industries)	199	0.061	0.048	1.472	0.141
drop Hardw (Green, 4 industries)	199	0.059	0.048	1.452	0.146
drop Smoke (Green, 4 industries)	199	0.064	0.043	1.396	0.163
drop Util (Brown, 4 industries)	199	0.068	0.056	1.530	0.126
drop Chems (Brown, 4 industries)	199	0.068	0.041	1.116	0.264
drop Other (Brown, 4 industries)	199	0.071	0.054	1.379	0.168
drop Agric (Brown, 4 industries)	199	0.069	0.054	1.562	0.118
drop Coal (Brown, 4 industries)	199	0.045	0.031	1.709	0.087

Table C.25: Team Short-Brown timing rule (Original attention signal, 80th percentile, 3/6-month holds, 5% residual vol target, team costs) re-run with the Brown leg swapped. team brown reproduces the team's numbers exactly.

brown leg	hold	period	n months	active months	ann net	sharpe net	alpha ff3 ann	alpha ff3 t hac6	b HML
team brown	hold3	post2010	199	96	0.001	0.021	0.001	0.109	-0.051
team brown	hold3	validation	151	74	0.013	0.412	0.013	1.468	-0.049
team brown	hold3	holdout	48	22	-0.038	-1.153	-0.040	-2.187	-0.034
team brown	hold3	last12	12	6	-0.088	-1.843	-0.113	-2.684	0.061
team brown	hold6	post2010	199	140	0.013	0.317	0.012	1.331	-0.061
team brown	hold6	validation	151	103	0.025	0.588	0.022	2.230	-0.064
team brown	hold6	holdout	48	37	-0.021	-0.492	-0.023	-1.657	-0.036
team brown	hold6	last12	12	11	-0.010	-0.149	-0.041	-0.502	0.022
team brown ex aero ships	hold3	post2010	199	96	0.004	0.106	0.003	0.281	-0.024
team brown ex aero ships	hold3	validation	151	74	0.010	0.296	0.009	0.924	-0.014
team brown ex aero ships	hold3	holdout	48	22	-0.018	-0.588	-0.019	-1.304	-0.035
team brown ex aero ships	hold3	last12	12	6	-0.064	-1.558	-0.057	-2.442	-0.081
team brown ex aero ships	hold6	post2010	199	140	0.018	0.394	0.015	1.456	-0.042
team brown ex aero ships	hold6	validation	151	103	0.022	0.486	0.019	1.536	-0.033
team brown ex aero ships	hold6	holdout	48	37	0.003	0.081	0.002	0.098	-0.055
team brown ex aero ships	hold6	last12	12	11	-0.002	-0.037	-0.011	-0.168	-0.111
epa brown	hold3	post2010	199	96	0.009	0.265	0.009	1.132	-0.055
epa brown	hold3	validation	151	74	0.017	0.448	0.017	1.753	-0.051
epa brown	hold3	holdout	48	22	-0.014	-0.569	-0.015	-1.295	-0.055
epa brown	hold3	last12	12	6	-0.022	-0.748	0.051	2.724	-0.355
epa brown	hold6	post2010	199	140	0.024	0.510	0.020	1.459	-0.062
epa brown	hold6	validation	151	103	0.031	0.660	0.029	1.748	-0.041
epa brown	hold6	holdout	48	37	0.001	0.022	-0.006	-0.606	-0.109
epa brown	hold6	last12	12	11	0.023	0.331	0.003	0.260	-0.208
epa41 brown	hold3	post2010	199	96	0.010	0.303	0.010	1.297	-0.060
epa41 brown	hold3	validation	151	74	0.018	0.487	0.017	1.877	-0.055
epa41 brown	hold3	holdout	48	22	-0.012	-0.462	-0.011	-0.890	-0.069
epa41 brown	hold3	last12	12	6	-0.024	-0.715	0.053	2.907	-0.356
epa41 brown	hold6	post2010	199	140	0.024	0.539	0.021	1.663	-0.069
epa41 brown	hold6	validation	151	103	0.029	0.647	0.027	1.745	-0.051
epa41 brown	hold6	holdout	48	37	0.008	0.188	0.005	0.449	-0.117
epa41 brown	hold6	last12	12	11	0.025	0.400	0.024	1.637	-0.239

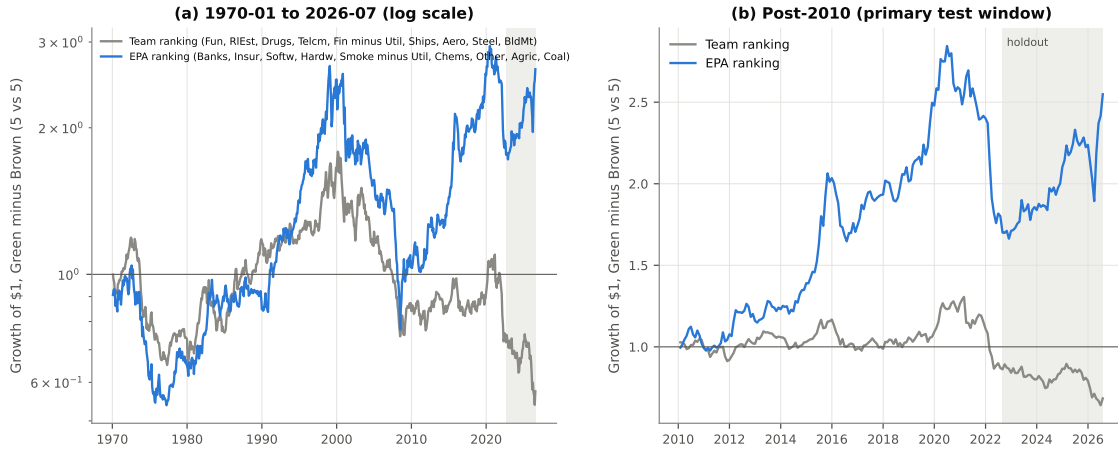


Figure C.8: Cumulative GB under the team and EPA rankings, from 1970 and from 2010. The EPA ranking reverses the sign of the spread after 2010, but its post-2010 FF5+UMD alpha is not significant.

C.6 M5: carbon-aware industry momentum

Industry momentum is the UMD premium in industry form, and a carbon constraint costs no detectable IR on it, with low power. The primary carbon test at $b = 0$ was weak (it cut long-book WACI by 11.6%); at $b = -1$ WACI falls 60% for an IR change of -0.005 (95% interval -0.110 to $+0.096$). The frozen pre-1970 run of the optimizer book is in Appendix E.7.

Table C.26: Industry momentum performance by period, net of 10 bp.

	Months	Gross ret %	Net ret %	Vol %	Net Sharpe	Max DD %	Turnover x/yr	Cost drag %
Full 1970-01 to 2026-07	679	9.77	8.52	18.00	0.47	-60.33	12.47	1.25
Post-2010 (2010-01 to 2026-07)	199	8.05	6.79	15.82	0.43	-29.48	12.69	1.27
Validation 2010-01 to 2022-07	151	9.59	8.35	15.60	0.54	-29.48	12.39	1.24
Holdout 2022-08 to 2026-07	48	3.23	1.86	16.58	0.11	-25.37	13.64	1.36
Pre-COVID 2010-01 to 2019-12	120	7.99	6.74	14.50	0.46	-29.48	12.58	1.26
COVID 2020-01 to 2021-12	24	15.52	14.44	19.92	0.72	-22.55	10.78	1.08
Inflation/rates 2022-01 to 2024-12	36	3.33	1.91	15.60	0.12	-21.27	14.14	1.41
Last 18m 2025-02 to 2026-07	18	5.17	3.90	19.49	0.20	-14.18	12.72	1.27
Last 12m 2025-08 to 2026-07	12	9.79	8.58	23.00	0.37	-14.18	12.11	1.21

Table C.27: Industry momentum alphas by factor model and window. No full-sample alpha of the equal-weight book is significant once UMD is in the model.

Period	Model	Alpha %/yr	t(alpha)	Mkt	SMB	HML	RMW	CMA	UMD	R2
Full 1970-2026	CAPM	9.45	3.92	-0.12						0.01
Full 1970-2026	FF3	10.92	4.60	-0.15	-0.14	-0.32				0.05
Full 1970-2026	FF5	10.25	3.96	-0.11	-0.16	-0.46	-0.10	0.40		0.06
Full 1970-2026	FF5+UMD	1.74	1.09	0.06	-0.11	0.05	-0.15	0.01	0.99	0.67
Post-2010	CAPM	7.44	1.83	-0.05						0.00
Post-2010	FF3	6.87	1.85	-0.02	-0.11	-0.29				0.05
Post-2010	FF5	7.06	2.07	0.02	-0.27	-0.40	-0.50	0.50		0.13
Post-2010	FF5+UMD	1.86	0.71	0.15	0.01	-0.06	-0.15	0.14	0.96	0.60

Table C.28: Carbon exclusion screens: long-leg WACI and Sharpe differences against the primary book.

Book	Sharpe full	Sharpe post-2010	Sharpe holdout	Alpha full %	t	Long WACI (X)	Short WACI (X)	dSharpe full	p boot full	dSharpe post-2010	p boot post-2010
primary	0.47	0.43	0.11	1.74	1.09	0.179	0.195				
A0X_long_covered_only	0.46	0.41	0.17	1.26	0.87	0.184	0.195	-0.01	0.726	-0.02	0.708
A5X_long_excl_top5	0.47	0.36	0.18	1.46	1.03	0.058	0.195	-0.01	0.839	-0.07	0.274
A8X_long_excl_top8	0.46	0.39	0.20	1.58	1.12	0.046	0.195	-0.02	0.679	-0.04	0.562
B0X_both_covered_only	0.47	0.36	0.14	1.58	1.27	0.184	0.200	0.00	0.984	-0.07	0.430
B5X_both_excl_top5	0.48	0.31	0.12	1.86	1.59	0.058	0.061	0.01	0.814	-0.12	0.258
B8X_both_excl_top8	0.48	0.32	0.05	2.10	1.68	0.046	0.048	0.01	0.911	-0.11	0.334
A5M_long_excl_top5	0.46	0.36	0.10	1.46	0.93	0.059	0.195	-0.02	0.291	-0.07	0.027
A8M_long_excl_top8	0.45	0.40	0.10	1.48	0.97	0.047	0.195	-0.02	0.364	-0.03	0.425
B5M_both_excl_top5	0.45	0.31	0.10	1.27	0.85	0.059	0.062	-0.02	0.356	-0.12	0.017
B8M_both_excl_top8	0.45	0.32	0.08	1.35	0.94	0.047	0.050	-0.03	0.481	-0.11	0.108

Table C.29: Grinold–Kahn carbon frontier: IR and alpha change against the unconstrained book, by carbon bound b . dIR is the IR change with its 95% bootstrap interval; convention X drops the 8 industries without an intensity and convention M assigns them the median. Only $b = -2$, partly infeasible, costs significantly.

Conv.	Bound b	IR	dIR	dIR 2.5%	dIR 97.5%	p boot	dAlpha %	t NW	Long WACI chg %
X	+1.00	0.56	-0.00	-0.00	0.00	0.358	-0.00	-1.60	-0.02
X	+0.50	0.56	0.00	-0.00	0.01	0.175	0.02	0.92	-0.07
X	+0.25	0.56	0.01	-0.00	0.02	0.251	0.04	0.93	-1.70
X	+0.00	0.57	0.01	-0.01	0.03	0.328	0.08	1.00	-11.56
X	-0.25	0.57	0.01	-0.03	0.05	0.576	0.14	0.98	-26.31
X	-0.50	0.56	0.01	-0.05	0.07	0.770	0.23	1.00	-40.45
X	-0.75	0.56	0.00	-0.08	0.08	0.919	0.29	0.96	-50.90
X	-1.00	0.55	-0.00	-0.11	0.10	0.930	0.34	0.89	-59.69
X	-1.25	0.53	-0.03	-0.16	0.10	0.707	0.31	0.66	-66.46
X	-1.50	0.48	-0.07	-0.24	0.09	0.375	0.23	0.40	-72.40
X	-2.00	0.27	-0.28	-0.55	-0.01	0.038	-0.49	-0.56	-81.70
M	+1.00	0.49	-0.00	-0.00	0.00	0.763	-0.00	-0.41	0.00
M	+0.50	0.49	0.00	-0.00	0.01	0.395	0.01	0.77	-0.26
M	+0.25	0.49	0.00	-0.01	0.02	0.492	0.03	0.70	-3.67
M	+0.00	0.49	0.00	-0.02	0.02	0.842	0.03	0.41	-14.19
M	-0.25	0.48	-0.01	-0.05	0.02	0.525	-0.03	-0.20	-26.43
M	-0.50	0.48	-0.01	-0.07	0.04	0.654	0.06	0.27	-39.02
M	-0.75	0.48	-0.01	-0.08	0.06	0.833	0.18	0.66	-48.44
M	-1.00	0.47	-0.02	-0.11	0.07	0.727	0.21	0.60	-55.78
M	-1.25	0.46	-0.03	-0.15	0.09	0.652	0.25	0.58	-61.91
M	-1.50	0.44	-0.05	-0.20	0.09	0.528	0.26	0.49	-66.89
M	-2.00	0.34	-0.15	-0.37	0.06	0.168	0.14	0.19	-75.38

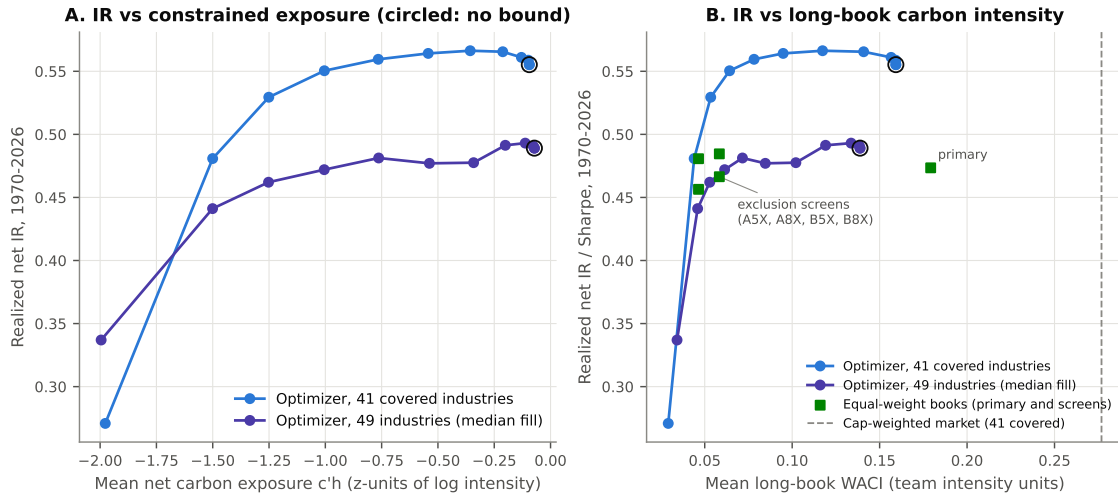


Figure C.9: The Grinold–Kahn carbon frontier of the optimized momentum book: realized IR against long-book carbon intensity as the bound $c'h \leq b$ tightens. The frontier is flat until the bound becomes extreme.

C.7 M6: shrinkage timing

Under its pre-registered cross-validation the ridge model collapses to the historical mean in every month, and attention makes forecasts worse at every fixed penalty.

Table C.30: Out-of-sample predictability of next-month Green-minus-Brown. Campbell-Thompson R_{OOS}^2 (percent) vs the expanding historical mean (or vs the nested no-attention model) and Clark-West t (NW6). Ridge penalty chosen each month by expanding-fold time-series CV. When CV picks the maximum penalty in every month the forecast equals the benchmark and the CW statistic is set to 0.

model	benchmark	period	start	end	n	r2 oos pct	cw t	cw p one sided	share lambda at max	label
A noVIX 1970 ridgecv	histmean	full oos	1990-01	2026-07	439	0.000	0.00	0.500	1.00	primary
A noVIX 1970 ridgecv	histmean	post2010	2010-01	2026-07	199	0.000	0.00	0.500	1.00	primary
A noVIX 1970 ridgecv	histmean	validation	2010-01	2022-07	151	0.000	0.00	0.500	1.00	exploratory
A noVIX 1970 ridgecv	histmean	holdout	2022-08	2026-07	48	0.000	0.00	0.500	1.00	exploratory
A noVIX 1970 ridgecv	histmean	covid	2020-01	2021-12	24	0.000	0.00	0.500	1.00	exploratory
A noVIX 1970 ridgecv	histmean	inflation rates	2022-01	2024-12	36	0.000	0.00	0.500	1.00	exploratory
A noVIX 1970 ridgecv	histmean	last18	2025-02	2026-07	18	0.000	0.00	0.500	1.00	exploratory
A noVIX 1970 ridgecv	histmean	last12	2025-08	2026-07	12	0.000	0.00	0.500	1.00	exploratory
B full 1990 ridgecv	histmean	full oos	2000-01	2026-07	319	0.000	0.00	0.500	1.00	primary
B full 1990 ridgecv	histmean	post2010	2010-01	2026-07	199	0.000	0.00	0.500	1.00	primary
B full 1990 ridgecv	histmean	validation	2010-01	2022-07	151	0.000	0.00	0.500	1.00	exploratory
B full 1990 ridgecv	histmean	holdout	2022-08	2026-07	48	0.000	0.00	0.500	1.00	exploratory
B full 1990 ridgecv	histmean	covid	2020-01	2021-12	24	0.000	0.00	0.500	1.00	exploratory
B full 1990 ridgecv	histmean	inflation rates	2022-01	2024-12	36	0.000	0.00	0.500	1.00	exploratory
B full 1990 ridgecv	histmean	last18	2025-02	2026-07	18	0.000	0.00	0.500	1.00	exploratory
B full 1990 ridgecv	histmean	last12	2025-08	2026-07	12	0.000	0.00	0.500	1.00	exploratory
B full 1990 ridgecv	B0 noATTN 1990 ridgecv	full oos	2000-01	2026-07	319	0.000	0.00	0.500		primary
B full 1990 ridgecv	B0 noATTN 1990 ridgecv	post2010	2010-01	2026-07	199	0.000	0.00	0.500		primary
B full 1990 ridgecv	B0 noATTN 1990 ridgecv	validation	2010-01	2022-07	151	0.000	0.00	0.500		exploratory
B full 1990 ridgecv	B0 noATTN 1990 ridgecv	holdout	2022-08	2026-07	48	0.000	0.00	0.500		exploratory
B full 1990 ridgecv	B0 noATTN 1990 ridgecv	covid	2020-01	2021-12	24	0.000	0.00	0.500		exploratory
B full 1990 ridgecv	B0 noATTN 1990 ridgecv	inflation rates	2022-01	2024-12	36	0.000	0.00	0.500		exploratory
B full 1990 ridgecv	B0 noATTN 1990 ridgecv	last18	2025-02	2026-07	18	0.000	0.00	0.500		exploratory
B full 1990 ridgecv	B0 noATTN 1990 ridgecv	last12	2025-08	2026-07	12	0.000	0.00	0.500		exploratory

Table C.31: Cross-validation design sensitivity. Primary: validate on all expanding 12-month blocks after the first 60 training rows. recent120/recent60: only blocks ending in the last 120/60 training months. first120: first fold trains on 120 rows. grid max1e3: grid without the 1e6 point. Share of monthly refits at the largest grid penalty, median penalty, R_{OOS}^2 (percent) and Clark-West t.

model	benchmark	cv design	period	n	share lambda at max	median lambda	r2 oos pct	cw t	cw p one sided
A noVIX 1970 ridgecv	histmean	expanding all blocks (primary)	full oos	439	1.00	1000000	0.000	0.00	0.500
A noVIX 1970 ridgecv	histmean	expanding all blocks (primary)	post2010	199	1.00	1000000	0.000	0.00	0.500
B full 1990 ridgecv	histmean	expanding all blocks (primary)	full oos	319	1.00	1000000	0.000	0.00	0.500
B full 1990 ridgecv	histmean	expanding all blocks (primary)	post2010	199	1.00	1000000	0.000	0.00	0.500
B full 1990 ridgecv	B0 noATTN 1990 ridgecv	expanding all blocks (primary)	full oos	319			0.000	0.00	0.500
B full 1990 ridgecv	B0 noATTN 1990 ridgecv	expanding all blocks (primary)	post2010	199			0.000	0.00	0.500
A noVIX 1970 ridgecv recent120	histmean	recent120	full oos	439	0.86	1000000	-0.206	-1.30	0.903
A noVIX 1970 ridgecv recent120	histmean	recent120	post2010	199	0.86	1000000	-0.410	-1.11	0.866
A noVIX 1970 ridgecv recent60	histmean	recent60	full oos	439	0.80	1000000	-0.542	-0.63	0.735
A noVIX 1970 ridgecv recent60	histmean	recent60	post2010	199	0.80	1000000	-0.325	-0.18	0.570
A noVIX 1970 ridgecv first120	histmean	first120	full oos	439	1.00	1000000	0.000	0.00	0.500
A noVIX 1970 ridgecv first120	histmean	first120	post2010	199	1.00	1000000	0.000	0.00	0.500
A noVIX 1970 ridgecv grid max1e3	histmean	grid max1e3	full oos	439	1.00	1000	-0.001	-1.06	0.855
A noVIX 1970 ridgecv grid max1e3	histmean	grid max1e3	post2010	199	1.00	1000	0.001	0.47	0.320
B full 1990 ridgecv recent120	histmean	recent120	full oos	319	0.81	1000000	0.070	0.50	0.307
B full 1990 ridgecv recent120	histmean	recent120	post2010	199	0.81	1000000	0.130	0.50	0.307
B full 1990 ridgecv recent60	histmean	recent60	full oos	319	0.52	1000000	-0.343	0.34	0.366
B full 1990 ridgecv recent60	histmean	recent60	post2010	199	0.52	1000000	-0.327	0.10	0.461
B full 1990 ridgecv first120	histmean	first120	full oos	319	0.99	1000000	-0.596	-1.04	0.850
B full 1990 ridgecv first120	histmean	first120	post2010	199	0.99	1000000	0.000	0.00	0.500
B full 1990 ridgecv grid max1e3	histmean	grid max1e3	full oos	319	1.00	1000	0.000	0.15	0.440
B full 1990 ridgecv grid max1e3	histmean	grid max1e3	post2010	199	1.00	1000	0.003	0.69	0.245
B0 noATTN 1990 ridgecv recent120	histmean	recent120	full oos	319	0.80	1000000	0.006	0.41	0.342
B0 noATTN 1990 ridgecv recent120	histmean	recent120	post2010	199	0.80	1000000	0.011	0.41	0.342
B0 noATTN 1990 ridgecv recent60	histmean	recent60	full oos	319	0.52	1000000	-0.389	0.31	0.377
B0 noATTN 1990 ridgecv recent60	histmean	recent60	post2010	199	0.52	1000000	-0.476	0.06	0.477
B0 noATTN 1990 ridgecv first120	histmean	first120	full oos	319	0.84	1000000	-0.498	-1.45	0.926
B0 noATTN 1990 ridgecv first120	histmean	first120	post2010	199	0.84	1000000	-0.138	-1.97	0.975
B0 noATTN 1990 ridgecv grid max1e3	histmean	grid max1e3	full oos	319	1.00	1000	0.001	0.51	0.306
B0 noATTN 1990 ridgecv grid max1e3	histmean	grid max1e3	post2010	199	1.00	1000	0.003	0.75	0.226
B full 1990 ridgecv recent120	B0 noATTN 1990 ridgecv recent120	recent120	full oos	319			0.064	0.73	0.233
B full 1990 ridgecv recent120	B0 noATTN 1990 ridgecv recent120	recent120	post2010	199			0.120	0.73	0.232
B full 1990 ridgecv recent60	B0 noATTN 1990 ridgecv recent60	recent60	full oos	319			0.045	0.38	0.353
B full 1990 ridgecv recent60	B0 noATTN 1990 ridgecv recent60	recent60	post2010	199			0.148	0.73	0.232
B full 1990 ridgecv first120	B0 noATTN 1990 ridgecv first120	first120	full oos	319			-0.098	-0.33	0.631
B full 1990 ridgecv first120	B0 noATTN 1990 ridgecv first120	first120	post2010	199			0.138	1.99	0.024
B full 1990 ridgecv grid max1e3	B0 noATTN 1990 ridgecv grid max1e3	grid max1e3	full oos	319			-0.001	-1.47	0.929
B full 1990 ridgecv grid max1e3	B0 noATTN 1990 ridgecv grid max1e3	grid max1e3	post2010	199			0.000	-0.14	0.554

Table C.32: Attention and the team's own target: next-month FF3 residual of the Brown leg (60-month rolling hedge, alpha and betas lagged one month). Out-of-sample R^2 (percent) and Clark-West t (NW6, one-sided p).

model	benchmark	period	start	end	n	r2 oos pct	cw t	cw p one sided
TEAMTGT ATTN ols	histmean	full oos	2000-01	2026-07	319	-0.738	-1.98	0.976
TEAMTGT ATTN ols	histmean	post2010	2010-01	2026-07	199	-0.253	-0.61	0.730
TEAMTGT ATTN ols	histmean	holdout	2022-08	2026-07	48	-0.778	-0.82	0.793
TEAMTGT B full 1990 ridgecv	histmean	full oos	2000-01	2026-07	319	-0.017	-1.10	0.865
TEAMTGT B full 1990 ridgecv	histmean	post2010	2010-01	2026-07	199	-0.046	-2.00	0.977
TEAMTGT B full 1990 ridgecv	histmean	holdout	2022-08	2026-07	48	0.000	0.00	0.500
TEAMTGT B full 1990 ridgecv	TEAMTGT B0 noATTN 1990 ridgecv	full oos	2000-01	2026-07	319	0.072	1.25	0.105
TEAMTGT B full 1990 ridgecv	TEAMTGT B0 noATTN 1990 ridgecv	post2010	2010-01	2026-07	199	0.066	1.11	0.134
TEAMTGT B full 1990 ridgecv	TEAMTGT B0 noATTN 1990 ridgecv	holdout	2022-08	2026-07	48	0.000	0.00	0.500
TEAMTGT B full 1990 fixed0	TEAMTGT B0 noATTN 1990 fixed0	full oos	2000-01	2026-07	319	-0.565	-0.60	0.726
TEAMTGT B full 1990 fixed0	TEAMTGT B0 noATTN 1990 fixed0	post2010	2010-01	2026-07	199	-0.662	-0.97	0.834
TEAMTGT B full 1990 fixed0	TEAMTGT B0 noATTN 1990 fixed0	holdout	2022-08	2026-07	48	0.238	0.83	0.204
TEAMTGT B full 1990 fixed1	TEAMTGT B0 noATTN 1990 fixed1	full oos	2000-01	2026-07	319	-0.264	-1.82	0.966
TEAMTGT B full 1990 fixed1	TEAMTGT B0 noATTN 1990 fixed1	post2010	2010-01	2026-07	199	-0.179	-1.55	0.939
TEAMTGT B full 1990 fixed1	TEAMTGT B0 noATTN 1990 fixed1	holdout	2022-08	2026-07	48	-0.133	-0.81	0.790

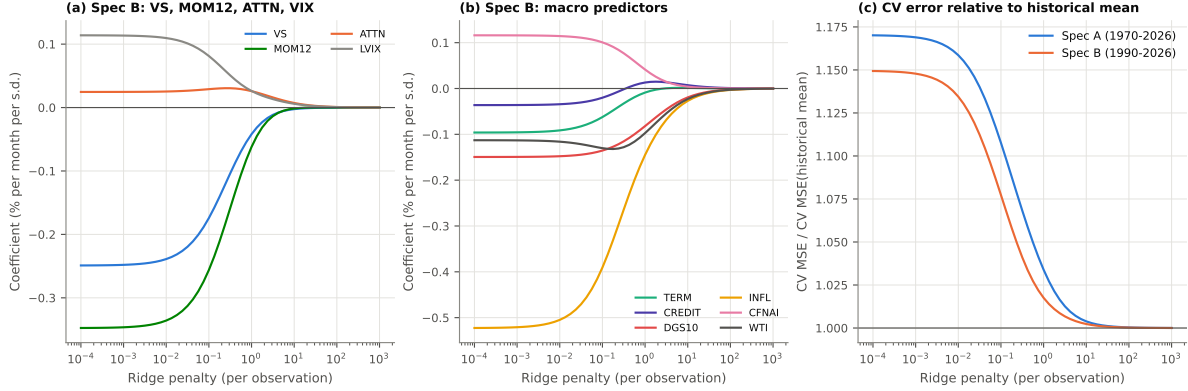


Figure C.10: Cross-validation error relative to the historical mean along the penalty path. It never falls below one, so the largest penalty, which returns the historical mean, always wins.

C.8 M8: the frozen 1994–2009 test

The rule meets one of four pre-registered components and fails. All 128 published numbers were rebuilt from raw files, and the corrections made after verification are output-only.

Timeline. The folder and the pre-registration stamp date from 03:29:54 PDT on 2026-09-26; the file was saved once at 03:30:46 (birth and modify times 4 ms apart) and has not changed (SHA-256 e926c3aeeef48555d3b895243cfceb2007a356654207eb10ed8a630de2f270b51). The code was first written at 03:35:43, a dry run on the seen 2010–2022 window ran at 03:36:35, and the primary run at 03:37:46; between the primary run and 03:41 only output formatting changed, which the verifier confirmed on compiled bytecode. A post-verification edit at 06:05 re-created `run.py` as a new file, so its on-disk birth time now reads 06:05:07; the original birth time is recorded in `corrections/pre_correction_0341/manifest.json`. Every later execution reproduced the first-run key.

Table C.33: Attribution of $D_t = R_t^T - \pi R_t^{AO}$, 1994-03 to 2009-12; NW(6) t, p from t(n-k), k = 15.

term	coef	t nw6	p nw6	t(n-k)	t nw12	coef (team)	t (team)
alpha (annualized)	-0.0018	-0.27		0.790	-0.29	0.0069	1.00
Mkt-RF	0.0517	0.99		0.325	0.99	-0.0432	-1.33
SMB	0.0349	1.52		0.130	1.47	0.0312	1.34
HML	0.1450	2.28		0.024	2.10	0.0176	0.44
RMW	-0.0280	-1.26		0.210	-1.27	-0.0062	-0.27
CMA	0.0251	0.82		0.411	0.86	0.0181	0.49
UMD	-0.0042	-0.37		0.708	-0.42	0.0160	1.23
BOND	0.0944	1.34		0.183	1.31	-0.0110	-0.26
WTI	-0.0010	-0.15		0.879	-0.16	-0.0075	-1.04
dVIX	0.0010	1.57		0.118	1.51	-0.0001	-0.28
dlogEMV	0.0043	1.74		0.084	1.83	0.0058	2.16
I x Mkt-RF	-0.0478	-0.83		0.409	-0.82	0.0469	1.36
I x HML	-0.1427	-2.36		0.019	-2.17	-0.0309	-0.73
I x BOND	-0.0889	-1.20		0.232	-1.18	-0.0452	-0.88
I x dVIX	-0.0008	-1.25		0.213	-1.21	0.0002	0.53

Table C.34: Decomposition of annualized mean(D), 1994-03 to 2009-12.

component	primary (share rule)	secondary (team signal)
alpha	-0.0018	0.0069
FF5+UMD	0.0089	0.0005
BOND (unconditional)	0.0028	-0.0003
WTI	-0.0001	-0.0008
Volatility (dVIX, dlogEMV, I x dVIX)	0.0000	0.0003
Conditional Mkt-RF and HML	-0.0069	0.0008
Conditional BOND	-0.0025	-0.0015
sum = mean(D)	0.0004	0.0059
check: 12 x mean(D)	0.0004	0.0059

Table C.35: Timing alpha with each Brown industry dropped, 1994-03 to 2009-12.

dropped	alpha ann	t nw6	p tnk	alpha (team)	t (team)
Util	-0.0061	-0.87	0.385	0.0004	0.06
Ships	0.0003	0.04	0.967	0.0068	0.88
Aero	0.0001	0.01	0.990	0.0135	2.02
Steel	-0.0019	-0.34	0.737	0.0034	0.51
BldMt	-0.0011	-0.17	0.869	0.0082	1.20

Table C.36: Crossings of the frozen EMV-share rule; EMV overall and VIX percentile ranks within the 1993-2009 data months.

data month	decision	share (%)	z	p80	EMV	EMV pct	VIX	VIX pct	extends	team $\pm 1m$
1995-08	1995-09	2.58	1.22	0.57	13.4	4	12.8	18	no	yes
1996-04	1996-05	1.91	0.75	0.57	15.3	20	16.6	40	no	yes
1996-11	1996-12	1.75	0.63	0.56	17.1	32	16.0	36	no	yes
1997-05	1997-06	1.78	0.71	0.57	17.7	38	19.9	52	yes	yes
1997-09	1997-10	2.34	1.19	0.58	19.1	49	23.8	72	yes	yes
1998-07	1998-08	1.90	1.17	0.57	27.6	83	19.9	53	no	yes
1999-03	1999-04	2.00	1.30	0.57	22.9	68	25.3	78	no	yes
1999-06	1999-07	1.46	0.77	0.57	20.6	57	23.6	70	yes	no
1999-08	1999-09	1.81	1.09	0.57	26.1	81	24.3	75	yes	yes
2000-09	2000-10	1.73	1.08	0.56	18.6	44	19.7	51	no	yes
2000-12	2001-01	2.60	1.75	0.57	34.1	93	26.5	84	yes	yes
2001-03	2001-04	1.51	0.76	0.58	31.1	89	28.5	89	yes	yes
2001-06	2001-07	1.90	1.18	0.59	17.1	31	20.9	59	yes	no
2001-08	2001-09	1.67	0.94	0.60	20.0	53	21.9	63	yes	no
2001-12	2002-01	1.46	0.77	0.60	24.4	73	23.7	71	yes	yes
2002-04	2002-05	2.28	1.41	0.60	20.1	54	19.9	52	yes	yes
2003-06	2003-07	1.85	1.07	0.61	20.1	54	20.4	57	no	no
2003-08	2003-09	1.53	0.78	0.63	17.7	37	19.3	49	yes	no
2004-02	2004-03	2.04	1.22	0.66	12.7	2	16.0	37	yes	no
2004-10	2004-11	1.93	1.21	0.64	15.5	21	15.0	31	no	no
2005-01	2005-02	1.51	0.75	0.66	17.3	34	13.4	23	yes	no
2005-05	2005-06	1.55	0.76	0.68	16.5	27	14.0	27	yes	no
2005-09	2005-10	1.90	1.12	0.69	15.1	18	12.6	18	yes	no
2005-12	2006-01	1.98	1.20	0.70	14.2	9	11.3	3	yes	no
2006-05	2006-06	1.80	1.05	0.70	17.6	37	14.5	29	yes	yes
2007-01	2007-02	2.18	1.50	0.69	14.5	12	11.0	1	no	yes
2007-04	2007-05	2.16	1.40	0.73	18.7	45	12.9	20	yes	yes
2007-09	2007-10	2.37	1.55	0.73	25.3	77	22.2	65	yes	yes
2007-12	2008-01	2.36	1.47	0.74	26.1	80	21.7	61	yes	no
2008-03	2008-04	1.65	0.82	0.75	28.8	85	27.1	85	yes	no
2008-06	2008-07	2.18	1.27	0.75	18.2	43	22.1	64	yes	yes
2009-01	2009-02	1.72	0.76	0.75	31.2	89	44.7	98	no	no
2009-05	2009-06	1.90	0.89	0.76	25.0	76	32.0	92	yes	yes

Table C.37: BOND factor checks (GS10 par bond, duration and convexity).

check	value	reference
calendar-year total return 1994	-0.0746	-0.0804
calendar-year total return 1999	-0.0671	-0.0825
calendar-year total return 2008	0.1940	0.2010
calendar-year total return 2009	-0.0671	-0.1112
calendar-year total return 2022	-0.1504	-0.1783
calendar-year total return 2023	0.0052	0.0388
corr(annual BOND, Damodaran) 1993-2025	0.9870	
mean GS10 1990-1999 (decimal)	0.0666	
modified duration 1990-1999 mean	7.2232	
modified duration 1993-2009 mean	7.7449	
modified duration 2010-2019 mean	8.8396	
max D - closed form $(1/y)(1-(1+y/2)^{-20})$	0.0000	0.0000
max monthly BOND total - exact repricing 1993-2009	0.0005	
corr(monthly BOND total, exact) 1993-2009	1.0000	
BOND excess mean 1993-2009, annualized	0.0329	
BOND excess vol 1993-2009, annualized	0.0643	

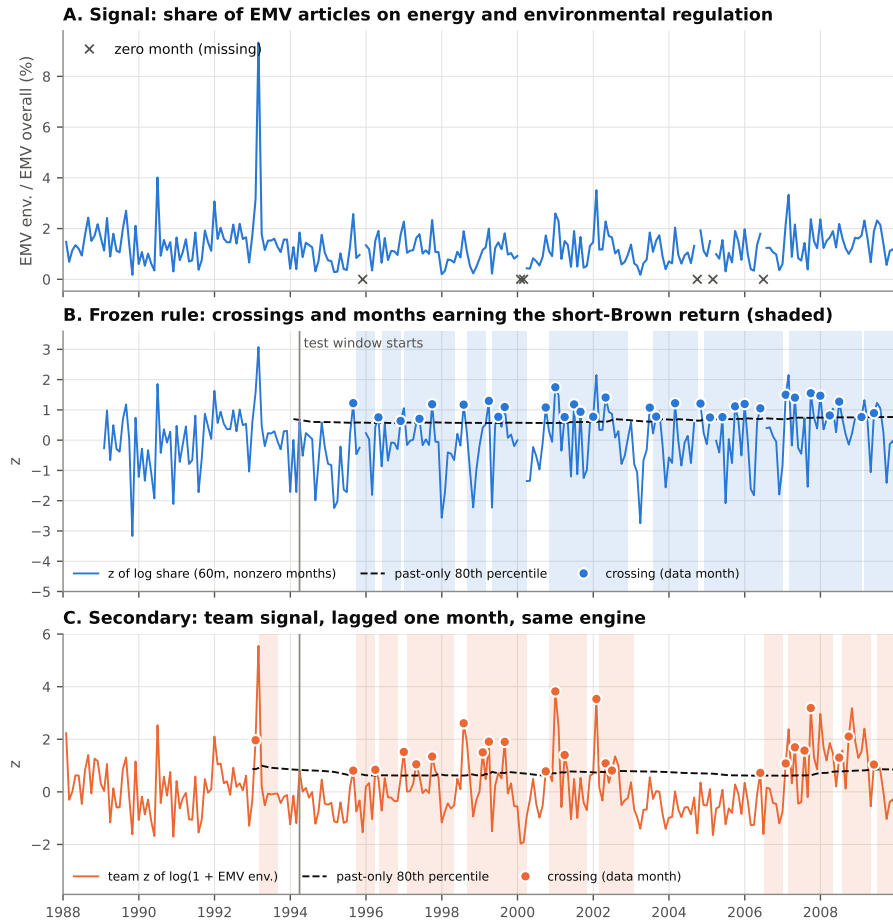


Figure C.11: The frozen rule's signal: the EMV share with zero months marked; its z-score, threshold, crossings and months in position; and our team's signal, lagged, for comparison. The share rule is in position in 142 of 190 months, close to an always-on short.

D Code snippets

The listings are read directly from the files that produced the results (line numbers are the files' own), so the code shown is the code that ran. Every module runs with `cd research && uv run python modules/<id>/run.py`.

D.1 The team trade, ported

The lagged hedge (`lib/team_pipeline.py`): betas estimated through t hedge month $t + 1$.

```

1 def rolling_factor_model(y, x, window=60):
2     """Trailing-window OLS of y on [1, x]. Returns (betas, hedged, epsilon, intercept).
3
4     betas/intercept at t use months t-window+1..t. hedged_t = y_t - beta_{t-1}'x_t and
5     epsilon_t = hedged_t - alpha_{t-1}. Same lstsq calls as the team (loop is numpy, not iloc)."""
6     data = pd.concat([y.rename("y"), x], axis=1, sort=True).dropna()
7     cols = list(x.columns)
8     Y = data["y"].to_numpy()
9     F = data[cols].to_numpy()
10    n, k = len(data), len(cols)
11    B = np.full((n, k), np.nan)
12    A = np.full(n, np.nan)
13    ones = np.ones(window)
14    for end in range(window - 1, n):
15        X = np.column_stack([ones, F[end - window + 1:end + 1]])
16        coef, *_ = np.linalg.lstsq(X, Y[end - window + 1:end + 1], rcond=None)
17        A[end], B[end] = coef[0], coef[1:]
18    betas = pd.DataFrame(B, index=data.index, columns=cols, dtype=float)
19    intercept = pd.Series(A, index=data.index, dtype=float)
20    beta_used, alpha_used = betas.shift(1), intercept.shift(1)
21    hedged = data["y"] - (beta_used * data[cols]).sum(axis=1, min_count=len(cols))
22    epsilon = hedged - alpha_used
23    return betas, hedged, epsilon, intercept

```

Past-only threshold, crossings and holds, and the 5% volatility target with a cap of one.

```

1 def expanding_threshold(series, q=.80, min_history=60):
2     """Past-only expanding quantile (threshold for month t uses values up to t-1)."""
3     return series.shift(1).expanding(min_periods=min_history).quantile(q)

1 def cross_and_holds(state, holds=(3, 6)):
2     """Returns (cross, {h: hold_state}). hold_state_t is True for the crossing month and the next h-1 months.
3     ↪ """
4     state = state.astype(bool)
5     cross = state & ~state.shift(1).fillna(False).astype(bool)
6     return cross, {h: cross.rolling(h, min_periods=1).max().astype(bool) for h in holds}

1 def _magnitude(residual, annual_vol_target=.05, vol_window=36, cap=1.0):
2     sigma = residual.rolling(vol_window, min_periods=vol_window).std(ddof=1)
3     return sigma, (annual_vol_target / (np.sqrt(12) * sigma)).clip(upper=cap)
4
5
6 def state_position(state, residual, direction, annual_vol_target=.05, vol_window=36, cap=1.0):
7     sigma, magnitude = _magnitude(residual, annual_vol_target, vol_window, cap)
8     position = pd.Series(np.nan, index=residual.index)
9     live = position.index >= sigma.first_valid_index()
10    position.loc[live] = (direction * magnitude).loc[live].where(state.reindex(position.index).fillna(False),
11    ↪ 0)
12    return position

```

The P&L engine: a position set at the end of t earns $t + 1$, and the cost of a trade at t is charged at $t + 1$.

```

1 def asset_strategy_returns(position, model, factors, factor_cols=TEAM_FACTOR_COLS, cost_rates=None,
2     ↪ leg_multiplier=1.0):
3     """Team P&L engine. position_t (set at end of t) earns model return and -position_t*beta_t overlay in t+1.
4     Cost of the trade at t is charged at t+1. Returns DataFrame with position, gross_return, cost (charged
5     ↪ that month),
6     cost_incurred (trade at t), net_return, turnover (asset + overlay), asset_turnover, overlay_turnover."""
7     factor_cols = list(factor_cols)
8     rates = cost_rates if cost_rates is not None else resolve_costs(factor_cols)
9     idx = position.index.intersection(model["return"].index).intersection(factors.index).intersection(model["
10     ↪ betas"].index)
11     h = position.reindex(idx)
12     idx = idx[idx >= h.first_valid_index()]
13     h = h.reindex(idx).fillna(0)
14     beta = model["betas"].reindex(idx)
15     f = factors[factor_cols].reindex(idx)
16     overlay = pd.DataFrame(-h.to_numpy()[ :, None] * beta.to_numpy(), index=idx, columns=beta.columns)
17     gross = h.shift(1) * model["return"].reindex(idx) + (overlay.shift(1) * f).sum(axis=1, min_count=len(
18     ↪ factor_cols))
19     asset_turnover = leg_multiplier * h.diff().abs().fillna(h.abs())
20     overlay_turnover = overlay.diff().abs().fillna(overlay.abs())
21     # group factors by rate in order of first appearance; team grouping = [Mkt-RF], [SMB, HML]
22     groups: dict = {}
23     for c in factor_cols:
24         groups.setdefault(rates[c], []).append(c)
25     cost = rates["asset"] * asset_turnover
26     for rate, cols in groups.items():
27         cost = cost + (rate * overlay_turnover[cols[0]] if len(cols) == 1 else rate * overlay_turnover[cols].
28         ↪ sum(axis=1))
29     charged = cost.shift(1).fillna(0)
30     return pd.DataFrame({"position": h, "gross_return": gross, "cost": charged, "net_return": gross - charged,
31         "turnover": asset_turnover + overlay_turnover.sum(axis=1), "cost_incurred": cost,
32         "asset_turnover": asset_turnover, "overlay_turnover": overlay_turnover.sum(axis=1)})

```

D.2 The frozen rule (M8)

The share signal, with zeros treated as missing and the off rule (M8, run.py).

```

1 def share_signal(env, overall):
2     """Frozen rule, data-month time t (C3-C8). Returns a DataFrame; 'cross' is in data-month time."""
3     idx = monthly_grid(env, overall)
4     env, overall = env.reindex(idx), overall.reindex(idx)
5     zero = env.eq(0) # C3
6     missing = zero | env.isna() | overall.isna() | overall.le(0)
7     s = (env / overall).where(~missing)
8     ls = np.log(s)
9     roll = ls.rolling(Z_WIN, min_periods=Z_MIN_NZ) # C5: >= 48 nonzero in trailing 60
10    mu, sd = roll.mean(), roll.std(ddof=1)
11    n_nz = ls.rolling(Z_WIN, min_periods=1).count()
12    zeros60 = zero.astype(float).rolling(Z_WIN, min_periods=1).sum()
13    off = zeros60 > MAX_ZEROS # C6
14    z = ((ls - mu) / sd.replace(0, np.nan)).where(~missing & ~off)
15    thr = expanding_threshold(z, TAIL_Q, TAIL_MIN) # C7 (NaN months skipped)
16    valid = z.notna() & thr.notna()
17    extreme = (z > thr) & valid
18    ev = extreme[valid]
19    cr = ev & ~ev.shift(1, fill_value=False) # C8: skip missing months
20    cross = cr.reindex(idx, fill_value=False).astype(bool)
21    return pd.DataFrame({"env": env, "overall": overall, "zero": zero, "share": s, "log_share": ls, "
22    ↪ n_nonzero60": n_nz,
23        "zeros60": zeros60, "off": off, "z": z, "threshold": thr, "valid": valid, "extreme":
24        ↪ extreme,
25        "cross": cross})

```

The publication lag and the hold.

```

1 def decision_hold(cross_data, grid):
2     """C4 + C9: shift crossings one month (publication lag), then the team's hold = rolling max of crossings.
        ↪ """
3     c = cross_data.reindex(grid, fill_value=False).astype(bool)
4     cross_dec = c.shift(1, fill_value=False).astype(bool)
5     hold = cross_dec.rolling(HOLD, min_periods=1).max().astype(bool)
6     return cross_dec, hold

```

The attribution regression of D_t with $k = 15$ and $t(n - k)$ p-values.

```

1 def attribution(net_T, net_A0, pos_T, pos_A0, fac, window):
2     """D_t = R^T - pi R^A0; regression on UNCOND + I_{t-1} x COND; NW(6) and NW(12); t(n-k) p-values;
        ↪ decomposition."""
3     w = pd.date_range(window[0], window[1], freq="ME")
4     hT = pos_T.shift(1).reindex(w)
5     hA = pos_A0.shift(1).reindex(w)
6     pi = hT.abs().mean() / hA.abs().mean() # C13
7     D = (net_T.reindex(w) - pi * net_A0.reindex(w)).rename("D")
8     I = hT.ne(0).astype(float) # C19
9     X = fac.reindex(w)[UNCOND].copy()
10    for c in COND:
11        X[f"I x {c}"] = I * X[c]
12    data = pd.concat([D, X], axis=1).dropna()
13    n, k = len(data), X.shape[1] + 1
14    fit6 = newey_west_regression(data["D"], data[X.columns], lags=NW_MAIN)
15    fit12 = newey_west_regression(data["D"], data[X.columns], lags=NW_CHECK)
16    df_ = n - k
17    tab = pd.DataFrame({"coef": fit6["coef"], "t_nw6": fit6["t_hac6"], "t_nw12": fit12["t_hac6"]})
18    tab["p_nw6_t(n-k)"] = 2 * stats.t.sf(tab["t_nw6"].abs(), df_)
19    tab["p_nw12_t(n-k)"] = 2 * stats.t.sf(tab["t_nw12"].abs(), df_)
20    means = data[X.columns].mean()
21    contrib = pd.Series({"alpha": fit6.loc["const", "coef"], **{c: fit6.loc[c, "coef"] * means[c] for c in X.
        ↪ columns}})
22    resid_mean = data["D"].mean() - contrib.sum()
23    a, t6 = fit6.loc["const", "coef"], fit6.loc["const", "t_hac6"]
24    return {"pi": pi, "D": D, "I": I, "X": X, "data": data, "n": n, "k": k, "df": df_, "table": tab,
25            "alpha_m": a, "alpha_ann": 12 * a, "t6": t6, "t12": fit12.loc["const", "t_hac6"],
26            "p6": 2 * stats.t.sf(abs(t6), df_), "p6_upper": stats.t.sf(t6, df_),
27            "p12": 2 * stats.t.sf(abs(fit12.loc["const", "t_hac6"]), df_),
28            "mean_D": data["D"].mean(), "contrib": contrib, "resid_mean": resid_mean, "fac_means": means}

```

The calendar shuffle: the same blocks, in random order, at random non-touching dates.

```

1 def shuffle_test(hold, eng: FastEngine, net_A0, pos_A0, fac, window, reps=N_SHUFFLE, seed=SEED):
2     """C24: move the same blocks (runs of hold inside decision months W0-1..E-1) to random non-touching dates.
        ↪ """
3     idx = eng.idx
4     w = pd.date_range(window[0], window[1], freq="ME")
5     d_lo = idx.get_loc(w[0]) - 1
6     d_hi = idx.get_loc(w[-1]) - 1
7     hold_arr = hold.reindex(idx, fill_value=False).to_numpy().astype(bool)
8     seg = hold_arr[d_lo:d_hi + 1]
9     blocks = runs(seg)
10    L = np.array([e - s + 1 for s, e in blocks])
11    N, m = len(seg), len(L)
12    free = N - L.sum() - (m - 1)
13    # fixed pieces of the regression on the window
14    wpos = np.array([idx.get_loc(t) for t in w])
15    hA = np.abs(pos_A0.reindex(idx).to_numpy()[wpos - 1])
16    nA = net_A0.reindex(idx).to_numpy()[wpos]
17    Xu = fac.reindex(w)[UNCOND].to_numpy()
18    Xc = fac.reindex(w)[COND].to_numpy()
19
20    def alpha_of(hold_bool):
21        h = eng.position(hold_bool)
22        nT = eng.net(h)[wpos]

```

```

23     hT = h[wpos - 1]
24     pi = np.abs(hT).mean() / hA.mean()
25     D = nT - pi * nA
26     I = (hT != 0).astype(float)
27     X = np.column_stack([Xu, I[:, None] * Xc])
28     ok = ~np.isnan(D) & ~np.isnan(X).any(1)
29     coef, se = nw_numpy(D[ok], X[ok], NW_MAIN)
30     return coef[0], coef[0] / se[0]
31
32     real_alpha, real_t = alpha_of(hold_arr)
33     if m == 0 or free < 0:
34         return {"blocks": L, "real_alpha": real_alpha, "draws": np.array([]), "p_upper": np.nan, "p_two": np.
            ↪ nan}
35     rng = np.random.default_rng(seed)
36     draws, tdraws = np.empty(reps), np.empty(reps)
37     for b in range(reps):
38         order = rng.permutation(m)
39         pos = np.sort(rng.choice(free + m, size=m, replace=False))
40         Ls = L[order]
41         starts = pos + np.r_[0, np.cumsum(Ls)[: -1]]
42         hb = hold_arr.copy()
43         hb[d_lo:d_hi + 1] = False
44         for s_, l_ in zip(starts, Ls):
45             hb[d_lo + s_: d_lo + s_ + l_] = True
46         draws[b], tdraws[b] = alpha_of(hb)
47     p_up = (1 + (draws >= real_alpha).sum()) / (reps + 1)
48     p_lo = (1 + (draws <= real_alpha).sum()) / (reps + 1)
49     return {"blocks": L, "N": N, "m": m, "free": free, "real_alpha": real_alpha, "real_t": real_t, "draws":
            ↪ draws,
50         "tdraws": tdraws, "p_upper": p_up, "p_lower": p_lo, "p_two": min(1.0, 2 * min(p_up, p_lo))}

```

The BOND factor (also used by M3): exact duration and convexity of a 10-year semiannual par bond.

```

1  def par_bond_dc(y, maturity=10, m=2):
2      """Price (per 100), modified duration (years) and convexity (years^2) of a par bond, m coupons a year,
            ↪ yield y."""
3      y = np.atleast_1d(np.asarray(y, float))
4      i = np.arange(1, maturity * m + 1)
5      cf = np.repeat((100 * y / m)[: , None], len(i), axis=1)
6      cf[:, -1] += 100
7      disc = (1 + y[:, None] / m) ** (-i)
8      P = (cf * disc).sum(1)
9      D = (cf * (i / m) * disc).sum(1) / P / (1 + y / m)
10     C = (cf * (i / m) * ((i + 1) / m) * disc).sum(1) / P / (1 + y / m) ** 2
11     return P, D, C

```

```

1  def build_bond(gs10, rf):
2      y = (gs10 / 100).rename("y10")
3      y0 = y.shift(1)
4      dy = y - y0
5      ok = y0.notna() & y.notna()
6      D = pd.Series(np.nan, y.index)
7      C = pd.Series(np.nan, y.index)
8      _, d_, c_ = par_bond_dc(y0[ok].to_numpy())
9      D[ok], C[ok] = d_, c_
10     r = y0 / 12 - D * dy + 0.5 * C * dy ** 2
11     ex = pd.Series(np.nan, y.index)
12     ex[ok] = exact_par_return(y0[ok].to_numpy(), y[ok].to_numpy())
13     out = pd.DataFrame({"y10": y, "dy": dy, "D_mod": D, "convexity": C, "ret_total": r, "ret_exact": ex})
14     out["RF"] = rf.reindex(out.index)
15     out["BOND"] = out["ret_total"] - out["RF"]
16     return out

```

D.3 Alpha versus beta (M3)

The Lewellen and Nagel (2006) identity: mean return equals conditional alpha plus static beta plus the beta-timing covariance (M3, m3lib.py).

```

1 def ln_decomposition(r: pd.Series, betas: pd.DataFrame, F: pd.DataFrame, start, end) -> dict:
2     """Exact identity over [start, end] with 1/n moments:
3     mean(r) = mean(alpha_c) + sum_k mean(beta_k) mean(f_k) + sum_k cov(beta_k, f_k), alpha_c,t = r_t - beta_t'
4         ↪ f_t.
5     Returns annualized (x12) components and per-factor timing covariances."""
6     cols = list(F.columns)
7     d = pd.concat([r.rename("r"), betas[cols].add_prefix("b_"), F], axis=1, sort=True).loc[start:end].dropna()
8     if len(d) < 3:
9         return {"n": len(d)}
10    B = d[[f"b_{c}" for c in cols]].to_numpy(); Fm = d[cols].to_numpy()
11    alpha_c = d["r"].to_numpy() - (B * Fm).sum(1)
12    mb, mf = B.mean(0), Fm.mean(0)
13    cov = ((B - mb) * (Fm - mf)).mean(0)
14    out = {"n": len(d), "mean_r": 12 * d["r"].mean(), "cond_alpha": 12 * alpha_c.mean(),
15          "static_beta": 12 * float(mb @ mf), "timing_cov": 12 * float(cov.sum())}
16    for j, c in enumerate(cols):
17        out[f"cov_{fname(c)}"] = 12 * cov[j]
18        out[f"meanbeta_{fname(c)}"] = mb[j]
19        out[f"static_{fname(c)}"] = 12 * mb[j] * mf[j]
20    out["identity_gap"] = out["mean_r"] - out["cond_alpha"] - out["static_beta"] - out["timing_cov"]
21    return out

```

D.4 Shrinkage timing (M6)

Walk-forward ridge with expanding-fold cross-validation, ties to the larger penalty, and the Campbell and Thompson (2008) and Clark and West (2007) statistics (M6, m6lib.py).

```

1 def walk_forward_ridge(X: pd.DataFrame, y: pd.Series, oos_start: str, grid=LAMBDA_GRID, fixed_lambda=None,
2                       first=CV_FIRST, block=CV_BLOCK, last_target=LAST_MONTH, recent=None):
3     """Monthly-refit expanding-window ridge. X rows = origins t (predictors known at t), y[t] = return in t+1.
4     At origin t: training rows are origins s < t (targets realized by t). Penalty chosen by expanding-fold
5     time-series CV inside the training rows (12-month validation blocks after the first 60 rows; each block is
6     predicted by a model fit, and standardized, on rows strictly before the block; pooled validation MSE; ties
7     go to the larger penalty). recent=m (CV-design sensitivity only): keep only the validation blocks that end
8     inside the last m training rows, i.e. validate on the most recent m months instead of all blocks.
9     Returns DataFrame indexed by TARGET month."""
10    d = pd.concat([X, y.rename("_y")], axis=1)
11    d = d.loc[d[X.columns].notna().all(axis=1)]
12    Xv, yv, rows = d[X.columns].to_numpy(float), d["_y"].to_numpy(float), d.index
13    grid = np.asarray(grid if fixed_lambda is None else [fixed_lambda], float)
14    cache, out = {}, []
15    oos_start = pd.Timestamp(oos_start)
16    for k, t in enumerate(rows):
17        tgt = t + ME1
18        if tgt < oos_start or tgt > pd.Timestamp(last_target):
19            continue
20        Xtr, ytr = Xv[:k], yv[:k]
21        if np.isnan(ytr).any():
22            raise ValueError(f"missing target inside training window at origin {t}")
23        if fixed_lambda is None and len(grid) > 1:
24            sse = np.zeros(len(grid))
25            for v in range(first, k, block):
26                e = min(v + block, k)
27                if recent is not None and e <= k - recent:
28                    continue
29                key = (v, e)
30                if key not in cache:
31                    mu, sd, a, B = _ridge_all(Xtr[:v], ytr[:v], grid)
32                    pred = a + ((Xtr[v:e] - mu) / sd) @ B

```

```

33         cache[key] = ((ytr[v:e, None] - pred) ** 2).sum(0)
34         sse += cache[key]
35         j = np.flatnonzero(sse <= sse.min() * (1 + 1e-12)).max()
36         lam = grid[j]
37     else:
38         lam = grid[0]
39     mu, sd, a, B = _ridge_all(Xtr, ytr, np.array([lam]))
40     b = B[:, 0]
41     f = a + ((Xv[k] - mu) / sd) @ b
42     rec = {"date": tgt, "origin": t, "y": yv[k], "forecast": f, "hist_mean": ytr.mean(), "lambda": lam,
43           "lambda_at_max": bool(lam >= grid.max()) and len(grid) > 1, "n_train": k}
44     rec.update({f"b_{c}": bb for c, bb in zip(X.columns, b)})
45     out.append(rec)
46     return pd.DataFrame(out).set_index("date")

```

```

1  def oos_stats(y, f_model, f_bench, lags=6):
2      """Campbell-Thompson OOS R2 of f_model vs f_bench and the Clark-West (2007) adjusted-MSPE test
3      (H0: equal MSPE of nested models; one-sided H1: model better). NW(lags) t-stat on the CW series."""
4      d = pd.concat([y.rename("y"), f_model.rename("m"), f_bench.rename("b")], axis=1).dropna()
5      em, eb = d["y"] - d["m"], d["y"] - d["b"]
6      r2 = 1 - (em ** 2).sum() / (eb ** 2).sum()
7      fcw = eb ** 2 - (em ** 2 - (d["b"] - d["m"]) ** 2)
8      if fcw.std() > 0 and len(fcw) > 3:
9          t = float(nw_ols(fcw, lags=min(lags, max(len(fcw) // 4, 1))).tvalues.iloc[0])
10     else:
11         t = np.nan
12     p1 = 1 - stats.norm.cdf(t) if np.isfinite(t) else np.nan
13     p2 = 2 * (1 - stats.norm.cdf(abs(t))) if np.isfinite(t) else np.nan
14     return {"n": len(d), "start": d.index.min(), "end": d.index.max(), "r2_oos": r2, "cw_t": t,
15           "cw_p_one_sided": p1, "cw_p_two_sided": p2,
16           "mse_model_x1e4": 1e4 * (em ** 2).mean(), "mse_bench_x1e4": 1e4 * (eb ** 2).mean(),
17           "corr_forecast_realized": d["m"].corr(d["y"]),
18           "dir_accuracy": float((np.sign(d["m"] - d["b"]) == np.sign(d["y"] - d["b"])).mean())}

```

D.5 The carbon frontier (M5)

The Grinold–Kahn problem: alpha minus cost, a tracking-error limit, dollar neutrality, a box and the carbon bound (M5, optimizer.py).

```

1  class GKProblem:
2      def __init__(self, n: int, kappa: float):
3          self.h = cp.Variable(n)
4          self.a, self.hp, self.cz = cp.Parameter(n), cp.Parameter(n), cp.Parameter(n)
5          self.u = cp.Parameter(n, nonneg=True); self.Lt = cp.Parameter((n, n)); self.b = cp.Parameter()
6          te2 = TE_ANN ** 2 / 12
7          base = [cp.sum_squares(self.Lt @ self.h) <= te2, cp.sum(self.h) == 0, self.h <= self.u, self.h >= -self.
8                  ↪ u]
9          self.main = cp.Problem(cp.Maximize(self.a @ self.h - kappa * cp.norm1(self.h - self.hp)),
10                                base + [self.cz @ self.h <= self.b])
11          self.mincarb = cp.Problem(cp.Minimize(self.cz @ self.h), base)
12          assert self.main.is_dcp(dpp=True) and self.mincarb.is_dcp(dpp=True)

```

D.6 Look-ahead discipline

The fuzz test kept in the unit-test suite (lib/test_team_pipeline.py).

```

1  def test_no_lookahead():
2      """Randomize every input after D; nothing in the monthly signals or strategy series up to D may change."""
3      T = load_team()
4      D = "2016-12-31"
5      rng = np.random.default_rng(0)
6      ind, ff3, mac = T["industries"].copy(), T["ff3"].copy(), T["macro"].copy()

```

```

7   m = ind.index > D; ind.loc[m] = rng.normal(0, .08, size=ind.loc[m].shape)
8   m = ff3.index > D; ff3.loc[m, ["Mkt-RF", "SMB", "HML"]] = rng.normal(0, .04, size=(m.sum(), 3))
9   m = mac.index > D
10  for c in ["attention", "rate10y", "wti", "cpi", "activity", "supply_chain"]:
11      mac.loc[m, c] = np.abs(rng.normal(mac[c].mean(), mac[c].std(), size=m.sum()))
12  r2, base = run_pipeline(industries=ind, factors=ff3, macro=mac, **CHEAP), cheap_run()
13  for n, df in base["strategies"].items():
14      for c in df.columns:
15          assert float((df[c].loc[:D] - r2["strategies"][n][c].loc[:D]).abs().max()) == 0.0, (n, c)
16  for k in ["raw", "pure", "w_raw", "w_pure", "threshold_raw", "threshold_pure"]:
17      assert float((base["signals"][k].loc[:D] - r2["signals"][k].loc[:D]).abs().max()) == 0.0, k

```

D.7 How the work was done and verified

The research ran as a set of modules, each built by an AI coding agent (Anthropic’s Claude) under my direction and then verified in two rounds by a second agent that re-implemented the module’s headline numbers from the raw files without importing the module’s own code; the shared team library and, for comparison only, M8’s BOND builder were the exceptions. Each verification record (modules/<id>/VERIFY.md) lists every check, the mismatches found and how each was resolved, and every fix adopted is reflected in this report. The ChatGPT exchanges followed the same division of labor: the prompts were written and the replies fact-checked with code, and the replies themselves came from a language model in the ChatGPT role (Appendix A).

D.8 Reproducibility

Every module is deterministic: seeds are 230 for the bootstraps and the M8 shuffle and 20260926 for M5 and for M7’s bootstraps, and each verifier confirmed that a rerun reproduces the module’s tables. Approximate run times: M1 20 to 60 seconds, M1b one minute, M2 2 to 4 minutes, M3 2.5 to 4 minutes, M4 one minute, M5 4 to 10.5 minutes, M6 one minute, M7 41 seconds. The pre-registration hashes are recorded in outputs/tables/M8_preregistration_hash.csv and outputs/tables/M7_preregistration_hash.csv. Each exhibit’s file name starts with the module that makes it; the report tables are built by report/make_tables.py and Appendix A by report/build_transcripts.py.

E Tests ledger and multiple testing

E.1 Ledger schema

Every module logs each inferential statistic once, in `outputs/tables/<id>_tests_ledger.csv`, with nine shared columns: `test_id`, `module`, `question`, `statistic_name`, `statistic`, `p_value_two_sided`, `n_obs`, `primary_or_exploratory` and `note`; M6 adds a one-sided p-value and the alternative for its Clark–West rows, and M7 adds both plus its family label. A hypothesis reported in two tables is logged once. M7 stacks the eight module ledgers into `M7_all_tests.csv` (24,404 rows with its own), with no exact duplicates.

E.2 Counts by ledger

Table E.1: Rows in each tests ledger, by label. Only primary rows form pre-specified families; M7’s one primary row is the frozen 1931–1969 test.

Ledger	Rows	Primary	Robustness	Exploratory	Other
M1	667	15	217	435	
M1b	1,900	40	1,102	588	170 (88 placebo, 82 reference)
M2	13,482	46	5,562	7,874	
M3	6,360	33	737	5,590	
M4	802	1	375	426	
M5	416	4	0	412	
M6	274	8	120	146	
M8	22	8	0	14	
Eight module ledgers	23,923	155	8,113	15,485	170
M7 (project-wide checks)	481	1	387	7	86 (86 descriptive)
All ledgers	24,404	156	8,500	15,492	256

E.3 Primary families within each module

Table E.2: Primary families. Each row is a pre-specified family with its correction and outcome. “Certified” in this report means a Holm survivor within a primary family, and nothing else.

Module	Family	Correction	Outcome
M1	Q2 HAC Wald of the team z on VIX and overall EMV; Q3 zero-rate Fisher test; Q4 crossings and high VIX; Q5 two correlations; Q6a two contemporaneous slopes; Q6b eight predictive ICs	Holm within Q5, Q6a and Q6b	$R^2 = 0.169$, $p = 8.8 \times 10^{-12}$; zeros $p = 1.3 \times 10^{-32}$; Q4 $p = 0.097$ (borderline); Q5 Holm 0.996 and 0.074; Q6a 0.62 and 0.43; Q6b smallest 0.62
M1b	40 tests: MCCC and CPU alphas (24) and ICs (16)	Holm and BH	Smallest Holm $p = 1.00$; no significant positive holdout alpha (largest $t = 0.37$)
M2	Q1, Q2, Q4: 14 alphas each; Q3: four HML loadings	Holm within each	0 positive survivors (smallest Holm 0.397); Q3 all four survive (largest 0.017)
M3	(i) two BOND loadings; (ii) six holdout alphas per baseline; (iii) seven beta-timing terms	Holm within each	(i) 0.019 and 0.549; (ii) smallest 0.155; (iii) 4 of 6 strategies survive, all negative
M4	One test: EPA 5-vs-5 post-2010 FF5+UMD alpha	none needed	$p = 0.129$
M5	Four tests: two alphas, the carbon IR change at $b = 0$, GB’s momentum loading	Holm	0.823, 0.823, 0.823, 0.211
M6	Six Clark–West tests (Q1 and Q2) plus two timing-minus-static tests	Holm over the six	All 1.00; timing minus static $p = 0.24$ and 0.65
M8	Four pass-bar components, all required	conjunction	1 of 4 met: FAIL

E.4 Project-wide correction (M7)

M7 implements the checks adopted from exchange 03 (`exchange/03_robustness_design/`; code `modules/M7_robustness_ledger/run.py`, about 41 seconds end to end, verified independently). Each of the 23,923 module tests belongs to one family, defined by the decision it could change (Table E.3): F, the frozen and hash-verified tests; P, the module primary alpha tests, corrected one-sided with Holm and BH; D, other primaries; X, placebo and reference rows; L, loadings and beta-timing terms, reported with confidence intervals only; R, the robustness alpha grids, corrected with BH and BY and read as distributions; and E, exploratory rows. Collapsing the 8 exact duplicates in P (Original and Pure holdout variants that coincide once the CPI gap is corrected) leaves 73 tests. The best one-sided p is 0.030 (M2, Pure 6m post-2010, $t = 1.89$); it would stay significant only if the effective number of tests were below 1.69. Uncorrected families, as nominal counts: D 15 of 73 rows with $p < 0.05$, X 33 of 170, L 1,452 of 4,850 and E 1,636 of 9,411. For reference, all 155 labelled primaries as one family have 4 Holm survivors, none of them an alpha: M1’s Fisher test on zero months, M1’s joint Wald test and two M2 HML loadings of the spread.

Table E.3: Project test census: module ledger rows by family and module. Loadings (L) and exploratory rows (E) are 60% of the census and enter no correction.

Family	Treatment	M1	M1b	M2	M3	M4	M5	M6	M8	Total
F	Holm, one-sided	0	0	0	0	0	0	0	1	1
P	Holm and BH, one-sided	0	24	42	12	1	2	0	0	81
D	raw p, no correction	15	16	4	21	0	2	8	7	73
X	calibration only	0	170	0	0	0	0	0	0	170
L	confidence intervals only	0	0	2231	2527	0	56	36	0	4850
R	BH and BY, one-sided, as distributions	0	736	8161	156	282	0	2	0	9337
E	no correction	652	954	3044	3644	519	356	228	14	9411
Total		667	1900	13482	6360	802	416	274	22	23923

23,923 rows from eight ledgers; no exact duplicates. F frozen and hash-verified; P module primary alpha tests; D other primaries; X placebo and reference; L loadings and beta-timing terms; R robustness alpha grids; E exploratory. Rows with two-sided p below 0.05: 4,184, of which 1,624 alpha (783 positive, 841 negative), 1,325 loading, 561 mean, 536 other and 138 beta-timing.

Table E.4: Families and survivors across the project. No module primary alpha survives Holm or BH; the one survivor in family F is the frozen 1931–1969 run, and the one BH survivor in family R is a CPU variant of the COVID rule that fails BY and does not beat the always-short position.

Family	Tests	Method	Smallest adjusted p	Survivors at 5%
F (frozen, hash-verified)	2	Holm, one-sided	0.015	1
P (module primary alphas)	73	Holm, one-sided	1.000	0
P (module primary alphas)	73	BH, one-sided	0.718	0
R-M2	8161	BH / BY, one-sided	0.475 / 1.000	0 / 0
R-other M1b	736	BH / BY, one-sided	0.049 / 0.349	1 / 0
R-other M3	156	BH / BY, one-sided	0.804 / 1.000	0 / 0
R-other M4	282	BH / BY, one-sided	0.125 / 0.777	0 / 0
R-other M6	2	BH / BY, one-sided	0.929 / 1.000	0 / 0
R-other pooled	1176	BH / BY, one-sided	0.078 / 0.594	0 / 0
R pooled	9337	BH / BY, one-sided	0.466 / 1.000	0 / 0
S-full (search)	58	Romano-Wolf; SPA	0.009; 0.009	15
S-post (search)	89	Romano-Wolf; SPA	0.150; 0.150	0

P: 81 labelled primary alpha rows, 8 exact duplicates collapsed; best one-sided p 0.030 (M2). R: robustness alpha grids, positive side only. Search families: FF5+UMD alphas of every saved or regenerated strategy series (58 over 1970-2026, 89 over 2010-2026). D (73 other primaries), X (170 placebo and reference rows), L (4,850 loadings) and E (9,411 exploratory rows) are not corrected.

Table E.5: Cristhian’s robustness grid (family R-M2) as a distribution: share of positive alpha t -statistics and median t by evaluation window. Gains sit in validation and its sub-windows; every window that starts in 2022 or later loses.

Window	Rows	Positive %	Median t	Strategy rows	Positive % (strat.)	Median t (strat.)
all	8161	54.0	0.13	8008	53.8	0.12
full, 1970	21	100.0	1.45	0		
full, live	1078	75.0	0.52	1078	75.0	0.52
post2010	1078	76.4	0.60	1057	76.6	0.61
validation	1099	88.9	1.14	1078	88.9	1.15
pre-COVID	1071	73.2	0.50	1050	72.9	0.49
covid	1071	70.7	0.44	1050	70.5	0.44
holdout	1078	2.7	-1.32	1057	2.3	-1.34
rates, 2022–24	1071	13.9	-0.73	1050	13.3	-0.74
last 18	297	0.0	-2.35	294	0.0	-2.35
last 12	297	20.2	-1.11	294	20.4	-1.11

Window is the third-from-last field of the test id. Strategy rows exclude the 153 Green-minus-Brown spread rows. One-sided BH and BY at 5%: 0 and 0 positive survivors in R-M2; 0 and 0 in R pooled.

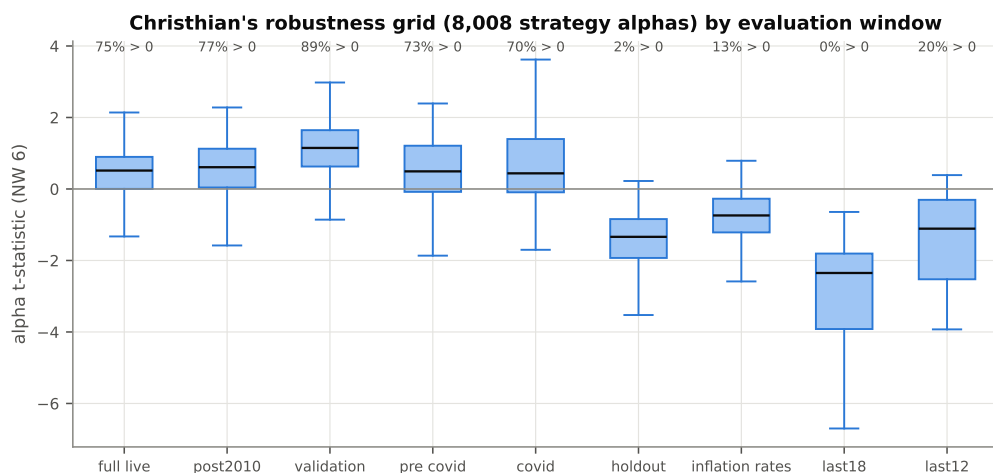


Figure E.1: Alpha t -statistics of Cristhian’s grid by window. The distribution sits to the right of zero in validation and to the left in the holdout and the last 18 months.

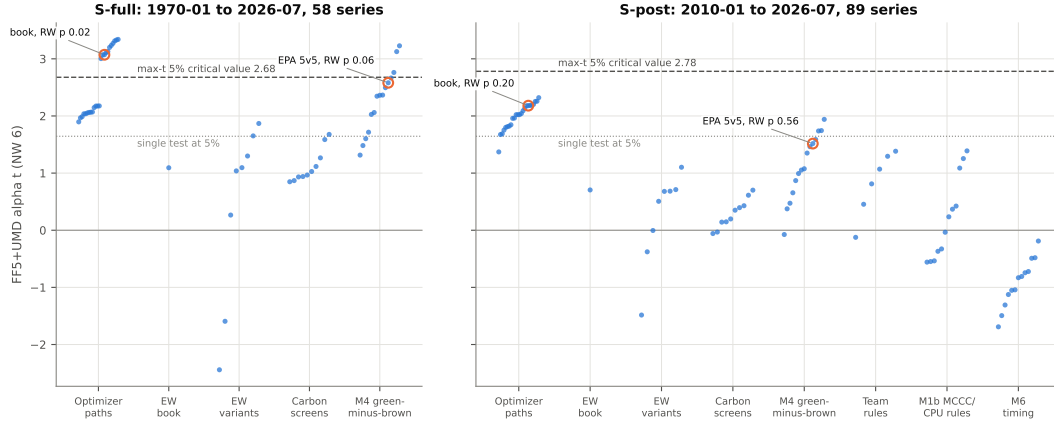
E.5 Search-adjusted tests and the deflated appraisal ratio

Two search families collect every strategy return series that could have been presented as the strategy: S-full, 58 series over 1970–2026 (24 optimizer paths, the equal-weight book and 8 variants, 10 carbon screens and 15 EPA spread variants), and S-post, 89 series over 2010–2026 (S-full plus our team’s six rules, 12 MCCC and CPU rules and 13 shrinkage series). Romano–Wolf stepdown (Romano and Wolf, 2005) and Hansen’s SPA (Hansen, 2005) use a stationary bootstrap with mean block 12 and 5,000 draws. Over 1970–2026, 15 members survive (11 optimizer paths and four EPA variants, not the EPA 5-vs-5 spread itself); over 2010–2026 none does. The deflated appraisal ratio (Bailey and López de Prado, 2014) deflates each candidate’s FF5+UMD appraisal ratio for the effective number of trials (Nyholt’s estimate from residual correlations: 42.4 and 74.8) and must reach 0.95 in both windows. Both candidates fail in both windows and under every sensitivity; the most lenient case, the book over 1970–2026 with the Li–Ji count and the variance floor, gives 0.897. M2’s grid is not in either search family, because its series are not saved, which makes both tests slightly lenient.

Table E.6: Search-adjusted tests of the best FF5+UMD alpha. Over 1970–2026 the best t of 3.34 clears the bootstrap max- t bar of 2.68; over 2010–2026 nothing does.

Family	Members	Months	Best member	Best t	RW adj. p	RW survivors	SPA p	Max- t 95%	Nyholt	Li-Ji	Sidak N
S-full	58	679	optimizer, $b = -1.00$	3.34	0.009	15	0.009	2.68	42.4	14.0	116
S-post	89	199	optimizer, $b = -0.25$	2.32	0.150	0	0.150	2.78	74.8	27.0	5

Stationary bootstrap, mean block 12, $B = 5,000$, seed 20260926; returns demeaned by each member's alpha, factors resampled jointly. Nyholt and Li-Ji effective numbers of tests from FF5+UMD residual correlations. Sidak N: number of independent tests at which the best one-sided p stops being significant at 5%. M2's grid is not in either family (its series are not saved), which makes both tests slightly lenient.

**Figure E.2:** Member t -statistics of the two search families by source, with the bootstrap max- t critical value. Several optimizer paths clear it over 1970–2026, and no member clears it over 2010–2026.**Table E.7:** Deflated appraisal ratio of the two candidates. The pass bar is a governing DSR of at least 0.95 in both windows; the book would pass only if the project had tried three or fewer independent strategies.

Candidate	Window	AR	t NW	N Nyholt	V x1000	SR0 x sqrt12	DSR iid	DSR NW	DSR (gov.)	Li-Ji	Raw M	V floor	Max N
book (unconstrained)	1970-2026	0.40	3.07	42.4	2.46	0.38	0.559	0.563	0.559	0.776	0.497	0.786	3
EPA 5v5 spread	1970-2026	0.38	2.58	42.4	2.46	0.38	0.488	0.488	0.488	0.709	0.427	0.720	2
book (unconstrained)	2010-2026	0.56	2.18	74.8	5.83	0.64	0.363	0.362	0.362	0.533	0.336	0.433	2
EPA 5v5 spread	2010-2026	0.41	1.52	74.8	5.83	0.64	0.171	0.182	0.171	0.302	0.155	0.222	0

AR: annualized FF5+UMD appraisal ratio. N: Nyholt effective number of trials from the family's residual correlations (S-full, 58 series; S-post, 89). V: cross-trial variance of the monthly AR over $K = \text{round}(N)$ clusters, floored at $1/(T-1)$. SR0: expected maximum AR under the null. Governing DSR: the smaller of the iid DSR and the DSR with z scaled by $t(\text{NW})/t(\text{OLS})$. The last four columns: governing DSR with $N = \text{Li-Ji}$, $N = \text{raw M}$, $V = 1/(T-1)$; and the largest N that still passes at the step-3 V.

E.6 The holdout read against a worthwhile alpha

Each holdout alpha is read against a margin of a quarter of its own residual volatility, a quarter of an appraisal-ratio unit: an alpha whose 90% upper bound lies below the margin rejects a worthwhile alpha, and one whose interval straddles it is inconclusive. The team rules are run as our team ran them (same-month timing, CPI gap uninterpolated), so Original and Pure differ here.

Table E.8: Holdout (2022-08 to 2026-07, 48 months) alphas read against the margin. Every attention rule rejects a worthwhile alpha under both factor models; the momentum books and the EPA spread are inconclusive, because 48 months cannot tell 0% from 3% for them.

Series	Model	Alpha %	SE %	90% CI low	90% CI high	Margin delta %	MDE80 %	Reading
optimizer book (unconstrained)	FF5+UMD	-0.34	2.27	-4.16	3.48	1.35	5.75	inconclusive
EW momentum book	FF5+UMD	-4.48	5.36	-13.51	4.54	2.35	13.58	inconclusive
EPA 5v5 spread	FF5+UMD	3.90	5.58	-5.49	13.29	3.18	14.14	inconclusive
Original 3m	FF5+UMD	-4.41	2.14	-8.00	-0.81	0.86	5.41	rejects a worthwhile alpha
Original 3m	FF3	-4.03	1.84	-7.13	-0.94	0.84	4.67	rejects a worthwhile alpha
Pure 3m	FF5+UMD	-2.09	1.04	-3.84	-0.35	0.62	2.63	rejects a worthwhile alpha
Pure 3m	FF3	-2.21	1.14	-4.13	-0.29	0.60	2.89	rejects a worthwhile alpha
Original 6m	FF5+UMD	-3.18	1.58	-5.85	-0.51	1.10	4.01	rejects a worthwhile alpha
Original 6m	FF3	-2.28	1.38	-4.59	0.03	1.12	3.48	rejects a worthwhile alpha
Pure 6m	FF5+UMD	-3.38	1.66	-6.19	-0.58	1.07	4.22	rejects a worthwhile alpha
Pure 6m	FF3	-2.71	1.40	-5.06	-0.37	1.06	3.53	rejects a worthwhile alpha
Continuous raw	FF5+UMD	-0.92	0.46	-1.69	-0.14	0.28	1.16	rejects a worthwhile alpha
Continuous raw	FF3	-0.76	0.38	-1.41	-0.12	0.28	0.97	rejects a worthwhile alpha
Continuous pure	FF5+UMD	-0.65	0.31	-1.17	-0.13	0.27	0.78	rejects a worthwhile alpha
Continuous pure	FF3	-0.54	0.30	-1.05	-0.03	0.26	0.77	rejects a worthwhile alpha
Ref.: Always-short Brown	FF5+UMD	-2.37	1.82	-5.44	0.70	1.29	4.62	rejects a worthwhile alpha

Alpha: annualized intercept, NW(6) standard error. 90% interval from $t(n-k)$. Margin delta = 0.25 times the annualized residual volatility of the same regression. MDE80: minimum detectable alpha at 80% power. Reading, first match: L > 0 and U < delta, positive but below the margin; L > 0, confirms; U < delta, rejects a worthwhile alpha; otherwise inconclusive.

E.7 The frozen 1931–1969 run of the momentum book

Specification and timeline. The book is the unconstrained Grinold–Kahn momentum book of M5, called exactly as M5 calls it (41 covered industries, IC 0.05, 5% ex-ante tracking error, 10 bp), with formation months 1931-06 to 1969-11 and return months 1931-07 to 1969-12 (462 months). The benchmark is FF3+UMD, because RMW and CMA start in 1963-07. When `run.py` first started, before any pre-1970 strategy return existed, it wrote the SHA-256 hashes of the adopted specification, `optimizer.py` and `m5lib.py` (08:49:39 PDT on 2026-09-26); the specification’s hash was taken from its text before the file was saved, and once it is saved every run re-hashes it and stops if the hash differs. Two development runs before the frozen run used only post-1970 formation months. The frozen run executed once, at 08:53:41 PDT, and every later execution reproduces its record. The pre-registered pass bar was an alpha with $t \geq 2.00$ and positive alphas in both halves; a pass moves the book to paper-trading, never to implementation, because its holdout alpha is negative.

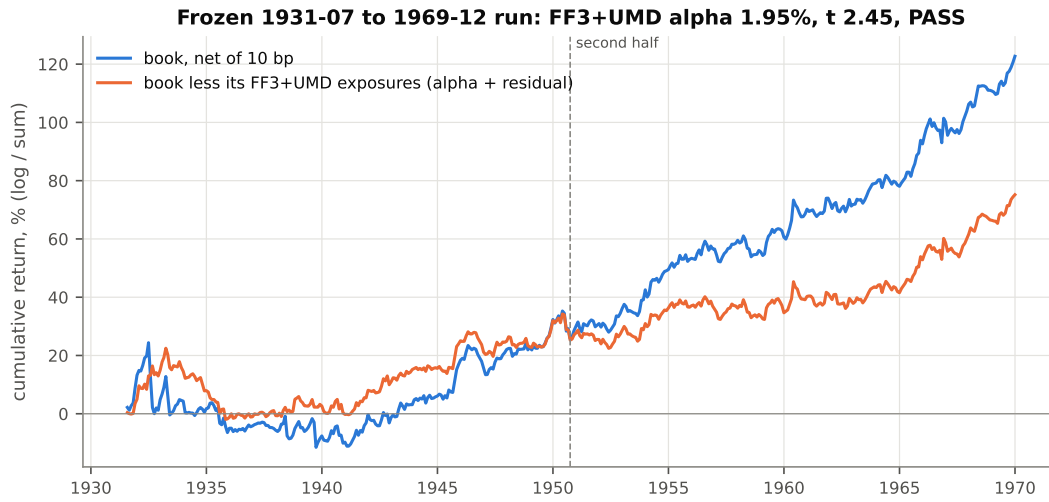
Result and caveats. The run passes: 1.95% a year ($t = 2.45$, one-sided $p = 0.007$; Holm 0.015 with the M8 test in family F), with 1.71% ($t = 1.44$) and 0.90% ($t = 0.83$) in the two halves. The window is unseen for this book, not for the field: UMD from 1927 is public, and industry trend rules have been studied over about a century (Zarattini and Antonacci, 2024). The second-half alpha is about half the first-half alpha, although the difference is far from significant (t about 0.5). After 1970 the alpha did not decay after 2010 (1.99%, $t = 2.31$, over 1970–2009; 3.12%, $t = 2.18$, over 2010–2026) but disappeared in the 2022–2026 holdout. At 25 bp the full-sample alpha stays positive (1.73%, $t = 2.43$). The carbon-constrained version ($b = -1$) differs from the unconstrained book by a net IR of -0.006 (95% interval -0.194 to $+0.228$) before 1970.

Table E.9: Frozen 1931–1969 run of the optimizer book (pre-registered). The primary test (i) passes, with both halves positive.

Statistic	Alpha % / value	t	One-sided p	Months
(i) FF3+UMD alpha, 1931-07 to 1969-12	1.95	2.45	0.007	462
(ii) first half, 1931-07 to 1950-09	1.71	1.44		231
(ii) second half, 1950-10 to 1969-12	0.90	0.83		231
S1 FF5+UMD alpha, 1963-07 to 1969-12	3.26	1.76		78
S4 FF3+UMD alpha at 25 bp	1.55	1.94		462
S5 mean rank IC (t against 0)	0.080	6.78		462

Verdict: PASS. 90% upper bound $U = 3.27\%$ against the margin $\delta = 1.28\%$ (a quarter of the residual volatility of 5.14%).

UMD beta 0.23. S2 ($b = -1$ minus unconstrained) net IR difference -0.006 , 95% interval $[-0.194, 0.228]$. S3: worst month -11.2% (1932-07), worst three months -21.6% , maximum drawdown -30.2% . Hashes of the spec, optimizer.py and m5lib.py were written before the run.

**Figure E.3:** The frozen run: cumulative net return of the book and of the book less its FF3+UMD exposures, 1931 to 1969. The hedged line rises in both halves, faster in the first.**Table E.10:** Cost stress on the optimizer book: FF5+UMD alpha of gross returns less c times turnover, holdings fixed. The full-sample alpha survives 25 bp; the holdout alpha is negative before any cost.

Cost bp	Full alpha %	Post-2010 alpha %	Holdout alpha %	Full t	Post-2010 t	Holdout t
0	2.47	3.42	-0.04	3.50	2.40	-0.02
10	2.17	3.12	-0.34	3.07	2.18	-0.15
25	1.73	2.68	-0.79	2.43	1.86	-0.35
50	1.00	1.95	-1.54	1.39	1.33	-0.67

Break-even cost: 84 bp (full), 117 bp (post-2010); the holdout gross alpha is negative. Annual turnover 2.92 (full). Pass bar: full-sample alpha above zero at 25 bp: PASS.

E.8 The verdict map

The map was fixed in the adopted specification before the frozen run: implement requires a hash-verified confirmatory alpha test, a Romano–Wolf adjusted p of at most 0.05 in both search families, a governing deflated appraisal ratio of at least 0.95 in both windows, and a positive holdout alpha that does not read “rejects”. The frozen run decided only between retiring the optimizer book and paper-trading it.

Table E.11: Final verdict by candidate. No candidate reaches “implement”; the optimizer book moves to paper-trading, and the report’s verdict stays “Do not implement”.

Candidate	Evidence for	Evidence against	Adjusted significance	Deflated Sharpe (AR)	Verdict
Optimizer book (unconstrained)	FF5+UMD alpha 2.17%, t 3.07, 1970-2026; RW adj. p 0.017 (1970-2026), 0.199 (2010-2026); pre-1970 FF3+UMD alpha 1.95%, t 2.45 (PASS)	holdout alpha -0.34% (inconclusive); DSR 0.559 / 0.362; validation-to-holdout decay	RW 0.017 / 0.199	0.559 / 0.362	Paper-trade (report verdict stays Do not implement)
EPA 5v5 spread	FF5+UMD alpha 4.72%, t 1.52, 2010-2026; RW adj. p 0.058 (1970-2026), 0.558 (2010-2026)	holdout alpha 3.90% (inconclusive); DSR 0.488 / 0.171; no confirmatory test; 2022 intensities applied back to 1970	RW 0.058 / 0.558	0.488 / 0.171	Do not implement
Attention thesis (Short-Brown rules)	validation-window alphas (team, M2 grid 89% positive)	M8 frozen FAIL; family P smallest Holm p 1.00; holdout rejects a worthwhile alpha for 6 of 6 rules	Holm 1.00, BH 0.72	not a candidate	Do not implement

Implement requires (a) a hash-verified confirmatory alpha test, (b) Romano–Wolf adjusted p at most 0.05 in both search families, (c) governing DSR at least 0.95 in both windows and (d) a positive holdout alpha whose reading is not ‘rejects’.
 Paper-trade: (a) holds and any of (b) to (d) fails. Do not implement: (a) fails.

E.9 Nominal positives, and why each is not a discovery

- **M5 COVID alpha of industry momentum**, 15.80% (NW $t = 3.18$, BH 0.022 on a normal p -value of 0.0015): 24 months and 7 parameters; the classical OLS t is 1.58 ($p = 0.132$).
- **M6 post-2010 yield-change forecasts** (one-sided $p = 0.018$ and 0.020): exploratory among 34 univariate forecasts; BH $q = 0.34$.
- **M1b paired COVID edge of the EMV rule over the volatility placebos** (t up to 3.22 under real-time timing): smallest Holm $p = 0.051$ within the 12 paired tests. Under team timing it vanishes against overall EMV (Original 3m +1.39%, $t = 0.87$) but stays nominally significant against VIX (Original 3m +4.08%, $t = 2.34$, $p = 0.030$; Pure 3m $t = 2.15$).
- **M7’s one BH survivor in the robustness grids**, a level-transform variant of the CPU Original 3m rule in COVID (6.83%, $t = 4.72$, 24 months): it fails BY, disappears when the grids are pooled, and M1b shows that CPU Original 3m does not beat the always-short position (+1.90%, $t = 1.02$).
- **M4 EPA-minus-team spread**, 8.2% a year ($t = 2.84$): Holm 0.071 over the 16 comparisons shown and 0.178 over 42.
- **M4 full-sample EPA alpha**, 4.5% ($t = 2.58$, 5 of 9 variants survive Holm): it comes from short CMA and UMD loadings on a 2022 sort applied back to 1970, it is absent after 2010, and the 5-vs-5 spread fails the search correction over 1970–2026 (Romano–Wolf $p = 0.058$) although four sister variants pass it.
- **M3 HML convexity**: 45 of the 48 pooled BH survivors among 1,080 timing tests are HML tests, but always-short Brown shows the same convexity (Treyner–Mazuy $t = 3.24$), so it is a property of the hedged Brown leg, not of attention (Treyner and Mazuy, 1966; Henriksson and

Merton, 1981).